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Econometrics I: OLS Setup and Derivation

Model setup

$$(y_i, x_{i1}, \dots, x_{ik}), \quad i = 1, \dots, n.$$

Assume that each observation is generated by

$$y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_k x_{ik} + \varepsilon_i.$$

Equivalently,

$$y_1 = \beta_1 x_{11} + \beta_2 x_{12} + \dots + \beta_k x_{1k} + \varepsilon_1,$$

$$y_2 = \beta_1 x_{21} + \beta_2 x_{22} + \dots + \beta_k x_{2k} + \varepsilon_2,$$

$$\vdots$$

$$y_n = \beta_1 x_{n1} + \beta_2 x_{n2} + \dots + \beta_k x_{nk} + \varepsilon_n.$$

Matrix form:

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1k} \\ x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nk} \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix},$$

where the dimensions are

$$\underbrace{\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}}_{n \times 1} = \underbrace{\begin{pmatrix} x_{11} & \cdots & x_{1k} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nk} \end{pmatrix}}_{n \times k} \underbrace{\begin{pmatrix} \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}}_{k \times 1} + \underbrace{\begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}}_{n \times 1}.$$

Define

$$y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad X = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1k} \\ x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nk} \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{pmatrix}, \quad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

Then

$$y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_k x_{ik} + \varepsilon_i \quad \Longleftrightarrow \quad y_i = (x_{i1} \ x_{i2} \ \cdots \ x_{ik}) \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{pmatrix} + \varepsilon_i.$$

Let

$$x_i = \begin{pmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{ik} \end{pmatrix}, \quad \text{so that} \quad y_i = x_i' \beta + \varepsilon_i.$$

Assumptions

$$y_i = x_i' \beta + \varepsilon_i.$$

1. **Linearity:** $Y = X\beta + \varepsilon$.
2. **Full rank:** $\text{rank}(X) = k$.
3. **Exogeneity:** $\mathbb{E}[\varepsilon \mid X] = 0$.

Law of iterated expectations:

$$\begin{aligned}\mathbb{E}[Y] &= \mathbb{E}[\mathbb{E}[Y \mid X]], \\ \mathbb{E}[\varepsilon] &= \mathbb{E}[\mathbb{E}[\varepsilon \mid X]] = \mathbb{E}[0] = 0.\end{aligned}$$

4. **Homoscedasticity and nonautocorrelation:**

$$\text{Var}(\varepsilon_i \mid X) = \sigma^2, \quad i = 1, \dots, n,$$

$$\text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0, \quad i \neq j.$$

5. X may be fixed and random.
6. **Normality:** $\varepsilon \mid X \sim N(0, \sigma^2 I)$.

OLS as least squares

Start from

$$y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_k x_{ik} + \varepsilon_i.$$

Candidate

$$b = (b_1, b_2, \dots, b_k).$$

Define residual

$$e_i = y_i - b_1 x_{i1} - b_2 x_{i2} - \dots - b_k x_{ik} \quad (\text{should be small}).$$

Define the OLS estimator:

$$\hat{\beta} = \arg \min_b \sum_{i=1}^n e_i^2 = \arg \min_b \sum_{i=1}^n \left(y_i - b_1 x_{i1} - b_2 x_{i2} - \dots - b_k x_{ik} \right)^2.$$

Example: simple linear regression

Example:

$$y_i = \beta_1 + \beta_2 x_i + \varepsilon_i.$$

Minimize

$$\min_{b_1, b_2} \sum_{i=1}^n (y_i - b_1 - b_2 x_i)^2.$$

F.O.C.:

$$\frac{\partial}{\partial b_1} \left[\sum_{i=1}^n (y_i - b_1 - b_2 x_i)^2 \right] = -2 \sum_{i=1}^n (y_i - b_1 - b_2 x_i) = 0 \quad (1)$$

$$\frac{\partial}{\partial b_2} \left[\sum_{i=1}^n (y_i - b_1 - b_2 x_i)^2 \right] = -2 \sum_{i=1}^n (y_i - b_1 - b_2 x_i) x_i = 0 \quad (2)$$

By (1),

$$\begin{aligned} \sum_{i=1}^n y_i &= n b_1 + b_2 \sum_{i=1}^n x_i, \\ \frac{1}{n} \sum_{i=1}^n y_i &= b_1 + b_2 \cdot \frac{1}{n} \sum_{i=1}^n x_i, \\ \bar{y} &= b_1 + b_2 \bar{x}, \\ b_1 &= \bar{y} - b_2 \bar{x}. \end{aligned} \quad (3)$$

Plug this into (2):

$$\begin{aligned} \sum_{i=1}^n (y_i - \bar{y} + b_2 \bar{x} - b_2 x_i) x_i &= 0, \\ \sum_{i=1}^n (y_i - \bar{y}) x_i - b_2 \sum_{i=1}^n (x_i - \bar{x}) x_i &\approx 0, \\ b_2 &= - \frac{\sum_{i=1}^n (y_i - \bar{y}) x_i}{\sum_{i=1}^n (x_i - \bar{x}) x_i}. \end{aligned} \quad (4)$$

Notice that

$$\sum_{i=1}^n (y_i - \bar{y}) \bar{x} = \bar{x} \sum_{i=1}^n (y_i - \bar{y}) = \bar{x} \left(\sum_{i=1}^n y_i - n \bar{y} \right) = 0.$$

Similarly,

$$\sum_{i=1}^n (x_i - \bar{x}) \bar{x} = 0.$$

So that

$$\begin{aligned} b_2 &= \frac{\sum_{i=1}^n (y_i - \bar{y}) x_i - \bar{x} \sum_{i=1}^n (y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x}) x_i - \bar{x} \sum_{i=1}^n (x_i - \bar{x})}, \\ b_2 &= \frac{\sum_{i=1}^n (y_i - \bar{y}) (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}. \end{aligned}$$

OLS in matrix form

$$\hat{\beta} = \arg \min_b \sum_{i=1}^n e_i^2 = \arg \min_b (y - Xb)'(y - Xb).$$

(For a scalar, $(a)' = a$.)

Expand:

$$\begin{aligned} \arg \min_b (y - Xb)'(y - Xb) &= \arg \min_b (y' - b' X')(y - Xb) \\ &= \arg \min_b (y' y - y' Xb - b' X' y + b' X' Xb) \\ &= \arg \min_b (y' y - 2y' Xb + b' X' Xb). \end{aligned}$$

Derivative preliminaries

Let z be $k \times 1$ and $f(z)$ be a scalar. Then

$$\frac{\partial f(z)}{\partial z} = \begin{pmatrix} \frac{\partial f(z)}{\partial z_1} \\ \vdots \\ \frac{\partial f(z)}{\partial z_k} \end{pmatrix} \quad (k \times 1).$$

Let a be $k \times 1$ and A be $k \times k$. Then

$$\frac{\partial(a'z)}{\partial z} = a, \quad \frac{\partial(z'Az)}{\partial z} = (A + A')z.$$

If $A' = A$, then

$$\frac{\partial(z'Az)}{\partial z} = 2Az.$$

Write

$$a = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_k \end{pmatrix}, \quad z = \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_k \end{pmatrix}, \quad a'z = a_1z_1 + a_2z_2 + \cdots + a_kz_k,$$

$$\frac{\partial(a'z)}{\partial z} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_k \end{pmatrix} = a.$$

Let

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kk} \end{pmatrix}, \quad \text{let } a_i = \begin{pmatrix} a_{i1} \\ a_{i2} \\ \vdots \\ a_{ik} \end{pmatrix}, \quad A = \begin{pmatrix} a'_1 \\ a'_2 \\ \vdots \\ a'_k \end{pmatrix}.$$

Then

$$Az = \begin{pmatrix} a'_1 \\ a'_2 \\ \vdots \\ a'_k \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_k \end{pmatrix} = \begin{pmatrix} a'_1 z \\ a'_2 z \\ \vdots \\ a'_k z \end{pmatrix}.$$

Also

$$\begin{aligned} z'Az &= \begin{pmatrix} z_1 & z_2 & \cdots & z_k \end{pmatrix} \begin{pmatrix} a'_1 z \\ a'_2 z \\ \vdots \\ a'_k z \end{pmatrix} = z_1 a'_1 z + z_2 a'_2 z + \cdots + z_k a'_k z, \\ &= z_1 \sum_{j=1}^k a_{1j} z_j + z_2 \sum_{j=1}^k a_{2j} z_j + \cdots + z_k \sum_{j=1}^k a_{kj} z_j. \end{aligned}$$

Then

$$\frac{\partial(z'Az)}{\partial z} = \begin{pmatrix} a'_1 z + z_1 a_{11} + z_2 a_{21} + \cdots + z_k a_{k1} \\ a'_2 z + z_1 a_{12} + z_2 a_{22} + \cdots + z_k a_{k2} \\ \vdots \\ a'_k z + z_1 a_{1k} + z_2 a_{2k} + \cdots + z_k a_{kk} \end{pmatrix} = Az + A'z = (A + A')z.$$

When $A = A'$, $(A + A')z = 2Az$.

Apply derivatives to OLS

Recall:

$$\hat{\beta} = \arg \min_b (y'y - 2y'Xb + b'X'Xb).$$

Using

$$\frac{\partial(a'z)}{\partial z} = a, \quad \frac{\partial(z'Az)}{\partial z} = (A + A')z,$$

we have

$$\begin{aligned} \frac{\partial(-2y'Xb)}{\partial b} &= -2X'y, \\ \frac{\partial(b'X'Xb)}{\partial b} &= [X'X + (X'X)']b = 2X'Xb. \end{aligned}$$

F.O.C.:

$$\begin{aligned} -2X'y + 2X'Xb &= 0, \\ X'Xb &= X'y. \end{aligned}$$

Multiply by $(X'X)^{-1}$:

$$\begin{aligned} (X'X)^{-1}X'Xb &= (X'X)^{-1}X'y, \\ \hat{\beta} &= (X'X)^{-1}X'y. \end{aligned}$$

OLS: Residual Properties and Projection Geometry

OLS identities

Start from the linear model:

$$y = X\beta + \varepsilon.$$

The OLS estimator is

$$\hat{\beta} = (X'X)^{-1}X'y.$$

Define fitted values and residuals:

$$\hat{y} = X\hat{\beta}, \quad e = y - X\hat{\beta}.$$

Normal equations (orthogonality of residuals to regressors):

$$\begin{aligned} X'e &= X'(y - X\hat{\beta}) \\ &= X'y - X'X\hat{\beta} \\ &= X'y - X'X(X'X)^{-1}X'y \\ &= 0. \end{aligned}$$

Also,

$$\hat{y} = X\hat{\beta}.$$

And the decomposition:

$$y = \hat{y} + e = X\hat{\beta} + (y - X\hat{\beta}) = y.$$

Design matrix with intercept

With an intercept, the regressor matrix is

$$X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1k} \\ 1 & x_{21} & \cdots & x_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{pmatrix},$$

and the regression can be written as

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \varepsilon_i.$$

From $X'e = 0$ we have the vector condition

$$X'e = 0.$$

Writing the dimensions explicitly:

$$\underbrace{X'}_{(k+1) \times n} \underbrace{e}_{n \times 1} = \underbrace{0}_{(k+1) \times 1}.$$

In particular, the first row corresponds to the constant regressor (a vector of ones). Let $\mathbf{1}$ denote the $n \times 1$ vector of ones. Then:

$$\mathbf{1}'e = 0.$$

Equivalently:

$$\sum_{i=1}^n e_i = 0.$$

Since

$$e_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik},$$

we have

$$\mathbf{1}'(y - X\hat{\beta}) = 0,$$

i.e.

$$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) = 0.$$

Therefore,

$$\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}_1 + \cdots + \hat{\beta}_k \bar{x}_k,$$

which implies

$$\bar{y} = \bar{\hat{y}}.$$

Projection (hat) matrix and residual-maker matrix

Define the projection (hat) matrix:

$$P = X(X'X)^{-1}X'.$$

(Here X is $n \times k$, $(X'X)^{-1}$ is $k \times k$, and X' is $k \times n$, so P is $n \times n$.)

Define the residual-maker matrix:

$$M = I - P.$$

Symmetry of P

$$\begin{aligned} P' &= [X(X'X)^{-1}X']' \\ &= X[(X'X)^{-1}]'X' \\ &= X[(X'X)']^{-1}X' \\ &= X(X'X)^{-1}X' \\ &= P. \end{aligned}$$

Idempotence of P

$$\begin{aligned} PP &= X(X'X)^{-1}X'X(X'X)^{-1}X' \\ &= X(X'X)^{-1}(X'X)(X'X)^{-1}X' \\ &= X(X'X)^{-1}X' \\ &= P. \end{aligned}$$

Fitted values and residuals as projections

$$\hat{y} = X\hat{\beta} = X(X'X)^{-1}X'y = Py.$$

$$e = y - \hat{y} = y - Py = (I - P)y = My.$$

Key algebraic properties

$$PX = X(X'X)^{-1}X'X = X.$$

$$MX = (I - P)X = X - PX = X - X = 0.$$

$$PM = P(I - P) = P - PP = 0.$$

$$MP = (I - P)P = P - PP = 0.$$

Orthogonality and decomposition

Since $\hat{y} = Py$ and $e = My$,

$$\hat{y}'e = y'P'My = y'PMMy = 0.$$

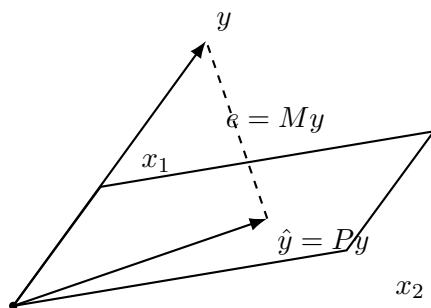
Also,

$$y = Iy = (M + P)y = Py + My = \hat{y} + e.$$

Pythagorean decomposition of sums of squares

$$\begin{aligned}
 \|y\|^2 &= y'y \\
 &= (Py + My)'(Py + My) \\
 &= (y'P' + y'M')(Py + My) \\
 &= y'P'Py + y'M'Py + y'P'My + y'M'My \\
 &= \hat{y}'\hat{y} + e'e \\
 &= \|\hat{y}\|^2 + \|e\|^2.
 \end{aligned}$$

Projection geometry (illustration)



Partitioned Regression and the Frisch–Waugh–Lovell (FWL) Theorem

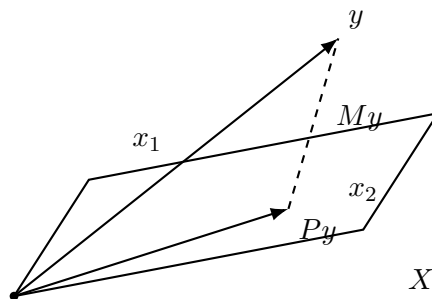
OLS basics and projection geometry

$$y = X\beta + \varepsilon.$$

$$\hat{\beta} = (X'X)^{-1}X'y.$$

$$P = X(X'X)^{-1}X', \quad M = I - P.$$

$$\hat{y} = X\hat{\beta} = Py, \quad e = y - \hat{y} = My.$$



Partition the regressors

Write the full model as

$$y = X\beta + \varepsilon,$$

and partition

$$X = (X_1 \ X_2), \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}.$$

In element form (with a column partition between X_1 and X_2):

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1k_1} & | & x_{1,k_1+1} & \cdots & x_{1k} \\ x_{21} & x_{22} & \cdots & x_{2k_1} & | & x_{2,k_1+1} & \cdots & x_{2k} \\ \vdots & \vdots & \ddots & \vdots & | & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nk_1} & | & x_{n,k_1+1} & \cdots & x_{nk} \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_{k_1} \\ \beta_{k_1+1} \\ \vdots \\ \beta_k \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

Recall again:

$$y = X\beta + \varepsilon, \quad X = (X_1 \ X_2), \quad \beta = (\beta_1, \beta_2)'$$

Then

$$\begin{aligned} y &= X\beta + \varepsilon \\ &= (X_1 \ X_2) \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} + \varepsilon \\ &= X_1\beta_1 + X_2\beta_2 + \varepsilon. \end{aligned}$$

Least squares and the normal equations

$$\min_b (y - Xb)'(y - Xb).$$

F.O.C.:

$$X'X\hat{\beta} = X'y.$$

Compute the block forms:

$$X'X = (X_1 \ X_2)'(X_1 \ X_2) = \begin{pmatrix} X_1' \\ X_2' \end{pmatrix} \begin{pmatrix} X_1 & X_2 \end{pmatrix} = \begin{pmatrix} X_1'X_1 & X_1'X_2 \\ X_2'X_1 & X_2'X_2 \end{pmatrix}.$$

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix}.$$

$$X'y = \begin{pmatrix} X_1' \\ X_2' \end{pmatrix} y = \begin{pmatrix} X_1'y \\ X_2'y \end{pmatrix}.$$

So the normal equations are

$$\begin{pmatrix} X_1'X_1 & X_1'X_2 \\ X_2'X_1 & X_2'X_2 \end{pmatrix} \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \begin{pmatrix} X_1'y \\ X_2'y \end{pmatrix}.$$

Equivalently, the system:

$$X_1'X_1\hat{\beta}_1 + X_1'X_2\hat{\beta}_2 = X_1'y \tag{1}$$

$$X_2'X_1\hat{\beta}_1 + X_2'X_2\hat{\beta}_2 = X_2'y. \tag{2}$$

Solve for $\hat{\beta}_2$ (FWL derivation)

From (1),

$$\begin{aligned}\hat{\beta}_1 &= (X_1'X_1)^{-1}(X_1'y - X_1'X_2\hat{\beta}_2) \\ &= (X_1'X_1)^{-1}X_1'(y - X_2\hat{\beta}_2).\end{aligned}\tag{3}$$

Plug (3) into (2):

$$X_2'X_1(X_1'X_1)^{-1}X_1'(y - X_2\hat{\beta}_2) + X_2'X_2\hat{\beta}_2 = X_2'y.$$

Let

$$P_1 = X_1(X_1'X_1)^{-1}X_1', \quad M_1 = I - P_1.$$

Then

$$X_2'P_1y - X_2'P_1X_2\hat{\beta}_2 + X_2'X_2\hat{\beta}_2 = X_2'y.$$

$$(X_2'X_2 - X_2'P_1X_2)\hat{\beta}_2 = X_2'y - X_2'P_1y.$$

$$X_2'(X_2 - P_1X_2)\hat{\beta}_2 = X_2'(y - P_1y).$$

$$X_2'(I - P_1)X_2\hat{\beta}_2 = X_2'(I - P_1)y.$$

$$X_2'M_1X_2\hat{\beta}_2 = X_2'M_1y.$$

Hence

$$\boxed{\hat{\beta}_2 = (X_2'M_1X_2)^{-1}X_2'M_1y}.$$

Also,

$$M_1' = M_1, \quad M_1M_1 = M_1.$$

So we can rewrite:

$$\begin{aligned}\hat{\beta}_2 &= (X_2'M_1X_2)^{-1}X_2'M_1y \\ &= (X_2'M_1M_1X_2)^{-1}X_2'M_1M_1y \\ &= [(M_1X_2)'M_1X_2]^{-1}(M_1X_2)'M_1y.\end{aligned}$$

(With the shorthand)

$$\tilde{X} = M_1X_2, \quad \tilde{y} = M_1y,$$

we have

$$\hat{\beta}_2 = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'\tilde{y}.$$

Therefore,

$$\hat{\beta}_2 \text{ is obtained by regressing } M_1y \text{ on } M_1X_2.$$

Frisch–Waugh–Lovell Theorem (statement)

Frisch–Waugh–Lovell Theorem. In the linear (least squares) regression of vector y on two sets of variables, X_1 and X_2 , the subvector $\hat{\beta}_2$ is the set of coefficients obtained when the residuals from a regression of y on X_1 alone are regressed on the set of residuals obtained when each column of X_2 is regressed on X_1 .

SST/SSR/SSE, R^2 , and the De-meaning Matrix

Setup

$$y_i = x_i' \beta + \varepsilon_i.$$

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2.$$

The $n \times 1$ vector of ones and M^0

Define

$$\ell_{n \times 1} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}.$$

Define the de-meaning (residual-maker for the intercept-only regression) matrix:

$$M^0 = I - \ell(\ell' \ell)^{-1} \ell'.$$

Compute:

$$\ell' \ell = (1 \ 1 \ \cdots \ 1) \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} = n, \quad (\ell' \ell)^{-1} = \frac{1}{n}.$$

How M^0 de-means a vector

Let

$$a_{n \times 1} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}.$$

Then

$$\ell' a = (1 \ 1 \ \cdots \ 1) \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \sum_{i=1}^n a_i.$$

Hence

$$(\ell' \ell)^{-1} \ell' a = \bar{a}, \quad \bar{a} = \frac{1}{n} \sum_{i=1}^n a_i.$$

And

$$\ell(\ell' \ell)^{-1} \ell' a = \begin{pmatrix} \bar{a} \\ \bar{a} \\ \vdots \\ \bar{a} \end{pmatrix}.$$

Therefore,

$$\begin{aligned} M^0 a &= \left[I - \ell(\ell' \ell)^{-1} \ell' \right] a \\ &= a - \ell(\ell' \ell)^{-1} \ell' a \\ &= \begin{pmatrix} a_1 - \bar{a} \\ a_2 - \bar{a} \\ \vdots \\ a_n - \bar{a} \end{pmatrix}. \end{aligned}$$

Apply M^0 to the OLS decomposition and derive $SST = SSR + SSE$

Let the OLS decomposition be

$$y = X\hat{\beta} + e, \quad \hat{y} = X\hat{\beta}.$$

Apply M^0 :

$$\begin{aligned} M^0 y &= M^0 X\hat{\beta} + M^0 e \\ &= M^0 X\hat{\beta} + e. \end{aligned}$$

Also,

$$SST = (M^0 y)' M^0 y.$$

Substitute $M^0 y = M^0 X\hat{\beta} + e$:

$$\begin{aligned} SST &= (M^0 y)' M^0 y \\ &= (M^0 X\hat{\beta} + e)' (M^0 X\hat{\beta} + e) \\ &= (M^0 X\hat{\beta})' M^0 X\hat{\beta} + 2(M^0 X\hat{\beta})' e + e' e \\ &= (M^0 X\hat{\beta})' M^0 X\hat{\beta} + e' e. \end{aligned}$$

In scalar form:

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n e_i^2.$$

Define

$$SST = SSR + SSE.$$

R^2 and adjusted R^2 (with full intermediate steps)

Step 0: Intercept and the de-meaning matrix (more intermediate steps). (a) **Define the intercept vector and its basic algebra.** Let $\ell \in \mathbb{R}^{n \times 1}$ be the vector of ones:

$$\ell \equiv \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}.$$

Then

$$\ell' \ell = \begin{pmatrix} 1 & 1 & \cdots & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} = \sum_{i=1}^n 1 = n, \quad (\ell' \ell)^{-1} = \frac{1}{n}.$$

(b) Define the sample mean using ℓ . For any $a \in \mathbb{R}^{n \times 1}$,

$$\ell' a = \begin{pmatrix} 1 & 1 & \cdots & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \sum_{i=1}^n a_i, \quad \bar{a} \equiv \frac{1}{n} \ell' a = \frac{1}{n} \sum_{i=1}^n a_i.$$

In particular,

$$\bar{y} = \frac{1}{n} \ell' y, \quad \bar{\hat{y}} = \frac{1}{n} \ell' \hat{y}.$$

(c) Projection onto the constant and residual-maker (de-meaning) matrix. The projection matrix onto the span of ℓ (i.e., intercept-only regression) is

$$P^0 \equiv \ell(\ell' \ell)^{-1} \ell' = \ell \left(\frac{1}{n} \right) \ell' = \frac{1}{n} \ell \ell'.$$

The associated residual-maker matrix is

$$M^0 \equiv I - P^0 = I - \ell(\ell' \ell)^{-1} \ell' = I - \frac{1}{n} \ell \ell'.$$

(d) Show explicitly that P^0 replaces a vector by its mean and M^0 de-means it. For any $a \in \mathbb{R}^{n \times 1}$,

$$P^0 a = \frac{1}{n} \ell \ell' a = \frac{1}{n} \ell (\ell' a) = \frac{1}{n} \ell \left(\sum_{i=1}^n a_i \right) = \left(\frac{1}{n} \sum_{i=1}^n a_i \right) \ell = \bar{a} \ell.$$

Therefore

$$M^0 a = (I - P^0) a = a - P^0 a = a - \bar{a} \ell.$$

Applying this to y and $\hat{y}$ gives

$$M^0 y = y - \bar{y} \ell, \quad M^0 \hat{y} = \hat{y} - \bar{\hat{y}} \ell.$$

(e) Symmetry of M^0 (show each step).

$$\begin{aligned} (M^0)' &= \left(I - \frac{1}{n} \ell \ell' \right)' \\ &= I' - \frac{1}{n} (\ell \ell')' \\ &= I - \frac{1}{n} \ell \ell' \quad (\text{since } I' = I \text{ and } (\ell \ell')' = \ell \ell') \\ &= M^0. \end{aligned}$$

(f) Idempotence of M^0 (expand and simplify).

$$\begin{aligned}
(M^0)^2 &= \left(I - \frac{1}{n}\ell\ell'\right) \left(I - \frac{1}{n}\ell\ell'\right) \\
&= I - \frac{1}{n}\ell\ell' - \frac{1}{n}\ell\ell' + \frac{1}{n^2}\ell\ell'\ell\ell' \\
&= I - \frac{2}{n}\ell\ell' + \frac{1}{n^2}\ell(\ell'\ell)\ell' \\
&= I - \frac{2}{n}\ell\ell' + \frac{1}{n^2}\ell(n)\ell' \quad (\text{since } \ell'\ell = n) \\
&= I - \frac{2}{n}\ell\ell' + \frac{1}{n}\ell\ell' \\
&= I - \frac{1}{n}\ell\ell' \\
&= M^0.
\end{aligned}$$

(g) M^0 removes constants (useful identity).

$$M^0\ell = \left(I - \frac{1}{n}\ell\ell'\right)\ell = \ell - \frac{1}{n}\ell(\ell'\ell) = \ell - \frac{1}{n}\ell(n) = 0.$$

Step 1: Use the OLS intercept property to link $\bar{y}$ and $\bar{\hat{y}}$. If the regression includes an intercept, then the normal equations imply

$$X'e = 0.$$

The first column of X is ℓ , so in particular

$$\ell'e = 0.$$

Since $e = y - \hat{y}$, we get

$$\ell'(y - \hat{y}) = 0 \implies \ell'y = \ell'\hat{y} \implies \bar{y} = \bar{\hat{y}}.$$

Therefore,

$$\hat{y} - \bar{y}\ell = \hat{y} - \bar{\hat{y}}\ell = M^0\hat{y}.$$

Step 2: Define SST, SSR, SSE in scalar form.

$$SST \equiv \sum_{i=1}^n (y_i - \bar{y})^2, \quad SSR \equiv \sum_{i=1}^n (\hat{y}_i - \bar{y})^2, \quad SSE \equiv \sum_{i=1}^n e_i^2 = e'e.$$

Step 3: Rewrite SST and SSR in matrix form using M^0 . Because $y - \bar{y}\ell = M^0y$,

$$SST = (y - \bar{y}\ell)'(y - \bar{y}\ell) = (M^0y)'(M^0y) = y'(M^0)'M^0y = y'M^0y.$$

Similarly, using $\bar{\hat{y}} = \bar{y}$ (from Step 1),

$$SSR = (\hat{y} - \bar{y}\ell)'(\hat{y} - \bar{y}\ell) = (M^0\hat{y})'(M^0\hat{y}) = \hat{y}'(M^0)'M^0\hat{y} = \hat{y}'M^0\hat{y}.$$

Step 4: Substitute $\hat{y} = X\hat{\beta}$ to get the quadratic form. Since $\hat{y} = X\hat{\beta}$,

$$SSR = \hat{y}'M^0\hat{y} = (X\hat{\beta})'M^0(X\hat{\beta}) = \hat{\beta}'X'M^0X\hat{\beta}.$$

So we have the key identities:

$$SST = y'M^0y, \quad SSR = \hat{\beta}'X'M^0X\hat{\beta}.$$

Step 5: Define R^2 and obtain the first equality. By definition,

$$R^2 \equiv \frac{SSR}{SST} = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2}.$$

Using the matrix forms from Steps 3–4,

$$R^2 = \frac{\hat{\beta}'X'M^0X\hat{\beta}}{y'M^0y}.$$

Step 6: Use $SST = SSR + SSE$ to get the “1 – .” form. From the OLS decomposition (with intercept),

$$SST = SSR + SSE.$$

Divide both sides by SST :

$$1 = \frac{SSR}{SST} + \frac{SSE}{SST} \implies \frac{SSR}{SST} = 1 - \frac{SSE}{SST}.$$

Therefore,

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{e'e}{y'M^0y}.$$

Step 7: Adjusted R^2 . Let k denote the number of estimated coefficients (including the intercept). Then

$$\bar{R}^2 \equiv 1 - \frac{SSE/(n-k)}{SST/(n-1)} = 1 - \frac{e'e/(n-k)}{y'M^0y/(n-1)}.$$

OLS Assumptions, Unbiasedness, and Specification Issues

Assumptions

1. **Linearity** $y = X\beta + \varepsilon$.
2. **Full rank** $\text{rank}(X) = k$.
3. **Exogeneity** $\mathbb{E}[\varepsilon | X] = 0$.

Law of iterated expectations:

$$\begin{aligned} \mathbb{E}[Y] &= \mathbb{E}[\mathbb{E}[Y | X]]. \\ \mathbb{E}[\varepsilon] &= \mathbb{E}[\mathbb{E}[\varepsilon | X]] = \mathbb{E}[0] = 0. \end{aligned}$$

4. **Homoscedasticity and nonautocorrelation**

$$\begin{aligned} \text{Var}(\varepsilon_i | X) &= \sigma^2, \quad i = 1, \dots, n, \\ \text{Cov}(\varepsilon_i, \varepsilon_j | X) &= 0, \quad i \neq j. \end{aligned}$$

5. X may be fixed and random.

Unbiasedness of OLS

Start from

$$y = X\beta + \varepsilon.$$

The OLS estimator is

$$\hat{\beta} = (X'X)^{-1}X'y.$$

Substitute $y = X\beta + \varepsilon$:

$$\begin{aligned}\hat{\beta} &= (X'X)^{-1}X'y \\ &= (X'X)^{-1}X'(X\beta + \varepsilon) \\ &= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'\varepsilon \\ &= \beta + (X'X)^{-1}X'\varepsilon.\end{aligned}$$

Take conditional expectation given X :

$$\begin{aligned}\mathbb{E}[\hat{\beta} \mid X] &= \beta + \mathbb{E}[(X'X)^{-1}X'\varepsilon \mid X] \\ &= \beta + (X'X)^{-1}X'\mathbb{E}[\varepsilon \mid X] \\ &= \beta.\end{aligned}$$

Then take unconditional expectation:

$$\mathbb{E}[\hat{\beta}] = \mathbb{E}[\mathbb{E}[\hat{\beta} \mid X]] = \mathbb{E}[\beta] = \beta.$$

1. Omitted Variable Bias

Suppose the correct model is

$$y = X_1\beta_1 + X_2\beta_2 + \varepsilon.$$

Regress y on X_1 (omit X_2) and denote the OLS estimator as $\hat{\beta}_1$:

$$\begin{aligned}\hat{\beta}_1 &= (X_1'X_1)^{-1}X_1'y \\ &= (X_1'X_1)^{-1}X_1'(X_1\beta_1 + X_2\beta_2 + \varepsilon) \\ &= \beta_1 + (X_1'X_1)^{-1}X_1'X_2\beta_2 + (X_1'X_1)^{-1}X_1'\varepsilon.\end{aligned}$$

Therefore,

$$\mathbb{E}[\hat{\beta}_1 \mid X] = \beta_1 + (X_1'X_1)^{-1}X_1'X_2\beta_2.$$

So $\hat{\beta}_1$ is biased unless:

$$(1) \beta_2 = 0 \quad \text{or} \quad (2) X_1'X_2 = 0.$$

2. Inclusion of Irrelevant Variables

Suppose the correct model is

$$y = X_1\beta_1 + \varepsilon.$$

Regress y on $X = (X_1, X_2)$ and denote the OLS estimator as $\hat{\beta}$:

$$\begin{aligned}\hat{\beta} &= (X'X)^{-1}X'y \\ &= (X'X)^{-1}X'(X_1\beta_1 + \varepsilon) \\ &= (X'X)^{-1}X'X_1\beta_1 + (X'X)^{-1}X'\varepsilon.\end{aligned}$$

Note the identity

$$(X'X)^{-1}X'X = I.$$

Embedding the true model in the expanded regression. Assume the true DGP is

$$y = X_1\beta_1 + \varepsilon,$$

but we (over-)estimate the expanded model with regressors

$$X = (X_1 \ X_2), \quad X_1 \in \mathbb{R}^{n \times k_1}, \ X_2 \in \mathbb{R}^{n \times k_2}, \quad X \in \mathbb{R}^{n \times (k_1 + k_2)}.$$

Step 1: Define the “true” coefficient vector in the expanded parameter space.

In the expanded regression

$$y = X\beta + \varepsilon, \quad X = (X_1 \ X_2),$$

the parameter vector β must conform to the number of columns of X :

$$X \in \mathbb{R}^{n \times (k_1 + k_2)} \implies \beta \in \mathbb{R}^{(k_1 + k_2) \times 1}.$$

Step 1.1 (dimension matching). Because X is partitioned into X_1 and X_2 , it is natural to partition β conformably:

$$\beta = \begin{pmatrix} \beta^{(1)} \\ \beta^{(2)} \end{pmatrix}, \quad \beta^{(1)} \in \mathbb{R}^{k_1 \times 1}, \ \beta^{(2)} \in \mathbb{R}^{k_2 \times 1}.$$

Step 1.2 (meaning of “irrelevant”). Assume the true DGP is

$$y = X_1\beta_1 + \varepsilon,$$

i.e., once X_1 is included, y has no systematic dependence on X_2 . In the expanded parameterization, this statement is equivalent to saying that the *population (true) coefficient on X_2 is zero*:

$$\beta^{(2)} = 0.$$

Step 1.3 (embed the true model into the expanded one). Therefore, the “true” coefficient vector in the expanded $(k_1 + k_2)$ -dimensional space is

$$\beta \equiv \begin{pmatrix} \beta_1 \\ 0 \end{pmatrix}, \quad \beta_1 \in \mathbb{R}^{k_1 \times 1}, \ 0 \in \mathbb{R}^{k_2 \times 1}.$$

Step 1.4 (verify $X\beta = X_1\beta_1$ explicitly). Using block multiplication,

$$X\beta = (X_1 \ X_2) \begin{pmatrix} \beta_1 \\ 0 \end{pmatrix} = X_1\beta_1 + X_2 \cdot 0 = X_1\beta_1.$$

Hence the true DGP can be written in the expanded form:

$$y = X_1\beta_1 + \varepsilon = X\beta + \varepsilon.$$

Step 2: Show explicitly that $X\beta = X_1\beta_1$.

Compute $X\beta$ using block-matrix multiplication:

$$X\beta = (X_1 \ X_2) \begin{pmatrix} \beta_1 \\ 0 \end{pmatrix} = X_1\beta_1 + X_2 \cdot 0 = X_1\beta_1.$$

Therefore, the true model can be written in the same expanded form:

$$y = X_1\beta_1 + \varepsilon = X\beta + \varepsilon.$$

Step 3: Write the OLS estimator in the expanded regression and substitute the true DGP.

The OLS estimator from regressing y on X is

$$\hat{\beta} = (X'X)^{-1}X'y.$$

Substitute $y = X\beta + \varepsilon$ (which holds because $X\beta = X_1\beta_1$):

$$\begin{aligned}\hat{\beta} &= (X'X)^{-1}X'(X\beta + \varepsilon) \\ &= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'\varepsilon.\end{aligned}$$

Step 4: Use the matrix identity $(X'X)^{-1}(X'X) = I$ (and check dimensions).

Since X has full column rank, $X'X$ is invertible and

$$(X'X)^{-1}(X'X) = I_{k_1+k_2}.$$

Thus

$$(X'X)^{-1}X'X\beta = I_{k_1+k_2}\beta = \beta.$$

So we obtain the key decomposition:

$$\hat{\beta} = \beta + (X'X)^{-1}X'\varepsilon = \begin{pmatrix} \beta_1 \\ 0 \end{pmatrix} + (X'X)^{-1}X'\varepsilon.$$

Step 5: Take conditional expectation given X (show each step).

$$\begin{aligned}\mathbb{E}[\hat{\beta} \mid X] &= \mathbb{E}[\beta + (X'X)^{-1}X'\varepsilon \mid X] \\ &= \beta + (X'X)^{-1}X'\mathbb{E}[\varepsilon \mid X].\end{aligned}$$

Under exogeneity $\mathbb{E}[\varepsilon \mid X] = 0$,

$$\mathbb{E}[\hat{\beta} \mid X] = \beta = \begin{pmatrix} \beta_1 \\ 0 \end{pmatrix}.$$

Step 6: Unconditional expectation via iterated expectations.

$$\mathbb{E}[\hat{\beta}] = \mathbb{E}[\mathbb{E}[\hat{\beta} \mid X]] = \mathbb{E}[\beta] = \beta.$$

Conclusion: Adding irrelevant regressors does not bias OLS (under exogeneity), but it can increase variance.

Econometrics I: OLS Assumptions, Variance of $\hat{\beta}$, and Unbiased Estimation of σ^2

Assumptions

1. **Linearity:** $y = X\beta + \varepsilon$.
2. **Full rank:** $\text{rank}(X) = k$.

3. **Exogeneity:** $\mathbb{E}[\varepsilon \mid X] = 0$.

Law of iterated expectations:

$$\begin{aligned}\mathbb{E}[Y] &= \mathbb{E}[\mathbb{E}[Y \mid X]]. \\ \mathbb{E}[\varepsilon] &= \mathbb{E}[\mathbb{E}[\varepsilon \mid X]] = \mathbb{E}[0] = 0.\end{aligned}$$

4. **Homoscedasticity and nonautocorrelation (more intermediate steps).**

Step 4.1 (componentwise statements). For each observation $i = 1, \dots, n$, homoscedasticity means the conditional variance is constant:

$$\text{Var}(\varepsilon_i \mid X) = \sigma^2, \quad i = 1, \dots, n.$$

For any two different observations $i \neq j$, nonautocorrelation means the conditional covariance is zero:

$$\text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0, \quad i \neq j.$$

Step 4.2 (rewrite variance/covariance in terms of conditional second moments).

Recall

$$\begin{aligned}\text{Var}(\varepsilon_i \mid X) &= \mathbb{E}[\varepsilon_i^2 \mid X] - (\mathbb{E}[\varepsilon_i \mid X])^2, \\ \text{Cov}(\varepsilon_i, \varepsilon_j \mid X) &= \mathbb{E}[\varepsilon_i \varepsilon_j \mid X] - \mathbb{E}[\varepsilon_i \mid X] \mathbb{E}[\varepsilon_j \mid X].\end{aligned}$$

Under exogeneity (Assumption 3), $\mathbb{E}[\varepsilon_i \mid X] = 0$ for all i , so these simplify to

$$\text{Var}(\varepsilon_i \mid X) = \mathbb{E}[\varepsilon_i^2 \mid X], \quad \text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = \mathbb{E}[\varepsilon_i \varepsilon_j \mid X] \quad (i \neq j).$$

Hence, homoscedasticity and nonautocorrelation can be restated as

$$\begin{aligned}\mathbb{E}[\varepsilon_i^2 \mid X] &= \sigma^2, \quad i = 1, \dots, n, \\ \mathbb{E}[\varepsilon_i \varepsilon_j \mid X] &= 0, \quad i \neq j.\end{aligned}$$

Step 4.3 (stack errors into a vector). Let

$$\varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix} \in \mathbb{R}^{n \times 1}.$$

Then $\varepsilon \varepsilon'$ is an $n \times n$ matrix whose (i, j) element is

$$(\varepsilon \varepsilon')_{ij} = \varepsilon_i \varepsilon_j.$$

Step 4.4 (take conditional expectation element-by-element). Define the conditional second-moment matrix

$$\mathbb{E}[\varepsilon \varepsilon' \mid X].$$

Its (i, j) element is

$$(\mathbb{E}[\varepsilon \varepsilon' \mid X])_{ij} = \mathbb{E}[(\varepsilon \varepsilon')_{ij} \mid X] = \mathbb{E}[\varepsilon_i \varepsilon_j \mid X].$$

Using the restated conditions from Step 4.2:

$$(\mathbb{E}[\varepsilon \varepsilon' \mid X])_{ii} = \mathbb{E}[\varepsilon_i^2 \mid X] = \sigma^2,$$

and for $i \neq j$,

$$(\mathbb{E}[\varepsilon\varepsilon' \mid X])_{ij} = \mathbb{E}[\varepsilon_i\varepsilon_j \mid X] = 0.$$

Step 4.5 (identify the matrix form). Therefore the conditional second-moment matrix has σ^2 on the diagonal and zeros off-diagonal:

$$\mathbb{E}[\varepsilon\varepsilon' \mid X] = \begin{pmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{pmatrix} = \sigma^2 I_n,$$

where I_n is the $n \times n$ identity matrix.

5. X may be fixed and random (conditioning explanation).

Step 5.1 (fixed design interpretation). In the classical “fixed- X ” view, X is treated as non-random. Then assumptions such as $\mathbb{E}[\varepsilon \mid X] = 0$ and $\mathbb{E}[\varepsilon\varepsilon' \mid X] = \sigma^2 I_n$ are interpreted as statements about ε given the realized X .

Step 5.2 (random design interpretation). In the “random- X ” view, X is random. The same assumptions are interpreted conditionally on X :

$$\mathbb{E}[\varepsilon \mid X] = 0, \quad \mathbb{E}[\varepsilon\varepsilon' \mid X] = \sigma^2 I_n,$$

and unconditional statements can be obtained by iterated expectations if needed.

Variance of $\hat{\beta}$ given X

Start from

$$y = X\beta + \varepsilon.$$

OLS estimator:

$$\hat{\beta} = (X'X)^{-1}X'y.$$

Substitute $y = X\beta + \varepsilon$:

$$\begin{aligned} \hat{\beta} &= (X'X)^{-1}X'(X\beta + \varepsilon) \\ &= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'\varepsilon \\ &= \beta + (X'X)^{-1}X'\varepsilon. \end{aligned}$$

Conditional expectation and deviation from the conditional mean (more steps)

Recall the decomposition derived earlier:

$$\hat{\beta} = \beta + (X'X)^{-1}X'\varepsilon. \tag{1}$$

Step A: Compute the conditional expectation $\mathbb{E}[\hat{\beta} \mid X]$. Start from (1) and take conditional expectation given X :

$$\mathbb{E}[\hat{\beta} \mid X] = \mathbb{E}[\beta + (X'X)^{-1}X'\varepsilon \mid X].$$

Use linearity of conditional expectation:

$$\mathbb{E}[\hat{\beta} \mid X] = \mathbb{E}[\beta \mid X] + \mathbb{E}[(X'X)^{-1}X'\varepsilon \mid X].$$

Since β is a non-random parameter (a constant), conditioning does not change it:

$$\mathbb{E}[\beta \mid X] = \beta.$$

Next, because $(X'X)^{-1}X'$ is a function of X only, it can be pulled outside the conditional expectation:

$$\mathbb{E}[(X'X)^{-1}X'\varepsilon \mid X] = (X'X)^{-1}X' \mathbb{E}[\varepsilon \mid X].$$

By the exogeneity assumption $\mathbb{E}[\varepsilon \mid X] = 0$, we obtain:

$$(X'X)^{-1}X' \mathbb{E}[\varepsilon \mid X] = (X'X)^{-1}X' \cdot 0 = 0.$$

Therefore,

$$\boxed{\mathbb{E}[\hat{\beta} \mid X] = \beta.} \tag{2}$$

Step B: Deviation from its conditional mean. By definition,

$$\hat{\beta} - \mathbb{E}[\hat{\beta} \mid X].$$

Substitute (1) for $\hat{\beta}$ and (2) for $\mathbb{E}[\hat{\beta} \mid X]$:

$$\begin{aligned} \hat{\beta} - \mathbb{E}[\hat{\beta} \mid X] &= [\beta + (X'X)^{-1}X'\varepsilon] - \beta \\ &= (X'X)^{-1}X'\varepsilon. \end{aligned}$$

Hence,

$$\boxed{\hat{\beta} - \mathbb{E}[\hat{\beta} \mid X] = (X'X)^{-1}X'\varepsilon.} \tag{3}$$

(Recall the transpose rule)

$$(AB)' = B'A'.$$

Conditional variance of $\hat{\beta}$ given X (more intermediate steps)

Recall from the previous step that

$$\hat{\beta} - \mathbb{E}[\hat{\beta} \mid X] = (X'X)^{-1}X'\varepsilon. \tag{1}$$

Let

$$A \equiv (X'X)^{-1}X'.$$

Then (1) becomes

$$\hat{\beta} - \mathbb{E}[\hat{\beta} \mid X] = A\varepsilon. \tag{2}$$

Step 1: Start from the definition of conditional variance. By definition, for a random vector z ,

$$\text{Var}(z \mid X) = \mathbb{E}[(z - \mathbb{E}[z \mid X])(z - \mathbb{E}[z \mid X])' \mid X].$$

Apply this to $z = \hat{\beta}$:

$$\text{Var}[\hat{\beta} \mid X] = \mathbb{E}[(\hat{\beta} - \mathbb{E}[\hat{\beta} \mid X])(\hat{\beta} - \mathbb{E}[\hat{\beta} \mid X])' \mid X]. \tag{3}$$

Step 2: Substitute the deviation expression. Using (2),

$$\text{Var}[\hat{\beta} \mid X] = \mathbb{E}[(A\varepsilon)(A\varepsilon)' \mid X]. \quad (4)$$

Step 3: Expand $(A\varepsilon)(A\varepsilon)'$ using transpose rules. First note the transpose identity

$$(AB)' = B'A'.$$

Thus

$$(A\varepsilon)' = \varepsilon'A'.$$

Therefore,

$$(A\varepsilon)(A\varepsilon)' = (A\varepsilon)(\varepsilon'A') = A(\varepsilon\varepsilon')A'. \quad (5)$$

Plug (5) into (4):

$$\text{Var}[\hat{\beta} \mid X] = \mathbb{E}[A(\varepsilon\varepsilon')A' \mid X]. \quad (6)$$

Step 4: Pull out matrices that are functions of X only. Since $A = (X'X)^{-1}X'$ depends only on X , conditioning on X treats A and A' as constants. Hence

$$\mathbb{E}[A(\varepsilon\varepsilon')A' \mid X] = A \mathbb{E}[\varepsilon\varepsilon' \mid X] A'. \quad (7)$$

So,

$$\text{Var}[\hat{\beta} \mid X] = A \mathbb{E}[\varepsilon\varepsilon' \mid X] A'. \quad (8)$$

Step 5: Substitute A and compute A' . We have

$$A = (X'X)^{-1}X'.$$

Compute A' :

$$A' = ((X'X)^{-1}X')' = X((X'X)^{-1})'. \quad (9)$$

Because $X'X$ is symmetric, $(X'X)' = X'X$, and therefore

$$((X'X)^{-1})' = ((X'X)')^{-1} = (X'X)^{-1}.$$

Hence,

$$A' = X(X'X)^{-1}. \quad (10)$$

Substitute (9)–(10) into (8):

$$\text{Var}[\hat{\beta} \mid X] = (X'X)^{-1}X' \mathbb{E}[\varepsilon\varepsilon' \mid X] X(X'X)^{-1}. \quad (11)$$

Step 6: Use homoscedasticity and nonautocorrelation. Under the assumption

$$\mathbb{E}[\varepsilon\varepsilon' \mid X] = \sigma^2 I, \quad (12)$$

plug (12) into (11):

$$\text{Var}[\hat{\beta} \mid X] = (X'X)^{-1}X'(\sigma^2 I)X(X'X)^{-1}. \quad (13)$$

Step 7: Simplify. Because σ^2 is a scalar,

$$X'(\sigma^2 I)X = \sigma^2 X'IX = \sigma^2 X'X.$$

So (13) becomes

$$\begin{aligned}\text{Var}[\hat{\beta} \mid X] &= (X'X)^{-1}(\sigma^2 X'X)(X'X)^{-1} \\ &= \sigma^2 (X'X)^{-1}(X'X)(X'X)^{-1} \\ &= \sigma^2 (X'X)^{-1}I(X'X)^{-1} \\ &= \sigma^2 (X'X)^{-1}.\end{aligned}$$

Therefore,

$$\boxed{\text{Var}[\hat{\beta} \mid X] = \sigma^2 (X'X)^{-1}.}$$

Residuals and the (biased) MLE-type estimator $\hat{\sigma}^2$

Define residuals:

$$e = y - X\hat{\beta}.$$

Define

$$\hat{\sigma}^2 = \frac{1}{n}e'e = \frac{1}{n}\sum_{i=1}^n e_i^2.$$

Let

$$M = I - X(X'X)^{-1}X'.$$

Then (using $MX = 0$),

$$e = My = M(X\beta + \varepsilon) = M\varepsilon.$$

Compute $\mathbb{E}[\hat{\sigma}^2 \mid X]$ using trace (more intermediate steps)

Step 0: Definitions and preliminaries. Recall

$$\hat{\sigma}^2 \equiv \frac{1}{n}e'e, \quad e \equiv y - X\hat{\beta}, \quad M \equiv I - X(X'X)^{-1}X'.$$

From $y = X\beta + \varepsilon$ and $MX = 0$, we have

$$e = My = M(X\beta + \varepsilon) = MX\beta + M\varepsilon = 0 + M\varepsilon = M\varepsilon.$$

Hence

$$e'e = (M\varepsilon)'(M\varepsilon) = \varepsilon'M'M\varepsilon.$$

Because M is symmetric and idempotent,

$$M' = M, \quad M^2 = M,$$

so

$$e'e = \varepsilon'M\varepsilon.$$

Step 1: Trace identities used. We will use the following facts:

1. For any scalar a , $\text{tr}(a) = a$.
2. For conformable matrices, $\text{tr}(AB) = \text{tr}(BA)$ (cyclic property).
3. $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$ (linearity).
4. $\text{tr}(cA) = c \text{tr}(A)$ for scalar c .
5. $\text{tr}(I_n) = n$.

Step 2: Start from the definition of $\hat{\sigma}^2$ and substitute $e'e = \varepsilon'M\varepsilon$.

$$\begin{aligned}\mathbb{E}[\hat{\sigma}^2 \mid X] &= \mathbb{E}\left[\frac{1}{n}e'e \mid X\right] \\ &= \frac{1}{n}\mathbb{E}[e'e \mid X] \\ &= \frac{1}{n}\mathbb{E}[\varepsilon'M\varepsilon \mid X].\end{aligned}\tag{1}$$

Step 3: Convert the quadratic form into a trace (explicitly). Since $\varepsilon'M\varepsilon$ is a 1×1 scalar,

$$\varepsilon'M\varepsilon = \text{tr}(\varepsilon'M\varepsilon).$$

So (1) becomes

$$\mathbb{E}[\hat{\sigma}^2 \mid X] = \frac{1}{n}\mathbb{E}[\text{tr}(\varepsilon'M\varepsilon) \mid X].\tag{2}$$

Step 4: Use cyclic property to move terms inside the trace. Apply $\text{tr}(AB) = \text{tr}(BA)$ repeatedly:

$$\text{tr}(\varepsilon'M\varepsilon) = \text{tr}(M\varepsilon\varepsilon').$$

(Reason: treat ε' as $1 \times n$, M as $n \times n$, and ε as $n \times 1$; then $\varepsilon'M\varepsilon$ is 1×1 , and cyclic permutation is valid.)

Thus (2) becomes

$$\mathbb{E}[\hat{\sigma}^2 \mid X] = \frac{1}{n}\mathbb{E}[\text{tr}(M\varepsilon\varepsilon') \mid X].\tag{3}$$

Step 5: Pull expectation inside the trace (linearity). Because trace is linear and expectation is linear,

$$\mathbb{E}[\text{tr}(M\varepsilon\varepsilon') \mid X] = \text{tr}\left(\mathbb{E}[M\varepsilon\varepsilon' \mid X]\right).$$

So

$$\mathbb{E}[\hat{\sigma}^2 \mid X] = \frac{1}{n}\text{tr}\left(\mathbb{E}[M\varepsilon\varepsilon' \mid X]\right).\tag{4}$$

Step 6: Take M out of the conditional expectation. Given X , the matrix $M = I - X(X'X)^{-1}X'$ is non-random (it is a function of X only), so

$$\mathbb{E}[M\varepsilon\varepsilon' \mid X] = M \mathbb{E}[\varepsilon\varepsilon' \mid X].$$

Substitute into (4):

$$\mathbb{E}[\hat{\sigma}^2 \mid X] = \frac{1}{n}\text{tr}\left(M \mathbb{E}[\varepsilon\varepsilon' \mid X]\right).\tag{5}$$

Step 7: Use homoscedasticity and nonautocorrelation. Under the assumption

$$\mathbb{E}[\varepsilon\varepsilon' \mid X] = \sigma^2 I,$$

(5) becomes

$$\mathbb{E}[\hat{\sigma}^2 \mid X] = \frac{1}{n} \text{tr}(M \cdot \sigma^2 I). \quad (6)$$

Step 8: Pull out the scalar σ^2 and simplify. Using $\text{tr}(cA) = c \text{tr}(A)$,

$$\text{tr}(M \cdot \sigma^2 I) = \sigma^2 \text{tr}(MI).$$

But $MI = M$, so

$$\text{tr}(MI) = \text{tr}(M).$$

Thus from (6),

$$\mathbb{E}[\hat{\sigma}^2 \mid X] = \frac{\sigma^2}{n} \text{tr}(M). \quad (7)$$

Step 9: Compute $\text{tr}(M)$ using $M = I - P$ with $P = X(X'X)^{-1}X'$. Write

$$M = I - X(X'X)^{-1}X' \equiv I - P.$$

Then by linearity of trace,

$$\text{tr}(M) = \text{tr}(I) - \text{tr}(P). \quad (8)$$

We have

$$\text{tr}(I) = n. \quad (9)$$

Now compute $\text{tr}(P)$:

$$\text{tr}(X(X'X)^{-1}X') = \text{tr}((X'X)^{-1}X'X) \quad (\text{cyclic property}).$$

But $(X'X)^{-1}X'X = I_k$, hence

$$\text{tr}((X'X)^{-1}X'X) = \text{tr}(I_k) = k. \quad (10)$$

Combine (8)–(10):

$$\text{tr}(M) = n - k. \quad (11)$$

Step 10: Conclude $\mathbb{E}[\hat{\sigma}^2 \mid X]$. Plug (11) into (7):

$$\mathbb{E}[\hat{\sigma}^2 \mid X] = \frac{\sigma^2}{n}(n - k).$$

Therefore,

$$\boxed{\mathbb{E}[\hat{\sigma}^2 \mid X] = \frac{n - k}{n} \sigma^2.}$$

Unbiased estimator s^2 for σ^2 (more intermediate steps)

Step 0: Recall the biased estimator $\hat{\sigma}^2$. We defined

$$\hat{\sigma}^2 \equiv \frac{1}{n} e' e.$$

From the previous derivation we obtained

$$\mathbb{E}[\hat{\sigma}^2 \mid X] = \frac{n-k}{n} \sigma^2. \quad (1)$$

This shows $\hat{\sigma}^2$ is biased downward because $\frac{n-k}{n} < 1$ when $k > 0$.

Step 1: Choose a scaling constant that removes the bias. We want an estimator of the form

$$s^2 \equiv c \cdot \hat{\sigma}^2$$

such that

$$\mathbb{E}[s^2 \mid X] = \sigma^2.$$

Using (1),

$$\mathbb{E}[s^2 \mid X] = \mathbb{E}[c \hat{\sigma}^2 \mid X] = c \mathbb{E}[\hat{\sigma}^2 \mid X] = c \cdot \frac{n-k}{n} \sigma^2.$$

To make this equal to σ^2 , choose c so that

$$c \cdot \frac{n-k}{n} = 1 \quad \implies \quad c = \frac{n}{n-k}. \quad (2)$$

Step 2: Define s^2 and rewrite it in equivalent forms. Using (2),

$$s^2 \equiv \frac{n}{n-k} \hat{\sigma}^2.$$

Since $\hat{\sigma}^2 = \frac{1}{n} e' e$, we can also write

$$s^2 = \frac{n}{n-k} \cdot \frac{1}{n} e' e = \frac{1}{n-k} e' e.$$

Thus,

$$\boxed{s^2 = \frac{1}{n-k} e' e = \frac{n}{n-k} \hat{\sigma}^2.} \quad (3)$$

Step 3: Verify unbiasedness directly. Starting from (3),

$$\begin{aligned} \mathbb{E}[s^2 \mid X] &= \mathbb{E}\left[\frac{n}{n-k} \hat{\sigma}^2 \mid X\right] \\ &= \frac{n}{n-k} \mathbb{E}[\hat{\sigma}^2 \mid X] \quad (\text{since } \frac{n}{n-k} \text{ is a constant given } X) \\ &= \frac{n}{n-k} \cdot \frac{n-k}{n} \sigma^2 \quad (\text{by (1)}) \\ &= 1 \cdot \sigma^2 \\ &= \sigma^2. \end{aligned}$$

Therefore,

$$\boxed{\mathbb{E}[s^2 \mid X] = \sigma^2.}$$

(Optional) Unconditional unbiasedness by iterated expectations.

$$\mathbb{E}[s^2] = \mathbb{E}[\mathbb{E}[s^2 | X]] = \mathbb{E}[\sigma^2] = \sigma^2.$$

Estimated variance of $\hat{\beta}$ and standard errors (more intermediate steps)

Step 0: Start from the true conditional variance formula. Previously we derived

$$\text{Var}[\hat{\beta} | X] = \sigma^2 (X'X)^{-1}. \quad (4)$$

Step 1: Plug-in principle (replace unknown σ^2 by its estimator). Since σ^2 is unknown, we estimate it by s^2 and form the plug-in estimator:

$$\widehat{\text{Var}}[\hat{\beta} | X] \equiv s^2 (X'X)^{-1}. \quad (5)$$

(Here $(X'X)^{-1}$ is computable from the data, and s^2 is computable from residuals.)

Step 2: Write componentwise variance for coefficient j . Let $\hat{\beta}_j$ denote the j -th element of $\hat{\beta}$. From (5), the estimated variance of $\hat{\beta}_j$ is the (j, j) diagonal entry:

$$\widehat{\text{Var}}(\hat{\beta}_j | X) = [s^2 (X'X)^{-1}]_{jj}. \quad (6)$$

Since s^2 is a scalar,

$$[s^2 (X'X)^{-1}]_{jj} = s^2 [(X'X)^{-1}]_{jj}. \quad (7)$$

Step 3: Standard errors are square roots of estimated variances. By definition,

$$se_j \equiv \sqrt{\widehat{\text{Var}}(\hat{\beta}_j | X)}.$$

Using (6) (or (7)),

$$se_j = \sqrt{[s^2 (X'X)^{-1}]_{jj}} = \sqrt{s^2 [(X'X)^{-1}]_{jj}}.$$

(Optional) Plug-in expression using $e'e$. Because $s^2 = \frac{1}{n-k} e'e$,

$$se_j = \sqrt{\frac{e'e}{n-k} [(X'X)^{-1}]_{jj}}.$$

Econometrics I: Gauss–Markov Theorem (BLUE)

Assumptions

1. **Linearity:**

$$y = X\beta + \varepsilon.$$

2. **Full rank:**

$$\text{rank}(X) = k.$$

3. **Exogeneity:**

$$\mathbb{E}[\varepsilon \mid X] = 0.$$

Law of iterated expectations:

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]].$$

In particular, applying it to ε :

$$\mathbb{E}[\varepsilon] = \mathbb{E}[\mathbb{E}[\varepsilon \mid X]] = \mathbb{E}[0] = 0.$$

4. **Homoscedasticity and nonautocorrelation:**

$$\text{Var}(\varepsilon_i \mid X) = \sigma^2, \quad i = 1, \dots, n,$$

$$\text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0, \quad i \neq j.$$

More process (how the matrix form arises):

Stack the errors:

$$\varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix} \in \mathbb{R}^{n \times 1}.$$

Then $\varepsilon\varepsilon'$ is an $n \times n$ matrix with (i, j) entry

$$(\varepsilon\varepsilon')_{ij} = \varepsilon_i\varepsilon_j.$$

Taking conditional expectations, the (i, j) entry of $\mathbb{E}[\varepsilon\varepsilon' \mid X]$ is

$$(\mathbb{E}[\varepsilon\varepsilon' \mid X])_{ij} = \mathbb{E}[\varepsilon_i\varepsilon_j \mid X].$$

Under Assumption 3, $\mathbb{E}[\varepsilon_i \mid X] = 0$, so

$$\text{Var}(\varepsilon_i \mid X) = \mathbb{E}[\varepsilon_i^2 \mid X], \quad \text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = \mathbb{E}[\varepsilon_i\varepsilon_j \mid X] \quad (i \neq j).$$

Hence,

$$\mathbb{E}[\varepsilon_i^2 \mid X] = \sigma^2 \quad (i = j), \quad \mathbb{E}[\varepsilon_i\varepsilon_j \mid X] = 0 \quad (i \neq j),$$

which means the conditional second-moment matrix has σ^2 on the diagonal and zeros off-diagonal:

$$\boxed{\mathbb{E}[\varepsilon\varepsilon' \mid X] = \sigma^2 I_n.}$$

5. **X may be fixed and random.**

More process (interpretation): If X is fixed, the conditioning is conceptual (treat X as given). If X is random, all assumptions are interpreted conditionally on the realized X .

Gauss–Markov Theorem

Statement. Under Assumptions 1–5, the least squares estimator $\hat{\beta}$ has the smallest variance among the class of estimators that are: (i) *linear in y* , and (ii) *unbiased* for β . Therefore, $\hat{\beta}$ is the **Best Linear Unbiased Estimator (BLUE)**.

Proof (with all steps preserved and fully expanded)

Step 1: Consider an arbitrary linear estimator. Let $\tilde{\beta}$ be any estimator that is linear in y :

$$\tilde{\beta} = Cy,$$

where C is a $(k \times n)$ matrix that may depend on X (but not on y).

Using the model $y = X\beta + \varepsilon$,

$$\tilde{\beta} = Cy = C(X\beta + \varepsilon) = CX\beta + C\varepsilon.$$

Step 2: Impose unbiasedness and derive the restriction $CX = I$. Compute the conditional expectation given X :

$$\mathbb{E}[\tilde{\beta} | X] = \mathbb{E}[Cy | X] = C \mathbb{E}[y | X].$$

From $y = X\beta + \varepsilon$ and $\mathbb{E}[\varepsilon | X] = 0$,

$$\mathbb{E}[y | X] = \mathbb{E}[X\beta + \varepsilon | X] = X\beta + \mathbb{E}[\varepsilon | X] = X\beta.$$

Hence

$$\mathbb{E}[Cy | X] = CX\beta.$$

Unbiasedness means $\mathbb{E}[\tilde{\beta} | X] = \beta$ for all β , i.e.

$$CX\beta = \beta \quad \text{for all } \beta.$$

This implies

$$\boxed{CX = I_k.}$$

Therefore we can rewrite

$$\tilde{\beta} = CX\beta + C\varepsilon = \beta + C\varepsilon.$$

Step 3: Compute the conditional variance of $\tilde{\beta}$. Since β is a constant parameter,

$$\text{Var}[\tilde{\beta} | X] = \text{Var}[\beta + C\varepsilon | X] = \text{Var}[C\varepsilon | X].$$

Using $\text{Var}(Az | X) = A \text{Var}(z | X) A'$,

$$\text{Var}[\tilde{\beta} | X] = C \text{Var}(\varepsilon | X) C'.$$

Under Assumption 4, $\text{Var}(\varepsilon | X) = \sigma^2 I_n$, hence

$$\text{Var}[\tilde{\beta} | X] = C(\sigma^2 I_n)C' = \sigma^2 CC'. \quad (\text{GM-1})$$

Step 4: Write the OLS estimator as a linear unbiased estimator. The OLS estimator is

$$\hat{\beta} = (X'X)^{-1}X'y.$$

So it is linear in y with

$$C_0 \equiv (X'X)^{-1}X'.$$

Check unbiasedness:

$$C_0X = (X'X)^{-1}X'X = I_k.$$

Thus OLS belongs to the admissible class.

Also,

$$\text{Var}[\hat{\beta} | X] = \sigma^2(X'X)^{-1} \quad (\text{GM-2})$$

(which follows from the standard variance derivation under Assumption 4).

Step 5: Compare $\text{Var}[\tilde{\beta} | X]$ to $\text{Var}[\hat{\beta} | X]$. Define

$$D \equiv C - (X'X)^{-1}X' = C - C_0.$$

Then

$$C = D + C_0.$$

Now compute DX :

$$DX = CX - C_0X = I_k - (X'X)^{-1}X'X = I_k - I_k = 0. \quad (\text{GM-3})$$

Taking transposes of (GM-3):

$$(DX)' = 0' \implies X'D' = 0.$$

Multiplying by $(X'X)^{-1}$:

$$(X'X)^{-1}X'D' = 0. \quad (\text{GM-4})$$

Step 6: Expand σ^2CC' using $C = D + C_0$ and show cross terms vanish. From (GM-1),

$$\text{Var}[\tilde{\beta} | X] = \sigma^2CC' = \sigma^2(D + C_0)(D + C_0)'$$

Expand:

$$(D + C_0)(D + C_0)' = DD' + DC_0' + C_0D' + C_0C_0'.$$

Compute C_0' :

$$C_0' = ((X'X)^{-1}X')' = X((X'X)^{-1})' = X(X'X)^{-1}.$$

Cross term 1: DC_0' .

$$DC_0' = D(X(X'X)^{-1}) = (DX)(X'X)^{-1} = 0 \cdot (X'X)^{-1} = 0 \quad \text{by (GM-3).}$$

Cross term 2: C_0D' .

$$C_0D' = (X'X)^{-1}X'D' = 0 \quad \text{by (GM-4).}$$

So only two terms remain:

$$(D + C_0)(D + C_0)' = DD' + C_0C_0'.$$

Now compute C_0C_0' :

$$C_0C_0' = ((X'X)^{-1}X')(X(X'X)^{-1}) = (X'X)^{-1}X'X(X'X)^{-1} = (X'X)^{-1}.$$

Therefore,

$$\text{Var}[\tilde{\beta} | X] = \sigma^2(DD' + (X'X)^{-1}) = \sigma^2DD' + \sigma^2(X'X)^{-1}.$$

Using (GM-2),

$$\boxed{\text{Var}[\tilde{\beta} | X] = \sigma^2DD' + \text{Var}[\hat{\beta} | X].}$$

Hence,

$$\boxed{\text{Var}[\tilde{\beta} | X] - \text{Var}[\hat{\beta} | X] = \sigma^2DD'.} \quad (\text{GM-5})$$

Step 7: Show $\sigma^2 DD'$ is nonnegative definite (positive semidefinite). Take any nonzero vector $q \neq 0$ (conformable with DD'). Then

$$q'(DD')q = q'D(D'q).$$

Let

$$v \equiv D'q.$$

Then

$$q'(DD')q = q'Dv = (D'q)'(D'q) = v'v \geq 0.$$

Multiplying by $\sigma^2 \geq 0$,

$$q'(\sigma^2 DD')q = \sigma^2 v'v \geq 0.$$

Therefore $\sigma^2 DD'$ is a nonnegative definite (positive semidefinite) matrix, and by (GM-5),

$$\text{Var}[\tilde{\beta} \mid X] - \text{Var}[\hat{\beta} \mid X] \text{ is nonnegative definite.}$$

Conclusion. Thus, among all linear unbiased estimators $\tilde{\beta} = Cy$ satisfying $CX = I_k$, the OLS estimator $\hat{\beta} = (X'X)^{-1}X'y$ has the smallest conditional variance. Hence, $\hat{\beta}$ is BLUE.

Probability Limits and Consistency of OLS

Assumptions (as used for consistency)

1. Linearity

$$y = X\beta + \varepsilon, \quad \text{equivalently } y_i = x_i'\beta + \varepsilon_i, \quad i = 1, \dots, n.$$

3. Exogeneity

$$\mathbb{E}[\varepsilon \mid X] = 0.$$

Law of iterated expectations (as written in the notes):

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]].$$

Apply to ε :

$$\mathbb{E}[\varepsilon] = \mathbb{E}[\mathbb{E}[\varepsilon \mid X]] = \mathbb{E}[0] = 0.$$

4. Homoscedasticity and nonautocorrelation

$$\text{Var}(\varepsilon_i \mid X) = \sigma^2, \quad i = 1, \dots, n,$$

$$\text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0, \quad i \neq j.$$

More process (derive the matrix form).

Step 4.1: Stack the error terms.

$$\varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix} \in \mathbb{R}^{n \times 1}.$$

Step 4.2: Recognize the entries of $\varepsilon\varepsilon'$. The (i, j) entry of $\varepsilon\varepsilon'$ is

$$(\varepsilon\varepsilon')_{ij} = \varepsilon_i\varepsilon_j.$$

Step 4.3: Take conditional expectations entry-by-entry. The (i, j) entry of $\mathbb{E}[\varepsilon\varepsilon' \mid X]$ is

$$(\mathbb{E}[\varepsilon\varepsilon' \mid X])_{ij} = \mathbb{E}[\varepsilon_i\varepsilon_j \mid X].$$

Step 4.4: Use Assumption 3 to simplify second moments. Since $\mathbb{E}[\varepsilon_i \mid X] = 0$ for all i , we have

$$\text{Var}(\varepsilon_i \mid X) = \mathbb{E}[\varepsilon_i^2 \mid X] - (\mathbb{E}[\varepsilon_i \mid X])^2 = \mathbb{E}[\varepsilon_i^2 \mid X],$$

and for $i \neq j$,

$$\text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = \mathbb{E}[\varepsilon_i\varepsilon_j \mid X] - \mathbb{E}[\varepsilon_i \mid X]\mathbb{E}[\varepsilon_j \mid X] = \mathbb{E}[\varepsilon_i\varepsilon_j \mid X].$$

Step 4.5: Plug in homoscedasticity and zero covariance. Therefore,

$$\mathbb{E}[\varepsilon_i^2 \mid X] = \sigma^2 \quad (i = j), \quad \mathbb{E}[\varepsilon_i\varepsilon_j \mid X] = 0 \quad (i \neq j).$$

Step 4.6: Conclude the matrix form. Hence $\mathbb{E}[\varepsilon\varepsilon' \mid X]$ has σ^2 on the diagonal and 0 off-diagonal:

$$\boxed{\mathbb{E}[\varepsilon\varepsilon' \mid X] = \sigma^2 I_n.}$$

7. Random sampling

$(y_i, x_i, \varepsilon_i)$, $i = 1, \dots, n$ is a sequence of i.i.d. observations, with $\mathbb{E}[x_i x_i'] = Q < \infty$ and Q^{-1} exists.

Definitions

Convergence in probability

A sequence of random variables $\{x_n\}$ is said to converge to a constant c *in probability* if, for every $\epsilon > 0$,

$$\Pr(|x_n - c| < \epsilon) \rightarrow 1 \quad \text{as } n \rightarrow \infty.$$

Equivalently,

$$\Pr(|x_n - c| \geq \epsilon) \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

and we write

$$x_n \xrightarrow{p} c.$$

Consistency

An estimator $\hat{\theta}_n$ of θ is called *consistent* if

$$\hat{\theta}_n \xrightarrow{p} \theta \quad \text{as } n \rightarrow \infty.$$

Theorems

A sufficient condition for convergence in probability

Theorem. If $\mathbb{E}[x_n] \rightarrow c$ and $\text{Var}(x_n) \rightarrow 0$ as $n \rightarrow \infty$, then $x_n \xrightarrow{p} c$.

More process (one standard proof).

Step T1: Consider the mean squared error around c .

$$\mathbb{E}[(x_n - c)^2] = \text{Var}(x_n) + (\mathbb{E}[x_n] - c)^2.$$

If $\text{Var}(x_n) \rightarrow 0$ and $\mathbb{E}[x_n] \rightarrow c$, then

$$\mathbb{E}[(x_n - c)^2] \rightarrow 0.$$

Step T2: Apply Markov's inequality to $(x_n - c)^2$. For any $\epsilon > 0$,

$$\Pr(|x_n - c| \geq \epsilon) = \Pr((x_n - c)^2 \geq \epsilon^2) \leq \frac{\mathbb{E}[(x_n - c)^2]}{\epsilon^2} \rightarrow 0.$$

Therefore $\Pr(|x_n - c| < \epsilon) \rightarrow 1$, i.e. $x_n \xrightarrow{p} c$.

(i.i.d.) Law of Large Numbers

Theorem (i.i.d. LLN). If $x_i, i = 1, \dots, n$ are i.i.d. with $\mathbb{E}[x_i] = \mu$ and $\mathbb{E}|x_i| < \infty$, then

$$\frac{1}{n} \sum_{i=1}^n x_i \xrightarrow{p} \mu \quad \text{as } n \rightarrow \infty.$$

Continuous mapping theorem

Theorem. If $x_n \xrightarrow{p} c$ and $f(\cdot)$ is continuous at c , then

$$f(x_n) \xrightarrow{p} f(c).$$

Operations on convergent random matrices

Theorem. Suppose X_n, Y_n, W_n are random matrices such that

$$X_n \xrightarrow{p} A, \quad Y_n \xrightarrow{p} B, \quad W_n \xrightarrow{p} C.$$

Then

$$X_n + Y_n \xrightarrow{p} A + B, \quad X_n Y_n \xrightarrow{p} AB,$$

and

$$W_n^{-1} \xrightarrow{p} C^{-1}, \quad \text{provided that } C^{-1} \text{ exists.}$$

More process (intuition of the inverse part). The inverse map $f(W) = W^{-1}$ is continuous at C whenever C is nonsingular (i.e. $\det(C) \neq 0$). Hence by the continuous mapping theorem,

$$W_n \xrightarrow{p} C \implies W_n^{-1} = f(W_n) \xrightarrow{p} f(C) = C^{-1}.$$

Consistency of OLS

Theorem (as in the notes). Under Assumptions 1, 3, 4, and 7, $\hat{\beta}$ is consistent for β .

Proof (all original steps kept, fully expanded)

Step 1: Start from the OLS formula.

$$\hat{\beta} = (X'X)^{-1}X'y.$$

Step 2: Substitute the linear model $y = X\beta + \varepsilon$.

$$\begin{aligned}\hat{\beta} &= (X'X)^{-1}X'(X\beta + \varepsilon) \\ &= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'\varepsilon \\ &= \beta + (X'X)^{-1}X'\varepsilon,\end{aligned}$$

where we used $(X'X)^{-1}X'X = I_k$.

Step 3: Show $\mathbb{E}[x_i\varepsilon_i] = 0$ using iterated expectations (as in the notes).

First note that by Assumption 3, $\mathbb{E}[\varepsilon_i | X] = 0$. Then

$$\begin{aligned}\mathbb{E}[x_i\varepsilon_i] &= \mathbb{E}[\mathbb{E}[x_i\varepsilon_i | X]] \\ &= \mathbb{E}[x_i \mathbb{E}[\varepsilon_i | X]] \quad (\text{since } x_i \text{ is measurable w.r.t. } X) \\ &= \mathbb{E}[x_i \cdot 0] \\ &= 0.\end{aligned}$$

Also (from Assumption 7),

$$\mathbb{E}[x_ix_i'] = Q, \quad Q < \infty, \quad Q^{-1} \text{ exists.}$$

Step 4: Express $\hat{\beta} - \beta$ in a form suitable for LLN. Start from

$$\hat{\beta} - \beta = (X'X)^{-1}X'\varepsilon.$$

Multiply and divide by n (exact algebra):

$$\hat{\beta} - \beta = \left(\frac{X'X}{n}\right)^{-1} \left(\frac{X'\varepsilon}{n}\right).$$

Now rewrite the two terms as sample averages:

Step 4.1: Expand $\frac{1}{n}X'X$ as a sum. Since X stacks rows x_i' , we have

$$X'X = \sum_{i=1}^n x_ix_i', \quad \Rightarrow \quad \frac{1}{n}X'X = \frac{1}{n} \sum_{i=1}^n x_ix_i'.$$

Step 4.2: Expand $\frac{1}{n}X'\varepsilon$ as a sum. Similarly,

$$X'\varepsilon = \sum_{i=1}^n x_i\varepsilon_i, \quad \Rightarrow \quad \frac{1}{n}X'\varepsilon = \frac{1}{n} \sum_{i=1}^n x_i\varepsilon_i.$$

Therefore,

$$\hat{\beta} - \beta = \left(\frac{1}{n} \sum_{i=1}^n x_ix_i'\right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n x_i\varepsilon_i\right). \quad (\star)$$

Step 5: Apply LLN to each sample average.

Step 5.1: LLN for $\frac{1}{n} \sum x_i x'_i$. By Assumption 7 (i.i.d. sampling) and $\mathbb{E}[x_i x'_i] = Q$, the LLN implies

$$\frac{1}{n} \sum_{i=1}^n x_i x'_i \xrightarrow{p} \mathbb{E}[x_i x'_i] = Q.$$

Step 5.2: LLN for $\frac{1}{n} \sum x_i \varepsilon_i$. Because (x_i, ε_i) are i.i.d. and $\mathbb{E}[x_i \varepsilon_i] = 0$ (Step 3), the LLN implies

$$\frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i \xrightarrow{p} \mathbb{E}[x_i \varepsilon_i] = 0.$$

Step 6: Take probability limits using continuous mapping + matrix operations.

Step 6.1: Invert the first limit. Since

$$\frac{1}{n} \sum_{i=1}^n x_i x'_i \xrightarrow{p} Q \quad \text{and} \quad Q^{-1} \text{ exists,}$$

by the “inverse” part of the matrix theorem (or continuous mapping),

$$\left(\frac{1}{n} \sum_{i=1}^n x_i x'_i \right)^{-1} \xrightarrow{p} Q^{-1}.$$

Step 6.2: Multiply the limits. From the matrix product rule, if $A_n \xrightarrow{p} A$ and $B_n \xrightarrow{p} B$, then $A_n B_n \xrightarrow{p} AB$. Apply this to $(\star)$:

$$\hat{\beta} - \beta = \underbrace{\left(\frac{1}{n} \sum_{i=1}^n x_i x'_i \right)^{-1}}_{\xrightarrow{p} Q^{-1}} \underbrace{\left(\frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i \right)}_{\xrightarrow{p} 0} \xrightarrow{p} Q^{-1} \cdot 0 = 0.$$

Step 7: Conclude consistency. Since $\hat{\beta} - \beta \xrightarrow{p} 0$, we have

$$\boxed{\hat{\beta} \xrightarrow{p} \beta.}$$

Convergence in Distribution and Asymptotic Normality of OLS

Assumptions

1. Linearity

$$y = X\beta + \varepsilon \quad (\text{equivalently } y_i = x'_i \beta + \varepsilon_i, \ i = 1, \dots, n).$$

3. Exogeneity

$$\mathbb{E}[\varepsilon \mid X] = 0.$$

Law of iterated expectations (as written in the notes):

$$\mathbb{E}[y] = \mathbb{E}[\mathbb{E}[y \mid X]].$$

Applying to ε :

$$\mathbb{E}[\varepsilon] = \mathbb{E}[\mathbb{E}[\varepsilon \mid X]] = \mathbb{E}[0] = 0.$$

4. Homoscedasticity and nonautocorrelation

$$\text{Var}(\varepsilon_i | X) = \sigma^2, \quad i = 1, \dots, n,$$

$$\text{Cov}(\varepsilon_i, \varepsilon_j | X) = 0, \quad i \neq j.$$

More process (derive the matrix form):

Step 4.1: Stack the errors.

$$\varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix} \in \mathbb{R}^{n \times 1}.$$

Step 4.2: Note the entries of $\varepsilon\varepsilon'$. The (i, j) element of $\varepsilon\varepsilon'$ is

$$(\varepsilon\varepsilon')_{ij} = \varepsilon_i \varepsilon_j.$$

Step 4.3: Take conditional expectation element-by-element. The (i, j) element of $\mathbb{E}[\varepsilon\varepsilon' | X]$ is

$$(\mathbb{E}[\varepsilon\varepsilon' | X])_{ij} = \mathbb{E}[\varepsilon_i \varepsilon_j | X].$$

Step 4.4: Use exogeneity to simplify second moments. Since $\mathbb{E}[\varepsilon_i | X] = 0$, we have

$$\text{Var}(\varepsilon_i | X) = \mathbb{E}[\varepsilon_i^2 | X] - (\mathbb{E}[\varepsilon_i | X])^2 = \mathbb{E}[\varepsilon_i^2 | X],$$

and for $i \neq j$,

$$\text{Cov}(\varepsilon_i, \varepsilon_j | X) = \mathbb{E}[\varepsilon_i \varepsilon_j | X] - \mathbb{E}[\varepsilon_i | X] \mathbb{E}[\varepsilon_j | X] = \mathbb{E}[\varepsilon_i \varepsilon_j | X].$$

Step 4.5: Plug in homoscedasticity and zero covariance. Thus

$$\mathbb{E}[\varepsilon_i^2 | X] = \sigma^2 \quad (i = j), \quad \mathbb{E}[\varepsilon_i \varepsilon_j | X] = 0 \quad (i \neq j).$$

Step 4.6: Conclude the matrix form. Hence $\mathbb{E}[\varepsilon\varepsilon' | X]$ has σ^2 on the diagonal and 0 off-diagonal:

$$\boxed{\mathbb{E}[\varepsilon\varepsilon' | X] = \sigma^2 I_n.}$$

7. Random sampling

$(y_i, x_i, \varepsilon_i)$, $i = 1, \dots, n$ is a sequence of independent and identically distributed observations,

with

$$\mathbb{E}[x_i x_i'] = Q < \infty, \quad Q^{-1} \text{ exists.}$$

Theorem (Consistency, as stated in the notes)

Under Assumptions 1, 3, 4, and 7, $\hat{\beta}$ is consistent for β , i.e.

$$\hat{\beta} \xrightarrow{p} \beta.$$

Definition: Convergence in distribution

Let $x_1, x_2, \dots$ be a sequence of random variables with distribution functions

$$F_n(z) = \Pr(x_n \leq z).$$

Let x be a random variable with distribution function $F(z)$. We say x_n converges to x **in distribution** if

$$x_n \xrightarrow{d} x \iff \lim_{n \rightarrow \infty} |F_n(z) - F(z)| = 0$$

for every z at which $F(z)$ is continuous.

Theorem (Slutsky-type results, as written in the notes)

(1) If $x_n \xrightarrow{d} x$ and $y_n \xrightarrow{p} c$ (a constant), then

$$x_n y_n \xrightarrow{d} cx.$$

Also,

$$x_n + y_n \xrightarrow{d} x + c, \quad \frac{x_n}{y_n} \xrightarrow{d} \frac{x}{c} \quad \text{if } c \neq 0.$$

(2) If $x_n \xrightarrow{d} x$ and $g(\cdot)$ is a continuous function, then

$$g(x_n) \xrightarrow{d} g(x).$$

Theorem (Central Limit Theorem for i.i.d. observations, as written in the notes)

Suppose $x_1, x_2, \dots$ are i.i.d. k -vectors with mean μ and variance matrix Σ :

$$\mathbb{E}[x_i] = \mu, \quad \text{Var}(x_i) = \Sigma.$$

Let

$$\bar{x}_n = \frac{1}{n} \sum_{i=1}^n x_i.$$

Then

$$\sqrt{n}(\bar{x}_n - \mu) \xrightarrow{d} N(0, \Sigma).$$

Asymptotic normality of OLS (full process, matching the notes)

Step 1: Start from the OLS formula.

$$\hat{\beta} = (X'X)^{-1}X'y.$$

Step 2: Substitute $y = X\beta + \varepsilon$.

$$\begin{aligned} \hat{\beta} &= (X'X)^{-1}X'(X\beta + \varepsilon) \\ &= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'\varepsilon \\ &= \beta + (X'X)^{-1}X'\varepsilon. \end{aligned}$$

So

$$\hat{\beta} - \beta = (X'X)^{-1}X'\varepsilon.$$

Step 3: Multiply by $\sqrt{n}$ and rewrite in “LLN $\times$ CLT” form.

$$\sqrt{n}(\hat{\beta} - \beta) = \sqrt{n}(X'X)^{-1}X'\varepsilon.$$

Insert $\frac{1}{n}$ and $\frac{1}{\sqrt{n}}$ exactly:

$$\sqrt{n}(\hat{\beta} - \beta) = \left(\frac{1}{n}X'X\right)^{-1} \left(\frac{1}{\sqrt{n}}X'\varepsilon\right).$$

Now rewrite each term as a sum over i .

Step 3.1: Expand $\frac{1}{n}X'X$. Since X stacks rows x'_i , we have

$$X'X = \sum_{i=1}^n x_i x'_i, \quad \Rightarrow \quad \frac{1}{n}X'X = \frac{1}{n} \sum_{i=1}^n x_i x'_i.$$

Step 3.2: Expand $\frac{1}{\sqrt{n}}X'\varepsilon$. Also

$$X'\varepsilon = \sum_{i=1}^n x_i \varepsilon_i, \quad \Rightarrow \quad \frac{1}{\sqrt{n}}X'\varepsilon = \frac{1}{\sqrt{n}} \sum_{i=1}^n x_i \varepsilon_i.$$

Equivalently (as written in the notes),

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n x_i \varepsilon_i = \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i \right).$$

Therefore,

$$\boxed{\sqrt{n}(\hat{\beta} - \beta) = \left(\frac{1}{n} \sum_{i=1}^n x_i x'_i\right)^{-1} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n x_i \varepsilon_i\right)}. \quad (\star)$$

Step 4: Apply i.i.d. LLN to the first term. By Assumption 7 and $\mathbb{E}[x_i x'_i] = Q$,

$$\frac{1}{n} \sum_{i=1}^n x_i x'_i \xrightarrow{p} \mathbb{E}[x_i x'_i] = Q.$$

Since Q^{-1} exists, by continuity of the inverse map (or Slutsky/continuous mapping),

$$\left(\frac{1}{n} \sum_{i=1}^n x_i x'_i\right)^{-1} \xrightarrow{p} Q^{-1}.$$

Step 5: Prepare the CLT for the second term by computing mean and variance. Define the $k \times 1$ random vector

$$u_i \equiv x_i \varepsilon_i.$$

Then the second factor in $(\star)$ is

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n x_i \varepsilon_i = \frac{1}{\sqrt{n}} \sum_{i=1}^n u_i = \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n u_i \right).$$

Step 5.1: Compute $\mathbb{E}[x_i \varepsilon_i]$ (as written in the notes).

$$\begin{aligned}\mathbb{E}[x_i \varepsilon_i] &= \mathbb{E}[\mathbb{E}[x_i \varepsilon_i \mid X]] \\ &= \mathbb{E}[x_i \mathbb{E}[\varepsilon_i \mid X]] \\ &= \mathbb{E}[x_i \cdot 0] \\ &= 0.\end{aligned}$$

So $\mathbb{E}[u_i] = 0$.

Step 5.2: Compute $\text{Var}(x_i \varepsilon_i)$ (fully expanded). Since $\mathbb{E}[u_i] = 0$,

$$\text{Var}(u_i) = \mathbb{E}[u_i u_i'].$$

Now

$$u_i u_i' = (x_i \varepsilon_i)(x_i \varepsilon_i)' = x_i \varepsilon_i \varepsilon_i' x_i' = x_i x_i' \varepsilon_i^2.$$

Thus

$$\text{Var}(u_i) = \mathbb{E}[x_i x_i' \varepsilon_i^2].$$

Use iterated expectations:

$$\begin{aligned}\mathbb{E}[x_i x_i' \varepsilon_i^2] &= \mathbb{E}[\mathbb{E}[x_i x_i' \varepsilon_i^2 \mid X]] \\ &= \mathbb{E}[x_i x_i' \mathbb{E}[\varepsilon_i^2 \mid X]].\end{aligned}$$

Under homoscedasticity,

$$\mathbb{E}[\varepsilon_i^2 \mid X] = \sigma^2.$$

Hence

$$\mathbb{E}[x_i x_i' \varepsilon_i^2] = \mathbb{E}[x_i x_i' \sigma^2] = \sigma^2 \mathbb{E}[x_i x_i'] = \sigma^2 Q.$$

Therefore,

$$\boxed{\text{Var}(x_i \varepsilon_i) = \sigma^2 Q.}$$

Step 6: Apply the i.i.d. multivariate CLT to the second term. Since $u_i = x_i \varepsilon_i$ are i.i.d. with mean 0 and variance $\sigma^2 Q$, the CLT implies

$$\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i \right) = \frac{1}{\sqrt{n}} \sum_{i=1}^n u_i \xrightarrow{d} N(0, \sigma^2 Q).$$

Equivalently (matching the notes line),

$$\boxed{\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i \right) \xrightarrow{d} N(0, \sigma^2 Q).}$$

Step 7: Combine the two limits using Slutsky. From Step 4,

$$\left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1} \xrightarrow{p} Q^{-1},$$

and from Step 6,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n x_i \varepsilon_i \xrightarrow{d} N(0, \sigma^2 Q).$$

By Slutsky (product form),

$$\left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n x_i \varepsilon_i \right) \xrightarrow{d} Q^{-1} Z, \quad Z \sim N(0, \sigma^2 Q).$$

Step 8: Identify the limiting distribution. If $Z \sim N(0, \Sigma)$ and A is a constant matrix, then

$$AZ \sim N(0, A\Sigma A').$$

Here $A = Q^{-1}$ and $\Sigma = \sigma^2 Q$, so

$$Q^{-1}Z \sim N(0, Q^{-1}(\sigma^2 Q)(Q^{-1})').$$

Since Q is symmetric and Q^{-1} exists (and is also symmetric),

$$Q^{-1}(\sigma^2 Q)(Q^{-1})' = \sigma^2 Q^{-1}QQ^{-1} = \sigma^2 Q^{-1}.$$

Therefore,

$$\boxed{\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2 Q^{-1}).}$$

Rule of thumb (as written in the notes).

$$n > 150.$$

A Consistent Estimator of the Asymptotic Variance for OLS

Assumptions

1. Linearity

$$y = X\beta + \varepsilon.$$

3. Exogeneity

$$\mathbb{E}[\varepsilon \mid X] = 0.$$

Law of iterated expectations (as written in the notes):

$$\mathbb{E}[y] = \mathbb{E}[\mathbb{E}[y \mid X]].$$

Apply it to ε :

$$\mathbb{E}[\varepsilon] = \mathbb{E}[\mathbb{E}[\varepsilon \mid X]] = \mathbb{E}[0] = 0.$$

4. Homoscedasticity and nonautocorrelation

$$\text{Var}(\varepsilon_i \mid X) = \sigma^2, \quad i = 1, \dots, n,$$

$$\text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0, \quad i \neq j,$$

equivalently,

$$\mathbb{E}[\varepsilon\varepsilon' \mid X] = \sigma^2 I_n.$$

7. Random sampling

$(y_i, x_i, \varepsilon_i)$, $i = 1, \dots, n$ is a sequence of independent and identically distributed observations,

with

$$\mathbb{E}[x_i x_i'] = Q < \infty, \quad Q^{-1} \text{ exists.}$$

Theorem (Asymptotic Normality of $\hat{\beta}$)

Assume Assumptions 1, 3, 4, and 7 hold. Then, as $n \rightarrow \infty$,

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2 Q^{-1}).$$

We need a **consistent estimator** of $\sigma^2 Q^{-1}$.

Candidate estimator

Candidate:

$$s^2 \left(\frac{X'X}{n} \right)^{-1} = s^2 \left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1}, \quad \text{where } s^2 = \frac{e'e}{n-k}.$$

Step 1: Show $\left(\frac{X'X}{n} \right)^{-1} \xrightarrow{p} Q^{-1}$

By i.i.d. LLN and $\mathbb{E}[x_i x_i'] = Q$,

$$\frac{1}{n} X'X = \frac{1}{n} \sum_{i=1}^n x_i x_i' \xrightarrow{p} \mathbb{E}[x_i x_i'] = Q.$$

Since Q^{-1} exists, the inverse map is continuous at Q , hence by continuous mapping (matrix inverse),

$$\left(\frac{1}{n} X'X \right)^{-1} \xrightarrow{p} Q^{-1}.$$

Step 2: Rewrite s^2 in a form suitable for probability limits

Step 2.1: Define the residual-maker matrix and express residuals. Define the hat matrix and residual-maker matrix:

$$P \equiv X(X'X)^{-1}X', \quad M \equiv I - P = I - X(X'X)^{-1}X'.$$

The OLS residual vector is

$$e \equiv y - X\hat{\beta}.$$

Using $\hat{y} = Py = X\hat{\beta}$, we have

$$e = y - \hat{y} = y - Py = (I - P)y = My.$$

Now substitute the model $y = X\beta + \varepsilon$:

$$e = My = M(X\beta + \varepsilon) = MX\beta + M\varepsilon.$$

But

$$MX = (I - P)X = X - PX = X - X = 0,$$

so $MX\beta = 0$ and therefore

$$\boxed{e = M\varepsilon.}$$

Step 2.2: Express $e'e$ as a quadratic form in ε .

$$e'e = (M\varepsilon)'(M\varepsilon) = \varepsilon'M'M\varepsilon.$$

Since M is symmetric and idempotent ($M' = M$ and $M^2 = M$),

$$\varepsilon'M'M\varepsilon = \varepsilon'M\varepsilon.$$

Hence

$$\boxed{e'e = \varepsilon'M\varepsilon.}$$

Step 2.3: Expand s^2 step-by-step. Start from

$$s^2 = \frac{e'e}{n-k}.$$

Substitute $e'e = \varepsilon'M\varepsilon$:

$$s^2 = \frac{1}{n-k} \varepsilon'M\varepsilon.$$

Substitute $M = I - X(X'X)^{-1}X'$:

$$s^2 = \frac{1}{n-k} \varepsilon' \left[I - X(X'X)^{-1}X' \right] \varepsilon.$$

Distribute $\varepsilon'(\cdot)\varepsilon$:

$$s^2 = \frac{1}{n-k} \left[\varepsilon'\varepsilon - \varepsilon'X(X'X)^{-1}X'\varepsilon \right].$$

Now rescale by n (exact algebra): multiply and divide inside by n :

$$s^2 = \frac{n}{n-k} \left[\frac{\varepsilon'\varepsilon}{n} - \frac{\varepsilon'X}{n} \left(\frac{X'X}{n} \right)^{-1} \frac{X'\varepsilon}{n} \right].$$

(This uses the identity $(X'X)^{-1} = \left(\frac{X'X}{n} \right)^{-1} \frac{1}{n}$ in the product, so the n 's cancel correctly in the middle term.)

Step 3: Show $s^2 \xrightarrow{p} \sigma^2$ by taking limits term-by-term

Step 3.1: $\frac{n}{n-k} \rightarrow 1$. As $n \rightarrow \infty$ with fixed k ,

$$\frac{n}{n-k} \rightarrow 1.$$

Step 3.2: Show $\frac{\varepsilon'\varepsilon}{n} \xrightarrow{p} \sigma^2$. First compute the moment needed for LLN. Using iterated expectations:

$$\mathbb{E}[\varepsilon_i^2] = \mathbb{E}[\mathbb{E}[\varepsilon_i^2 | X]] = \mathbb{E}[\text{Var}(\varepsilon_i | X) + (\mathbb{E}[\varepsilon_i | X])^2].$$

By Assumption 4, $\text{Var}(\varepsilon_i | X) = \sigma^2$, and by Assumption 3, $\mathbb{E}[\varepsilon_i | X] = 0$, hence

$$\mathbb{E}[\varepsilon_i^2] = \mathbb{E}[\sigma^2 + 0] = \sigma^2.$$

Also

$$\mathbb{E}[|\varepsilon_i^2|] = \mathbb{E}[\varepsilon_i^2] = \sigma^2 < \infty.$$

Therefore, by i.i.d. LLN,

$$\frac{\varepsilon'\varepsilon}{n} = \frac{1}{n} \sum_{i=1}^n \varepsilon_i^2 \xrightarrow{p} \mathbb{E}[\varepsilon_i^2] = \sigma^2.$$

Step 3.3: Show $\frac{\varepsilon'X}{n} \xrightarrow{p} 0$ and $\frac{X'\varepsilon}{n} \xrightarrow{p} 0$. First compute $\mathbb{E}[x_i\varepsilon_i]$:

$$\mathbb{E}[x_i\varepsilon_i] = \mathbb{E}[\mathbb{E}[x_i\varepsilon_i \mid X]] = \mathbb{E}[x_i \mathbb{E}[\varepsilon_i \mid X]] = \mathbb{E}[x_i \cdot 0] = 0.$$

To apply LLN componentwise, fix $\ell \in \{1, \dots, k\}$ and consider $x_{i\ell}\varepsilon_i$. We need $\mathbb{E}|x_{i\ell}\varepsilon_i| < \infty$. By Cauchy–Schwarz,

$$\mathbb{E}[|x_{i\ell}\varepsilon_i|] \leq (\mathbb{E}[\varepsilon_i^2])^{1/2} (\mathbb{E}[x_{i\ell}^2])^{1/2} < \infty,$$

where $\mathbb{E}[\varepsilon_i^2] = \sigma^2 < \infty$ and $\mathbb{E}[x_{i\ell}^2] < \infty$ is implied by $\mathbb{E}[x_i x_i'] = Q < \infty$.

Hence, by i.i.d. LLN (componentwise),

$$\frac{\varepsilon'X}{n} = \frac{1}{n} \sum_{i=1}^n x_i' \varepsilon_i \xrightarrow{p} \mathbb{E}[x_i' \varepsilon_i] = 0',$$

and similarly,

$$\frac{X'\varepsilon}{n} = \frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i \xrightarrow{p} \mathbb{E}[x_i \varepsilon_i] = 0.$$

Step 3.4: Show $\frac{X'X}{n} \xrightarrow{p} Q$ and $\left(\frac{X'X}{n}\right)^{-1} \xrightarrow{p} Q^{-1}$. For $\ell, m \in \{1, \dots, k\}$, by Cauchy–Schwarz,

$$\mathbb{E}[|x_{i\ell}x_{im}|] \leq (\mathbb{E}[x_{i\ell}^2])^{1/2} (\mathbb{E}[x_{im}^2])^{1/2} < \infty.$$

Thus LLN applies element-by-element to $x_i x_i'$, giving

$$\frac{X'X}{n} = \frac{1}{n} \sum_{i=1}^n x_i x_i' \xrightarrow{p} \mathbb{E}[x_i x_i'] = Q.$$

Since Q^{-1} exists,

$$\left(\frac{X'X}{n}\right)^{-1} \xrightarrow{p} Q^{-1}.$$

Step 3.5: Combine the limits to get $s^2 \xrightarrow{p} \sigma^2$. We have shown:

$$\frac{n}{n-k} \rightarrow 1, \quad \frac{\varepsilon'\varepsilon}{n} \xrightarrow{p} \sigma^2, \quad \frac{\varepsilon'X}{n} \xrightarrow{p} 0', \quad \left(\frac{X'X}{n}\right)^{-1} \xrightarrow{p} Q^{-1}, \quad \frac{X'\varepsilon}{n} \xrightarrow{p} 0.$$

Therefore,

$$s^2 = \frac{n}{n-k} \left[\frac{\varepsilon'\varepsilon}{n} - \frac{\varepsilon'X}{n} \left(\frac{X'X}{n}\right)^{-1} \frac{X'\varepsilon}{n} \right] \xrightarrow{p} 1 \cdot [\sigma^2 - 0' \cdot Q^{-1} \cdot 0] = \sigma^2.$$

Step 4: Conclude consistency of $s^2 \left(\frac{X'X}{n}\right)^{-1}$

Since

$$s^2 \xrightarrow{p} \sigma^2 \quad \text{and} \quad \left(\frac{X'X}{n}\right)^{-1} \xrightarrow{p} Q^{-1},$$

by the product rule (Slutsky / continuous mapping for products),

$$\boxed{s^2 \left(\frac{X'X}{n}\right)^{-1} \xrightarrow{p} \sigma^2 Q^{-1}.}$$

Econometrics I: Delta Method and Asymptotic Distribution of Functions of $\hat{\beta}$

Background asymptotic normality (given)

We start from the (previously derived) asymptotic normality of OLS:

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2 Q^{-1}).$$

Theorem (Delta method: multivariate / vector-valued version)

Theorem (Delta method). Suppose Z_n is a sequence of k -vector random variables such that

$$\sqrt{n}(Z_n - \mu) \xrightarrow{d} N(0, \Sigma) \quad \text{as } n \rightarrow \infty.$$

Let $c(z)$ be a J -vector valued continuously differentiable function, and let

$$C(z) \equiv \frac{\partial c(z)}{\partial z'}$$

be the $J \times k$ Jacobian matrix. Then

$$\sqrt{n}[c(Z_n) - c(\mu)] \xrightarrow{d} N(0, C(\mu) \Sigma C(\mu)').$$

Proof (fully expanded). Because $c(\cdot)$ is continuously differentiable at μ , we can use the first-order multivariate Taylor expansion around μ :

$$c(Z_n) = c(\mu) + C(\mu)(Z_n - \mu) + r_n,$$

where the remainder r_n satisfies

$$\|r_n\| = o(\|Z_n - \mu\|) \quad \text{as } Z_n \rightarrow \mu.$$

Equivalently (in probability form),

$$r_n = o_p(\|Z_n - \mu\|).$$

Subtract $c(\mu)$ from both sides:

$$c(Z_n) - c(\mu) = C(\mu)(Z_n - \mu) + r_n.$$

Multiply by $\sqrt{n}$:

$$\sqrt{n}[c(Z_n) - c(\mu)] = C(\mu) \sqrt{n}(Z_n - \mu) + \sqrt{n}r_n.$$

Now use the fact that $Z_n - \mu = O_p(n^{-1/2})$ when $\sqrt{n}(Z_n - \mu)$ has a non-degenerate limit. Hence

$$\|Z_n - \mu\| = O_p(n^{-1/2}) \implies r_n = o_p(n^{-1/2}) \implies \sqrt{n}r_n = o_p(1).$$

Therefore,

$$\sqrt{n}[c(Z_n) - c(\mu)] = C(\mu) \sqrt{n}(Z_n - \mu) + o_p(1).$$

Since

$$\sqrt{n}(Z_n - \mu) \xrightarrow{d} N(0, \Sigma),$$

we have by Slutsky's theorem that adding $o_p(1)$ does not affect the limit:

$$C(\mu) \sqrt{n}(Z_n - \mu) + o_p(1) \xrightarrow{d} C(\mu) Z, \quad Z \sim N(0, \Sigma).$$

Finally, if $Z \sim N(0, \Sigma)$ and A is a constant matrix, then $AZ \sim N(0, A\Sigma A')$. Taking $A = C(\mu)$ yields

$$C(\mu) Z \sim N(0, C(\mu)\Sigma C(\mu)'),$$

which proves the theorem.

Theorem (Delta method: scalar version)**Theorem (Delta method).** Suppose Y_n is a sequence of random variables such that

$$\sqrt{n}(Y_n - \theta) \xrightarrow{d} N(0, \sigma^2) \quad \text{as } n \rightarrow \infty.$$

Let g be a function and assume $g'(\theta)$ exists and is not 0. Then

$$\sqrt{n}[g(Y_n) - g(\theta)] \xrightarrow{d} N(0, \sigma^2 [g'(\theta)]^2).$$

Proof (matching the notes, with added steps). Step 1 (first-order Taylor expansion).A first-order Taylor expansion of $g(\cdot)$ around θ gives

$$g(Y_n) \approx g(\theta) + g'(\theta)(Y_n - \theta).$$

More formally, write

$$g(Y_n) = g(\theta) + g'(\theta)(Y_n - \theta) + r_n, \quad \text{where } r_n = o_p(|Y_n - \theta|).$$

Step 2 (multiply by $\sqrt{n}$). Subtract $g(\theta)$ and multiply by $\sqrt{n}$:

$$\sqrt{n}[g(Y_n) - g(\theta)] = g'(\theta) \sqrt{n}(Y_n - \theta) + \sqrt{n} r_n.$$

Since $\sqrt{n}(Y_n - \theta)$ has a non-degenerate limit, we have $|Y_n - \theta| = O_p(n^{-1/2})$, so

$$r_n = o_p(n^{-1/2}) \implies \sqrt{n} r_n = o_p(1).$$

Thus

$$\sqrt{n}[g(Y_n) - g(\theta)] = g'(\theta) \sqrt{n}(Y_n - \theta) + o_p(1).$$

Step 3 (apply the given limit + Slutsky). Let $Z \sim N(0, \sigma^2)$ denote the limit of $\sqrt{n}(Y_n - \theta)$:

$$\sqrt{n}(Y_n - \theta) \xrightarrow{d} Z.$$

Then, by Slutsky,

$$g'(\theta) \sqrt{n}(Y_n - \theta) + o_p(1) \xrightarrow{d} g'(\theta) Z.$$

Because $Z \sim N(0, \sigma^2)$ and multiplying a normal by a constant scales the variance by the square:

$$g'(\theta) Z \sim N(0, \sigma^2 [g'(\theta)]^2).$$

(Expectation/variance intuition as in the notes).

First-order approximation:

$$g(Y_n) \approx g(\theta) + g'(\theta)(Y_n - \theta).$$

Expectation:

$$\mathbb{E}[g(Y_n)] \approx \mathbb{E}[g(\theta) + g'(\theta)(Y_n - \theta)] = g(\theta) + g'(\theta) \mathbb{E}[Y_n - \theta] = g(\theta),$$

since $\mathbb{E}[Y_n - \theta] = 0$ asymptotically (consistent with $\sqrt{n}(Y_n - \theta)$ centering at 0).*Variance:*

$$\text{Var}[g(Y_n)] = \mathbb{E}\left([g(Y_n) - \mathbb{E}[g(Y_n)]]^2\right) \approx \mathbb{E}\left([g'(\theta)(Y_n - \theta)]^2\right) = [g'(\theta)]^2 \mathbb{E}[(Y_n - \theta)^2].$$

Since $\sqrt{n}(Y_n - \theta) \Rightarrow N(0, \sigma^2)$, we have $\mathbb{E}[(Y_n - \theta)^2] \approx \sigma^2/n$, so the asymptotic variance of $\sqrt{n}(g(Y_n) - g(\theta))$ becomes $\sigma^2 [g'(\theta)]^2$.

Theorem (Asymptotic distribution of a function of $\hat{\beta}$)

Suppose

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2 Q^{-1}).$$

For a $k \times 1$ vector b , let $f(b)$ be a J -vector continuously differentiable function, and let

$$C(b) = \frac{\partial f(b)}{\partial b'}$$

be the $J \times k$ Jacobian matrix. Then

$$\sqrt{n}(f(\hat{\beta}) - f(\beta)) \xrightarrow{d} N(0, C(\beta) \sigma^2 Q^{-1} C(\beta)').$$

Proof (fully expanded; direct application of the delta method). Set in the multivariate delta method:

$$Z_n = \hat{\beta}, \quad \mu = \beta, \quad \Sigma = \sigma^2 Q^{-1}, \quad c(\cdot) = f(\cdot).$$

Then the Jacobian is exactly

$$C(b) = \frac{\partial f(b)}{\partial b'}.$$

Since $\sqrt{n}(Z_n - \mu) = \sqrt{n}(\hat{\beta} - \beta) \Rightarrow N(0, \sigma^2 Q^{-1})$, the delta method yields

$$\sqrt{n}[f(\hat{\beta}) - f(\beta)] \Rightarrow N(0, C(\beta) (\sigma^2 Q^{-1}) C(\beta)') = N(0, C(\beta) \sigma^2 Q^{-1} C(\beta)').$$

Consistent estimation of the asymptotic covariance matrix

The asymptotic covariance matrix is

$$C(\beta) \sigma^2 Q^{-1} C(\beta)'.$$

A consistent estimator is obtained by plug-in:

$$\hat{\Omega} \equiv C(\hat{\beta}) s^2 \left(\frac{X'X}{n} \right)^{-1} C(\hat{\beta})'.$$

Why this is consistent (fully expanded).

Step 1: (Consistency of $\hat{\beta}$.) Under the standard assumptions,

$$\hat{\beta} \xrightarrow{p} \beta.$$

Step 2: (Continuity of $C(\cdot)$.) Since $f(\cdot)$ is continuously differentiable, $C(\cdot)$ is continuous at β . By the continuous mapping theorem,

$$C(\hat{\beta}) \xrightarrow{p} C(\beta).$$

Step 3: (Consistency of s^2 .) With $s^2 = \frac{e'e}{n-k}$ and the usual assumptions,

$$s^2 \xrightarrow{p} \sigma^2.$$

Step 4: (LLN for $X'X/n$.) By i.i.d. sampling and $\mathbb{E}[x_i x_i'] = Q$,

$$\frac{X'X}{n} = \frac{1}{n} \sum_{i=1}^n x_i x_i' \xrightarrow{p} Q.$$

Step 5: (Invert and combine.) Since Q^{-1} exists, the inverse map is continuous at Q , so

$$\left(\frac{X'X}{n} \right)^{-1} \xrightarrow{p} Q^{-1}.$$

Using the product rule for probability limits repeatedly,

$$C(\hat{\beta}) s^2 \left(\frac{X'X}{n} \right)^{-1} C(\hat{\beta})' \xrightarrow{p} C(\beta) \sigma^2 Q^{-1} C(\beta)'.$$

Econometrics I: Confidence Intervals for Regression Coefficients

Recap: sampling variability of $\hat{\beta}$ and standard errors

Start from the linear regression model

$$y = X\beta + \varepsilon.$$

The OLS estimator is

$$\hat{\beta} = (X'X)^{-1} X'y.$$

Substitute $y = X\beta + \varepsilon$:

$$\begin{aligned} \hat{\beta} &= (X'X)^{-1} X'(X\beta + \varepsilon) \\ &= (X'X)^{-1} X'X\beta + (X'X)^{-1} X'\varepsilon \\ &= \beta + (X'X)^{-1} X'\varepsilon. \end{aligned}$$

Hence,

$$\boxed{\hat{\beta} = \beta + (X'X)^{-1} X'\varepsilon.}$$

Under homoscedasticity and no autocorrelation,

$$\mathbb{E}[\varepsilon \varepsilon' \mid X] = \sigma^2 I_n.$$

Then the conditional variance of $\hat{\beta}$ is

$$\text{Var}[\hat{\beta} \mid X] = \sigma^2 (X'X)^{-1}.$$

Define the residuals and the usual estimator of σ^2 :

$$e \equiv y - X\hat{\beta}, \quad s^2 \equiv \frac{e'e}{n-k}.$$

The standard error of the j -th coefficient is

$$se_j \equiv \sqrt{[s^2 (X'X)^{-1}]_{jj}}.$$

Confidence intervals: general idea

A confidence interval (CI) for a parameter θ has the form

$$\theta = \hat{\theta} \pm (\text{sampling variability}).$$

Below we derive CIs for β_j under two setups.

(1) Exact CI under normal errors

Assumption. Assume

$$\varepsilon \mid X \sim N(0, \sigma^2 I_n).$$

Step 1: Distribution of $\hat{\beta} - \beta$ given X . From the decomposition,

$$\hat{\beta} - \beta = (X'X)^{-1}X'\varepsilon.$$

Because $\varepsilon \mid X$ is multivariate normal and $(X'X)^{-1}X'$ is a nonrandom matrix given X , a linear transformation of a normal is normal:

$$\hat{\beta} - \beta \mid X \sim N\left(0, (X'X)^{-1}X'(\sigma^2 I_n)X(X'X)^{-1}\right).$$

Simplify the covariance:

$$(X'X)^{-1}X'(\sigma^2 I_n)X(X'X)^{-1} = \sigma^2(X'X)^{-1}X'X(X'X)^{-1} = \sigma^2(X'X)^{-1}.$$

Therefore,

$$\boxed{\hat{\beta} - \beta \mid X \sim N(0, \sigma^2(X'X)^{-1}).}$$

Step 2: Distribution of the j -th component. Taking the j -th element yields

$$\hat{\beta}_j - \beta_j \mid X \sim N\left(0, [\sigma^2(X'X)^{-1}]_{jj}\right).$$

Standardize by its (conditional) standard deviation:

$$\frac{\hat{\beta}_j - \beta_j}{\sqrt{[\sigma^2(X'X)^{-1}]_{jj}}} \mid X \sim N(0, 1).$$

Step 3: Replace σ^2 by s^2 and obtain a t statistic. Define

$$t \equiv \frac{\hat{\beta}_j - \beta_j}{\sqrt{[s^2(X'X)^{-1}]_{jj}}} = \frac{\hat{\beta}_j - \beta_j}{se_j}.$$

More process (why $t \sim t_{n-k}$ under normality). Let

$$P \equiv X(X'X)^{-1}X', \quad M \equiv I - P.$$

Then $e = My$ and, substituting $y = X\beta + \varepsilon$,

$$e = M(X\beta + \varepsilon) = MX\beta + M\varepsilon = 0 + M\varepsilon = M\varepsilon.$$

Hence

$$e'e = \varepsilon' M' M \varepsilon.$$

Since M is symmetric and idempotent, $M' = M$ and $M^2 = M$, so

$$e'e = \varepsilon' M \varepsilon.$$

Under $\varepsilon \mid X \sim N(0, \sigma^2 I_n)$, the quadratic form

$$\frac{e'e}{\sigma^2} = \frac{\varepsilon' M \varepsilon}{\sigma^2}$$

has a chi-square distribution with $\text{rank}(M) = n - k$ degrees of freedom:

$$\frac{e'e}{\sigma^2} \mid X \sim \chi_{n-k}^2.$$

Equivalently,

$$\frac{(n-k)s^2}{\sigma^2} = \frac{e'e}{\sigma^2} \mid X \sim \chi_{n-k}^2.$$

Moreover, under normality, $\hat{\beta}$ (a function of $P\varepsilon$) is independent of e (a function of $M\varepsilon$) because P and M project onto orthogonal subspaces. Hence the standardized ratio is t :

$$t = \frac{\frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{[(X'X)^{-1}]_{jj}}}}{\sqrt{\frac{(n-k)s^2}{\sigma^2} \cdot \frac{1}{n-k}}} \mid X \sim t_{n-k}.$$

Therefore, we write

$$t = \frac{\hat{\beta}_j - \beta_j}{se_j} \sim t_{n-k}.$$

Step 4: Use a t critical value to build an exact 95% CI. Let $t_{0.975, n-k}$ denote the 97.5th percentile of the $t(n-k)$ distribution. Then

$$\Pr(|t| \leq t_{0.975, n-k}) = 0.95.$$

Equivalently,

$$-t_{0.975, n-k} \leq \frac{\hat{\beta}_j - \beta_j}{se_j} \leq t_{0.975, n-k}.$$

Multiply through by $se_j > 0$:

$$-se_j \cdot t_{0.975, n-k} \leq \hat{\beta}_j - \beta_j \leq se_j \cdot t_{0.975, n-k}.$$

Now rearrange to isolate β_j .

First multiply by -1 (which flips inequalities):

$$-\hat{\beta}_j - se_j \cdot t_{0.975, n-k} \leq -\beta_j \leq -\hat{\beta}_j + se_j \cdot t_{0.975, n-k}.$$

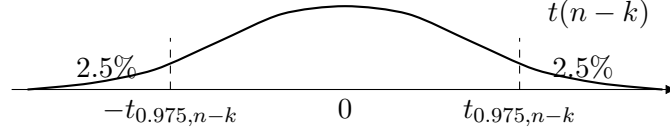
Multiply by -1 again (flip back) and add $\hat{\beta}_j$:

$$\hat{\beta}_j + se_j \cdot t_{0.975, n-k} \geq \beta_j \geq \hat{\beta}_j - se_j \cdot t_{0.975, n-k}.$$

Hence an **exact** 95% confidence interval is

$$\left[\hat{\beta}_j - se_j \cdot t_{0.975, n-k}, \hat{\beta}_j + se_j \cdot t_{0.975, n-k} \right].$$

(Sketch: $t(n-k)$ with two 2.5% tails).



(2) Asymptotic CI without normality

Assumption (no normality). Do *not* assume $\varepsilon \mid X \sim N(0, \sigma^2 I_n)$.

Step 1: Start from asymptotic normality. Under standard regularity conditions (e.g. i.i.d. sampling, exogeneity, finite second moments),

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2 Q^{-1}), \quad Q \equiv \mathbb{E}[x_i x_i'].$$

Thus for the j -th component,

$$\sqrt{n}(\hat{\beta}_j - \beta_j) \xrightarrow{d} N(0, [\sigma^2 Q^{-1}]_{jj}).$$

Standardizing yields

$$\frac{\sqrt{n}(\hat{\beta}_j - \beta_j)}{\sqrt{[\sigma^2 Q^{-1}]_{jj}}} \xrightarrow{d} N(0, 1).$$

Step 2: Show the standard error consistently estimates the asymptotic standard deviation. We use the probability limits (as in the notes):

$$s^2 \xrightarrow{p} \sigma^2, \quad \left(\frac{X'X}{n} \right)^{-1} \xrightarrow{p} Q^{-1}.$$

Hence (by continuous mapping on the square-root and selecting the (j, j) element),

$$\sqrt{[s^2(X'X/n)^{-1}]_{jj}} \xrightarrow{p} \sqrt{[\sigma^2 Q^{-1}]_{jj}}.$$

Now connect this to $se_j = \sqrt{[s^2(X'X)^{-1}]_{jj}}$ by scaling. Use the identity

$$(X'X)^{-1} = \frac{1}{n} \left(\frac{X'X}{n} \right)^{-1}.$$

Therefore,

$$s^2(X'X)^{-1} = s^2 \cdot \frac{1}{n} \left(\frac{X'X}{n} \right)^{-1} = \frac{1}{n} s^2 \left(\frac{X'X}{n} \right)^{-1}.$$

Taking the (j, j) element and square roots,

$$se_j = \sqrt{[s^2(X'X)^{-1}]_{jj}} = \sqrt{\frac{1}{n} [s^2(X'X/n)^{-1}]_{jj}} = \frac{1}{\sqrt{n}} \sqrt{[s^2(X'X/n)^{-1}]_{jj}}.$$

Multiply both sides by $\sqrt{n}$:

$$\sqrt{n} se_j = \sqrt{[s^2(X'X/n)^{-1}]_{jj}} \xrightarrow{p} \sqrt{[\sigma^2 Q^{-1}]_{jj}}.$$

Step 3: Asymptotic distribution of the t -statistic. Combine

$$\sqrt{n}(\hat{\beta}_j - \beta_j) \xrightarrow{d} N(0, [\sigma^2 Q^{-1}]_{jj})$$

and

$$\sqrt{n} se_j \xrightarrow{p} \sqrt{[\sigma^2 Q^{-1}]_{jj}}$$

using Slutsky:

$$\frac{\sqrt{n}(\hat{\beta}_j - \beta_j)}{\sqrt{n} se_j} = \frac{\hat{\beta}_j - \beta_j}{se_j} \xrightarrow{d} N(0, 1).$$

Define

$$t \equiv \frac{\hat{\beta}_j - \beta_j}{se_j}.$$

Let $z_{0.975}$ be the 97.5th percentile of $N(0, 1)$. Then, asymptotically,

$$\Pr(|t| \leq z_{0.975}) \rightarrow 0.95.$$

That is,

$$\left| \frac{\hat{\beta}_j - \beta_j}{se_j} \right| \leq z_{0.975} \quad (\text{approximately for large } n).$$

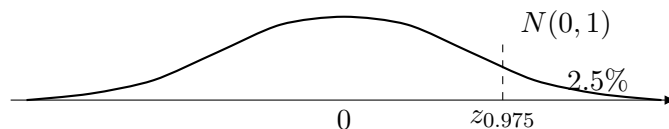
Equivalently,

$$\hat{\beta}_j + se_j \cdot z_{0.975} \geq \beta_j \geq \hat{\beta}_j - se_j \cdot z_{0.975}.$$

Hence an **asymptotic** 95% confidence interval is

$$\left[\hat{\beta}_j - se_j \cdot z_{0.975}, \hat{\beta}_j + se_j \cdot z_{0.975} \right].$$

(Sketch: $N(0, 1)$ with right-tail 2.5% at $z_{0.975}$).



Out-of-Sample Prediction Error and Its Variance

Model and prediction target

Step 0: Sample regression model. For $i = 1, \dots, n$, assume the linear model

$$y_i = x_i' \beta + \varepsilon_i,$$

or in matrix form

$$y = X\beta + \varepsilon.$$

Step 1: A new observation to be predicted. Consider predicting the value of y^0 associated with the regression vector x^0 :

$$y^0 = x^{0'} \beta + \varepsilon^0.$$

Step 2: The OLS predictor. The OLS point prediction for y^0 is

$$\hat{y}^0 = x^{0'}\hat{\beta}.$$

Prediction error

Step 3: Define the prediction error. Define

$$e^0 \equiv \hat{y}^0 - y^0.$$

Substitute $\hat{y}^0 = x^{0'}\hat{\beta}$ and $y^0 = x^{0'}\beta + \varepsilon^0$:

$$\begin{aligned} e^0 &= x^{0'}\hat{\beta} - (x^{0'}\beta + \varepsilon^0) \\ &= x^{0'}\hat{\beta} - x^{0'}\beta - \varepsilon^0 \\ &= x^{0'}(\hat{\beta} - \beta) - \varepsilon^0. \end{aligned} \tag{PE-1}$$

Step 4: Substitute the OLS estimation error. Recall the OLS estimator:

$$\hat{\beta} = (X'X)^{-1}X'y.$$

Using $y = X\beta + \varepsilon$,

$$\begin{aligned} \hat{\beta} &= (X'X)^{-1}X'(X\beta + \varepsilon) \\ &= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'\varepsilon \\ &= \beta + (X'X)^{-1}X'\varepsilon, \end{aligned}$$

so

$$\hat{\beta} - \beta = (X'X)^{-1}X'\varepsilon. \tag{PE-2}$$

Substitute (PE-2) into (PE-1):

$$\boxed{e^0 = x^{0'}(X'X)^{-1}X'\varepsilon - \varepsilon^0.} \tag{PE-3}$$

Conditional mean of the prediction error

Step 5: Compute $\mathbb{E}[e^0 \mid X, x^0]$ (keep all intermediate lines). Start from (PE-1):

$$\mathbb{E}[e^0 \mid X, x^0] = \mathbb{E}[x^{0'}(\hat{\beta} - \beta) - \varepsilon^0 \mid X, x^0].$$

Use the substitution (PE-2):

$$\mathbb{E}[e^0 \mid X, x^0] = \mathbb{E}[x^{0'}(X'X)^{-1}X'\varepsilon - \varepsilon^0 \mid X, x^0].$$

Separate expectations (linearity):

$$\mathbb{E}[e^0 \mid X, x^0] = \mathbb{E}[x^{0'}(X'X)^{-1}X'\varepsilon \mid X, x^0] - \mathbb{E}[\varepsilon^0 \mid X, x^0].$$

Pull out terms that are functions of (X, x^0) only:

$$\mathbb{E}[e^0 \mid X, x^0] = x^{0'}(X'X)^{-1}X' \mathbb{E}[\varepsilon \mid X, x^0] - \mathbb{E}[\varepsilon^0 \mid X, x^0].$$

Under exogeneity for the sample errors, $\mathbb{E}[\varepsilon \mid X] = 0$, hence also $\mathbb{E}[\varepsilon \mid X, x^0] = 0$. Assume the new shock has mean zero conditional on (X, x^0) , i.e. $\mathbb{E}[\varepsilon^0 \mid X, x^0] = 0$. Therefore,

$$\boxed{\mathbb{E}[e^0 \mid X, x^0] = x^{0'}(X'X)^{-1}X' \cdot 0 - 0 = 0.} \tag{PE-Mean}$$

Conditional variance of the prediction error

Step 6: Start from the definition of conditional variance. By definition,

$$\text{Var}(e^0 \mid X, x^0) = \mathbb{E}[(e^0)^2 \mid X, x^0] - \left(\mathbb{E}[e^0 \mid X, x^0]\right)^2.$$

Using (PE-Mean), $\mathbb{E}[e^0 \mid X, x^0] = 0$, so

$$\text{Var}(e^0 \mid X, x^0) = \mathbb{E}[(e^0)^2 \mid X, x^0].$$

Since e^0 is scalar, $(e^0)^2 = e^0 e^{0'}$, hence

$$\text{Var}(e^0 \mid X, x^0) = \mathbb{E}[e^0 e^{0'} \mid X, x^0]. \quad (\text{PE-Var-Def})$$

Step 7: Expand $e^0 e^{0'}$ (keep the same structure as the notes). From (PE-1),

$$e^0 = x^{0'}(\hat{\beta} - \beta) - \varepsilon^0, \quad e^{0'} = (\hat{\beta} - \beta)' x^0 - \varepsilon^{0'}.$$

Therefore,

$$e^0 e^{0'} = \left(x^{0'}(\hat{\beta} - \beta) - \varepsilon^0\right) \left((\hat{\beta} - \beta)' x^0 - \varepsilon^{0'}\right).$$

Multiply out:

$$e^0 e^{0'} = x^{0'}(\hat{\beta} - \beta)(\hat{\beta} - \beta)' x^0 - x^{0'}(\hat{\beta} - \beta)\varepsilon^{0'} - \varepsilon^0(\hat{\beta} - \beta)' x^0 + \varepsilon^0 \varepsilon^{0'}.$$

Because ε^0 is scalar, $\varepsilon^{0'} = \varepsilon^0$, and $x^{0'}(\hat{\beta} - \beta)$ is also scalar and equals $(\hat{\beta} - \beta)' x^0$, so the two middle terms are equal. Hence

$$e^0 e^{0'} = x^{0'}(\hat{\beta} - \beta)(\hat{\beta} - \beta)' x^0 - 2\varepsilon^0(\hat{\beta} - \beta)' x^0 + (\varepsilon^0)^2. \quad (\text{PE-Expand1})$$

Step 8: Substitute $\hat{\beta} - \beta = (X'X)^{-1}X'\varepsilon$ and expand again. Using (PE-2),

$$(\hat{\beta} - \beta) = (X'X)^{-1}X'\varepsilon, \quad (\hat{\beta} - \beta)' = \varepsilon'X(X'X)^{-1}.$$

Plug into the first (quadratic) term in (PE-Expand1):

$$\begin{aligned} x^{0'}(\hat{\beta} - \beta)(\hat{\beta} - \beta)' x^0 &= x^{0'} \left((X'X)^{-1}X'\varepsilon \right) \left(\varepsilon'X(X'X)^{-1} \right) x^0 \\ &= x^{0'}(X'X)^{-1}X'\varepsilon\varepsilon'X(X'X)^{-1}x^0. \end{aligned}$$

Plug into the cross term in (PE-Expand1):

$$(\hat{\beta} - \beta)' x^0 = (\varepsilon'X(X'X)^{-1})x^0 = \varepsilon'X(X'X)^{-1}x^0.$$

So

$$-2\varepsilon^0(\hat{\beta} - \beta)' x^0 = -2\varepsilon^0\varepsilon'X(X'X)^{-1}x^0.$$

Therefore,

$$e^0 e^{0'} = x^{0'}(X'X)^{-1}X'\varepsilon\varepsilon'X(X'X)^{-1}x^0 - 2\varepsilon^0\varepsilon'X(X'X)^{-1}x^0 + (\varepsilon^0)^2. \quad (\text{PE-Expand2})$$

Step 9: Take conditional expectation given (X, x^0) . From (PE-Var-Def) and (PE-Expand2),

$$\text{Var}(e^0 \mid X, x^0) = \mathbb{E}[e^0 e^{0'} \mid X, x^0].$$

Thus,

$$\text{Var}(e^0 \mid X, x^0) = \mathbb{E}\left[x^{0'}(X'X)^{-1}X'\varepsilon\varepsilon'X(X'X)^{-1}x^0 - 2\varepsilon^0\varepsilon'X(X'X)^{-1}x^0 + (\varepsilon^0)^2 \mid X, x^0\right].$$

Use linearity of conditional expectation and pull out terms that depend only on (X, x^0) :

$$\begin{aligned} \text{Var}(e^0 \mid X, x^0) &= x^{0'}(X'X)^{-1}X'\mathbb{E}[\varepsilon\varepsilon' \mid X, x^0]X(X'X)^{-1}x^0 \\ &\quad - 2\mathbb{E}[\varepsilon^0\varepsilon' \mid X, x^0]X(X'X)^{-1}x^0 + \mathbb{E}[(\varepsilon^0)^2 \mid X, x^0]. \end{aligned} \quad (\text{PE-Var1})$$

Step 10: Impose standard assumptions for prediction. Assume (homoscedasticity in the sample)

$$\mathbb{E}[\varepsilon\varepsilon' \mid X, x^0] = \mathbb{E}[\varepsilon\varepsilon' \mid X] = \sigma^2 I_n,$$

assume the new shock has variance σ^2 conditional on (X, x^0) ,

$$\mathbb{E}[(\varepsilon^0)^2 \mid X, x^0] = \sigma^2,$$

and assume the new shock is conditionally uncorrelated with the sample shocks,

$$\mathbb{E}[\varepsilon^0\varepsilon' \mid X, x^0] = 0'.$$

Plug these into (PE-Var1):

$$\text{Var}(e^0 \mid X, x^0) = x^{0'}(X'X)^{-1}X'(\sigma^2 I_n)X(X'X)^{-1}x^0 - 2 \cdot 0' \cdot X(X'X)^{-1}x^0 + \sigma^2.$$

The middle term is zero, so

$$\text{Var}(e^0 \mid X, x^0) = x^{0'}(X'X)^{-1}X'(\sigma^2 I_n)X(X'X)^{-1}x^0 + \sigma^2. \quad (\text{PE-Var2})$$

Step 11: Simplify the matrix product. Because σ^2 is a scalar,

$$X'(\sigma^2 I_n)X = \sigma^2 X'IX = \sigma^2 X'X.$$

Substitute into (PE-Var2):

$$\begin{aligned} \text{Var}(e^0 \mid X, x^0) &= x^{0'}(X'X)^{-1}(\sigma^2 X'X)(X'X)^{-1}x^0 + \sigma^2 \\ &= \sigma^2 x^{0'}(X'X)^{-1}(X'X)(X'X)^{-1}x^0 + \sigma^2 \\ &= \sigma^2 x^{0'}(X'X)^{-1}x^0 + \sigma^2. \end{aligned}$$

Therefore,

$$\boxed{\text{Var}(e^0 \mid X, x^0) = \sigma^2 + \sigma^2 x^{0'}(X'X)^{-1}x^0 = \sigma^2 \left(1 + x^{0'}(X'X)^{-1}x^0\right).} \quad (\text{PE-Final})$$

Step 12: Rewrite using the normalization by n (as in the notes). Use the identity

$$(X'X)^{-1} = \frac{1}{n} \left(\frac{X'X}{n} \right)^{-1} \quad (\text{since } X'X = n \cdot (X'X/n)).$$

Then

$$\sigma^2 x^{0'} (X'X)^{-1} x^0 = \sigma^2 x^{0'} \left[\frac{1}{n} \left(\frac{X'X}{n} \right)^{-1} \right] x^0 = \frac{1}{n} \sigma^2 x^{0'} \left(\frac{X'X}{n} \right)^{-1} x^0.$$

Hence an equivalent expression is

$$\boxed{\text{Var}(e^0 \mid X, x^0) = \sigma^2 + \frac{1}{n} \sigma^2 x^{0'} \left(\frac{X'X}{n} \right)^{-1} x^0.}$$

Hypothesis Testing — Exact and Asymptotic t -Tests

H_0 and H_1

We write

$$H_0 \quad (\text{null hypothesis}), \quad H_1 \quad (\text{alternative hypothesis}).$$

Hypothesis testing (procedure)

1. Construct a statistic T that is a function of the data.
2. Find the distribution of T when H_0 is true.
3. Set the rejection rule: we reject H_0 when $T \in C$, where C is called the region of rejections.

(1) Exact t -test

Assumption (normality). Assume

$$\varepsilon \mid X \sim N(0, \sigma^2 I_n).$$

Null hypothesis.

$$H_0 : \beta_j = \beta_j^0.$$

Consider a test statistic (as written in the notes).

$$t \equiv \frac{\hat{\beta}_j - \beta_j^0}{se_j} \sim t(n - k).$$

More process (define se_j and derive the $t(n - k)$ distribution). Step 1 (OLS estimator and its sampling distribution under normality). Start from the linear model

$$y = X\beta + \varepsilon.$$

OLS:

$$\hat{\beta} = (X'X)^{-1} X'y.$$

Substitute $y = X\beta + \varepsilon$:

$$\hat{\beta} = (X'X)^{-1}X'(X\beta + \varepsilon) = \beta + (X'X)^{-1}X'\varepsilon.$$

Hence

$$\hat{\beta} - \beta = (X'X)^{-1}X'\varepsilon.$$

Because $\varepsilon | X \sim N(0, \sigma^2 I_n)$ and $(X'X)^{-1}X'$ is nonrandom given X ,

$$\hat{\beta} - \beta | X \sim N\left(0, (X'X)^{-1}X'(\sigma^2 I_n)X(X'X)^{-1}\right) = N(0, \sigma^2(X'X)^{-1}).$$

So the j -th component satisfies

$$\hat{\beta}_j - \beta_j | X \sim N\left(0, \sigma^2[(X'X)^{-1}]_{jj}\right).$$

Under $H_0 : \beta_j = \beta_j^0$,

$$\hat{\beta}_j - \beta_j^0 | X \sim N\left(0, \sigma^2[(X'X)^{-1}]_{jj}\right),$$

and therefore the standardized statistic

$$Z \equiv \frac{\hat{\beta}_j - \beta_j^0}{\sigma \sqrt{[(X'X)^{-1}]_{jj}}} \Big| X \sim N(0, 1).$$

Step 2 (residual variance estimator and chi-square distribution). Define the hat matrix and residual-maker matrix:

$$P = X(X'X)^{-1}X', \quad M = I - P.$$

Define residuals:

$$e \equiv y - X\hat{\beta} = y - \hat{y} = (I - P)y = My.$$

Substitute $y = X\beta + \varepsilon$:

$$e = M(X\beta + \varepsilon) = MX\beta + M\varepsilon.$$

Since $MX = (I - P)X = X - PX = X - X = 0$, we have

$$e = M\varepsilon.$$

Thus

$$e'e = (M\varepsilon)'(M\varepsilon) = \varepsilon'M'M\varepsilon.$$

Because M is symmetric and idempotent ($M' = M$, $M^2 = M$),

$$e'e = \varepsilon'M\varepsilon.$$

Under $\varepsilon | X \sim N(0, \sigma^2 I_n)$, the quadratic form satisfies

$$\frac{e'e}{\sigma^2} = \frac{\varepsilon'M\varepsilon}{\sigma^2} \Big| X \sim \chi_{n-k}^2,$$

because $\text{rank}(M) = n - k$.

Define

$$s^2 \equiv \frac{e'e}{n - k}.$$

Then

$$\frac{(n-k)s^2}{\sigma^2} = \frac{e'e}{\sigma^2} \mid X \sim \chi_{n-k}^2.$$

Step 3 (define se_j and obtain the Student t ratio). Define the standard error of $\hat{\beta}_j$ as

$$se_j \equiv \sqrt{[s^2(X'X)^{-1}]_{jj}} = s\sqrt{[(X'X)^{-1}]_{jj}}.$$

Now write

$$t = \frac{\hat{\beta}_j - \beta_j^0}{se_j} = \frac{\hat{\beta}_j - \beta_j^0}{s\sqrt{[(X'X)^{-1}]_{jj}}} = \frac{\frac{\hat{\beta}_j - \beta_j^0}{\sigma\sqrt{[(X'X)^{-1}]_{jj}}}}{\frac{s}{\sigma}}.$$

Using

$$Z = \frac{\hat{\beta}_j - \beta_j^0}{\sigma\sqrt{[(X'X)^{-1}]_{jj}}} \sim N(0, 1), \quad \frac{(n-k)s^2}{\sigma^2} \sim \chi_{n-k}^2,$$

and (under normality) the independence between $\hat{\beta}$ and e (projection onto orthogonal subspaces), we get

$$t \sim t_{n-k}.$$

This matches the notes:

$$t = \frac{\hat{\beta}_j - \beta_j^0}{se_j} \sim t(n-k).$$

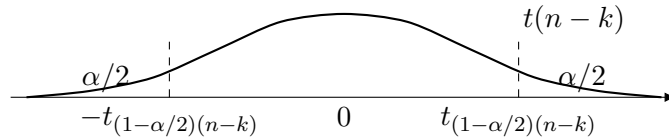
Critical value and rejection rule (as written in the notes). Let $t_{(1-\alpha/2)(n-k)}$ denote the $100(1-\alpha/2)$ -percentile of $t(n-k)$. Reject $H_0 : \beta_j = \beta_j^0$ if

$$|t| \geq t_{(1-\alpha/2)(n-k)}.$$

Then

$$\Pr(\text{reject } H_0 \mid H_0 \text{ is correct}) = \alpha.$$

Sketch (two-sided rejection region under $t(n-k)$).



(2) Asymptotic t -test

Null hypothesis.

$$H_0 : \beta_j = \beta_j^0.$$

Asymptotic normality line (as written in the notes).

$$\sqrt{n}(\hat{\beta}_j - \beta_j^0) \xrightarrow{d} N\left(0, [\sigma^2 Q^{-1}]_{jj}\right).$$

Standardization line (as written in the notes).

$$\frac{\sqrt{n}(\hat{\beta}_j - \beta_j^0)}{\sqrt{[s^2(X'X/n)^{-1}]_{jj}}} \xrightarrow{d} N(0, 1).$$

More process (why the statistic equals $(\hat{\beta}_j - \beta_j^0)/se_j$). Recall the identity

$$(X'X)^{-1} = \frac{1}{n} \left(\frac{X'X}{n} \right)^{-1}.$$

Therefore,

$$s^2(X'X)^{-1} = s^2 \cdot \frac{1}{n} \left(\frac{X'X}{n} \right)^{-1} = \frac{1}{n} s^2 \left(\frac{X'X}{n} \right)^{-1}.$$

Taking the (j, j) element and square roots:

$$se_j = \sqrt{[s^2(X'X)^{-1}]_{jj}} = \sqrt{\frac{1}{n} [s^2(X'X/n)^{-1}]_{jj}} = \frac{1}{\sqrt{n}} \sqrt{[s^2(X'X/n)^{-1}]_{jj}}.$$

Multiply both sides by $\sqrt{n}$:

$$\sqrt{n} se_j = \sqrt{[s^2(X'X/n)^{-1}]_{jj}}.$$

Hence the standardized statistic can be rewritten exactly as

$$\frac{\sqrt{n}(\hat{\beta}_j - \beta_j^0)}{\sqrt{[s^2(X'X/n)^{-1}]_{jj}}} = \frac{\sqrt{n}(\hat{\beta}_j - \beta_j^0)}{\sqrt{n} se_j} = \frac{\hat{\beta}_j - \beta_j^0}{se_j} \equiv t.$$

So the notes' line

$$\frac{\hat{\beta}_j - \beta_j^0}{se_j} = t \xrightarrow{d} N(0, 1)$$

follows.

Critical value definition (as written in the notes). Let $z_{(1-\alpha/2)}$ denote the $100(1 - \alpha/2)$ -percentile of $N(0, 1)$.

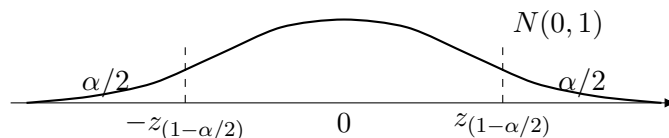
Rejection rule (as written in the notes). Reject $H_0 : \beta_j = \beta_j^0$ if

$$|t| > z_{(1-\alpha/2)}.$$

Size statement (as written in the notes). Then

$$\Pr(\text{reject } H_0 \mid H_0 \text{ is correct}) \rightarrow \alpha \quad \text{as } n \rightarrow \infty.$$

Sketch (two-sided rejection region under $N(0, 1)$).



Joint (Linear) Hypothesis Testing and the F -Test

1. Model and a joint null hypothesis

Model (as written in the notes).

$$y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_4 x_{i4} + \beta_3 x_{i3} + \beta_5 x_{i5} + \varepsilon_i.$$

Equivalently, in matrix form,

$$y = X\beta + \varepsilon, \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \end{pmatrix}.$$

Null hypothesis (as written in the notes).

$$H_0 : \beta_2 + \beta_3 = 1, \quad \beta_4 = \beta_5.$$

Rewrite the null as linear restrictions $R\beta = q$ (as written in the notes). That is,

$$H_0 : R\beta = q,$$

where (matching the notes exactly)

$$R = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{pmatrix}, \quad q = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Check that $R\beta = q$ matches the two restrictions (extra process, no deletion). Compute $R\beta$ explicitly:

$$R\beta = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \end{pmatrix} = \begin{pmatrix} \beta_2 + \beta_3 \\ \beta_4 - \beta_5 \end{pmatrix}.$$

So $R\beta = q$ is

$$\begin{pmatrix} \beta_2 + \beta_3 \\ \beta_4 - \beta_5 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \iff \beta_2 + \beta_3 = 1, \beta_4 - \beta_5 = 0 (\Leftrightarrow \beta_4 = \beta_5).$$

Hypotheses (as written in the notes).

$$H_0 : R\beta - q = 0, \quad H_1 : R\beta - q \neq 0.$$

Let the number of linear restrictions be

$$J \equiv \text{rank}(R).$$

(In the example above, $J = 2$.)

2. Distributional assumption for the exact F -test (as written in the notes)

$$\varepsilon \mid X \sim N(0, \sigma^2 I_n), \quad \frac{\varepsilon}{\sigma} \mid X \sim N(0, I_n).$$

3. Restricted least squares and residual sums of squares

Unrestricted OLS estimator and residuals. The unrestricted OLS estimator is

$$\hat{\beta} = \arg \min_b \|y - Xb\|^2, \quad e \equiv y - X\hat{\beta}.$$

Restricted estimator (as written in the notes). Let $\hat{\beta}_x$ be the solution to the constrained least squares problem:

$$\min_b \|y - Xb\|^2 \quad \text{s.t.} \quad Rb = q.$$

$\hat{\beta}_x$ is the restricted estimate of β .

Define restricted residuals (as written in the notes):

$$e_x \equiv y - X\hat{\beta}_x.$$

Why $e'_x e_x \geq e'e$ (extra process, no deletion). The unrestricted estimator $\hat{\beta}$ minimizes $\|y - Xb\|^2$ over *all* b . The restricted estimator $\hat{\beta}_x$ minimizes $\|y - Xb\|^2$ over a *subset* of b satisfying $Rb = q$. Hence the minimum over the restricted set cannot be smaller than the unrestricted minimum:

$$e'_x e_x = \min_{b: Rb=q} \|y - Xb\|^2 \geq \min_b \|y - Xb\|^2 = e'e.$$

Key intuition (as written in the notes). If $H_0 : R\beta - q = 0$ is true, then $e'_x e_x - e'e$ should be small.

4. The F statistic (as written in the notes)

Define the F statistic:

$$F[J, n - k] = \frac{(e'_x e_x - e'e)/J}{e'e/(n - k)}.$$

Here:

- $J = \text{rank}(R)$ is the number of independent restrictions,
- k is the number of regressors in the unrestricted model (columns of X),
- $n - k$ is the unrestricted residual degrees of freedom.

5. Computing the restricted residuals via a transformed regression (as written in the notes, fully expanded)

Step 5.1: Use $\text{rank}(R) = J$ to partition R (as in the notes). Assume

$$\text{rank}(R) = J.$$

Then we can (possibly after reordering the columns of β) partition

$$R = [R_1 \ R_2], \quad R_2 \text{ is nonsingular,}$$

where

$$R_1 \in \mathbb{R}^{J \times (k-J)}, \quad R_2 \in \mathbb{R}^{J \times J}.$$

Step 5.2: Partition β and X conformably (extra process, consistent with the notes). Write

$$\beta = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}, \quad \beta_1 \in \mathbb{R}^{(k-J) \times 1}, \quad \beta_2 \in \mathbb{R}^{J \times 1},$$

and partition the design matrix as

$$X = [X_1 \ X_2], \quad X_1 \in \mathbb{R}^{n \times (k-J)}, \quad X_2 \in \mathbb{R}^{n \times J}.$$

Then

$$y = X\beta + \varepsilon = X_1\beta_1 + X_2\beta_2 + \varepsilon.$$

Step 5.3: Rewrite the restriction $R\beta = q$ (as in the notes). Since $R = [R_1 \ R_2]$ and $\beta = (\beta_1', \beta_2')'$,

$$R\beta = q \iff R_1\beta_1 + R_2\beta_2 = q.$$

Solve for β_2 using that R_2 is nonsingular:

$$R_2\beta_2 = q - R_1\beta_1 \implies \boxed{\beta_2 = R_2^{-1}(q - R_1\beta_1)}.$$

Step 5.4: Substitute $\beta_2 = R_2^{-1}(q - R_1\beta_1)$ into the model (as in the notes). Start from

$$y = X_1\beta_1 + X_2\beta_2 + \varepsilon.$$

Substitute:

$$\begin{aligned} y &= X_1\beta_1 + X_2R_2^{-1}(q - R_1\beta_1) + \varepsilon \\ &= X_1\beta_1 + X_2R_2^{-1}q - X_2R_2^{-1}R_1\beta_1 + \varepsilon \\ &= (X_1 - X_2R_2^{-1}R_1)\beta_1 + X_2R_2^{-1}q + \varepsilon. \end{aligned}$$

Rearrange (as in the notes):

$$y - X_2R_2^{-1}q = (X_1 - X_2R_2^{-1}R_1)\beta_1 + \varepsilon.$$

Step 5.5: Define transformed variables (as in the notes). Define

$$\tilde{y} \equiv y - X_2R_2^{-1}q, \quad Z_1 \equiv X_1 - X_2R_2^{-1}R_1.$$

Then the transformed regression is

$$\boxed{\tilde{y} = Z_1\beta_1 + \varepsilon.}$$

Step 5.6: Obtain the restricted estimator by OLS in the transformed regression (as in the notes).

$\hat{\beta}_x$ is obtained by regressing $\tilde{y}$ on Z_1 .

More explicitly, the restricted estimator for β_1 is

$$\hat{\beta}_{1,x} = (Z_1' Z_1)^{-1} Z_1' \tilde{y}.$$

Then recover β_2 using the restriction:

$$\hat{\beta}_{2,x} = R_2^{-1}(q - R_1 \hat{\beta}_{1,x}), \quad \hat{\beta}_x = \begin{pmatrix} \hat{\beta}_{1,x} \\ \hat{\beta}_{2,x} \end{pmatrix}.$$

Step 5.7: Projection matrices and residual-maker matrices (as in the notes). Define

$$P_x \equiv Z_1(Z_1' Z_1)^{-1} Z_1', \quad M_x \equiv I - P_x.$$

Then the restricted residuals can be written as

$$e_x = M_x \tilde{y}.$$

Substitute $\tilde{y} = Z_1 \beta_1 + \varepsilon$:

$$\begin{aligned} e_x &= M_x (Z_1 \beta_1 + \varepsilon) \\ &= M_x Z_1 \beta_1 + M_x \varepsilon. \end{aligned}$$

Because $M_x Z_1 = (I - P_x) Z_1 = Z_1 - P_x Z_1 = Z_1 - Z_1 = 0$, we get (as written in the notes)

$$\boxed{e_x = M_x \varepsilon.}$$

Step 5.8: Unrestricted residuals in matrix form (as in the notes). Define

$$P \equiv X(X' X)^{-1} X', \quad M \equiv I - P.$$

Then

$$e = My = M(X\beta + \varepsilon) = MX\beta + M\varepsilon = 0 + M\varepsilon,$$

so

$$\boxed{e = M\varepsilon.}$$

6. Rewriting the F statistic as quadratic forms in ε (as in the notes, fully expanded)

Step 6.1: Replace $e_x' e_x$ and $e' e$ by quadratic forms. Using $e_x = M_x \varepsilon$ and $e = M\varepsilon$,

$$\begin{aligned} e_x' e_x &= (M_x \varepsilon)' (M_x \varepsilon) = \varepsilon' M_x' M_x \varepsilon = \varepsilon' M_x \varepsilon, \\ e' e &= (M \varepsilon)' (M \varepsilon) = \varepsilon' M' M \varepsilon = \varepsilon' M \varepsilon, \end{aligned}$$

where we used that M_x and M are symmetric and idempotent.

Substitute into the F formula:

$$F = \frac{(\varepsilon' M_x \varepsilon - \varepsilon' M \varepsilon) / J}{\varepsilon' M \varepsilon / (n - k)}.$$

Step 6.2: Factor the difference (as in the notes).

$$\varepsilon' M_x \varepsilon - \varepsilon' M \varepsilon = \varepsilon' (M_x - M) \varepsilon.$$

Hence

$$F = \frac{\varepsilon' (M_x - M) \varepsilon / J}{\varepsilon' M \varepsilon / (n - k)}.$$

Step 6.3: Standardize by σ (as in the notes). Since $\varepsilon \mid X \sim N(0, \sigma^2 I_n)$, define

$$u \equiv \frac{\varepsilon}{\sigma} \mid X \sim N(0, I_n).$$

Then $\varepsilon = \sigma u$ and

$$\varepsilon' (M_x - M) \varepsilon = (\sigma u)' (M_x - M) (\sigma u) = \sigma^2 u' (M_x - M) u,$$

$$\varepsilon' M \varepsilon = (\sigma u)' M (\sigma u) = \sigma^2 u' M u.$$

So the σ^2 cancels in the ratio:

$$F = \frac{u' (M_x - M) u / J}{u' M u / (n - k)}.$$

Written exactly as in the notes:

$$F = \frac{((\varepsilon' / \sigma) (M_x - M) (\varepsilon / \sigma)) / J}{((\varepsilon' / \sigma) M (\varepsilon / \sigma)) / (n - k)}.$$

7. Why the statistic is $F_{J, n-k}$ under H_0 (fully expanded, consistent with the notes)

Key matrix ranks (extra process, no deletion). Unrestricted model uses k regressors, so $\text{rank}(P) = k$ and

$$\text{rank}(M) = n - k.$$

Restricted model imposes J independent restrictions, effectively leaving $k - J$ free parameters, so the transformed regression uses $k - J$ regressors (columns of Z_1), hence

$$\text{rank}(P_x) = k - J, \quad \text{rank}(M_x) = n - (k - J) = n - k + J.$$

Therefore the difference in residual degrees of freedom is

$$\text{rank}(M_x) - \text{rank}(M) = (n - k + J) - (n - k) = J.$$

Quadratic-form distributions (extra process, no deletion). Under H_0 and normality, $u = \varepsilon / \sigma \mid X \sim N(0, I_n)$.

- The denominator:

$$u' M u \sim \chi_{n-k}^2,$$

because M is symmetric idempotent with rank $n - k$.

- The numerator:

$$u' (M_x - M) u \sim \chi_J^2,$$

because $(M_x - M)$ is symmetric idempotent with rank J in the nested (restricted vs unrestricted) setup.

- Independence: since these quadratic forms correspond to orthogonal projections in the nested-regression geometry, the numerator and denominator quadratic forms are independent under normality.

Hence

$$\frac{u'(M_x - M)u/J}{u'Mu/(n-k)} \sim F_{J, n-k}.$$

Final result (matching the notes).

$$F[J, n-k] = \frac{(e'_x e_x - e' e)/J}{e' e/(n-k)} \sim F_{J, n-k} \quad \text{under } H_0.$$

Econometrics I: Exact vs Asymptotic Tests; Wald Tests for Linear and Nonlinear Restrictions

0. Setup and notation

Linear regression model.

$$y = X\beta + \varepsilon, \quad y \in \mathbb{R}^{h \times 1}, \quad X \in \mathbb{R}^{h \times k}, \quad \beta \in \mathbb{R}^{k \times 1}, \quad \varepsilon \in \mathbb{R}^{h \times 1}.$$

(Here h denotes the sample size; in many texts $h = n$.)

OLS estimator.

$$\hat{\beta} = (X'X)^{-1}X'y.$$

Residuals and residual variance estimator.

$$e \equiv y - X\hat{\beta}, \quad s^2 \equiv \frac{e'e}{h-k}.$$

Asymptotic quantities. Let x'_i be the i -th row of X (so $x_i \in \mathbb{R}^{k \times 1}$), and define

$$Q \equiv \mathbb{E}[x_i x'_i].$$

Under standard regularity conditions (i.i.d. sampling, exogeneity, finite moments, Q nonsingular),

$$\sqrt{h}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2 Q^{-1}), \quad s^2 \xrightarrow{p} \sigma^2, \quad \left(\frac{X'X}{h} \right)^{-1} \xrightarrow{p} Q^{-1}.$$

1. Exact tests under normality

Normality assumption. Assume

$$\varepsilon | X \sim N(0, \sigma^2 I_h).$$

1.1 Exact t test for a single coefficient

Null.

$$H_0 : \beta_j = \beta_j^0.$$

Standard error and t statistic. Define

$$se_j \equiv \sqrt{[s^2(X'X)^{-1}]_{jj}}.$$

Then the exact t statistic is

$$t \equiv \frac{\hat{\beta}_j - \beta_j^0}{se_j}, \quad t \sim t(h-k).$$

1.2 Exact F test for linear joint restrictions

Null. Let R be a $J \times k$ matrix with $\text{rank}(R) = J$ and $q \in \mathbb{R}^{J \times 1}$. Consider

$$H_0 : R\beta - q = 0.$$

Restricted and unrestricted residual sums of squares. Let $\hat{\beta}_x$ be the restricted least-squares estimator:

$$\hat{\beta}_x = \arg \min_b \|y - Xb\|^2 \quad \text{s.t.} \quad Rb = q, \quad e_x \equiv y - X\hat{\beta}_x.$$

Unrestricted residuals are $e = y - X\hat{\beta}$.

F statistic (exact under normality).

$$F[J, h-k] = \frac{(e'_x e_x - e' e)/J}{e' e / (h-k)}, \quad F[J, h-k] \sim F_{J, h-k} \quad \text{under } H_0.$$

2. Asymptotic tests without assuming normality

No normality. Do *not* assume $\varepsilon | X \sim N(0, \sigma^2 I_h)$.

2.1 Asymptotic t test for $H_0 : \beta_j = \beta_j^0$

Step 1: Start from asymptotic normality of $\hat{\beta}$. From

$$\sqrt{h}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2 Q^{-1}),$$

the j -th component satisfies

$$\sqrt{h}(\hat{\beta}_j - \beta_j) \xrightarrow{d} N\left(0, \sigma^2 [Q^{-1}]_{jj}\right).$$

Under $H_0 : \beta_j = \beta_j^0$,

$$\sqrt{h}(\hat{\beta}_j - \beta_j^0) \xrightarrow{d} N\left(0, \sigma^2 [Q^{-1}]_{jj}\right).$$

Step 2: Relate se_j to the asymptotic standard deviation. Use

$$(X'X)^{-1} = \frac{1}{h} \left(\frac{X'X}{h} \right)^{-1} \Rightarrow s^2(X'X)^{-1} = \frac{1}{h} s^2 \left(\frac{X'X}{h} \right)^{-1}.$$

Hence

$$se_j = \sqrt{[s^2(X'X)^{-1}]_{jj}} = \frac{1}{\sqrt{h}} \sqrt{[s^2 \left(\frac{X'X}{h} \right)^{-1}]_{jj}},$$

so

$$\sqrt{h} se_j = \sqrt{[s^2 \left(\frac{X'X}{h} \right)^{-1}]_{jj}} \xrightarrow{p} \sqrt{[\sigma^2 Q^{-1}]_{jj}}.$$

Step 3: Form the t statistic and apply Slutsky. Define

$$t \equiv \frac{\hat{\beta}_j - \beta_j^0}{se_j} = \frac{\sqrt{h}(\hat{\beta}_j - \beta_j^0)}{\sqrt{h} se_j}.$$

Then, by Slutsky,

$$t \xrightarrow{d} N(0, 1).$$

3. Asymptotic Wald test for linear restrictions $H_0 : R\beta - q = 0$

3.1 Distribution of $R\hat{\beta} - q$ under H_0

Null.

$$H_0 : R\beta - q = 0.$$

Step 1: Rewrite the sample restriction error under H_0 . If H_0 is correct, then $q = R\beta$, and thus

$$R\hat{\beta} - q = R\hat{\beta} - R\beta = R(\hat{\beta} - \beta).$$

Step 2: Multiply by $\sqrt{h}$ and use asymptotic normality.

$$\sqrt{h}(R\hat{\beta} - q) = R\sqrt{h}(\hat{\beta} - \beta).$$

Since $\sqrt{h}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2 Q^{-1})$, a linear transformation yields

$$\boxed{\sqrt{h}(R\hat{\beta} - q) \xrightarrow{d} N\left(0, R\sigma^2 Q^{-1}R'\right).}$$

3.2 Consistent estimation of the asymptotic covariance

Step 3: Use probability limits for s^2 and $(X'X/h)^{-1}$.

$$s^2 \xrightarrow{p} \sigma^2, \quad \left(\frac{X'X}{h}\right)^{-1} \xrightarrow{p} Q^{-1}.$$

Therefore,

$$R s^2 \left(\frac{X'X}{h}\right)^{-1} R' \xrightarrow{p} R \sigma^2 Q^{-1} R'.$$

3.3 Wald statistic and χ^2 limit

Step 4: Define the Wald statistic as a quadratic form in $\sqrt{h}(R\hat{\beta} - q)$.

$$W \equiv \sqrt{h}(R\hat{\beta} - q)' \left[R s^2 \left(\frac{X'X}{h}\right)^{-1} R' \right]^{-1} \sqrt{h}(R\hat{\beta} - q).$$

Step 5: Show the algebraic simplification to a form without $\sqrt{h}$. Let

$$\widehat{\Sigma}_R \equiv R s^2 \left(\frac{X'X}{h} \right)^{-1} R'.$$

Since $\left(\frac{X'X}{h} \right)^{-1} = h(X'X)^{-1}$,

$$\widehat{\Sigma}_R = R s^2 \left(h(X'X)^{-1} \right) R' = h \cdot \left(R s^2 (X'X)^{-1} R' \right).$$

Hence

$$\widehat{\Sigma}_R^{-1} = \frac{1}{h} \left(R s^2 (X'X)^{-1} R' \right)^{-1}.$$

Now expand the quadratic form:

$$\begin{aligned} W &= (\sqrt{h}(R\hat{\beta} - q))' \widehat{\Sigma}_R^{-1} (\sqrt{h}(R\hat{\beta} - q)) \\ &= h (R\hat{\beta} - q)' \widehat{\Sigma}_R^{-1} (R\hat{\beta} - q) \\ &= h (R\hat{\beta} - q)' \left[\frac{1}{h} \left(R s^2 (X'X)^{-1} R' \right)^{-1} \right] (R\hat{\beta} - q) \\ &= (R\hat{\beta} - q)' \left(R s^2 (X'X)^{-1} R' \right)^{-1} (R\hat{\beta} - q). \end{aligned}$$

So equivalently

$$W = (R\hat{\beta} - q)' \left(R s^2 (X'X)^{-1} R' \right)^{-1} (R\hat{\beta} - q).$$

Step 6: Limiting distribution. Let

$$Z_h \equiv \sqrt{h}(R\hat{\beta} - q).$$

Under H_0 , we have

$$Z_h \xrightarrow{d} Z, \quad Z \sim N(0, \Sigma_R), \quad \Sigma_R \equiv R\sigma^2 Q^{-1} R'.$$

Also, $\widehat{\Sigma}_R \xrightarrow{p} \Sigma_R$. A standard result: if $Z \sim N(0, \Sigma_R)$ with Σ_R nonsingular, then

$$Z' \Sigma_R^{-1} Z \sim \chi^2(J).$$

By Slutsky (replacing Σ_R by $\widehat{\Sigma}_R$),

$$W \xrightarrow{d} \chi^2(J).$$

Critical value and asymptotic size. Use the $100(1 - \alpha)$ -percentile of $\chi^2(J)$ as the critical value. Reject H_0 when

$$W > \chi_{J, 1-\alpha}^2.$$

As $h \rightarrow \infty$,

$$\Pr(\text{reject } H_0 \mid H_0 \text{ is correct}) \rightarrow \alpha.$$

4. Wald test for nonlinear restrictions $H_0 : c(\beta) = 0$

4.1 Nonlinear hypothesis and Jacobian

Nonlinear null and alternative.

$$H_0 : c(\beta) = 0, \quad H_1 : c(\beta) \neq 0,$$

where

$$c(\cdot) : \mathbb{R}^{k \times 1} \rightarrow \mathbb{R}^{J \times 1}, \quad \beta \in \mathbb{R}^{k \times 1}.$$

Define the Jacobian (gradient matrix). Let

$$G(b) \equiv \frac{\partial c(b)}{\partial b'}, \quad G(b) \in \mathbb{R}^{J \times k}.$$

4.2 Asymptotic distribution of $\sqrt{h} c(\hat{\beta})$ under H_0

Step 1: Use a first-order Taylor expansion (delta-method step). Assume $c(\cdot)$ is continuously differentiable in a neighborhood of β . Then

$$c(\hat{\beta}) = c(\beta) + G(\beta)(\hat{\beta} - \beta) + r_h,$$

where the remainder satisfies

$$\|r_h\| = o(\|\hat{\beta} - \beta\|) \Rightarrow r_h = o_p(\|\hat{\beta} - \beta\|).$$

Since $\hat{\beta} - \beta = O_p(h^{-1/2})$ (because $\sqrt{h}(\hat{\beta} - \beta)$ has a non-degenerate limit),

$$r_h = o_p(h^{-1/2}) \Rightarrow \sqrt{h} r_h = o_p(1).$$

Step 2: Impose $H_0 : c(\beta) = 0$ and multiply by $\sqrt{h}$. Under H_0 , $c(\beta) = 0$, so

$$\sqrt{h} c(\hat{\beta}) = \sqrt{h}(c(\hat{\beta}) - c(\beta)) = G(\beta)\sqrt{h}(\hat{\beta} - \beta) + \sqrt{h} r_h.$$

Thus

$$\sqrt{h} c(\hat{\beta}) = G(\beta)\sqrt{h}(\hat{\beta} - \beta) + o_p(1).$$

Step 3: Apply asymptotic normality of $\hat{\beta}$. Since

$$\sqrt{h}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2 Q^{-1}),$$

a linear transformation gives

$$\boxed{\sqrt{h} c(\hat{\beta}) = \sqrt{h}(c(\hat{\beta}) - c(\beta)) \xrightarrow{d} N\left(0, G(\beta) \sigma^2 Q^{-1} G(\beta)'\right)}.$$

4.3 Estimating the asymptotic covariance and constructing the nonlinear Wald statistic

Step 4: Plug-in estimator for the asymptotic covariance. Estimate $\sigma^2 Q^{-1}$ by $s^2(X'X/h)^{-1}$ and $G(\beta)$ by $G(\hat{\beta})$:

$$\hat{\Sigma}_c \equiv G(\hat{\beta}) s^2 \left(\frac{X'X}{h} \right)^{-1} G(\hat{\beta})'.$$

Step 5: Wald statistic (quadratic form). Define

$$W \equiv \sqrt{h} c(\hat{\beta})' \hat{\Sigma}_c^{-1} \sqrt{h} c(\hat{\beta}).$$

Step 6: Simplify the $\sqrt{h}$ form to the version without $\sqrt{h}$. Since $\left(\frac{X'X}{h}\right)^{-1} = h(X'X)^{-1}$,

$$\hat{\Sigma}_c = G(\hat{\beta}) s^2 (h(X'X)^{-1}) G(\hat{\beta})' = h \cdot \left(G(\hat{\beta}) s^2 (X'X)^{-1} G(\hat{\beta})' \right),$$

so

$$\hat{\Sigma}_c^{-1} = \frac{1}{h} \left(G(\hat{\beta}) s^2 (X'X)^{-1} G(\hat{\beta})' \right)^{-1}.$$

Then

$$\begin{aligned} W &= (\sqrt{h} c(\hat{\beta}))' \hat{\Sigma}_c^{-1} (\sqrt{h} c(\hat{\beta})) \\ &= h c(\hat{\beta})' \hat{\Sigma}_c^{-1} c(\hat{\beta}) \\ &= h c(\hat{\beta})' \left[\frac{1}{h} \left(G(\hat{\beta}) s^2 (X'X)^{-1} G(\hat{\beta})' \right)^{-1} \right] c(\hat{\beta}) \\ &= c(\hat{\beta})' \left(G(\hat{\beta}) s^2 (X'X)^{-1} G(\hat{\beta})' \right)^{-1} c(\hat{\beta}). \end{aligned}$$

Thus equivalently

$$\boxed{W = c(\hat{\beta})' \left(G(\hat{\beta}) s^2 (X'X)^{-1} G(\hat{\beta})' \right)^{-1} c(\hat{\beta}).}$$

Step 7: Limiting distribution and critical value. Under $H_0 : c(\beta) = 0$, we have

$$\sqrt{h} c(\hat{\beta}) \xrightarrow{d} N\left(0, G(\beta) \sigma^2 Q^{-1} G(\beta)'\right), \quad \hat{\Sigma}_c \xrightarrow{p} G(\beta) \sigma^2 Q^{-1} G(\beta)'. \quad \square$$

Therefore, by the same quadratic-form argument as in the linear case,

$$\boxed{W \xrightarrow{d} \chi^2(J).}$$

Use the $100(1 - \alpha)$ -percentile of $\chi^2(J)$ as the critical value and reject H_0 when

$$W > \chi_{J, 1-\alpha}^2.$$

As $h \rightarrow \infty$,

$$\Pr(\text{reject } H_0 \mid H_0 \text{ is correct}) \rightarrow \alpha.$$

Non-nested Model Comparison (J Test) and Model Selection (AIC/BIC)

1. Competing (non-nested) models

(Original lines; kept unchanged).

$$H_0 \text{ (model 0)} : Y = X\beta + \varepsilon_0$$

$$H_1 \text{ (model 1)} : Y = Z\gamma + \varepsilon_1$$

More process: dimensions and what “non-nested” means. Let

$$Y \in \mathbb{R}^{n \times 1}, \quad X \in \mathbb{R}^{n \times k_X}, \quad Z \in \mathbb{R}^{n \times k_Z}, \quad \beta \in \mathbb{R}^{k_X \times 1}, \quad \gamma \in \mathbb{R}^{k_Z \times 1}.$$

Model 0 uses regressors X ; model 1 uses regressors Z . They are *non-nested* if neither model is a special case of the other, i.e.

- X is not (in general) contained in the column space of Z , and
- Z is not (in general) contained in the column space of X .

Equivalently, there is no parameter restriction that turns model 0 into model 1 (or vice versa).

2. An “encompassing” (mixture) representation

(Original line; kept unchanged).

$$Y = (1 - \lambda)X\beta + \lambda Z\gamma + \varepsilon$$

More process: show how λ connects the two models. Take $\lambda \in \mathbb{R}$ as a scalar. Then:

- If $\lambda = 0$, the equation becomes

$$Y = (1 - 0)X\beta + 0 \cdot Z\gamma + \varepsilon = X\beta + \varepsilon,$$

which corresponds to model 0 (up to notation).

- If $\lambda = 1$, the equation becomes

$$Y = (1 - 1)X\beta + 1 \cdot Z\gamma + \varepsilon = Z\gamma + \varepsilon,$$

which corresponds to model 1 (up to notation).

So testing $\lambda = 0$ can be interpreted as testing whether the $Z\gamma$ component is needed when $X\beta$ is the baseline.

(Original sentence; kept unchanged). If regress Y on X and Z , we can estimate $(1 - \lambda)\beta$ and $\lambda\gamma$.

More process: why only the products are identified in that direct regression. Rewrite the mixture model as a standard linear regression:

$$Y = X \underbrace{((1 - \lambda)\beta)}_{\theta_X} + Z \underbrace{(\lambda\gamma)}_{\theta_Z} + \varepsilon,$$

where

$$\theta_X \equiv (1 - \lambda)\beta, \quad \theta_Z \equiv \lambda\gamma.$$

Regressing Y on (X, Z) identifies θ_X and θ_Z as coefficients on (X, Z) , but does not separately identify λ and (β, γ) without additional restrictions. The J test proceeds differently: it uses an auxiliary regression built from fitted values to test whether the rival model contributes explanatory power.

3. J test (Davidson–MacKinnon type) procedure

(Original steps; kept unchanged). J test:

(1) Estimate γ by regressing Y on Z .

Let $\hat{\gamma}$ denote the estimate.

(2) Regress Y on X and $Z\hat{\gamma}$.

(3) Test $H_0 : \lambda = 0$.

3.1 Fully expanded version of the J test (test model 0 against model 1)

Step (1): Estimate model 1 and form the fitted component

Run the OLS regression (model 1):

$$Y = Z\gamma + \varepsilon_1.$$

The OLS estimator is

$$\hat{\gamma} = (Z'Z)^{-1}Z'Y.$$

Define fitted values from model 1:

$$\hat{Y}_1 \equiv Z\hat{\gamma}.$$

Thus $\hat{Y}_1$ is a constructed regressor (generated from the first-stage regression).

Step (2): Auxiliary regression that augments model 0 with $\hat{Y}_1$

Run the auxiliary regression:

$$Y = X\beta + \lambda\hat{Y}_1 + u \quad \text{where} \quad \hat{Y}_1 = Z\hat{\gamma}.$$

Equivalently, the regressor matrix is $(X, \hat{Y}_1)$ and the coefficient on $\hat{Y}_1$ is λ .

Step (3): Hypotheses and test

$$H_0 : \lambda = 0, \quad H_1 : \lambda \neq 0.$$

Interpretation:

- Under H_0 (model 0 is adequate), adding $\hat{Y}_1$ should not improve the explanation of Y beyond X ; hence λ should be 0.
- If λ is significantly different from 0, then the rival model's fitted component contains information that is not captured by X , which is evidence against model 0 in this J-test direction.

Test statistic (fully explicit)

Let $\hat{\lambda}$ be the OLS estimate from the auxiliary regression and $se(\hat{\lambda})$ its standard error. The usual t statistic is

$$t_\lambda \equiv \frac{\hat{\lambda} - 0}{se(\hat{\lambda})} = \frac{\hat{\lambda}}{se(\hat{\lambda})}.$$

Exact vs asymptotic reference distribution (more process).

- If one additionally assumes normality and classical homoscedastic conditions for the auxiliary regression, t_λ is often compared to a Student- t critical value with the auxiliary regression degrees of freedom.
- Without normality (and in large samples), an asymptotic approximation is used:

$$t_\lambda \xrightarrow{d} N(0,1) \quad \text{under } H_0.$$

- In practice, since $\hat{Y}_1$ is a generated regressor, inference is typically justified by large-sample theory; robust (heteroskedasticity-consistent) standard errors are often used for the auxiliary regression.

3.2 The reverse J test direction (test model 1 against model 0)

Because the models are non-nested, the J test is not inherently symmetric: one can also test model 1 as the null by reversing roles.

Reverse procedure (added process, no deletion).

- (1) Estimate β by regressing Y on X , and obtain $\hat{\beta} = (X'X)^{-1}X'Y$.
- (2) Form fitted values $\hat{Y}_0 \equiv X\hat{\beta}$.
- (3) Regress Y on Z and $\hat{Y}_0$:

$$Y = Z\gamma + \delta\hat{Y}_0 + v,$$

and test $H_0 : \delta = 0$.

Interpretation of outcomes (added process).

- It is possible that model 0 is rejected against model 1 *and* model 1 is rejected against model 0, suggesting neither model encompasses the other well.
- It is also possible that only one direction rejects, suggesting that one model provides additional explanatory content beyond the other.

4. Model selection criteria: AIC and BIC

(Original lines; kept unchanged). Model selection

Let k be the number of explanatory variables,

$$AIC(k) = \log \left(\frac{e'e}{n} \right) + 2\frac{k}{n}$$

$$BIC(k) = \log \left(\frac{e'e}{n} \right) + k\frac{\log n}{n}$$

4.1 More process: define $e'e$ and connect to likelihood

Residual sum of squares. For a candidate model with k regressors and OLS residual vector $e \in \mathbb{R}^{n \times 1}$,

$$e \equiv Y - X\hat{\beta}, \quad e'e = \sum_{i=1}^n e_i^2$$

is the residual sum of squares (RSS or SSE).

Gaussian likelihood derivation (fully expanded; constants shown then dropped). Assume for this derivation that errors are Gaussian:

$$\varepsilon \mid X \sim N(0, \sigma^2 I_n).$$

Then the log-likelihood of the sample (conditional on X) can be written as

$$\ell(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} (Y - X\beta)'(Y - X\beta).$$

At the OLS estimator $\hat{\beta}$, we have $(Y - X\hat{\beta}) = e$, so

$$\ell(\hat{\beta}, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} e'e.$$

Maximize over σ^2 :

$$\frac{\partial \ell(\hat{\beta}, \sigma^2)}{\partial \sigma^2} = -\frac{n}{2} \frac{1}{\sigma^2} + \frac{1}{2} \frac{e'e}{(\sigma^2)^2} = 0 \implies \hat{\sigma}_{\text{MLE}}^2 = \frac{e'e}{n}.$$

Plugging in gives

$$\ell(\hat{\beta}, \hat{\sigma}_{\text{MLE}}^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log\left(\frac{e'e}{n}\right) - \frac{n}{2}.$$

Therefore

$$-2\ell(\hat{\beta}, \hat{\sigma}_{\text{MLE}}^2) = n \log(2\pi) + n \log\left(\frac{e'e}{n}\right) + n.$$

Dividing by n ,

$$\frac{-2\ell(\hat{\beta}, \hat{\sigma}_{\text{MLE}}^2)}{n} = \log(2\pi) + \log\left(\frac{e'e}{n}\right) + 1.$$

Up to the additive constant $\log(2\pi) + 1$ (which does not affect model ranking across different k), the fit part is exactly

$$\log\left(\frac{e'e}{n}\right).$$

AIC and BIC add different penalties for model complexity.

4.2 Penalty terms and the criteria (per-observation forms)

AIC (per observation, matching your notes). The standard AIC is often written as

$$\text{AIC} = -2\ell(\hat{\theta}) + 2k,$$

where k is the number of estimated parameters. Dividing by n and dropping constants yields your notes' form:

$$\text{AIC}(k) = \log\left(\frac{e'e}{n}\right) + 2\frac{k}{n}.$$

BIC (per observation, matching your notes). The standard BIC is often written as

$$\text{BIC} = -2\ell(\hat{\theta}) + k \log n.$$

Dividing by n and dropping constants yields your notes' form:

$$\text{BIC}(k) = \log \left(\frac{e'e}{n} \right) + k \frac{\log n}{n}.$$

5. Large-sample statements and prediction

(Original statements; kept unchanged). As $n \rightarrow \infty$

$$\Pr(\text{The correct model minimizes BIC}) \rightarrow 1$$

As $n \rightarrow \infty$

$$\Pr(\text{The correct model minimizes AIC}) \rightarrow 1$$

AIC minimizes prediction error.

More process: how to interpret these claims.

- The phrase “minimizes BIC” means: among the candidate models indexed by k , choose the one with the smallest $\text{BIC}(k)$.
- “Probability $\rightarrow 1$ ” means that, in repeated samples, the selection rule chooses the correct model with probability approaching 1 as sample size grows.
- “AIC minimizes prediction error” means: AIC is constructed to favor models that perform well for out-of-sample prediction (or, more precisely, to target good approximation quality in terms of predictive discrepancy / Kullback–Leibler loss).

(Optional remark; included only as context, while preserving the notes). Many econometrics references emphasize that BIC is consistent for selecting the true finite-dimensional model (when it is in the candidate set), whereas AIC is often discussed as prediction-oriented and not necessarily consistent for selecting the true model. The statements above preserve the notes' wording, and this remark only clarifies common interpretations.

Dummy Variables, Difference-in-Differences, and Structural Breaks

1 Dummy Variables

Definition

A *dummy variable* takes values 1 or 0.

Example: Gender indicator

Let

$$\text{female}_i = \begin{cases} 1 & \text{if individual } i \text{ is female,} \\ 0 & \text{if individual } i \text{ is male,} \end{cases} \quad \text{male}_i = \begin{cases} 1 & \text{if individual } i \text{ is male,} \\ 0 & \text{if individual } i \text{ is female.} \end{cases}$$

Then for each i ,

$$\text{female}_i + \text{male}_i = 1.$$

This illustrates the key constraint: if we include an intercept, we cannot include *all* mutually exclusive dummies because they sum to the intercept (“dummy variable trap”).

Example: Restaurant ratings (5 = best, 1 = worst)

Let $R \in \{1, 2, 3, 4, 5\}$ denote the rating and let z denote other controls. Consider the **wrong regression** (as in the notes):

$$y = \beta_1 + \beta_2 R + z' \gamma + \varepsilon. \quad (\text{DV-Wrong})$$

Fully expanded implication of the wrong regression. Take conditional expectation given (R, z) and use $\mathbb{E}[\varepsilon \mid R, z] = 0$:

$$\mathbb{E}[y \mid R, z] = \beta_1 + \beta_2 R + z' \gamma.$$

Hence the “step” from one rating to the next is constant:

$$\mathbb{E}[y \mid R = r, z] - \mathbb{E}[y \mid R = r - 1, z] = (\beta_1 + \beta_2 r + z' \gamma) - (\beta_1 + \beta_2(r - 1) + z' \gamma) = \beta_2,$$

for every $r \in \{2, 3, 4, 5\}$. In particular,

$$\mathbb{E}[y \mid R = 4, z] - \mathbb{E}[y \mid R = 3, z] = \beta_2 \quad \text{and} \quad \mathbb{E}[y \mid R = 2, z] - \mathbb{E}[y \mid R = 1, z] = \beta_2,$$

so the wrong regression (??) *assumes these two differences are the same*.

Correct approach: use rating dummies (as in the notes). Define four dummies (baseline $R = 1$ omitted):

$$R_1 \equiv \mathbf{1}\{R = 2\}, \quad R_2 \equiv \mathbf{1}\{R = 3\}, \quad R_3 \equiv \mathbf{1}\{R = 4\}, \quad R_4 \equiv \mathbf{1}\{R = 5\}.$$

Then the **regression model** (as in the notes) is

$$y = \beta_1 + \beta_2 R_1 + \beta_3 R_2 + \beta_4 R_3 + \beta_5 R_4 + z' \gamma + \varepsilon. \quad (\text{DV-Right})$$

Fully expanded conditional means under the dummy specification. Because exactly one of $\{R_1, R_2, R_3, R_4\}$ is 1 when $R \in \{2, 3, 4, 5\}$ (and all are 0 when $R = 1$),

$$\mathbb{E}[y \mid R = 1, z] = \beta_1 + z' \gamma,$$

$$\mathbb{E}[y \mid R = 2, z] = \beta_1 + \beta_2 + z' \gamma,$$

$$\mathbb{E}[y \mid R = 3, z] = \beta_1 + \beta_3 + z' \gamma,$$

$$\mathbb{E}[y \mid R = 4, z] = \beta_1 + \beta_4 + z' \gamma,$$

$$\mathbb{E}[y \mid R = 5, z] = \beta_1 + \beta_5 + z' \gamma.$$

Now the differences are *not* forced to be equal:

$$\mathbb{E}[y \mid R = 4, z] - \mathbb{E}[y \mid R = 3, z] = (\beta_4 - \beta_3), \quad \mathbb{E}[y \mid R = 2, z] - \mathbb{E}[y \mid R = 1, z] = \beta_2.$$

So the dummy specification allows arbitrary nonlinearity across rating categories.

2 Difference in Differences (DID)

A reminder form (as in the notes)

A simple regression with a treatment dummy can be written as

$$y_i = x_i' \beta + \delta D_i + \varepsilon_i,$$

where $D_i = 1$ indicates “treated” and $D_i = 0$ indicates “control”. DID extends this logic to *before/after* comparisons.

Natural experimental approach (two periods, two groups)

Definitions (as in the notes).

- y_{it} : outcome of individual i in year (time) t .
- D_i : treatment dummy (= 1 if i is in the treatment group).
- T_t : time dummy (= 1 if $t = 2$; i.e., post period).

DID regression model (as in the notes). For $t = 1, 2$,

$$y_{it} = \beta_1 + \beta_2 T_t + \beta_3 D_i + \delta T_t D_i + \varepsilon_{it}. \quad (\text{DID-Model})$$

Derivation of the DID estimand (fully expanded)

Step 1: Define the within-individual change. Define

$$\Delta y_i \equiv y_{i2} - y_{i1}.$$

Step 2: Write y_{i2} and y_{i1} from the model. When $t = 2$, $T_2 = 1$:

$$y_{i2} = \beta_1 + \beta_2(1) + \beta_3 D_i + \delta(1) \cdot D_i + \varepsilon_{i2} = \beta_1 + \beta_2 + \beta_3 D_i + \delta D_i + \varepsilon_{i2}.$$

When $t = 1$, $T_1 = 0$:

$$y_{i1} = \beta_1 + \beta_2(0) + \beta_3 D_i + \delta(0) \cdot D_i + \varepsilon_{i1} = \beta_1 + \beta_3 D_i + \varepsilon_{i1}.$$

Step 3: Subtract to obtain Δy_i .

$$\begin{aligned} \Delta y_i &= y_{i2} - y_{i1} \\ &= (\beta_1 + \beta_2 + \beta_3 D_i + \delta D_i + \varepsilon_{i2}) - (\beta_1 + \beta_3 D_i + \varepsilon_{i1}) \\ &= \beta_2 + \delta D_i + (\varepsilon_{i2} - \varepsilon_{i1}). \end{aligned} \quad (\text{DID-Delta})$$

Step 4: Take expectations by treatment group (as in the notes). For treated ($D_i = 1$):

$$\begin{aligned} \mathbb{E}[\Delta y \mid D = 1] &= \mathbb{E}[y_{i2} \mid D_i = 1] - \mathbb{E}[y_{i1} \mid D_i = 1] \\ &= \mathbb{E}[\beta_2 + \delta \cdot 1 + (\varepsilon_{i2} - \varepsilon_{i1}) \mid D_i = 1] \\ &= \beta_2 + \delta + \mathbb{E}[\varepsilon_{i2} - \varepsilon_{i1} \mid D_i = 1]. \end{aligned}$$

For control ($D_i = 0$):

$$\begin{aligned} \mathbb{E}[\Delta y \mid D = 0] &= \mathbb{E}[y_{i2} \mid D_i = 0] - \mathbb{E}[y_{i1} \mid D_i = 0] \\ &= \mathbb{E}[\beta_2 + \delta \cdot 0 + (\varepsilon_{i2} - \varepsilon_{i1}) \mid D_i = 0] \\ &= \beta_2 + \mathbb{E}[\varepsilon_{i2} - \varepsilon_{i1} \mid D_i = 0]. \end{aligned}$$

Step 5: Difference the differences (as in the notes).

$$\begin{aligned}\mathbb{E}[\Delta y \mid D = 1] - \mathbb{E}[\Delta y \mid D = 0] &= \left(\beta_2 + \delta + \mathbb{E}[\varepsilon_{i2} - \varepsilon_{i1} \mid D_i = 1] \right) - \left(\beta_2 + \mathbb{E}[\varepsilon_{i2} - \varepsilon_{i1} \mid D_i = 0] \right) \\ &= \delta + \left(\mathbb{E}[\varepsilon_{i2} - \varepsilon_{i1} \mid D_i = 1] - \mathbb{E}[\varepsilon_{i2} - \varepsilon_{i1} \mid D_i = 0] \right).\end{aligned}$$

Under the DID identifying condition (“parallel trends” in mean errors),

$$\mathbb{E}[\varepsilon_{i2} - \varepsilon_{i1} \mid D_i = 1] = \mathbb{E}[\varepsilon_{i2} - \varepsilon_{i1} \mid D_i = 0],$$

the bracketed term equals 0, so

$$\boxed{\mathbb{E}[\Delta y \mid D = 1] - \mathbb{E}[\Delta y \mid D = 0] = \delta.}$$

3 Structural Break

Two-sample (two-regime) setup (as in the notes)

Let

$$Y_1 = X_1\beta_1 + \varepsilon_1, \quad Y_2 = X_2\beta_2 + \varepsilon_2,$$

with dimensions

$$Y_1 : n_1 \times 1, \quad Y_2 : n_2 \times 1.$$

We test

$$H_0 : \beta_1 = \beta_2 \quad \text{vs} \quad H_1 : \beta_1 \neq \beta_2.$$

Unrestricted model (allows $\beta_1 \neq \beta_2$) (as in the notes)

Step 1: Stack the two samples. Define

$$Y \equiv \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix}, \quad \beta \equiv \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}, \quad \varepsilon \equiv \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \end{pmatrix}.$$

Define the block-diagonal regressor matrix

$$X \equiv \begin{pmatrix} X_1 & 0 \\ 0 & X_2 \end{pmatrix}.$$

Then the unrestricted stacked model is

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = \begin{pmatrix} X_1 & 0 \\ 0 & X_2 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \end{pmatrix}. \quad (\text{SB-Unres})$$

Step 2: OLS under the block-diagonal design (as in the notes).

$$\begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = (X'X)^{-1}X'Y.$$

Because X is block diagonal,

$$X'X = \begin{pmatrix} X_1'X_1 & 0 \\ 0 & X_2'X_2 \end{pmatrix}, \quad (X'X)^{-1} = \begin{pmatrix} (X_1'X_1)^{-1} & 0 \\ 0 & (X_2'X_2)^{-1} \end{pmatrix},$$

and

$$X'Y = \begin{pmatrix} X_1'Y_1 \\ X_2'Y_2 \end{pmatrix}.$$

Therefore

$$\boxed{\begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \begin{pmatrix} (X_1'X_1)^{-1}X_1'Y_1 \\ (X_2'X_2)^{-1}X_2'Y_2 \end{pmatrix}}. \quad (\text{SB-BetaHat})$$

Step 3: Unrestricted SSE decomposition (as in the notes). Let $e_1 \equiv Y_1 - X_1\hat{\beta}_1$ and $e_2 \equiv Y_2 - X_2\hat{\beta}_2$. Then

$$e'e = e_1'e_1 + e_2'e_2. \quad (\text{SB-SSE-U})$$

Restricted model (imposes $\beta_1 = \beta_2$) (as in the notes)

Under H_0 , write $\beta_1 = \beta_2 \equiv \beta$. Stack the two equations to obtain the restricted (pooled) model

$$\underbrace{\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix}}_Y = \underbrace{\begin{pmatrix} X_1 \\ X_2 \end{pmatrix}}_{X_R} \beta + \underbrace{\begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \end{pmatrix}}_{\varepsilon}. \quad (\text{SB-Res})$$

Let $\hat{\beta}_R$ denote the pooled OLS estimator from (SB-Res),

$$\hat{\beta}_R = (X_R'X_R)^{-1}X_R'Y,$$

and define the restricted residual vector

$$e_R \equiv Y - X_R\hat{\beta}_R.$$

The restricted sum of squared errors (SSE) is therefore

$$\text{SSE}_R \equiv e_R'e_R.$$

Chow (structural break) F statistic (as in the notes)

Normality assumption used for the exact F -test. If

$$\varepsilon \sim N(0, \sigma^2 I),$$

then the Chow statistic compares restricted and unrestricted SSEs.

Step 1: Identify the number of restrictions. Unrestricted model estimates $2k$ parameters (both β_1 and β_2), while restricted estimates k parameters (common β). Hence the number of restrictions is

$$J = 2k - k = k.$$

Step 2: Write the Chow F statistic (exactly as in the notes).

$$\boxed{F = \frac{[e_R'e_R - (e_1'e_1 + e_2'e_2)]/k}{(e_1'e_1 + e_2'e_2)/(n_1 + n_2 - 2k)}}. \quad (\text{Chow-F})$$

Here the denominator degrees of freedom is $n_1 + n_2 - 2k$ (unrestricted residual df), and under H_0 the statistic follows an F distribution:

$$F \sim F_{k, n_1+n_2-2k} \quad \text{under } H_0.$$

Asymptotic (Wald) approach (as in the notes, fully expanded)

Independence across the two samples (as in the notes). Assume (X_1, ε_1) and (X_2, ε_2) are independent. Assume for convenience (as in the notes) that $n_1 = n_2 = n$.

Step 1: Asymptotic distribution of the two OLS estimators. Let

$$Q_1 \equiv \mathbb{E}[x_{1i}x'_{1i}], \quad Q_2 \equiv \mathbb{E}[x_{2i}x'_{2i}].$$

Then, asymptotically,

$$\sqrt{n} \begin{pmatrix} \hat{\beta}_1 - \beta_1 \\ \hat{\beta}_2 - \beta_2 \end{pmatrix} \xrightarrow{d} N \left(0, \begin{pmatrix} \sigma^2 Q_1^{-1} & 0 \\ 0 & \sigma^2 Q_2^{-1} \end{pmatrix} \right). \quad (\text{SB-AN})$$

Step 2: Express $H_0 : \beta_1 = \beta_2$ as linear restrictions. Define the stacked parameter $\theta \equiv (\beta'_1, \beta'_2)'$ and take

$$R \equiv (I_k \quad -I_k), \quad q \equiv 0.$$

Then

$$H_0 : \beta_1 = \beta_2 \iff H_0 : R\theta - q = \beta_1 - \beta_2 = 0.$$

Step 3: Asymptotic distribution of $\sqrt{n}(R\hat{\theta} - q)$ (as in the notes). Let $\hat{\theta} \equiv (\hat{\beta}'_1, \hat{\beta}'_2)'$. Then

$$\sqrt{n}(R\hat{\theta} - q) = \sqrt{n}R(\hat{\theta} - \theta) \xrightarrow{d} N \left(0, R \begin{pmatrix} \sigma^2 Q_1^{-1} & 0 \\ 0 & \sigma^2 Q_2^{-1} \end{pmatrix} R' \right). \quad (\text{SB-R})$$

Step 4: Wald statistic based on the difference $\hat{\beta}_1 - \hat{\beta}_2$ (as in the notes). Since $R\hat{\theta} - q = \hat{\beta}_1 - \hat{\beta}_2$, a Wald statistic is

$$W = n(\hat{\beta}_1 - \hat{\beta}_2)' \left[\hat{\Sigma}_1 + \hat{\Sigma}_2 \right]^{-1} (\hat{\beta}_1 - \hat{\beta}_2), \quad (\text{SB-W})$$

where $\hat{\Sigma}_1$ and $\hat{\Sigma}_2$ are consistent estimators of the asymptotic covariance matrices of $\sqrt{n}(\hat{\beta}_1 - \beta_1)$ and $\sqrt{n}(\hat{\beta}_2 - \beta_2)$:

$$\Sigma_1 = \sigma^2 Q_1^{-1}, \quad \Sigma_2 = \sigma^2 Q_2^{-1}, \quad \hat{\Sigma}_1 \approx s_1^2 \left(\frac{X'_1 X_1}{n} \right)^{-1}, \quad \hat{\Sigma}_2 \approx s_2^2 \left(\frac{X'_2 X_2}{n} \right)^{-1},$$

with

$$s_1^2 = \frac{e'_1 e_1}{n - k}, \quad s_2^2 = \frac{e'_2 e_2}{n - k}.$$

Under $H_0 : \beta_1 = \beta_2$, the Wald statistic satisfies (as in the notes)

$$\boxed{W \xrightarrow{d} \chi^2(\text{rank}(R)) = \chi^2(k).}$$

Endogeneity, Instruments, and Why OLS Fails

4 Classical OLS Assumptions (as written)

1. **Linearity**

$$y = X\beta + \varepsilon.$$

2. **Full rank**

$$\text{rank}(X) = k.$$

3. **Exogeneity**

$$\mathbb{E}[\varepsilon \mid X] = 0.$$

More process: what exogeneity implies for $\mathbb{E}[x_i\varepsilon_i]$. Let x_i denote the regressor vector for observation i (so x_i is measurable w.r.t. X). Then, using the law of iterated expectations (LIE),

$$\mathbb{E}[x_i\varepsilon_i] = \mathbb{E}[\mathbb{E}[x_i\varepsilon_i \mid x_i]].$$

Inside the conditional expectation, treat x_i as given (a constant):

$$\mathbb{E}[x_i\varepsilon_i \mid x_i] = x_i \mathbb{E}[\varepsilon_i \mid x_i].$$

Under exogeneity, $\mathbb{E}[\varepsilon_i \mid x_i] = 0$, hence

$$\mathbb{E}[x_i\varepsilon_i \mid x_i] = x_i \cdot 0 = 0,$$

and therefore

$$\boxed{\mathbb{E}[x_i\varepsilon_i] = \mathbb{E}[0] = 0.}$$

Law of iterated expectations (as written).

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]].$$

Applying LIE to ε :

$$\mathbb{E}[\varepsilon] = \mathbb{E}[\mathbb{E}[\varepsilon \mid X]] = \mathbb{E}[0] = 0.$$

4. **Homoscedasticity and nonautocorrelation**

$$\text{Var}(\varepsilon_i \mid X) = \sigma^2, \quad i = 1, \dots, n,$$

$$\text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0, \quad i \neq j.$$

5. **X may be fixed and random.**

5 Endogeneity and Instrumental Variables (as written, fully expanded)

Endogeneity (contrast to exogeneity)

Assumptions (as written).

1. **Linearity** $y = X\beta + \varepsilon.$

2. **Full rank** $\text{rank}(X) = k.$

3. **Endogeneity** $\mathbb{E}[\varepsilon_i \mid x_i] = \eta \neq 0.$

Introduce instruments

Additional variables (as written). Assume there is an additional vector of variables z_i with

$$z_i \in \mathbb{R}^{L \times 1}, \quad L > k.$$

Instrument conditions (as written).

- (1) **Exogeneity:** z_i is uncorrelated with the disturbance ε_i .
- (2) **Relevance:** z_i is correlated with the explanatory variable x_i .

We call z_i *instrumental variables*.

Homoscedasticity conditional on instruments (as written).

$$\mathbb{E}[\varepsilon_i^2 \mid z_i] = \sigma^2.$$

Random sampling (as written).

$$(x_i, z_i, \varepsilon_i) \sim \text{i.i.d.}$$

6 Moment Conditions for (x_i, z_i)

Step 0 (dimensions and notation). Let

$$x_i \in \mathbb{R}^{k \times 1}, \quad z_i \in \mathbb{R}^{L \times 1}, \quad L \geq k,$$

and assume $(x_i, z_i, \varepsilon_i)$ are i.i.d. across i with finite second moments.

Step 1 (second-moment matrices: definitions). Define the population second moments (as written in the notes):

$$\begin{aligned} \mathbb{E}[x_i x_i'] &= Q_{xx} < \infty, & \text{rank}(Q_{xx}) &= k. \\ \mathbb{E}[z_i z_i'] &= Q_{zz} < \infty, & \text{rank}(Q_{zz}) &= L. \\ \mathbb{E}[z_i x_i'] &= Q_{zx} < \infty, & \text{rank}(Q_{zx}) &= k, & Q_{zx} &\in \mathbb{R}^{L \times k}. \end{aligned}$$

Step 2 (what the rank conditions mean; why they are imposed).

- $\text{rank}(Q_{xx}) = k$ means Q_{xx} is full rank (nonsingular), so Q_{xx}^{-1} exists. This is the population analog of “no perfect multicollinearity” among regressors.
- $\text{rank}(Q_{zz}) = L$ means Q_{zz} is full rank (nonsingular), so Q_{zz}^{-1} exists. This is the population analog of “no perfect multicollinearity” among instruments.
- $\text{rank}(Q_{zx}) = k$ means Q_{zx} has full *column* rank, i.e. its k columns are linearly independent. This is the population **instrument relevance / identification** condition: the instruments contain enough independent variation correlated with x_i to identify all k components of β .

Step 3 (sample analogs; LLN convergence that will be used later). Let the data matrices be

$$X \equiv \begin{pmatrix} x'_1 \\ x'_2 \\ \vdots \\ x'_n \end{pmatrix} \in \mathbb{R}^{n \times k}, \quad Z \equiv \begin{pmatrix} z'_1 \\ z'_2 \\ \vdots \\ z'_n \end{pmatrix} \in \mathbb{R}^{n \times L}, \quad \varepsilon \equiv \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

Then

$$\frac{1}{n} X'X = \frac{1}{n} \sum_{i=1}^n x_i x'_i, \quad \frac{1}{n} Z'Z = \frac{1}{n} \sum_{i=1}^n z_i z'_i, \quad \frac{1}{n} Z'X = \frac{1}{n} \sum_{i=1}^n z_i x'_i.$$

By i.i.d. LLN and finite second moments,

$$\frac{1}{n} X'X \xrightarrow{p} \mathbb{E}[x_i x'_i] = Q_{xx}, \quad \frac{1}{n} Z'Z \xrightarrow{p} \mathbb{E}[z_i z'_i] = Q_{zz}, \quad \frac{1}{n} Z'X \xrightarrow{p} \mathbb{E}[z_i x'_i] = Q_{zx}.$$

Since $\text{rank}(Q_{xx}) = k$ and $\text{rank}(Q_{zz}) = L$, the inverse maps are continuous at Q_{xx} and Q_{zz} , so

$$\left(\frac{1}{n} X'X \right)^{-1} \xrightarrow{p} Q_{xx}^{-1}, \quad \left(\frac{1}{n} Z'Z \right)^{-1} \xrightarrow{p} Q_{zz}^{-1}.$$

Step 4 (instrument exogeneity $\Rightarrow$ orthogonality moment).

Exogeneity of instruments (as written).

$$\mathbb{E}[\varepsilon_i \mid z_i] = 0.$$

Derive the implied moment restriction $\mathbb{E}[z_i \varepsilon_i] = 0$ (all steps). Using the law of iterated expectations (LIE),

$$\mathbb{E}[z_i \varepsilon_i] = \mathbb{E}[\mathbb{E}[z_i \varepsilon_i \mid z_i]].$$

Inside the conditional expectation, treat z_i as given (a constant):

$$\mathbb{E}[z_i \varepsilon_i \mid z_i] = z_i \mathbb{E}[\varepsilon_i \mid z_i].$$

By instrument exogeneity, $\mathbb{E}[\varepsilon_i \mid z_i] = 0$, hence

$$\mathbb{E}[z_i \varepsilon_i \mid z_i] = z_i \cdot 0 = 0,$$

so

$$\boxed{\mathbb{E}[z_i \varepsilon_i] = \mathbb{E}[0] = 0.}$$

Step 5 (population IV moment condition for β ; explicit algebra). From the structural equation $y_i = x'_i \beta + \varepsilon_i$, multiply both sides by z_i and take expectations:

$$\mathbb{E}[z_i y_i] = \mathbb{E}[z_i x'_i \beta] + \mathbb{E}[z_i \varepsilon_i].$$

Because β is constant,

$$\mathbb{E}[z_i x'_i \beta] = \mathbb{E}[z_i x'_i] \beta = Q_{zx} \beta.$$

And from Step 4, $\mathbb{E}[z_i \varepsilon_i] = 0$. Therefore,

$$\boxed{\mathbb{E}[z_i y_i] = Q_{zx} \beta.} \quad \text{(IV-moment)}$$

Equivalently,

$$\boxed{\mathbb{E}[z_i (y_i - x'_i \beta)] = 0.}$$

Step 6 (why $\text{rank}(Q_{zx}) = k$ is an identification condition). Equation (??) is a system of L equations in k unknowns ($L \geq k$). If $\text{rank}(Q_{zx}) = k$ (full column rank), then β is identified. One convenient way to solve (when Q_{zz} is nonsingular) is to left-multiply by $Q'_{zx}Q_{zz}^{-1}$:

$$Q'_{zx}Q_{zz}^{-1}\mathbb{E}[z_i y_i] = Q'_{zx}Q_{zz}^{-1}Q_{zx}\beta.$$

If $Q'_{zx}Q_{zz}^{-1}Q_{zx}$ is nonsingular (implied by $\text{rank}(Q_{zx}) = k$ and $\text{rank}(Q_{zz}) = L$), then

$$\boxed{\beta = (Q'_{zx}Q_{zz}^{-1}Q_{zx})^{-1}Q'_{zx}Q_{zz}^{-1}\mathbb{E}[z_i y_i].}$$

(This is the population analog of 2SLS.)

7 Why OLS Fails Under Endogeneity (more detailed process added)

(1) OLS is biased (as written, with more intermediate lines)

Start from the OLS decomposition.

$$\hat{\beta} = (X'X)^{-1}X'y.$$

Substitute $y = X\beta + \varepsilon$:

$$\begin{aligned}\hat{\beta} &= (X'X)^{-1}X'(X\beta + \varepsilon) \\ &= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'\varepsilon \\ &= \beta + (X'X)^{-1}X'\varepsilon.\end{aligned}$$

So

$$\boxed{\hat{\beta} = \beta + (X'X)^{-1}X'\varepsilon.}$$

Compute the conditional expectation given X (with explicit linearity steps).

$$\begin{aligned}\mathbb{E}[\hat{\beta} \mid X] &= \mathbb{E}[\beta + (X'X)^{-1}X'\varepsilon \mid X] \\ &= \mathbb{E}[\beta \mid X] + \mathbb{E}[(X'X)^{-1}X'\varepsilon \mid X] \quad (\text{linearity of conditional expectation}) \\ &= \beta + \mathbb{E}[(X'X)^{-1}X'\varepsilon \mid X] \quad (\text{since } \beta \text{ is constant}) \\ &= \beta + (X'X)^{-1}X'\mathbb{E}[\varepsilon \mid X] \quad (\text{pull out functions of } X).\end{aligned}$$

Under endogeneity (as written), $\mathbb{E}[\varepsilon \mid X] = \eta \neq 0$, hence

$$\boxed{\mathbb{E}[\hat{\beta} \mid X] = \beta + (X'X)^{-1}X'\eta \neq \beta.}$$

(Optional) Unconditional bias (extra process). Taking expectations again,

$$\mathbb{E}[\hat{\beta}] = \mathbb{E}[\mathbb{E}[\hat{\beta} \mid X]] = \mathbb{E}[\beta + (X'X)^{-1}X'\eta] = \beta + \mathbb{E}[(X'X)^{-1}X'\eta] \neq \beta,$$

unless special cases make the added term equal to zero.

(2) OLS is inconsistent (as written, with fully expanded probability limits)

Endogeneity in moment form (as written).

$$\mathbb{E}[x_i \varepsilon_i] \neq 0.$$

Rewrite $\hat{\beta}$ in a “sample moments” form. From

$$\hat{\beta} - \beta = (X'X)^{-1}X'\varepsilon,$$

multiply and divide by n :

$$\hat{\beta} - \beta = \left(\frac{X'X}{n} \right)^{-1} \left(\frac{X'\varepsilon}{n} \right).$$

Now expand each term as a sum over i .

Expand $X'X/n$. Since X stacks rows x'_i ,

$$X'X = \sum_{i=1}^n x_i x'_i, \quad \Rightarrow \quad \frac{X'X}{n} = \frac{1}{n} \sum_{i=1}^n x_i x'_i.$$

Expand $X'\varepsilon/n$. Similarly,

$$X'\varepsilon = \sum_{i=1}^n x_i \varepsilon_i, \quad \Rightarrow \quad \frac{X'\varepsilon}{n} = \frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i.$$

Therefore,

$$\boxed{\hat{\beta} = \beta + \left(\frac{1}{n} \sum_{i=1}^n x_i x'_i \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i \right)}. \quad (\text{OLS-moment})$$

Take probability limits term-by-term (LLN). Assume i.i.d. sampling and finite second moments so that LLN applies.

First term:

$$\frac{1}{n} \sum_{i=1}^n x_i x'_i \xrightarrow{p} \mathbb{E}[x_i x'_i] = Q_{xx}.$$

If $\text{rank}(Q_{xx}) = k$, then Q_{xx} is nonsingular, and by continuous mapping (matrix inverse),

$$\left(\frac{1}{n} \sum_{i=1}^n x_i x'_i \right)^{-1} \xrightarrow{p} Q_{xx}^{-1}.$$

Second term:

$$\frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i \xrightarrow{p} \mathbb{E}[x_i \varepsilon_i] \neq 0.$$

Combine limits (product rule / Slutsky). Using (OLS-moment),

$$\hat{\beta} \xrightarrow{p} \beta + Q_{xx}^{-1} \mathbb{E}[x_i \varepsilon_i].$$

So the probability limit differs from β whenever $\mathbb{E}[x_i \varepsilon_i] \neq 0$:

$$\boxed{\hat{\beta} \text{ is inconsistent when } \mathbb{E}[x_i \varepsilon_i] \neq 0.}$$

Notes' final conclusion line (kept; not deleted). The notes write the concluding inequality as

$$\hat{\beta} \xrightarrow{p} \beta + Q_{xx}\mathbb{E}[x_i\varepsilon_i] \neq \beta,$$

expressing that the probability limit differs from β when $\mathbb{E}[x_i\varepsilon_i] \neq 0$. (If Q_{xx} is defined as $\mathbb{E}[x_ix_i']$, the standard algebra yields the Q_{xx}^{-1} form above; the core message is the same: the plim is not β under endogeneity.)

Bias, Consistency, and Identification: How Exogeneity and Relevance Matter

Key distinction.

- **Bias** is a finite-sample expectation concept:

$$\text{Bias}(\hat{\theta}) \equiv \mathbb{E}[\hat{\theta}] - \theta \quad (\text{or conditional bias } \mathbb{E}[\hat{\theta} | X] - \theta).$$

- **Consistency / asymptotic bias** is a probability-limit concept:

$$\hat{\theta} \xrightarrow{p} \theta \iff \hat{\theta} \xrightarrow{p} \theta = \theta.$$

If $\hat{\theta} \not\xrightarrow{p} \theta$, we say the estimator has *asymptotic bias* (is inconsistent).

- **Identification** is a population concept: do the model + moment conditions pin down a unique θ ?

1. OLS: Exogeneity is the unbiasedness (and consistency) condition

Start from the OLS decomposition:

$$\hat{\beta} = \beta + (X'X)^{-1}X'\varepsilon.$$

Take conditional expectation given X :

$$\mathbb{E}[\hat{\beta} | X] - \beta = (X'X)^{-1}X'\mathbb{E}[\varepsilon | X].$$

Therefore:

$$\mathbb{E}[\varepsilon | X] = 0 \implies \mathbb{E}[\hat{\beta} | X] = \beta \quad (\text{OLS is conditionally unbiased}).$$

If instead $\mathbb{E}[\varepsilon | X] \neq 0$ (endogeneity), then

$$\mathbb{E}[\hat{\beta} | X] \neq \beta \implies \text{OLS is biased, and typically also inconsistent.}$$

2. IV/2SLS: Two instrument conditions and what they control

Consider the structural equation

$$y_i = x_i'\beta + \varepsilon_i,$$

and an instrument vector $z_i \in \mathbb{R}^{L \times 1}$.

Instrument exogeneity (validity).

$$\mathbb{E}[\varepsilon_i | z_i] = 0 \iff \mathbb{E}[z_i \varepsilon_i] = 0.$$

This delivers the population moment condition:

$$\mathbb{E}[z_i(y_i - x_i' \beta)] = 0.$$

Interpretation: exogeneity ensures the “instrument-generated” variation is uncorrelated with the error. If exogeneity fails ($\mathbb{E}[z_i \varepsilon_i] \neq 0$), then IV/2SLS is generally **inconsistent**:

$$\xrightarrow{p} \hat{\beta}_{IV} \neq \beta \quad (\text{asymptotic bias}).$$

Instrument relevance (strength / identification). Let

$$Q_{zx} \equiv \mathbb{E}[z_i x_i'] \in \mathbb{R}^{L \times k}.$$

A standard relevance/identification condition is

$$\text{rank}(Q_{zx}) = k.$$

Interpretation: instruments must be correlated with the endogenous regressors in enough independent directions to identify all components of β . If relevance fails (rank deficiency), β is **not identified** (the IV problem is underidentified).

Weak relevance (weak instruments). If relevance holds only weakly (correlation is very small), then even with valid instruments:

- the IV estimator can have **large variance** (imprecise);
- the IV estimator can exhibit **substantial finite-sample bias** (often toward OLS);
- standard asymptotic approximations can perform poorly (weak-IV inference issues).

Summary (what relates to bias?)

- **Exogeneity** $\Rightarrow$ *validity*: controls whether the estimator targets the correct parameter (no asymptotic bias / consistency). For OLS, it is also the direct unbiasedness condition.
- **Relevance** $\Rightarrow$ *identification/strength*: controls whether β is identifiable and how noisy the estimator is. Weak relevance can create large *finite-sample bias* and poor inference even when exogeneity holds.

OLS and IV as Method-of-Moments Estimators

Assumptions (OLS)

1. Linearity

$$y = X\beta + \varepsilon.$$

2. Full rank

$$\text{rank}(X) = k.$$

3. Exogeneity

$$\mathbb{E}[\varepsilon | X] = 0.$$

Law of iterated expectations (as written).

$$\mathbb{E}[y] = \mathbb{E}[\mathbb{E}[y \mid X]].$$

Apply it to the disturbance:

$$\mathbb{E}[\varepsilon] = \mathbb{E}[\mathbb{E}[\varepsilon \mid X]] = \mathbb{E}[0] = 0.$$

4. Homoscedasticity and nonautocorrelation

$$\text{Var}(\varepsilon_i \mid X) = \sigma^2, \quad i = 1, \dots, n,$$

$$\text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0, \quad i \neq j.$$

Equivalently (matrix form),

$$\mathbb{E}[\varepsilon\varepsilon' \mid X] = \sigma^2 I_n.$$

5. X may be fixed and random.

Exogeneity implies an orthogonality moment

Step 1: Work observation-by-observation. Let x_i denote the $k \times 1$ regressor vector for observation i , and let ε_i be the corresponding error. From exogeneity, we have

$$\mathbb{E}[\varepsilon_i \mid x_i] = 0.$$

Step 2: Use iterated expectations to show $\mathbb{E}[\varepsilon_i x_i] = 0$ (as written, with full steps).

$$\begin{aligned} \mathbb{E}[\varepsilon_i x_i] &= \mathbb{E}[\mathbb{E}[\varepsilon_i x_i \mid x_i]] \\ &= \mathbb{E}[x_i \mathbb{E}[\varepsilon_i \mid x_i]] \quad (x_i \text{ is fixed given } x_i) \\ &= \mathbb{E}[x_i \cdot 0] \\ &= 0. \end{aligned}$$

Equivalently (same statement with factors swapped),

$$\mathbb{E}[x_i \varepsilon_i] = 0.$$

Step 3: Substitute $\varepsilon_i = y_i - x_i' \beta$ (as written). Since $y_i = x_i' \beta + \varepsilon_i$, we have $\varepsilon_i = y_i - x_i' \beta$, so

$$\mathbb{E}[x_i \varepsilon_i] = \mathbb{E}[x_i (y_i - x_i' \beta)] = 0.$$

This is the population moment condition that motivates the OLS method-of-moments estimator.

OLS as a Method-of-Moments (MOM) Estimator

Method-of-moments idea (as written). A method of moment estimator b_{MOM} sets the sample analogue to 0:

$$\frac{1}{n} \sum_{i=1}^n x_i (y_i - x_i' b_{\text{MOM}}) = 0. \quad (\text{MOM-OLS-1})$$

Step 1: Expand the sample moment (as written, fully expanded). Distribute the product inside the sum:

$$\frac{1}{n} \sum_{i=1}^n x_i y_i - \frac{1}{n} \sum_{i=1}^n x_i x'_i b_{\text{MOM}} = 0. \quad (\text{MOM-OLS-2})$$

Step 2: Factor out b_{MOM} from the second term (as written). Because b_{MOM} does not depend on i ,

$$\frac{1}{n} \sum_{i=1}^n x_i x'_i b_{\text{MOM}} = \left(\frac{1}{n} \sum_{i=1}^n x_i x'_i \right) b_{\text{MOM}}.$$

So (MOM-OLS-2) becomes

$$\left(\frac{1}{n} \sum_{i=1}^n x_i x'_i \right) b_{\text{MOM}} = \left(\frac{1}{n} \sum_{i=1}^n x_i y_i \right). \quad (\text{MOM-OLS-3})$$

Step 3: Solve for b_{MOM} (add fully intermediate steps). Assuming $\left(\frac{1}{n} \sum_{i=1}^n x_i x'_i \right)$ is invertible, left-multiply both sides of (MOM-OLS-3) by its inverse:

$$\left(\frac{1}{n} \sum_{i=1}^n x_i x'_i \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n x_i x'_i \right) b_{\text{MOM}} = \left(\frac{1}{n} \sum_{i=1}^n x_i x'_i \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n x_i y_i \right).$$

Since $A^{-1}A = I$, this yields

$$b_{\text{MOM}} = \left(\frac{1}{n} \sum_{i=1}^n x_i x'_i \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n x_i y_i \right). \quad (\text{MOM-OLS-4})$$

Step 4: Rewrite in matrix notation and identify OLS (as written, with full mapping).

Let

$$X = \begin{pmatrix} x'_1 \\ x'_2 \\ \vdots \\ x'_n \end{pmatrix}, \quad y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}.$$

Then

$$X'X = \sum_{i=1}^n x_i x'_i, \quad X'y = \sum_{i=1}^n x_i y_i.$$

So (MOM-OLS-4) is equivalently

$$b_{\text{MOM}} = (X'X)^{-1}X'y \equiv b_{\text{OLS}}.$$

Assumptions (IV)

1. Linearity

$$y = X\beta + \varepsilon.$$

2. Full rank

$$\text{rank}(X) = k.$$

3. Endogeneity

$$\mathbb{E}[\varepsilon_i \mid x_i] = \eta \neq 0.$$

4. Homoscedasticity (conditional on instruments)

$$\mathbb{E}[\varepsilon_i^2 \mid z_i] = \sigma^2.$$

5. Random sampling

$$(x_i, z_i, \varepsilon_i) \sim \text{i.i.d.}$$

6. Moments of x_i and z_i

$$\begin{aligned} \mathbb{E}[x_i x_i'] &= Q_{xx} < \infty, & \text{rank}(Q_{xx}) &= k, \\ \mathbb{E}[z_i z_i'] &= Q_{zz} < \infty, & \text{rank}(Q_{zz}) &= L, \\ \mathbb{E}[z_i x_i'] &= Q_{zx} < \infty, & \text{rank}(Q_{zx}) &= k, & Q_{zx} &\in \mathbb{R}^{L \times k}. \end{aligned}$$

7. Exogeneity of instruments

$$\mathbb{E}[\varepsilon_i \mid z_i] = 0.$$

Instrument exogeneity implies $\mathbb{E}[z_i \varepsilon_i] = 0$ (as written, fully expanded)

$$\mathbb{E}[z_i \varepsilon_i] = \mathbb{E}[\mathbb{E}[z_i \varepsilon_i \mid z_i]] = \mathbb{E}[z_i \mathbb{E}[\varepsilon_i \mid z_i]] = \mathbb{E}[z_i \cdot 0] = 0.$$

IV moment condition (as written, with substitution)

Since $\varepsilon_i = y_i - x_i' \beta$, we have

$$\mathbb{E}[z_i \varepsilon_i] = \mathbb{E}[z_i (y_i - x_i' \beta)] = 0.$$

This is the population IV orthogonality condition.

IV as a Method-of-Moments (MOM) Estimator

Method-of-moments idea (as written). A method-of-moments IV estimator b_{IV} sets the sample analogue to 0:

$$\frac{1}{n} \sum_{i=1}^n z_i (y_i - x_i' b_{\text{IV}}) = 0. \quad (\text{MOM-IV-1})$$

Step 1: Expand the sample moment (as written, fully expanded). Distribute inside the sum:

$$\frac{1}{n} \sum_{i=1}^n z_i y_i - \frac{1}{n} \sum_{i=1}^n z_i x_i' b_{\text{IV}} = 0. \quad (\text{MOM-IV-2})$$

Step 2: Factor out b_{IV} (as written). Because b_{IV} does not depend on i ,

$$\frac{1}{n} \sum_{i=1}^n z_i x_i' b_{\text{IV}} = \left(\frac{1}{n} \sum_{i=1}^n z_i x_i' \right) b_{\text{IV}}.$$

So (MOM-IV-2) becomes

$$\left(\frac{1}{n} \sum_{i=1}^n z_i x_i' \right) b_{\text{IV}} = \left(\frac{1}{n} \sum_{i=1}^n z_i y_i \right). \quad (\text{MOM-IV-3})$$

Step 3: Multiply both sides by n (pure algebra; keeps the same condition).

$$\left(\sum_{i=1}^n z_i x'_i \right) b_{IV} = \sum_{i=1}^n z_i y_i. \quad (\text{MOM-IV-4})$$

Step 4: Just-identified case (as written): if $L = k$. If $L = k$, then $\sum_{i=1}^n z_i x'_i$ is a $k \times k$ matrix, and assuming it is invertible, left-multiply both sides of (MOM-IV-4) by its inverse:

$$\left(\sum_{i=1}^n z_i x'_i \right)^{-1} \left(\sum_{i=1}^n z_i x'_i \right) b_{IV} = \left(\sum_{i=1}^n z_i x'_i \right)^{-1} \left(\sum_{i=1}^n z_i y_i \right).$$

Hence,

$$b_{IV} = \left(\sum_{i=1}^n z_i x'_i \right)^{-1} \left(\sum_{i=1}^n z_i y_i \right). \quad (\text{MOM-IV-5})$$

Step 5: Rewrite in matrix notation (as written). Let

$$Z = \begin{pmatrix} z'_1 \\ z'_2 \\ \vdots \\ z'_n \end{pmatrix} \in \mathbb{R}^{n \times L}.$$

Then

$$Z'X = \sum_{i=1}^n z_i x'_i, \quad Z'y = \sum_{i=1}^n z_i y_i.$$

So (MOM-IV-5) becomes

$$\boxed{b_{IV} = (Z'X)^{-1}Z'y} \quad (\text{when } L = k).$$

Comparison line (as written).

$$b_{IV} = (Z'X)^{-1}Z'y, \quad b_{OLS} = (X'X)^{-1}X'y.$$

Just-identified vs Overidentified IV (what it means)

Consider the structural equation with endogenous regressors:

$$y_i = x_{1i}\beta_1 + \cdots + x_{ki}\beta_k + \varepsilon_i,$$

and suppose there are k *endogenous* regressors collected in the vector

$$x_i \in \mathbb{R}^{k \times 1}.$$

Let the instrument vector be

$$z_i \in \mathbb{R}^{L \times 1},$$

where L is the number of (excluded) instruments available.¹

¹Typically we may also have exogenous controls that can serve as their own instruments; the counting below is about *excluded* instruments relative to endogenous regressors.

Step 1: The IV moment conditions. Instrument exogeneity implies the population orthogonality conditions

$$\mathbb{E}[z_i \varepsilon_i] = 0.$$

Since $\varepsilon_i = y_i - x_i' \beta$, this becomes

$$\mathbb{E}[z_i (y_i - x_i' \beta)] = 0.$$

This is a system of L moment conditions (one for each component of z_i).

Step 2: Count equations vs unknowns. There are:

$$\text{Number of moment conditions} = L, \quad \text{Number of unknown coefficients in } \beta = k.$$

Definition (just-identified). The model is **just-identified** if

$$L = k.$$

Intuition: the number of valid moment conditions exactly matches the number of parameters to be identified. In the just-identified case, there is *exactly enough* information to solve for β .

Definition (overidentified). The model is **overidentified** if

$$L > k.$$

Intuition: there are *more* moment conditions than unknown parameters. The moment conditions generally cannot all be satisfied exactly in finite samples, so IV/2SLS chooses $\hat{\beta}$ that makes them “as close to zero as possible” in a least-squares (GMM) sense.

Definition (underidentified). The model is **underidentified** if

$$L < k.$$

Intuition: there are not enough moment conditions to pin down β ; the parameter is not identified.

Step 3: Why overidentification enables an instrument-validity test.

- If $L = k$ (just-identified), the sample moment equations can be solved exactly, leaving *no extra restrictions*. Therefore, we *cannot* test instrument validity from “leftover” information (no overidentifying restrictions).
- If $L > k$ (overidentified), there are extra moment conditions beyond what is needed to identify β . Those extra conditions generate *overidentifying restrictions*, which can be tested using a Hansen J test (or Sargan test under homoskedasticity).

Step 4: Concrete examples. Example A (just-identified). One endogenous regressor ($k = 1$): x_i . One excluded instrument ($L = 1$): z_i .

$$k = 1, L = 1 \Rightarrow \text{just-identified.}$$

Example B (overidentified). One endogenous regressor ($k = 1$): x_i . Two excluded instruments ($L = 2$): (z_{1i}, z_{2i}) .

$$k = 1, L = 2 \Rightarrow \text{overidentified.}$$

Now you can (in principle) run an overidentification test (Hansen J).

Example C (underidentified). Two endogenous regressors ($k = 2$): (x_{1i}, x_{2i}) . One excluded instrument ($L = 1$).

$$k = 2, L = 1 \Rightarrow \text{underidentified (cannot identify both coefficients).}$$

Step 5: Identification rank condition (beyond counting). Counting L vs k is necessary but not sufficient. A standard relevance/identification condition is

$$\text{rank}(\mathbb{E}[z_i x_i']) = k.$$

This ensures the instruments move the endogenous regressors in enough independent directions to identify β .

Econometrics I: IV Assumptions; Consistency and Asymptotic Distribution (Fully Expanded)

8 Assumptions (IV)

1. Linearity

$$y = X\beta + \varepsilon.$$

2. Full rank

$$\text{rank}(X) = k.$$

3. Endogeneity (OLS exogeneity fails)

$$\mathbb{E}[\varepsilon_i \mid x_i] = \eta \neq 0.$$

4. Homoscedasticity conditional on instruments

$$\mathbb{E}[\varepsilon_i^2 \mid z_i] = \sigma^2.$$

5. Random sampling

$$(x_i, z_i, \varepsilon_i) \text{ are i.i.d. across } i = 1, \dots, n.$$

6. Moments of x_i and z_i (as written)

$$\mathbb{E}[x_i x_i'] = Q_{xx} < \infty, \quad \text{rank}(Q_{xx}) = k.$$

$$\mathbb{E}[z_i z_i'] = Q_{zz} < \infty, \quad \text{rank}(Q_{zz}) = L.$$

$$\mathbb{E}[z_i x_i'] = Q_{zx} < \infty, \quad \text{rank}(Q_{zx}) = k, \quad Q_{zx} \in \mathbb{R}^{L \times k}.$$

(Typically $L \geq k$; the just-identified case is $L = k$.)

7. Exogeneity of instruments (validity)

$$\mathbb{E}[\varepsilon_i \mid z_i] = 0.$$

9 Just-identified IV estimator ($L = k$) and consistency

Step 1: Define the just-identified IV estimator (as written). When $L = k$ (so that $Z'X$ is $k \times k$), define

$$b_{IV} \equiv (Z'X)^{-1}Z'y. \quad (\text{IV-1})$$

Step 2: Substitute the structural model $y = X\beta + \varepsilon$.

$$\begin{aligned} b_{IV} &= (Z'X)^{-1}Z'y \\ &= (Z'X)^{-1}Z'(X\beta + \varepsilon) \\ &= (Z'X)^{-1}Z'X\beta + (Z'X)^{-1}Z'\varepsilon. \end{aligned} \quad (\text{IV-2})$$

Since $(Z'X)^{-1}Z'X = I_k$ in the just-identified case ($Z'X$ is square and invertible),

$$\boxed{b_{IV} = \beta + (Z'X)^{-1}Z'\varepsilon.} \quad (\text{IV-3})$$

Step 3: Rewrite $(Z'X)^{-1}Z'\varepsilon$ as sample sums (as written). Note that

$$Z'X = \sum_{i=1}^n z_i x'_i, \quad Z'\varepsilon = \sum_{i=1}^n z_i \varepsilon_i.$$

So (IV-3) becomes

$$b_{IV} = \beta + \left(\sum_{i=1}^n z_i x'_i \right)^{-1} \left(\sum_{i=1}^n z_i \varepsilon_i \right). \quad (\text{IV-4})$$

Multiply and divide by n inside the product (pure algebra):

$$\left(\sum_{i=1}^n z_i x'_i \right)^{-1} \left(\sum_{i=1}^n z_i \varepsilon_i \right) = \left(\frac{1}{n} \sum_{i=1}^n z_i x'_i \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n z_i \varepsilon_i \right).$$

Hence

$$\boxed{b_{IV} = \beta + \left(\frac{1}{n} \sum_{i=1}^n z_i x'_i \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n z_i \varepsilon_i \right).} \quad (\text{IV-5})$$

Step 4: Take probability limits (LLN + continuity). By i.i.d. LLN and finite moments,

$$\frac{1}{n} \sum_{i=1}^n z_i x'_i \xrightarrow{p} \mathbb{E}[z_i x'_i] = Q_{zx}. \quad (\text{IV-6})$$

Because $\text{rank}(Q_{zx}) = k$ and we are in the just-identified case ($L = k$), Q_{zx} is invertible, so

$$\left(\frac{1}{n} \sum_{i=1}^n z_i x'_i \right)^{-1} \xrightarrow{p} Q_{zx}^{-1}. \quad (\text{IV-7})$$

Also by LLN,

$$\frac{1}{n} \sum_{i=1}^n z_i \varepsilon_i \xrightarrow{p} \mathbb{E}[z_i \varepsilon_i]. \quad (\text{IV-8})$$

Step 5: Show $\mathbb{E}[z_i \varepsilon_i] = 0$ from instrument exogeneity (as written, fully expanded). Using the law of iterated expectations,

$$\mathbb{E}[z_i \varepsilon_i] = \mathbb{E}[\mathbb{E}[z_i \varepsilon_i \mid z_i]].$$

Inside the conditional expectation z_i is fixed, so

$$\mathbb{E}[z_i \varepsilon_i \mid z_i] = z_i \mathbb{E}[\varepsilon_i \mid z_i].$$

By instrument exogeneity $\mathbb{E}[\varepsilon_i \mid z_i] = 0$, hence

$$\mathbb{E}[z_i \varepsilon_i \mid z_i] = z_i \cdot 0 = 0,$$

and therefore

$$\boxed{\mathbb{E}[z_i \varepsilon_i] = \mathbb{E}[0] = 0.} \quad (\text{IV-9})$$

Step 6: Conclude consistency (as written). Combine (IV-5)–(IV-9) using Slutsky/product rule:

$$b_{IV} \xrightarrow{p} \beta + Q_{zx}^{-1} \mathbb{E}[z_i \varepsilon_i] = \beta + Q_{zx}^{-1} \cdot 0 = \beta.$$

Hence

$$\boxed{b_{IV} \xrightarrow{p} \beta, \quad \text{i.e. } b_{IV} \text{ is consistent.}} \quad (\text{IV-Cons})$$

10 Asymptotic distribution of b_{IV} (as written, fully expanded)

Step 1: Start from the exact expression for $b_{IV} - \beta$. From (IV-4),

$$b_{IV} - \beta = \left(\sum_{i=1}^n z_i x_i' \right)^{-1} \left(\sum_{i=1}^n z_i \varepsilon_i \right). \quad (\text{AN-1})$$

Step 2: Multiply by $\sqrt{n}$ and rewrite as “LLN factor $\times$ CLT factor” (as written).

$$\sqrt{n}(b_{IV} - \beta) = \sqrt{n} \left(\sum_{i=1}^n z_i x_i' \right)^{-1} \left(\sum_{i=1}^n z_i \varepsilon_i \right).$$

Insert $1/n$ and $1/\sqrt{n}$ exactly:

$$\boxed{\sqrt{n}(b_{IV} - \beta) = \left(\frac{1}{n} \sum_{i=1}^n z_i x_i' \right)^{-1} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n z_i \varepsilon_i \right).} \quad (\text{AN-2})$$

Also note the identity (as written in the notes):

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n z_i \varepsilon_i = \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n z_i \varepsilon_i \right). \quad (\text{AN-3})$$

Step 3: LLN for the first factor (as written).

$$\frac{1}{n} \sum_{i=1}^n z_i x_i' \xrightarrow{p} Q_{zx}. \quad (\text{AN-4})$$

Hence (by continuity of inversion),

$$\left(\frac{1}{n} \sum_{i=1}^n z_i x_i' \right)^{-1} \xrightarrow{p} Q_{zx}^{-1}. \quad (\text{AN-5})$$

Step 4: Mean and variance of the summand $z_i \varepsilon_i$. First compute the mean:

$$\mathbb{E}[z_i \varepsilon_i] = 0 \quad (\text{from (IV-9)}). \quad (\text{AN-6})$$

Now compute the variance matrix (as written, with all intermediate steps):

$$\text{Var}(z_i \varepsilon_i) = \mathbb{E}[(z_i \varepsilon_i)(z_i \varepsilon_i)'] - \mathbb{E}[z_i \varepsilon_i] \mathbb{E}[z_i \varepsilon_i]'$$

Using $\mathbb{E}[z_i \varepsilon_i] = 0$, this reduces to

$$\text{Var}(z_i \varepsilon_i) = \mathbb{E}[(z_i \varepsilon_i)(z_i \varepsilon_i)']$$

Expand the product:

$$(z_i \varepsilon_i)(z_i \varepsilon_i)' = z_i \varepsilon_i \varepsilon_i' z_i' = \varepsilon_i^2 z_i z_i'$$

Therefore,

$$\text{Var}(z_i \varepsilon_i) = \mathbb{E}[\varepsilon_i^2 z_i z_i']. \quad (\text{AN-7})$$

Apply iterated expectations conditioning on z_i :

$$\mathbb{E}[\varepsilon_i^2 z_i z_i'] = \mathbb{E}[\mathbb{E}[\varepsilon_i^2 z_i z_i' \mid z_i]]$$

Inside the conditional expectation, $z_i z_i'$ is fixed, so

$$\mathbb{E}[\varepsilon_i^2 z_i z_i' \mid z_i] = z_i z_i' \mathbb{E}[\varepsilon_i^2 \mid z_i]$$

By homoscedasticity conditional on instruments, $\mathbb{E}[\varepsilon_i^2 \mid z_i] = \sigma^2$, hence

$$\mathbb{E}[\varepsilon_i^2 z_i z_i'] = \mathbb{E}[z_i z_i' \sigma^2] = \sigma^2 \mathbb{E}[z_i z_i'] = \sigma^2 Q_{zz}.$$

Thus

$$\boxed{\text{Var}(z_i \varepsilon_i) = \sigma^2 Q_{zz}}. \quad (\text{AN-8})$$

Step 5: CLT for the second factor (as written). Since $(z_i \varepsilon_i)$ are i.i.d. with mean 0 and variance $\sigma^2 Q_{zz}$, the multivariate CLT gives

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n z_i \varepsilon_i = \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n z_i \varepsilon_i - \mathbb{E}[z_i \varepsilon_i] \right) \xrightarrow{d} N(0, \sigma^2 Q_{zz}). \quad (\text{AN-9})$$

Step 6: Combine (Slutsky) to get asymptotic normality of b_{IV} . From (AN-5) and (AN-9), by Slutsky:

$$\sqrt{n}(b_{IV} - \beta) = \left(\frac{1}{n} \sum_{i=1}^n z_i x_i' \right)^{-1} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n z_i \varepsilon_i \right) \xrightarrow{d} Q_{zx}^{-1} W,$$

where $W \sim N(0, \sigma^2 Q_{zz})$. Therefore,

$$Q_{zx}^{-1} W \sim N(0, Q_{zx}^{-1} (\sigma^2 Q_{zz}) (Q_{zx}^{-1})').$$

Equivalently, writing $(Q_{zx}^{-1})' = (Q_{xz}^{-1})$ when $Q_{xz} = Q_{zx}'$,

$$\boxed{\sqrt{n}(b_{IV} - \beta) \xrightarrow{d} N(0, \sigma^2 Q_{zx}^{-1} Q_{zz} Q_{xz}^{-1})}. \quad (\text{AN-Final})$$

11 Consistency does not imply unbiasedness (as written)

Step 1: Re-state the exact decomposition. From (IV-3),

$$b_{IV} = \beta + (Z'X)^{-1} Z' \varepsilon. \quad (\text{Bias-1})$$

Step 2: Take conditional expectation given (X, Z) (as written, with intermediate steps).

$$\begin{aligned} \mathbb{E}[b_{IV} \mid X, Z] &= \mathbb{E}[\beta + (Z'X)^{-1} Z' \varepsilon \mid X, Z] \\ &= \beta + \mathbb{E}[(Z'X)^{-1} Z' \varepsilon \mid X, Z] \quad (\beta \text{ is constant}) \\ &= \beta + (Z'X)^{-1} Z' \mathbb{E}[\varepsilon \mid X, Z] \quad ((Z'X)^{-1} Z' \text{ depends only on } X, Z). \end{aligned} \quad (\text{Bias-2})$$

In general, $\mathbb{E}[\varepsilon \mid X, Z]$ need not be 0, so

$$\boxed{\mathbb{E}[b_{IV} \mid X, Z] \neq \beta \text{ in general.}} \quad (\text{Bias-3})$$

Thus b_{IV} can be *consistent* even if it is not (*finite-sample*) *unbiased*.

Instrumental Variables: Assumptions, IV vs 2SLS, and the $L = k$ Equivalence (Fully Expanded)

12 Assumptions (IV)

1. Linearity

$$y = X\beta + \varepsilon.$$

2. Full rank

$$\text{rank}(X) = k.$$

3. Endogeneity (OLS exogeneity fails)

$$\mathbb{E}[\varepsilon_i \mid x_i] = \eta \neq 0.$$

4. Homoscedasticity conditional on instruments

$$\mathbb{E}[\varepsilon_i^2 \mid z_i] = \sigma^2.$$

5. Random sampling

$$(x_i, z_i, \varepsilon_i) \sim \text{i.i.d.}$$

6. Moments of x_i and z_i

$$\begin{aligned}\mathbb{E}[x_i x_i'] &= Q_{xx} < \infty, & \text{rank}(Q_{xx}) &= k, \\ \mathbb{E}[z_i z_i'] &= Q_{zz} < \infty, & \text{rank}(Q_{zz}) &= L, \\ \mathbb{E}[z_i x_i'] &= Q_{zx} < \infty, & \text{rank}(Q_{zx}) &= k, & Q_{zx} &\in \mathbb{R}^{L \times k}, & (L \geq k).\end{aligned}$$

7. Exogeneity of instruments

$$\mathbb{E}[\varepsilon_i | z_i] = 0.$$

13 Just-identified IV estimator ($L = k$)

When $L = k$,

$$b_{IV} = (Z'X)^{-1}Z'y.$$

14 Overidentified case ($L > k$): 2SLS construction and algebra

Step 1 (projection matrix onto the column space of Z). When $L > k$, define the projection (hat) matrix

$$P_Z \equiv Z(Z'Z)^{-1}Z'.$$

Step 2 (first stage: project X onto Z). Define the fitted (projected) regressor matrix

$$\hat{X} \equiv P_Z X.$$

Using the definition of P_Z ,

$$\hat{X} = P_Z X = Z(Z'Z)^{-1}Z'X.$$

Step 3 (second stage: regress y on $\hat{X}$). Define the 2SLS estimator as the OLS estimator using $\hat{X}$:

$$b_{2SLS} \equiv (\hat{X}'\hat{X})^{-1}\hat{X}'y.$$

Step 4 (rewrite $\hat{X}'\hat{X}$ and $\hat{X}'y$ using P_Z). Because $\hat{X} = P_Z X$,

$$\hat{X}'\hat{X} = (P_Z X)'(P_Z X) = X'P_Z'P_Z X, \quad \hat{X}'y = (P_Z X)'y = X'P_Z'y.$$

Now show the two key projection properties:

(i) **Symmetry:** since $Z'Z$ is symmetric,

$$P_Z' = (Z(Z'Z)^{-1}Z')' = Z((Z'Z)^{-1})'Z' = Z(Z'Z)^{-1}Z' = P_Z.$$

(ii) **Idempotence:**

$$\begin{aligned}P_Z^2 &= (Z(Z'Z)^{-1}Z')(Z(Z'Z)^{-1}Z') \\ &= Z(Z'Z)^{-1}(Z'Z)(Z'Z)^{-1}Z' \\ &= Z(Z'Z)^{-1}Z' \\ &= P_Z.\end{aligned}$$

Hence $P_Z'P_Z = P_ZP_Z = P_Z$, and therefore

$$\hat{X}'\hat{X} = X'P_ZX, \quad \hat{X}'y = X'P_Zy.$$

Step 5 (the standard “single-equation” 2SLS formula; original line kept). Thus

$$\boxed{b_{2SLS} = (X'P_ZX)^{-1}X'P_Zy.}$$

Step 6 (substitute $P_Z = Z(Z'Z)^{-1}Z'$; original process kept). Substitute into the two components:

(a) For the “ $X'P_ZX$ ” term:

$$X'P_ZX = X'Z(Z'Z)^{-1}Z'X.$$

(b) For the “ $X'P_Zy$ ” term:

$$X'P_Zy = X'Z(Z'Z)^{-1}Z'y.$$

Therefore,

$$\boxed{b_{2SLS} = (X'Z(Z'Z)^{-1}Z'X)^{-1}X'Z(Z'Z)^{-1}Z'y.} \quad (*)$$

15 When $L = k$: show that 2SLS reduces to IV (fully expanded)

(Original note kept). When $L = k$, $X'Z$ is invertible (equivalently $Z'X$ is invertible).

Step 1 (rewrite the inverse term in $(*)$ using an explicit matrix identity). Let

$$A \equiv X'Z \in \mathbb{R}^{k \times k}, \quad B \equiv Z'Z \in \mathbb{R}^{k \times k}.$$

Then the “ $X'Z(Z'Z)^{-1}Z'X$ ” term can be written as

$$X'Z(Z'Z)^{-1}Z'X = AB^{-1}A', \quad \text{where } A' = Z'X.$$

If A is invertible, then

$$(AB^{-1}A')^{-1} = (A')^{-1}BA^{-1}.$$

(Verification by multiplication:)

$$\begin{aligned} ((A')^{-1}BA^{-1})(AB^{-1}A') &= (A')^{-1}B(A^{-1}A)B^{-1}A' \\ &= (A')^{-1}BB^{-1}A' \\ &= (A')^{-1}A' \\ &= I_k. \end{aligned}$$

Step 2 (apply the identity to $(*)$). Using $(AB^{-1}A')^{-1} = (A')^{-1}BA^{-1}$, equation $(*)$ becomes

$$\begin{aligned} b_{2SLS} &= (AB^{-1}A')^{-1}AB^{-1}Z'y \\ &= ((A')^{-1}BA^{-1})AB^{-1}Z'y \\ &= (A')^{-1}B(A^{-1}A)B^{-1}Z'y \\ &= (A')^{-1}BI_kB^{-1}Z'y \\ &= (A')^{-1}Z'y. \end{aligned}$$

Since $A' = Z'X$, we have

$$\boxed{b_{2SLS} = (Z'X)^{-1}Z'y.}$$

Step 3 (match the IV formula; original conclusion line kept). But when $L = k$,

$$b_{IV} = (Z'X)^{-1}Z'y,$$

so

$$\boxed{b_{2SLS} = b_{IV} \quad \text{when } L = k.}$$

$$b_{2SLS} = (X'P_ZX)^{-1}X'P_Zy.$$

Note: Relationship between “simple IV” and 2SLS

Let $X \in \mathbb{R}^{n \times k}$ denote the matrix of endogenous regressors and $Z \in \mathbb{R}^{n \times L}$ denote the matrix of excluded instruments.

Step 1: The “simple IV” (method-of-moments) formula. From the sample moment condition

$$\frac{1}{n}Z'(y - Xb) = 0,$$

we obtain

$$\frac{1}{n}Z'y = \frac{1}{n}Z'Xb.$$

If $L = k$ and $Z'X$ is invertible (square $k \times k$), then we can solve exactly:

$$\boxed{b_{IV} = (Z'X)^{-1}Z'y.}$$

Key point: if $L \neq k$, then $Z'X$ is not a square matrix, so $(Z'X)^{-1}$ does *not* exist and this “simple IV” expression is not defined.

Step 2: 2SLS exists for both $L = k$ and $L > k$. Define the projection matrix onto the column space of Z :

$$P_Z \equiv Z(Z'Z)^{-1}Z'.$$

Then the 2SLS estimator is

$$\boxed{b_{2SLS} = (X'P_ZX)^{-1}X'P_Zy.}$$

This formula is well-defined when Z has full column rank and the usual identification conditions hold, even when $L > k$ (overidentified).

Step 3: Show equality when $L = k$ (just-identified). When $L = k$ and $Z'X$ is invertible, one can algebraically simplify 2SLS:

$$b_{2SLS} = (X'P_ZX)^{-1}X'P_Zy = (Z'X)^{-1}Z'y = b_{IV}.$$

Hence

$$\boxed{L = k \implies b_{2SLS} = b_{IV}.}$$

Step 4: Overidentified case ($L > k$). When $L > k$, the moment conditions $Z'(y - Xb) = 0$ give L equations in k unknowns. In general, they cannot all be solved exactly in finite samples, so 2SLS chooses b that satisfies them “as well as possible” (in a least-squares / GMM sense). In this case:

$$\boxed{L > k \implies b_{IV} = (Z'X)^{-1}Z'y \text{ is not defined, but } b_{2SLS} \text{ is defined.}}$$

(Optional) With exogenous controls. If the regression includes exogenous controls W , then W can be included among instruments. After partialling out W , the same logic applies to the remaining excluded instruments and endogenous regressors: “just-identified” means *number of excluded instruments equals number of endogenous regressors*.

Econometrics: 2SLS (IV) Assumptions, Consistency, and Asymptotic Distribution

16 Assumptions (IV)

1. **Linearity**

$$y = X\beta + \varepsilon.$$

2. **Full rank**

$$\text{rank}(X) = k.$$

3. **Endogeneity (OLS exogeneity fails)**

$$\mathbb{E}[\varepsilon_i \mid x_i] = \eta \neq 0.$$

4. **Homoscedasticity conditional on instruments**

$$\mathbb{E}[\varepsilon_i^2 \mid z_i] = \sigma^2.$$

5. **Random sampling**

$$(x_i, z_i, \varepsilon_i) \text{ are i.i.d. across } i = 1, \dots, n.$$

6. **Moments of x_i and z_i**

$$\mathbb{E}[x_i x_i'] = Q_{xx} < \infty, \quad \text{rank}(Q_{xx}) = k,$$

$$\mathbb{E}[z_i z_i'] = Q_{zz} < \infty, \quad \text{rank}(Q_{zz}) = L,$$

$$\mathbb{E}[z_i x_i'] = Q_{zx} < \infty, \quad \text{rank}(Q_{zx}) = k, \quad Q_{zx} \in \mathbb{R}^{L \times k}.$$

(Consequently, $Q_{xz} \equiv \mathbb{E}[x_i z_i'] = Q_{zx}' \in \mathbb{R}^{k \times L}$.) We typically have $L \geq k$.

7. **Exogeneity of instruments**

$$\mathbb{E}[\varepsilon_i \mid z_i] = 0.$$

17 2SLS estimator and consistency

Step 0: Define matrices and notation

Let

$$X = \begin{pmatrix} x_1' \\ x_2' \\ \vdots \\ x_n' \end{pmatrix} \in \mathbb{R}^{n \times k}, \quad Z = \begin{pmatrix} z_1' \\ z_2' \\ \vdots \\ z_n' \end{pmatrix} \in \mathbb{R}^{n \times L}, \quad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

Define the projection matrix onto the column space of Z :

$$P_Z \equiv Z(Z'Z)^{-1}Z'.$$

Step 1: Write the 2SLS estimator (as in the notes)

$$b_{2SLS} = (X'P_ZX)^{-1}X'P_Zy. \quad (2SLS-1)$$

Step 2: Substitute $y = X\beta + \varepsilon$ and simplify (as in the notes)

$$\begin{aligned} b_{2SLS} &= (X'P_ZX)^{-1}X'P_Zy \\ &= (X'P_ZX)^{-1}X'P_Z(X\beta + \varepsilon) \\ &= (X'P_ZX)^{-1}X'P_ZX\beta + (X'P_ZX)^{-1}X'P_Z\varepsilon. \end{aligned}$$

Since $(X'P_ZX)^{-1}X'P_ZX = I_k$,

$$\boxed{b_{2SLS} = \beta + (X'P_ZX)^{-1}X'P_Z\varepsilon.} \quad (2SLS-2)$$

Step 3: Expand $(X'P_ZX)^{-1}X'P_Z\varepsilon$ in terms of Z (as in the notes)

Because $P_Z = Z(Z'Z)^{-1}Z'$,

$$(X'P_ZX)^{-1}X'P_Z\varepsilon = (X'Z(Z'Z)^{-1}Z'X)^{-1}X'Z(Z'Z)^{-1}Z'\varepsilon. \quad (2SLS-3)$$

Now use the identities

$$X'Z = \sum_{i=1}^n x_i z'_i, \quad Z'Z = \sum_{i=1}^n z_i z'_i, \quad Z'X = \sum_{i=1}^n z_i x'_i, \quad Z'\varepsilon = \sum_{i=1}^n z_i \varepsilon_i.$$

Then (2SLS-3) becomes

$$\begin{aligned} (X'P_ZX)^{-1}X'P_Z\varepsilon &= \left[\left(\sum_{i=1}^n x_i z'_i \right) \left(\sum_{i=1}^n z_i z'_i \right)^{-1} \left(\sum_{i=1}^n z_i x'_i \right) \right]^{-1} \\ &\quad \cdot \left(\sum_{i=1}^n x_i z'_i \right) \left(\sum_{i=1}^n z_i z'_i \right)^{-1} \left(\sum_{i=1}^n z_i \varepsilon_i \right). \end{aligned} \quad (2SLS-4)$$

Divide and multiply by n in every sum (pure algebra):

$$\begin{aligned} (X'P_ZX)^{-1}X'P_Z\varepsilon &= \left[\left(\frac{1}{n} \sum_{i=1}^n x_i z'_i \right) \left(\frac{1}{n} \sum_{i=1}^n z_i z'_i \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n z_i x'_i \right) \right]^{-1} \\ &\quad \cdot \left(\frac{1}{n} \sum_{i=1}^n x_i z'_i \right) \left(\frac{1}{n} \sum_{i=1}^n z_i z'_i \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n z_i \varepsilon_i \right). \end{aligned} \quad (2SLS-5)$$

Step 4: Take probability limits to show consistency

By i.i.d. LLN and the finite-moment assumptions:

$$\frac{1}{n} \sum_{i=1}^n x_i z'_i \xrightarrow{p} \mathbb{E}[x_i z'_i] = Q_{xz}, \quad \frac{1}{n} \sum_{i=1}^n z_i z'_i \xrightarrow{p} \mathbb{E}[z_i z'_i] = Q_{zz}, \quad \frac{1}{n} \sum_{i=1}^n z_i x'_i \xrightarrow{p} \mathbb{E}[z_i x'_i] = Q_{zx}.$$

Also,

$$\frac{1}{n} \sum_{i=1}^n z_i \varepsilon_i \xrightarrow{p} \mathbb{E}[z_i \varepsilon_i].$$

Now show $\mathbb{E}[z_i \varepsilon_i] = 0$ from instrument exogeneity (all steps):

$$\mathbb{E}[z_i \varepsilon_i] = \mathbb{E}[\mathbb{E}[z_i \varepsilon_i \mid z_i]] = \mathbb{E}[z_i \mathbb{E}[\varepsilon_i \mid z_i]] = \mathbb{E}[z_i \cdot 0] = 0. \quad (2SLS\text{-}Exog)$$

Therefore, taking limits in (2SLS-5):

$$\begin{aligned} (X'P_Z X)^{-1} X'P_Z \varepsilon &\xrightarrow{p} (Q_{xz} Q_{zz}^{-1} Q_{zx})^{-1} Q_{xz} Q_{zz}^{-1} \mathbb{E}[z_i \varepsilon_i] \\ &= (Q_{xz} Q_{zz}^{-1} Q_{zx})^{-1} Q_{xz} Q_{zz}^{-1} \cdot 0 \\ &= 0. \end{aligned}$$

Hence, from $b_{2SLS} = \beta + (X'P_Z X)^{-1} X'P_Z \varepsilon$,

$$\boxed{b_{2SLS} \xrightarrow{p} \beta.}$$

18 Asymptotic distribution of b_{2SLS}

Step 1: Start from $\sqrt{n}(b_{2SLS} - \beta)$

From (2SLS-2),

$$b_{2SLS} - \beta = (X'P_Z X)^{-1} X'P_Z \varepsilon.$$

Using the normalized-sum form (2SLS-5),

$$\begin{aligned} \sqrt{n}(b_{2SLS} - \beta) &= \sqrt{n} \left[\left(\frac{1}{n} \sum_{i=1}^n x_i z'_i \right) \left(\frac{1}{n} \sum_{i=1}^n z_i z'_i \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n z_i \varepsilon_i \right) \right]^{-1} \\ &\quad \cdot \left(\frac{1}{n} \sum_{i=1}^n x_i z'_i \right) \left(\frac{1}{n} \sum_{i=1}^n z_i z'_i \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n z_i \varepsilon_i \right). \end{aligned} \quad (AN-1)$$

Group $\sqrt{n}$ with the last average (as in the notes):

$$\begin{aligned} \sqrt{n}(b_{2SLS} - \beta) &= \left[\left(\frac{1}{n} \sum_{i=1}^n x_i z'_i \right) \left(\frac{1}{n} \sum_{i=1}^n z_i z'_i \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n z_i \varepsilon_i \right) \right]^{-1} \\ &\quad \cdot \left(\frac{1}{n} \sum_{i=1}^n x_i z'_i \right) \left(\frac{1}{n} \sum_{i=1}^n z_i z'_i \right)^{-1} \left(\sqrt{n} \frac{1}{n} \sum_{i=1}^n z_i \varepsilon_i \right). \end{aligned} \quad (AN-2)$$

Step 2: LLN for the deterministic matrix factors

By LLN,

$$\frac{1}{n} \sum_{i=1}^n x_i z'_i \xrightarrow{p} Q_{xz}, \quad \frac{1}{n} \sum_{i=1}^n z_i z'_i \xrightarrow{p} Q_{zz}, \quad \frac{1}{n} \sum_{i=1}^n z_i \varepsilon_i \xrightarrow{p} Q_{zx}.$$

Therefore (by continuous mapping + matrix inversion),

$$\left[\left(\frac{1}{n} \sum_{i=1}^n x_i z_i' \right) \left(\frac{1}{n} \sum_{i=1}^n z_i z_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n z_i x_i' \right) \right]^{-1} \xrightarrow{p} (Q_{xz} Q_{zz}^{-1} Q_{zx})^{-1},$$

and

$$\left(\frac{1}{n} \sum_{i=1}^n x_i z_i' \right) \left(\frac{1}{n} \sum_{i=1}^n z_i z_i' \right)^{-1} \xrightarrow{p} Q_{xz} Q_{zz}^{-1}.$$

Step 3: CLT for $\sqrt{n} \frac{1}{n} \sum z_i \varepsilon_i$

First compute the variance of the summand $z_i \varepsilon_i$ (as in the notes, with full steps). Because $\mathbb{E}[z_i \varepsilon_i] = 0$,

$$\text{Var}(z_i \varepsilon_i) = \mathbb{E}[(z_i \varepsilon_i)(z_i \varepsilon_i)'].$$

Expand:

$$(z_i \varepsilon_i)(z_i \varepsilon_i)' = z_i \varepsilon_i \varepsilon_i z_i' = \varepsilon_i^2 z_i z_i'.$$

So

$$\text{Var}(z_i \varepsilon_i) = \mathbb{E}[\varepsilon_i^2 z_i z_i'].$$

Apply iterated expectations:

$$\mathbb{E}[\varepsilon_i^2 z_i z_i'] = \mathbb{E}[\mathbb{E}[\varepsilon_i^2 z_i z_i' \mid z_i]] = \mathbb{E}[z_i z_i' \mathbb{E}[\varepsilon_i^2 \mid z_i]].$$

By homoscedasticity conditional on instruments, $\mathbb{E}[\varepsilon_i^2 \mid z_i] = \sigma^2$, hence

$$\mathbb{E}[\varepsilon_i^2 z_i z_i'] = \mathbb{E}[z_i z_i' \sigma^2] = \sigma^2 \mathbb{E}[z_i z_i'] = \sigma^2 Q_{zz}.$$

Therefore

$$\boxed{\text{Var}(z_i \varepsilon_i) = \sigma^2 Q_{zz}.} \quad (\text{AN-Var})$$

By the multivariate CLT,

$$\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n z_i \varepsilon_i - \mathbb{E}[z_i \varepsilon_i] \right) \xrightarrow{d} N(0, \sigma^2 Q_{zz}).$$

Since $\mathbb{E}[z_i \varepsilon_i] = 0$, this is

$$\boxed{\sqrt{n} \frac{1}{n} \sum_{i=1}^n z_i \varepsilon_i \xrightarrow{d} N(0, \sigma^2 Q_{zz}).} \quad (\text{AN-CLT})$$

Step 4: Combine limits (Slutsky) and simplify the asymptotic variance

From (AN-2), LLN limits, and (AN-CLT), Slutsky implies

$$\sqrt{n}(b_{2SLS} - \beta) \xrightarrow{d} (Q_{xz} Q_{zz}^{-1} Q_{zx})^{-1} Q_{xz} Q_{zz}^{-1} W, \quad W \sim N(0, \sigma^2 Q_{zz}).$$

A linear transformation of a normal is normal, so the limit is

$$N\left(0, (Q_{xz} Q_{zz}^{-1} Q_{zx})^{-1} Q_{xz} Q_{zz}^{-1} (\sigma^2 Q_{zz}) (Q_{zz}^{-1})' Q_{zx}' ((Q_{xz} Q_{zz}^{-1} Q_{zx})^{-1})'\right).$$

Since Q_{zz} is symmetric, Q_{zz}^{-1} is symmetric, so $(Q_{zz}^{-1})' = Q_{zz}^{-1}$. Also $Q'_{zx} = Q_{xz}$. Thus the covariance becomes

$$\sigma^2 (Q_{xz} Q_{zz}^{-1} Q_{zx})^{-1} Q_{xz} Q_{zz}^{-1} Q_{zz} Q_{zz}^{-1} Q_{zx} (Q_{xz} Q_{zz}^{-1} Q_{zx})^{-1}.$$

Because $Q_{zz}^{-1} Q_{zz} Q_{zz}^{-1} = Q_{zz}^{-1}$, this simplifies to

$$\sigma^2 (Q_{xz} Q_{zz}^{-1} Q_{zx})^{-1} \underbrace{(Q_{xz} Q_{zz}^{-1} Q_{zx})}_{Q_{xz} Q_{zz}^{-1} Q_{zx}} (Q_{xz} Q_{zz}^{-1} Q_{zx})^{-1} = \sigma^2 (Q_{xz} Q_{zz}^{-1} Q_{zx})^{-1}.$$

Therefore, the final asymptotic normality statement is

$$\boxed{\sqrt{n}(b_{2SLS} - \beta) \xrightarrow{d} N(0, \sigma^2 (Q_{xz} Q_{zz}^{-1} Q_{zx})^{-1})}.$$

OLS vs IV: Asymptotic Variance Comparison and the Hausman Test

19 Notation and asymptotic distributions

Population second moments. Let

$$Q_{xx} = \mathbb{E}[x_i x_i'], \quad Q_{zx} = \mathbb{E}[z_i x_i'], \quad Q_{zz} = \mathbb{E}[z_i z_i'], \quad Q_{xz} = \mathbb{E}[x_i z_i'] = Q'_{zx}.$$

Asymptotic distributions (as written).

$$\begin{aligned} \sqrt{n}(b_{OLS} - \beta) &\xrightarrow{d} N(0, \sigma^2 Q_{xx}^{-1}), \\ \sqrt{n}(b_{IV} - \beta) &\xrightarrow{d} N(0, \sigma^2 Q_{zx}^{-1} Q_{zz} Q_{xz}^{-1}). \end{aligned}$$

20 A matrix inequality used in the efficiency comparison

Inverse-ordering fact (as written). If $A - B$ is positive semidefinite, then $B^{-1} - A^{-1}$ is positive semidefinite (and positive definite if $A - B$ is positive definite).

One way to see it (explicit steps). Assume A and B are symmetric positive definite and $A - B \succeq 0$. Then

$$A - B \succeq 0 \implies B^{-1/2}(A - B)B^{-1/2} \succeq 0.$$

But

$$B^{-1/2}(A - B)B^{-1/2} = B^{-1/2}AB^{-1/2} - I.$$

So $B^{-1/2}AB^{-1/2} \succeq I$, and inverting reverses the inequality:

$$(B^{-1/2}AB^{-1/2})^{-1} \preceq I.$$

Now use

$$(B^{-1/2}AB^{-1/2})^{-1} = B^{1/2}A^{-1}B^{1/2},$$

so

$$B^{1/2}A^{-1}B^{1/2} \preceq I.$$

Multiply on both sides by $B^{-1/2}$ (left and right):

$$A^{-1} \preceq B^{-1} \iff B^{-1} - A^{-1} \succeq 0.$$

21 Show the variance gap is positive semidefinite

Step 1: Rewrite the matrix inside the inverse

Start from the matrix difference (as written):

$$Q_{xx} - \left(Q_{zx}^{-1} Q_{zz} Q_{xz} \right)^{-1}.$$

Use the inverse-of-a-product identity $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$:

$$\left(Q_{zx}^{-1} Q_{zz} Q_{xz} \right)^{-1} = \left(Q_{xz} \right)^{-1} Q_{zz}^{-1} \left(Q_{zx} \right)^{-1} = Q_{xz} Q_{zz}^{-1} Q_{zx}.$$

Hence

$$Q_{xx} - \left(Q_{zx}^{-1} Q_{zz} Q_{xz} \right)^{-1} = Q_{xx} - Q_{xz} Q_{zz}^{-1} Q_{zx}.$$

Step 2: Express each population moment as a probability limit

Use sample analogs:

$$\xrightarrow{p}_{n \rightarrow \infty} \frac{X'X}{n} = Q_{xx}, \quad \xrightarrow{p}_{n \rightarrow \infty} \frac{X'Z}{n} = Q_{xz}, \quad \xrightarrow{p}_{n \rightarrow \infty} \frac{Z'Z}{n} = Q_{zz}, \quad \xrightarrow{p}_{n \rightarrow \infty} \frac{Z'X}{n} = Q_{zx}.$$

Therefore,

$$Q_{xx} - Q_{xz} Q_{zz}^{-1} Q_{zx} = \xrightarrow{p}_{n \rightarrow \infty} \frac{X'X}{n} - \xrightarrow{p}_{n \rightarrow \infty} \frac{X'Z}{n} \left(\xrightarrow{p}_{n \rightarrow \infty} \frac{Z'Z}{n} \right)^{-1} \xrightarrow{p}_{n \rightarrow \infty} \frac{Z'X}{n}.$$

Step 3: Combine into one probability limit

Because products of convergent sequences converge to the product of the limits (under standard regularity),

$$\xrightarrow{p}_{n \rightarrow \infty} \frac{X'Z}{n} \left(\xrightarrow{p}_{n \rightarrow \infty} \frac{Z'Z}{n} \right)^{-1} \xrightarrow{p}_{n \rightarrow \infty} \frac{Z'X}{n} = \xrightarrow{p}_{n \rightarrow \infty} \left[\frac{X'Z}{n} \left(\frac{Z'Z}{n} \right)^{-1} \frac{Z'X}{n} \right].$$

So

$$Q_{xx} - Q_{xz} Q_{zz}^{-1} Q_{zx} = \xrightarrow{p}_{n \rightarrow \infty} \left[\frac{X'X}{n} - \frac{X'Z}{n} \left(\frac{Z'Z}{n} \right)^{-1} \frac{Z'X}{n} \right].$$

Multiply the second term inside by n/n to match the notes' next line:

$$\frac{X'Z}{n} \left(\frac{Z'Z}{n} \right)^{-1} \frac{Z'X}{n} = \frac{1}{n} X'Z (Z'Z)^{-1} Z'X.$$

Hence

$$Q_{xx} - Q_{xz} Q_{zz}^{-1} Q_{zx} = \xrightarrow{p}_{n \rightarrow \infty} \frac{1}{n} \left[X'X - X'Z (Z'Z)^{-1} Z'X \right].$$

Step 4: Introduce projection matrices

Define

$$P_Z \equiv Z(Z'Z)^{-1}Z', \quad M_Z \equiv I - P_Z.$$

Then

$$X'Z(Z'Z)^{-1}Z'X = X'P_ZX,$$

so

$$X'X - X'Z(Z'Z)^{-1}Z'X = X'X - X'P_ZX = X'(I - P_Z)X = X'M_ZX.$$

Therefore

$$Q_{xx} - Q_{xz}Q_{zz}^{-1}Q_{zx} \xrightarrow{p}_{n \rightarrow \infty} \frac{1}{n}X'M_ZX.$$

Step 5: Show $X'M_ZX$ is positive semidefinite

First note P_Z is symmetric and idempotent, so M_Z is also symmetric and idempotent:

$$P_Z' = P_Z, \quad P_Z^2 = P_Z \implies M_Z' = M_Z, \quad M_Z^2 = M_Z.$$

Then

$$X'M_ZX = X'M_Z' M_ZX = (M_ZX)'(M_ZX).$$

Equivalently, the notes' identity:

$$X'M_ZX = (X'M_Z)(X'M_Z)'$$

For any nonzero conformable vector $r \neq 0$, define

$$v \equiv r(X'M_Z),$$

then

$$rX'M_ZXr' = r(X'M_Z)(X'M_Z)'r' = vv' = \sum_j v_j^2 \geq 0.$$

Hence $X'M_ZX \succeq 0$, and therefore

$$Q_{xx} - Q_{xz}Q_{zz}^{-1}Q_{zx} \succeq 0.$$

Step 6: Translate this into a variance comparison

We have shown

$$Q_{xx} - \left(Q_{zx}^{-1}Q_{zz}Q_{xz}^{-1}\right)^{-1} \succeq 0.$$

Apply the inverse-ordering fact with

$$A = Q_{xx}, \quad B = \left(Q_{zx}^{-1}Q_{zz}Q_{xz}^{-1}\right)^{-1}.$$

Then

$$B^{-1} - A^{-1} \succeq 0 \implies \left(Q_{zx}^{-1}Q_{zz}Q_{xz}^{-1}\right) - Q_{xx}^{-1} \succeq 0.$$

Multiplying by $\sigma^2 \geq 0$ preserves the ordering:

$$\sigma^2 Q_{zx}^{-1}Q_{zz}Q_{xz}^{-1} - \sigma^2 Q_{xx}^{-1} \succeq 0.$$

Thus

$$\text{Var}_{asy}(b_{OLS}) \preceq \text{Var}_{asy}(b_{IV}),$$

i.e.

$$\text{Asy.var}(b_{OLS}) \leq \text{Asy.var}(b_{IV}) \quad (\text{in PSD ordering}).$$

22 Hausman test

Null hypothesis (as written).

$$H_0 : \mathbb{E}[\varepsilon_i \mid x_i] = 0.$$

Difference of estimators (as written). Let

$$d \equiv b_{IV} - b_{OLS}.$$

Under H_0 , both estimators are consistent for β , so

$$d \xrightarrow{p} 0.$$

If we can derive

$$\sqrt{n}d \xrightarrow{d} N(0, V) \quad \text{and estimate } V \text{ by } \widehat{V},$$

then we can test H_0 by a Wald statistic.

Wald statistic (as written). Let $r \equiv \dim(d)$ (number of components being tested). Define

$$W = (\sqrt{n}d)' \widehat{V}^{-1} (\sqrt{n}d) = n d' \widehat{V}^{-1} d \xrightarrow{d} \chi^2(r).$$

23 Hausman's principle and the variance formula for d

Hausman's principle (as written). Let b_E be an estimator of β with

$$\sqrt{n}(b_E - \beta) \xrightarrow{d} N(0, V_E).$$

Suppose b_E is efficient in the following sense: for any other estimator b of β such that

$$\sqrt{n}(b - \beta) \xrightarrow{d} N(0, V),$$

we have $V \succeq V_E$.

Let b_I be an inefficient estimator of β , namely

$$\sqrt{n}(b_I - \beta) \xrightarrow{d} N(0, \Sigma), \quad \Sigma \succeq V_E.$$

Then

$$\text{Cov}_{asy}(b_E, b_I) = \text{Var}_{asy}(b_E).$$

Scalar proof (as written, with all intermediate steps)

Let β be scalar. Consider an estimator

$$\widehat{\beta} = \alpha b_I + (1 - \alpha) b_E = b_E + \alpha(b_I - b_E),$$

for a constant α .

Then

$$\sqrt{n}(\widehat{\beta} - \beta) \xrightarrow{d} N(0, \Omega),$$

where

$$\Omega = \text{Var}_{asy}[b_E + \alpha(b_I - b_E)].$$

Expand:

$$\Omega = \text{Var}_{asy}(b_E) + \alpha^2 \text{Var}_{asy}(b_I - b_E) + 2\alpha \text{Cov}_{asy}(b_E, b_I - b_E).$$

Write $\text{Var}_{asy}(b_E) = V_E$:

$$\Omega = V_E + 2\alpha \text{Cov}_{asy}(b_E, b_I - b_E) + \alpha^2 \text{Var}_{asy}(b_I - b_E).$$

Differentiate with respect to α and set to zero to minimize:

$$\frac{\partial \Omega}{\partial \alpha} = 2 \text{Cov}_{asy}(b_E, b_I - b_E) + 2\alpha \text{Var}_{asy}(b_I - b_E) = 0.$$

Hence the minimizer is

$$\alpha^* = -\frac{\text{Cov}_{asy}(b_E, b_I - b_E)}{\text{Var}_{asy}(b_I - b_E)}.$$

If $\alpha^* \neq 0$, then $\hat{\beta}$ with $\alpha = \alpha^*$ has a smaller asymptotic variance than $\hat{\beta}$ with $\alpha = 0$. But $\alpha = 0$ yields $\hat{\beta} = b_E$, contradicting b_E being efficient. Therefore $\alpha^* = 0$, implying

$$\text{Cov}_{asy}(b_E, b_I - b_E) = 0.$$

Expand the left-hand side:

$$\text{Cov}_{asy}(b_E, b_I - b_E) = \text{Cov}_{asy}(b_E, b_I) - \text{Cov}_{asy}(b_E, b_E) = \text{Cov}_{asy}(b_E, b_I) - \text{Var}_{asy}(b_E).$$

So

$$\text{Cov}_{asy}(b_E, b_I) - \text{Var}_{asy}(b_E) = 0 \implies \text{Cov}_{asy}(b_E, b_I) = \text{Var}_{asy}(b_E).$$

24 Apply Hausman's principle to b_{OLS} and b_{IV}

Under $H_0 : \mathbb{E}[\varepsilon_i | x_i] = 0$, OLS is consistent and efficient (relative to IV), so take

$$b_E = b_{OLS}, \quad b_I = b_{IV}.$$

Then

$$\text{Cov}_{asy}(b_{OLS}, b_{IV}) = \text{Var}_{asy}(b_{OLS}).$$

25 Derive $V = \text{Var}_{asy}(b_{IV} - b_{OLS})$ and the test statistic

Variance of the difference (as written, with expansions). Let

$$V = \text{Var}_{asy}(b_{IV} - b_{OLS}).$$

Then

$$\begin{aligned} V &= \text{Var}_{asy}(b_{IV}) + \text{Var}_{asy}(b_{OLS}) - 2\text{Cov}_{asy}(b_{IV}, b_{OLS}) \\ &= \text{Var}_{asy}(b_{IV}) + \text{Var}_{asy}(b_{OLS}) - 2\text{Var}_{asy}(b_{OLS}) \\ &= \text{Var}_{asy}(b_{IV}) - \text{Var}_{asy}(b_{OLS}). \end{aligned}$$

Consistent estimation. Let

$$\hat{V}_{IV} \xrightarrow{p} \text{Var}_{asy}(b_{IV}), \quad \hat{V}_{OLS} \xrightarrow{p} \text{Var}_{asy}(b_{OLS}).$$

Then a consistent estimator of V is

$$\hat{V} = \hat{V}_{IV} - \hat{V}_{OLS}.$$

Final Wald statistic (as written).

$$W = n d'(\hat{V}_{IV} - \hat{V}_{OLS})^{-1} d \xrightarrow{d} \chi^2(r).$$

Measurement Error and IV Using a Second Noisy Measure

26 Measurement error in the regressor and attenuation bias

True model.

$$y = \beta_0 + \beta_1 x^* + \varepsilon.$$

Assume x^* is not observed and is uncorrelated with ε .

Observable proxy. Let x be an observable measure of x^* . Assume x^* and x are related by

$$x = x^* + u,$$

where u is independent of (y, x^*, ε) , $\mathbb{E}[u] = 0$, and $\text{Var}(u) = \delta_u^2$.

Rewrite the model in terms of x . From $x^* = x - u$,

$$\begin{aligned} y &= \beta_0 + \beta_1(x - u) + \varepsilon \\ &= \beta_0 + \beta_1 x + \varepsilon - \beta_1 u. \end{aligned}$$

Define the composite regression error (when regressing y on x) as

$$v \equiv \varepsilon - \beta_1 u, \quad \text{so that} \quad y = \beta_0 + \beta_1 x + v.$$

Endogeneity induced by measurement error. Compute the covariance between x and the composite error:

$$\begin{aligned} \text{Cov}(x, v) &= \text{Cov}(x^* + u, \varepsilon - \beta_1 u) \\ &= \text{Cov}(x^*, \varepsilon) - \beta_1 \text{Cov}(x^*, u) + \text{Cov}(u, \varepsilon) - \beta_1 \text{Cov}(u, u) \\ &= 0 - 0 + 0 - \beta_1 \text{Var}(u) \\ &= -\beta_1 \delta_u^2, \end{aligned}$$

using the assumptions that x^* is uncorrelated with ε and u is independent of (x^*, ε) .

OLS slope estimator. Let b_1 denote the OLS slope from regressing y on x (with intercept):

$$b_1 = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}.$$

Use $\sum_{i=1}^n (x_i - \bar{x}) = 0$ to rewrite the numerator:

$$\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) = \sum_{i=1}^n y_i (x_i - \bar{x}) - \bar{y} \sum_{i=1}^n (x_i - \bar{x}) = \sum_{i=1}^n y_i (x_i - \bar{x}),$$

so

$$b_1 = \frac{\sum_{i=1}^n y_i (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}.$$

Substitute $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i - \beta_1 u_i$.

$$\begin{aligned} b_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x})(\beta_0 + \beta_1 x_i + \varepsilon_i - \beta_1 u_i)}{\sum_{i=1}^n (x_i - \bar{x})^2} \\ &= \frac{\beta_0 \sum_{i=1}^n (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} + \frac{\beta_1 \sum_{i=1}^n (x_i - \bar{x})x_i}{\sum_{i=1}^n (x_i - \bar{x})^2} + \frac{\sum_{i=1}^n (x_i - \bar{x})\varepsilon_i}{\sum_{i=1}^n (x_i - \bar{x})^2} - \beta_1 \frac{\sum_{i=1}^n (x_i - \bar{x})u_i}{\sum_{i=1}^n (x_i - \bar{x})^2}. \end{aligned}$$

Now simplify each deterministic term:

$$\sum_{i=1}^n (x_i - \bar{x}) = 0,$$

and

$$\begin{aligned} \sum_{i=1}^n (x_i - \bar{x})x_i &= \sum_{i=1}^n (x_i - \bar{x})((x_i - \bar{x}) + \bar{x}) \\ &= \sum_{i=1}^n (x_i - \bar{x})^2 + \bar{x} \sum_{i=1}^n (x_i - \bar{x}) \\ &= \sum_{i=1}^n (x_i - \bar{x})^2. \end{aligned}$$

Therefore,

$$b_1 = \beta_1 + \frac{\sum_{i=1}^n (x_i - \bar{x})\varepsilon_i}{\sum_{i=1}^n (x_i - \bar{x})^2} - \beta_1 \frac{\sum_{i=1}^n (x_i - \bar{x})u_i}{\sum_{i=1}^n (x_i - \bar{x})^2}. \quad (1)$$

Probability limits of the components.

Denominator.

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 &= \frac{1}{n} \sum_{i=1}^n x_i^2 - \left(\frac{1}{n} \sum_{i=1}^n x_i \right)^2 \\ &\xrightarrow{p} \mathbb{E}[x_i^2] - (\mathbb{E}[x_i])^2 \\ &= \text{Var}(x_i). \end{aligned}$$

First numerator term.

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})\varepsilon_i &= \frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i - \left(\frac{1}{n} \sum_{i=1}^n x_i \right) \left(\frac{1}{n} \sum_{i=1}^n \varepsilon_i \right) \\ &\xrightarrow{p} \mathbb{E}[x_i \varepsilon_i] - \mathbb{E}[x_i] \mathbb{E}[\varepsilon_i]. \end{aligned}$$

Since $x_i = x_i^* + u_i$ and x_i^* is uncorrelated with ε_i while u_i is independent of ε_i ,

$$\mathbb{E}[x_i \varepsilon_i] = \mathbb{E}[x_i^* \varepsilon_i] + \mathbb{E}[u_i \varepsilon_i] = 0 + 0 = 0, \quad \mathbb{E}[\varepsilon_i] = 0,$$

so

$$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})\varepsilon_i \xrightarrow{p} 0.$$

Second numerator term.

$$\begin{aligned}\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}) u_i &= \frac{1}{n} \sum_{i=1}^n x_i u_i - \left(\frac{1}{n} \sum_{i=1}^n x_i \right) \left(\frac{1}{n} \sum_{i=1}^n u_i \right) \\ &\xrightarrow{p} \mathbb{E}[x_i u_i] - \mathbb{E}[x_i] \mathbb{E}[u_i].\end{aligned}$$

Because $\mathbb{E}[u_i] = 0$, this reduces to $\mathbb{E}[x_i u_i]$. Using $x_i = x_i^* + u_i$,

$$\begin{aligned}\mathbb{E}[x_i u_i] &= \mathbb{E}[(x_i^* + u_i) u_i] \\ &= \mathbb{E}[x_i^* u_i] + \mathbb{E}[u_i^2] \\ &= 0 + \delta_u^2 \\ &= \delta_u^2,\end{aligned}$$

since u_i is independent of x_i^* .

Limit of b_1 . Divide the limits in (1):

$$b_1 \xrightarrow{p} \beta_1 - \beta_1 \frac{\delta_u^2}{\text{Var}(x_i)}.$$

Let $\text{Var}(x_i^*) = \delta^2$. Then, since $x_i = x_i^* + u_i$ with independence,

$$\text{Var}(x_i) = \text{Var}(x_i^*) + \text{Var}(u_i) = \delta^2 + \delta_u^2.$$

Hence

$$b_1 \xrightarrow{p} \beta_1 \frac{\delta^2}{\delta^2 + \delta_u^2} < \beta_1.$$

27 Using a second noisy measure as an instrument

A second measure of x^* . Suppose z is another measure of x^* :

$$z = x^* + w,$$

where w is independent of (y, x^*, ε, u) , $\mathbb{E}[w] = 0$, and $\text{Var}(w) = \delta_w^2$.

Use z as an IV for x . Consider the moment conditions (intercept and instrument orthogonality):

$$\sum_{i=1}^n (y_i - b_0 - b_1 x_i) = 0, \quad \sum_{i=1}^n (y_i - b_0 - b_1 x_i) z_i = 0.$$

From the first moment condition,

$$b_0 = \bar{y} - b_1 \bar{x}.$$

Substitute into the second:

$$\begin{aligned}0 &= \sum_{i=1}^n [(y_i - \bar{y}) - b_1 (x_i - \bar{x})] z_i \\ &= \sum_{i=1}^n (y_i - \bar{y}) z_i - b_1 \sum_{i=1}^n (x_i - \bar{x}) z_i.\end{aligned}$$

Therefore,

$$b_1 \sum_{i=1}^n (x_i - \bar{x}) z_i = \sum_{i=1}^n (y_i - \bar{y}) z_i.$$

Because $\sum_{i=1}^n (y_i - \bar{y}) = 0$ and $\sum_{i=1}^n (x_i - \bar{x}) = 0$, we can center z without changing either sum:

$$\sum_{i=1}^n (y_i - \bar{y}) z_i = \sum_{i=1}^n (y_i - \bar{y}) (z_i - \bar{z}), \quad \sum_{i=1}^n (x_i - \bar{x}) z_i = \sum_{i=1}^n (x_i - \bar{x}) (z_i - \bar{z}).$$

Hence

$$b_1 = \frac{\sum_{i=1}^n (y_i - \bar{y}) (z_i - \bar{z})}{\sum_{i=1}^n (x_i - \bar{x}) (z_i - \bar{z})}. \quad (2)$$

Substitute $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i - \beta_1 u_i$ into (2).

$$\begin{aligned} b_1 &= \frac{\sum_{i=1}^n (z_i - \bar{z}) (\beta_0 + \beta_1 x_i + \varepsilon_i - \beta_1 u_i)}{\sum_{i=1}^n (x_i - \bar{x}) (z_i - \bar{z})} \\ &= \frac{\beta_0 \sum_{i=1}^n (z_i - \bar{z})}{\sum_{i=1}^n (x_i - \bar{x}) (z_i - \bar{z})} + \beta_1 \frac{\sum_{i=1}^n (z_i - \bar{z}) x_i}{\sum_{i=1}^n (x_i - \bar{x}) (z_i - \bar{z})} + \frac{\sum_{i=1}^n (z_i - \bar{z}) \varepsilon_i}{\sum_{i=1}^n (x_i - \bar{x}) (z_i - \bar{z})} - \beta_1 \frac{\sum_{i=1}^n (z_i - \bar{z}) u_i}{\sum_{i=1}^n (x_i - \bar{x}) (z_i - \bar{z})}. \end{aligned}$$

Since $\sum_{i=1}^n (z_i - \bar{z}) = 0$ and $\sum_{i=1}^n (z_i - \bar{z}) x_i = \sum_{i=1}^n (z_i - \bar{z}) (x_i - \bar{x})$,

$$b_1 = \beta_1 + \frac{\sum_{i=1}^n (z_i - \bar{z}) \varepsilon_i}{\sum_{i=1}^n (x_i - \bar{x}) (z_i - \bar{z})} - \beta_1 \frac{\sum_{i=1}^n (z_i - \bar{z}) u_i}{\sum_{i=1}^n (x_i - \bar{x}) (z_i - \bar{z})}. \quad (3)$$

Probability limits.

Denominator.

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}) (z_i - \bar{z}) &= \frac{1}{n} \sum_{i=1}^n [x_i z_i - \bar{x} z_i - x_i \bar{z} + \bar{x} \bar{z}] \\ &= \frac{1}{n} \sum_{i=1}^n x_i z_i - \left(\frac{1}{n} \sum_{i=1}^n x_i \right) \left(\frac{1}{n} \sum_{i=1}^n z_i \right) \\ &\xrightarrow{P} \mathbb{E}[x_i z_i] - \mathbb{E}[x_i] \mathbb{E}[z_i] \\ &= \text{Cov}(x_i, z_i) \\ &= \text{Cov}(x_i^* + u_i, x_i^* + w_i) \\ &= \text{Var}(x_i^*) \\ &= \delta^2. \end{aligned}$$

First numerator term.

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n (z_i - \bar{z}) \varepsilon_i &= \frac{1}{n} \sum_{i=1}^n z_i \varepsilon_i - \left(\frac{1}{n} \sum_{i=1}^n z_i \right) \left(\frac{1}{n} \sum_{i=1}^n \varepsilon_i \right) \\ &\xrightarrow{P} \mathbb{E}[z_i \varepsilon_i] - \mathbb{E}[z_i] \mathbb{E}[\varepsilon_i]. \end{aligned}$$

Because $z_i = x_i^* + w_i$, x_i^* is uncorrelated with ε_i and w_i is independent of ε_i ,

$$\mathbb{E}[z_i \varepsilon_i] = \mathbb{E}[x_i^* \varepsilon_i] + \mathbb{E}[w_i \varepsilon_i] = 0 + 0 = 0, \quad \mathbb{E}[\varepsilon_i] = 0,$$

so

$$\frac{1}{n} \sum_{i=1}^n (z_i - \bar{z}) \varepsilon_i \xrightarrow{p} 0.$$

Second numerator term.

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n (z_i - \bar{z}) u_i &= \frac{1}{n} \sum_{i=1}^n z_i u_i - \left(\frac{1}{n} \sum_{i=1}^n z_i \right) \left(\frac{1}{n} \sum_{i=1}^n u_i \right) \\ &\xrightarrow{p} \mathbb{E}[z_i u_i] - \mathbb{E}[z_i] \mathbb{E}[u_i]. \end{aligned}$$

Since $\mathbb{E}[u_i] = 0$, this reduces to $\mathbb{E}[z_i u_i]$. Using $z_i = x_i^* + w_i$ and independence of u_i from (x_i^*, w_i) ,

$$\mathbb{E}[z_i u_i] = \mathbb{E}[x_i^* u_i] + \mathbb{E}[w_i u_i] = 0 + 0 = 0.$$

Hence

$$\frac{1}{n} \sum_{i=1}^n (z_i - \bar{z}) u_i \xrightarrow{p} 0.$$

Limit of the IV slope. Using (3) and the limits above,

$$b_1 \xrightarrow{p} \beta_1 + \frac{0}{\delta^2} - \beta_1 \frac{0}{\delta^2} = \beta_1,$$

so

$$b_1 \xrightarrow{p} \beta_1.$$

Heteroskedasticity: Conditional Variance of OLS and Asymptotic Distribution

28 Assumptions

1. Linearity

$$Y = X\beta + \varepsilon.$$

2. Full rank

$$\text{rank}(X) = k.$$

3. Exogeneity

$$\mathbb{E}[\varepsilon \mid X] = 0.$$

Law of iterated expectations.

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]].$$

Applying it to the disturbance:

$$\mathbb{E}[\varepsilon] = \mathbb{E}[\mathbb{E}[\varepsilon \mid X]] = \mathbb{E}[0] = 0.$$

4. Heteroskedasticity and nonautocorrelation

$$\text{Var}(\varepsilon_i \mid X) = \sigma_i^2, \quad i = 1, \dots, n,$$

$$\text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0, \quad i \neq j.$$

5. X may be fixed and random.

29 Conditional second-moment matrix under heteroskedasticity

Stack the errors as

$$\varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

Define

$$\Sigma \equiv \mathbb{E}[\varepsilon\varepsilon' \mid X].$$

Diagonal entries. The (i, i) element of Σ is

$$\Sigma_{ii} = \mathbb{E}[\varepsilon_i^2 \mid X].$$

Using $\mathbb{E}[\varepsilon_i \mid X] = 0$ (from Assumption 3),

$$\text{Var}(\varepsilon_i \mid X) = \mathbb{E}[\varepsilon_i^2 \mid X] - (\mathbb{E}[\varepsilon_i \mid X])^2 = \mathbb{E}[\varepsilon_i^2 \mid X],$$

so

$$\Sigma_{ii} = \mathbb{E}[\varepsilon_i^2 \mid X] = \text{Var}(\varepsilon_i \mid X) = \sigma_i^2.$$

Off-diagonal entries. For $i \neq j$, the (i, j) element of Σ is

$$\Sigma_{ij} = \mathbb{E}[\varepsilon_i\varepsilon_j \mid X].$$

Using $\mathbb{E}[\varepsilon_i \mid X] = \mathbb{E}[\varepsilon_j \mid X] = 0$,

$$\text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = \mathbb{E}[\varepsilon_i\varepsilon_j \mid X] - \mathbb{E}[\varepsilon_i \mid X]\mathbb{E}[\varepsilon_j \mid X] = \mathbb{E}[\varepsilon_i\varepsilon_j \mid X],$$

so

$$\Sigma_{ij} = \mathbb{E}[\varepsilon_i\varepsilon_j \mid X] = \text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0 \quad (i \neq j).$$

Therefore,

$$\Sigma = \begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n^2 \end{pmatrix} = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2).$$

30 OLS decomposition and conditional variance under Σ

Let the OLS estimator be

$$b \equiv (X'X)^{-1}X'Y.$$

Substitute $Y = X\beta + \varepsilon$:

$$\begin{aligned} b &= (X'X)^{-1}X'(X\beta + \varepsilon) \\ &= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'\varepsilon \\ &= \beta + (X'X)^{-1}X'\varepsilon. \end{aligned}$$

Thus

$$b - \beta = (X'X)^{-1}X'\varepsilon.$$

Conditional expectation

$$\begin{aligned}\mathbb{E}[b \mid X] &= \mathbb{E}[\beta + (X'X)^{-1}X'\varepsilon \mid X] \\ &= \beta + (X'X)^{-1}X'\mathbb{E}[\varepsilon \mid X] \\ &= \beta,\end{aligned}$$

so b is unbiased conditional on X .

Conditional variance

Start from the definition:

$$\text{Var}(b \mid X) = \mathbb{E}[(b - \beta)(b - \beta)' \mid X].$$

Substitute $b - \beta = (X'X)^{-1}X'\varepsilon$:

$$\begin{aligned}\text{Var}(b \mid X) &= \mathbb{E}\left[\left((X'X)^{-1}X'\varepsilon\right)\left((X'X)^{-1}X'\varepsilon\right)' \mid X\right] \\ &= \mathbb{E}\left[(X'X)^{-1}X'\varepsilon\varepsilon'X((X'X)^{-1})' \mid X\right].\end{aligned}$$

Because $X'X$ is symmetric, $((X'X)^{-1})' = (X'X)^{-1}$, hence

$$\text{Var}(b \mid X) = \mathbb{E}\left[(X'X)^{-1}X'\varepsilon\varepsilon'X(X'X)^{-1} \mid X\right].$$

Pull out the matrices that depend only on X :

$$\begin{aligned}\text{Var}(b \mid X) &= (X'X)^{-1}X' \mathbb{E}[\varepsilon\varepsilon' \mid X] X(X'X)^{-1} \\ &= (X'X)^{-1}X'\Sigma X(X'X)^{-1}.\end{aligned}$$

31 An i.i.d. heteroskedastic setup and the sandwich asymptotic variance

Additional assumption

Assume (x_i, ε_i) are i.i.d. across $i = 1, \dots, n$ and

$$\text{Var}(\varepsilon_i \mid x_i) = g(x_i) \quad \text{with } g(\cdot) \text{ unknown.}$$

Assume the following moments exist:

$$Q_x \equiv \mathbb{E}[x_i x_i'] < \infty, \quad Q_x^{-1} \text{ exists,}$$

$$Q_x^\varepsilon \equiv \mathbb{E}[x_i x_i' \varepsilon_i^2] < \infty.$$

Consistency (step-by-step)

From the OLS decomposition,

$$b - \beta = (X'X)^{-1}X'\varepsilon.$$

Rewrite by multiplying and dividing by n :

$$b - \beta = \left(\frac{X'X}{n}\right)^{-1} \left(\frac{X'\varepsilon}{n}\right).$$

Using the sample-sum representations,

$$\frac{X'X}{n} = \frac{1}{n} \sum_{i=1}^n x_i x_i', \quad \frac{X'\varepsilon}{n} = \frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i.$$

By LLN,

$$\frac{1}{n} \sum_{i=1}^n x_i x_i' \xrightarrow{p} \mathbb{E}[x_i x_i'] = Q_x,$$

and (using $\mathbb{E}[\varepsilon_i | x_i] = 0$ implied by exogeneity in this i.i.d. setting),

$$\mathbb{E}[x_i \varepsilon_i] = \mathbb{E}[\mathbb{E}[x_i \varepsilon_i | x_i]] = \mathbb{E}[x_i \mathbb{E}[\varepsilon_i | x_i]] = \mathbb{E}[x_i \cdot 0] = 0,$$

so again by LLN,

$$\frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i \xrightarrow{p} \mathbb{E}[x_i \varepsilon_i] = 0.$$

By continuity of inversion at Q_x ,

$$\left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1} \xrightarrow{p} Q_x^{-1}.$$

Therefore,

$$b - \beta = \left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i \right) \xrightarrow{p} Q_x^{-1} \cdot 0 = 0.$$

Asymptotic normality (step-by-step)

Write

$$\sqrt{n}(b - \beta) = \sqrt{n}(X'X)^{-1} X'\varepsilon.$$

Insert the normalized sums:

$$\sqrt{n}(b - \beta) = \left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n x_i \varepsilon_i \right).$$

The first factor satisfies

$$\left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1} \xrightarrow{p} Q_x^{-1}.$$

For the second factor, define $u_i \equiv x_i \varepsilon_i$. Then $\mathbb{E}[u_i] = \mathbb{E}[x_i \varepsilon_i] = 0$ and

$$\text{Var}(u_i) = \mathbb{E}[u_i u_i'] = \mathbb{E}[(x_i \varepsilon_i)(x_i \varepsilon_i)'] = \mathbb{E}[x_i x_i' \varepsilon_i^2] = Q_x^\varepsilon.$$

By the multivariate CLT,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n x_i \varepsilon_i = \frac{1}{\sqrt{n}} \sum_{i=1}^n u_i \xrightarrow{d} N(0, Q_x^\varepsilon).$$

By Slutsky, if $A_n \xrightarrow{p} A$ and $W_n \xrightarrow{d} W$, then $A_n W_n \xrightarrow{d} AW$, so

$$\sqrt{n}(b - \beta) \xrightarrow{d} Q_x^{-1} W, \quad W \sim N(0, Q_x^\varepsilon).$$

Finally, if $W \sim N(0, \Omega)$, then $AW \sim N(0, A\Omega A')$. Hence

$$\sqrt{n}(b - \beta) \xrightarrow{d} N(0, Q_x^{-1} Q_x^\varepsilon (Q_x^{-1})').$$

Because Q_x is symmetric, Q_x^{-1} is symmetric, so $(Q_x^{-1})' = Q_x^{-1}$ and

$$\boxed{\sqrt{n}(b - \beta) \xrightarrow{d} N(0, Q_x^{-1} Q_x^\varepsilon Q_x^{-1})}.$$

Heteroskedasticity: Asymptotic Variance and Its Estimation

32 Assumptions

Let the linear regression model be

$$Y = X\beta + \varepsilon, \quad Y \in \mathbb{R}^{n \times 1}, \quad X \in \mathbb{R}^{n \times k}, \quad \beta \in \mathbb{R}^{k \times 1}, \quad \varepsilon \in \mathbb{R}^{n \times 1}.$$

(Here the sample size may be denoted by n ; in some notes it is written as h .)

1. Linearity

$$Y = X\beta + \varepsilon.$$

2. Full rank

$$\text{rank}(X) = k.$$

3. Exogeneity

$$\mathbb{E}[\varepsilon \mid X] = 0.$$

Law of iterated expectations:

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]].$$

In particular,

$$\mathbb{E}[\varepsilon] = \mathbb{E}[\mathbb{E}[\varepsilon \mid X]] = \mathbb{E}[0] = 0.$$

4. Heteroskedasticity and nonautocorrelation

$$\text{Var}(\varepsilon_i \mid X) = \sigma_i^2, \quad i = 1, \dots, n,$$

$$\text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0, \quad i \neq j.$$

5. X may be fixed and random.

6. i.i.d. sampling and moments (written in the notes' style)

Assume (x_i, ε_i) , $i = 1, \dots, n$, are i.i.d. and

$$\text{Var}(\varepsilon_i \mid x_i) = g(x_i), \quad g(\cdot) \text{ unknown.}$$

Define

$$Q_x \equiv \mathbb{E}[x_i x_i'] < \infty, \quad Q_x^{-1} \text{ exists,}$$

and

$$Q_x^\varepsilon \equiv \mathbb{E}[x_i x_i' \varepsilon_i^2] < \infty.$$

33 OLS and the asymptotic variance under heteroskedasticity

Let the OLS estimator be

$$b \equiv (X'X)^{-1}X'Y.$$

Substitute $Y = X\beta + \varepsilon$:

$$\begin{aligned} b &= (X'X)^{-1}X'(X\beta + \varepsilon) \\ &= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'\varepsilon \\ &= \beta + (X'X)^{-1}X'\varepsilon. \end{aligned}$$

Hence

$$b - \beta = (X'X)^{-1}X'\varepsilon.$$

Conditional expectation

$$\begin{aligned} \mathbb{E}[b \mid X] &= \mathbb{E}[\beta + (X'X)^{-1}X'\varepsilon \mid X] \\ &= \beta + (X'X)^{-1}X'\mathbb{E}[\varepsilon \mid X] \\ &= \beta. \end{aligned}$$

Consistency (as written in the notes)

$$b \xrightarrow{p} \beta.$$

Asymptotic normality (sandwich form, as written)

$$\sqrt{n}(b - \beta) \xrightarrow{d} N(0, Q_x^{-1}Q_x^\varepsilon Q_x^{-1}).$$

34 How to estimate the asymptotic variance

Estimating Q_x

By the law of large numbers,

$$\frac{X'X}{n} = \frac{1}{n} \sum_{i=1}^n x_i x_i' \xrightarrow{p} \mathbb{E}[x_i x_i'] = Q_x.$$

Therefore we use

$$\hat{Q}_x \equiv \frac{X'X}{n}$$

to estimate Q_x .

Understanding and estimating Q_x^ε

By definition,

$$Q_x^\varepsilon = \mathbb{E}[x_i x_i' \varepsilon_i^2].$$

Use iterated expectations:

$$\begin{aligned} Q_x^\varepsilon &= \mathbb{E}[x_i x_i' \varepsilon_i^2] \\ &= \mathbb{E}\left[\mathbb{E}[x_i x_i' \varepsilon_i^2 \mid x_i]\right] \\ &= \mathbb{E}\left[x_i x_i' \mathbb{E}[\varepsilon_i^2 \mid x_i]\right]. \end{aligned}$$

If (special case) $\mathbb{E}[\varepsilon_i^2 \mid x_i] = \sigma^2$ is constant, then

$$Q_x^\varepsilon = \mathbb{E}[x_i x_i' \sigma^2] = \sigma^2 \mathbb{E}[x_i x_i'] = \sigma^2 Q_x.$$

In general, we estimate Q_x^ε consistently by

$$\hat{Q}_x^\varepsilon \equiv \frac{1}{n} \sum_{i=1}^n x_i x_i' e_i^2, \quad e_i \equiv y_i - x_i' b,$$

where b is the OLS estimator.

Plug-in (sandwich) estimator

Using $\hat{Q}_x$ and $\hat{Q}_x^\varepsilon$, estimate the asymptotic covariance matrix by

$$\hat{\Omega} \equiv \hat{Q}_x^{-1} \hat{Q}_x^\varepsilon \hat{Q}_x^{-1}.$$

35 Consistency of $\hat{Q}_x^\varepsilon$ in a scalar example

Suppose

$$y_i = \beta x_i + \varepsilon_i \quad (i = 1, \dots, n),$$

and define

$$\hat{Q}_x^\varepsilon = \frac{1}{n} \sum_{i=1}^n x_i^2 (y_i - \hat{\beta} x_i)^2,$$

where $\hat{\beta}$ is the OLS estimator from regressing y_i on x_i .

Algebraic expansion

Start from

$$\hat{Q}_x^\varepsilon = \frac{1}{n} \sum_{i=1}^n x_i^2 (y_i - \hat{\beta} x_i)^2.$$

Substitute $y_i = \beta x_i + \varepsilon_i$:

$$\hat{Q}_x^\varepsilon = \frac{1}{n} \sum_{i=1}^n x_i^2 (\beta x_i + \varepsilon_i - \hat{\beta} x_i)^2.$$

Collect terms:

$$\hat{Q}_x^\varepsilon = \frac{1}{n} \sum_{i=1}^n x_i^2 [(\beta - \hat{\beta}) x_i + \varepsilon_i]^2.$$

Expand the square:

$$[(\beta - \hat{\beta}) x_i + \varepsilon_i]^2 = (\beta - \hat{\beta})^2 x_i^2 + \varepsilon_i^2 + 2(\beta - \hat{\beta}) x_i \varepsilon_i.$$

Multiply by x_i^2 :

$$x_i^2 [(\beta - \hat{\beta}) x_i + \varepsilon_i]^2 = x_i^4 (\beta - \hat{\beta})^2 + x_i^2 \varepsilon_i^2 + 2x_i^3 (\beta - \hat{\beta}) \varepsilon_i.$$

Therefore,

$$\hat{Q}_x^\varepsilon = \frac{1}{n} \sum_{i=1}^n x_i^4 (\beta - \hat{\beta})^2 + \frac{1}{n} \sum_{i=1}^n x_i^2 \varepsilon_i^2 + \frac{2}{n} \sum_{i=1}^n x_i^3 (\beta - \hat{\beta}) \varepsilon_i.$$

Probability limits of each term

For the first term,

$$\frac{1}{n} \sum_{i=1}^n x_i^4 (\beta - \hat{\beta})^2 = (\beta - \hat{\beta})^2 \left(\frac{1}{n} \sum_{i=1}^n x_i^4 \right).$$

Under the maintained assumptions,

$$\frac{1}{n} \sum_{i=1}^n x_i^4 \xrightarrow{p} \mathbb{E}[x_i^4], \quad \beta - \hat{\beta} \xrightarrow{p} 0,$$

so

$$\frac{1}{n} \sum_{i=1}^n x_i^4 (\beta - \hat{\beta})^2 \xrightarrow{p} 0.$$

For the third term,

$$\frac{2}{n} \sum_{i=1}^n x_i^3 (\beta - \hat{\beta}) \varepsilon_i = 2(\beta - \hat{\beta}) \left(\frac{1}{n} \sum_{i=1}^n x_i^3 \varepsilon_i \right).$$

Under LLN (finite moment condition),

$$\frac{1}{n} \sum_{i=1}^n x_i^3 \varepsilon_i \xrightarrow{p} \mathbb{E}[x_i^3 \varepsilon_i], \quad \beta - \hat{\beta} \xrightarrow{p} 0,$$

so

$$\frac{2}{n} \sum_{i=1}^n x_i^3 (\beta - \hat{\beta}) \varepsilon_i \xrightarrow{p} 0.$$

Hence

$$\hat{Q}_x^\varepsilon \xrightarrow{p} \xrightarrow{p}_{n \rightarrow \infty} \left(\frac{1}{n} \sum_{i=1}^n x_i^2 \varepsilon_i^2 \right) = \mathbb{E}[x_i^2 \varepsilon_i^2] = \mathbb{E}[x_i x_i' \varepsilon_i^2] = Q_x^\varepsilon.$$

Therefore,

$$\hat{Q}_x^\varepsilon \xrightarrow{p} Q_x^\varepsilon.$$

Heteroskedasticity and Weighted Least Squares

Assumptions

1. Linearity

$$Y = X\beta + \varepsilon.$$

2. Full rank

$$\text{rank}(X) = k.$$

3. Exogeneity

$$\mathbb{E}[\varepsilon \mid X] = 0.$$

Law of iterated expectations:

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]].$$

In particular,

$$\mathbb{E}[\varepsilon] = \mathbb{E}[\mathbb{E}[\varepsilon \mid X]] = \mathbb{E}[0] = 0.$$

4. Homoscedasticity (benchmark form)

$$\mathbb{E}[\varepsilon_i^2 \mid z_i] = \sigma^2.$$

4'. Heteroskedasticity and nonautocorrelation

$$\text{Var}(\varepsilon_i \mid X) = \sigma_i^2, \quad i = 1, \dots, n,$$

$$\text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0, \quad i \neq j.$$

5. X may be fixed and random.

Matrix form of the conditional second moment

Define

$$\mathbb{E}[\varepsilon\varepsilon' \mid X] \equiv \Sigma.$$

Under Assumption 4',

$$\Sigma = \begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n^2 \end{pmatrix} = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2).$$

Gauss–Markov theorem (benchmark statement)

Under Assumptions 1–5 (with the homoscedastic benchmark), the least squares estimator $\hat{\beta}$ has the smallest variance among the class of linear (in Y) and unbiased estimators of β . $\hat{\beta}$ is called the best linear unbiased estimator (BLUE).

Heteroskedasticity setup to be transformed

We work with

$$Y = X\beta + \varepsilon, \quad \mathbb{E}[\varepsilon \mid X] = 0, \quad \mathbb{E}[\varepsilon\varepsilon' \mid X] = \Sigma.$$

Define $\Sigma^{-1/2}$ and verify $\Sigma^{-1/2}\Sigma\Sigma^{-1/2} = I$

Define

$$\Sigma^{-1/2} \equiv \text{diag}\left(\frac{1}{\sigma_1}, \frac{1}{\sigma_2}, \dots, \frac{1}{\sigma_n}\right) = \begin{pmatrix} \frac{1}{\sigma_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{\sigma_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\sigma_n} \end{pmatrix}.$$

Compute:

$$\Sigma^{-1/2}\Sigma\Sigma^{-1/2} = \begin{pmatrix} \frac{1}{\sigma_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{\sigma_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\sigma_n} \end{pmatrix} \begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n^2 \end{pmatrix} \begin{pmatrix} \frac{1}{\sigma_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{\sigma_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\sigma_n} \end{pmatrix}.$$

First multiply the left two matrices:

$$\begin{pmatrix} \frac{1}{\sigma_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{\sigma_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\sigma_n} \end{pmatrix} \begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n^2 \end{pmatrix} = \begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n \end{pmatrix}.$$

Then multiply by the right $\Sigma^{-1/2}$:

$$\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n \end{pmatrix} \begin{pmatrix} \frac{1}{\sigma_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{\sigma_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\sigma_n} \end{pmatrix} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \equiv I_n.$$

Thus

$$\Sigma^{-1/2} \Sigma \Sigma^{-1/2} = I_n.$$

Transform the model and check the transformed error moments

Define the transformed error

$$\varepsilon_* \equiv \Sigma^{-1/2} \varepsilon.$$

Compute its conditional mean:

$$\mathbb{E}[\varepsilon_* \mid X] = \mathbb{E}[\Sigma^{-1/2} \varepsilon \mid X] = \Sigma^{-1/2} \mathbb{E}[\varepsilon \mid X] = \Sigma^{-1/2} \cdot 0 = 0.$$

Compute its conditional second moment:

$$\mathbb{E}[\varepsilon_* \varepsilon_*' \mid X] = \mathbb{E}[(\Sigma^{-1/2} \varepsilon)(\Sigma^{-1/2} \varepsilon)' \mid X].$$

Use $(AB)' = B' A'$:

$$(\Sigma^{-1/2} \varepsilon)' = \varepsilon' (\Sigma^{-1/2})',$$

and since $\Sigma^{-1/2}$ is diagonal, $(\Sigma^{-1/2})' = \Sigma^{-1/2}$, hence

$$\mathbb{E}[\varepsilon_* \varepsilon_*' \mid X] = \mathbb{E}[\Sigma^{-1/2} \varepsilon \varepsilon' \Sigma^{-1/2} \mid X] = \Sigma^{-1/2} \mathbb{E}[\varepsilon \varepsilon' \mid X] \Sigma^{-1/2} = \Sigma^{-1/2} \Sigma \Sigma^{-1/2} = I_n.$$

Transform Y and X

Start from

$$Y = X\beta + \varepsilon.$$

Multiply both sides by $\Sigma^{-1/2}$:

$$\Sigma^{-1/2} Y = \Sigma^{-1/2} X\beta + \Sigma^{-1/2} \varepsilon.$$

Define

$$Y_* \equiv \Sigma^{-1/2} Y, \quad X_* \equiv \Sigma^{-1/2} X, \quad \varepsilon_* \equiv \Sigma^{-1/2} \varepsilon,$$

so the transformed model is

$$Y_* = X_* \beta + \varepsilon_*, \quad \mathbb{E}[\varepsilon_* \mid X] = 0, \quad \mathbb{E}[\varepsilon_* \varepsilon_*' \mid X] = I_n.$$

Write Y_* explicitly:

$$\Sigma^{-1/2} Y = \begin{pmatrix} \frac{1}{\sigma_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{\sigma_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\sigma_n} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} \frac{y_1}{\sigma_1} \\ \frac{y_2}{\sigma_2} \\ \vdots \\ \frac{y_n}{\sigma_n} \end{pmatrix}.$$

Weighted least squares (WLS) estimator

Define the least squares estimator in the transformed regression:

$$\hat{\beta}_{WLS} \equiv (X_*' X_*)^{-1} X_*' Y_*.$$

Substitute $X_* = \Sigma^{-1/2} X$ and $Y_* = \Sigma^{-1/2} Y$:

$$\hat{\beta}_{WLS} = ((\Sigma^{-1/2} X)' (\Sigma^{-1/2} X))^{-1} (\Sigma^{-1/2} X)' (\Sigma^{-1/2} Y).$$

Apply transpose and regroup:

$$(\Sigma^{-1/2} X)' = X' (\Sigma^{-1/2})' = X' \Sigma^{-1/2},$$

so

$$X_*' X_* = X' \Sigma^{-1/2} \Sigma^{-1/2} X, \quad X_*' Y_* = X' \Sigma^{-1/2} \Sigma^{-1/2} Y.$$

Since $\Sigma^{-1/2} \Sigma^{-1/2} = \Sigma^{-1}$ (diagonal entries $(1/\sigma_i)(1/\sigma_i) = 1/\sigma_i^2$),

$$X_*' X_* = X' \Sigma^{-1} X, \quad X_*' Y_* = X' \Sigma^{-1} Y.$$

Therefore,

$$\boxed{\hat{\beta}_{WLS} = (X' \Sigma^{-1} X)^{-1} X' \Sigma^{-1} Y.}$$

Equivalent weighted objective (same estimator)

Consider the weighted sum of squares:

$$S(\beta) \equiv (Y - X\beta)' \Sigma^{-1} (Y - X\beta).$$

Expand:

$$S(\beta) = Y' \Sigma^{-1} Y - 2\beta' X' \Sigma^{-1} Y + \beta' X' \Sigma^{-1} X \beta.$$

Differentiate w.r.t. β and set to zero:

$$\frac{\partial S(\beta)}{\partial \beta} = -2X' \Sigma^{-1} Y + 2X' \Sigma^{-1} X \beta = 0.$$

Thus the normal equations are

$$X' \Sigma^{-1} X \beta = X' \Sigma^{-1} Y,$$

and solving yields

$$\hat{\beta} = (X' \Sigma^{-1} X)^{-1} X' \Sigma^{-1} Y,$$

which matches the transformed-regression formula above.

Generalized Least Squares (GLS), WLS, and Feasible GLS

36 Assumptions

1. Linearity

$$y = X\beta + \varepsilon.$$

2. **Full rank**

$$\text{rank}(X) = k.$$

3. **Exogeneity**

$$\mathbb{E}[\varepsilon \mid X] = 0.$$

Law of iterated expectations:

$$\mathbb{E}[Y] = \mathbb{E}(\mathbb{E}[Y \mid X]).$$

Apply it to the disturbance:

$$\mathbb{E}[\varepsilon] = \mathbb{E}(\mathbb{E}[\varepsilon \mid X]) = \mathbb{E}[0] = 0.$$

4. **Conditional second moments**

$$\mathbb{E}[\varepsilon\varepsilon' \mid X] = \Sigma.$$

5. **X may be fixed and random.**

37 GLS transformation

Start from the model:

$$y = X\beta + \varepsilon.$$

Existence of $\Sigma^{-1/2}$ and whitening

Assume Σ is positive definite. Then there exists a symmetric matrix $\Sigma^{1/2}$ such that

$$\Sigma^{1/2}\Sigma^{1/2} = \Sigma.$$

Define

$$\Sigma^{-1/2} \equiv (\Sigma^{1/2})^{-1}.$$

Then

$$\Sigma^{-1/2}\Sigma\Sigma^{-1/2} = (\Sigma^{1/2})^{-1}(\Sigma^{1/2}\Sigma^{1/2})(\Sigma^{1/2})^{-1} = ((\Sigma^{1/2})^{-1}\Sigma^{1/2})(\Sigma^{1/2}(\Sigma^{1/2})^{-1}) = I.$$

Transform the regression

Multiply both sides of $y = X\beta + \varepsilon$ by $\Sigma^{-1/2}$:

$$\Sigma^{-1/2}y = \Sigma^{-1/2}X\beta + \Sigma^{-1/2}\varepsilon.$$

Define

$$y_* \equiv \Sigma^{-1/2}y, \quad X_* \equiv \Sigma^{-1/2}X, \quad \varepsilon_* \equiv \Sigma^{-1/2}\varepsilon.$$

Then

$$y_* = X_*\beta + \varepsilon_*.$$

Transformed error moments

Compute the conditional mean:

$$\mathbb{E}[\varepsilon_* \mid X] = \mathbb{E}[\Sigma^{-1/2}\varepsilon \mid X] = \Sigma^{-1/2}\mathbb{E}[\varepsilon \mid X] = \Sigma^{-1/2} \cdot 0 = 0.$$

Compute the conditional second moment:

$$\begin{aligned}\mathbb{E}[\varepsilon_*\varepsilon'_* \mid X] &= \mathbb{E}[(\Sigma^{-1/2}\varepsilon)(\Sigma^{-1/2}\varepsilon)' \mid X] \\ &= \mathbb{E}[\Sigma^{-1/2}\varepsilon\varepsilon'(\Sigma^{-1/2})' \mid X] \\ &= \Sigma^{-1/2}\mathbb{E}[\varepsilon\varepsilon' \mid X](\Sigma^{-1/2})' \\ &= \Sigma^{-1/2}\Sigma(\Sigma^{-1/2})'.\end{aligned}$$

If $\Sigma^{-1/2}$ is chosen symmetric (which is standard when Σ is symmetric p.d.), then $(\Sigma^{-1/2})' = \Sigma^{-1/2}$ and

$$\mathbb{E}[\varepsilon_*\varepsilon'_* \mid X] = \Sigma^{-1/2}\Sigma\Sigma^{-1/2} = I.$$

38 GLS estimator from OLS on the transformed model

OLS on the transformed regression $y_* = X_*\beta + \varepsilon_*$ yields

$$\hat{\beta}_{GLS} = (X'_*X_*)^{-1}X'_*y_*.$$

Substitute $X_* = \Sigma^{-1/2}X$ and $y_* = \Sigma^{-1/2}y$:

$$\hat{\beta}_{GLS} = ((\Sigma^{-1/2}X)'(\Sigma^{-1/2}X))^{-1}(\Sigma^{-1/2}X)'(\Sigma^{-1/2}y).$$

Use $(AB)' = B'A'$:

$$(\Sigma^{-1/2}X)' = X'(\Sigma^{-1/2})', \quad (\Sigma^{-1/2}y)' = y'(\Sigma^{-1/2})'.$$

If $\Sigma^{-1/2}$ is symmetric, $(\Sigma^{-1/2})' = \Sigma^{-1/2}$, so

$$X'_*X_* = X'\Sigma^{-1/2}\Sigma^{-1/2}X = X'\Sigma^{-1}X,$$

$$X'_*y_* = X'\Sigma^{-1/2}\Sigma^{-1/2}y = X'\Sigma^{-1}y.$$

Therefore

$$\boxed{\hat{\beta}_{GLS} = (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}y.}$$

39 Equivalent GLS objective function

Consider the weighted sum of squares:

$$S(b) \equiv (y - Xb)' \Sigma^{-1} (y - Xb).$$

Expand:

$$\begin{aligned}S(b) &= (y' - b'X')\Sigma^{-1}(y - Xb) \\ &= y'\Sigma^{-1}y - y'\Sigma^{-1}Xb - b'X'\Sigma^{-1}y + b'X'\Sigma^{-1}Xb \\ &= y'\Sigma^{-1}y - 2b'X'\Sigma^{-1}y + b'X'\Sigma^{-1}Xb,\end{aligned}$$

using that the middle two terms are scalars and equal.

Differentiate with respect to b and set to zero:

$$\frac{\partial S(b)}{\partial b} = -2X'\Sigma^{-1}y + 2X'\Sigma^{-1}Xb = 0.$$

Thus the normal equations are

$$X'\Sigma^{-1}Xb = X'\Sigma^{-1}y,$$

and solving gives

$$b = (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}y,$$

which matches $\hat{\beta}_{GLS}$ above.

40 WLS as a special case of GLS

If Σ is diagonal,

$$\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_n^2), \quad \Sigma^{-1} = \text{diag}\left(\frac{1}{\sigma_1^2}, \dots, \frac{1}{\sigma_n^2}\right),$$

then the GLS objective

$$(y - Xb)'\Sigma^{-1}(y - Xb)$$

equals

$$\sum_{i=1}^n \frac{(y_i - x_i'b)^2}{\sigma_i^2}.$$

Minimizing this is weighted least squares (WLS).

41 Feasible GLS (FGLS)

In practice Σ is unknown. Let $\hat{\Sigma}$ be a consistent estimator of Σ . Define the feasible GLS estimator:

$$\boxed{\hat{\beta}_{FGLS} = (X'\hat{\Sigma}^{-1}X)^{-1}X'\hat{\Sigma}^{-1}y.}$$

When Σ is diagonal (the WLS case), $\hat{\Sigma}$ corresponds to estimated weights.

Kronecker Product $\otimes$

Definition

For an $m \times n$ matrix $A = (a_{ij})$ and an $m' \times n'$ matrix $B = (b_{ij})$, the *Kronecker product* $A \otimes B$ is the $(mm') \times (nn')$ block matrix defined by

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix}.$$

Example

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}.$$

Then

$$A \otimes B = \begin{pmatrix} aB & bB \\ cB & dB \end{pmatrix}.$$

Compute each block:

$$aB = \begin{pmatrix} ae & af \\ ag & ah \end{pmatrix}, \quad bB = \begin{pmatrix} be & bf \\ bg & bh \end{pmatrix}, \quad cB = \begin{pmatrix} ce & cf \\ cg & ch \end{pmatrix}, \quad dB = \begin{pmatrix} de & df \\ dg & dh \end{pmatrix}.$$

Therefore

$$A \otimes B = \begin{pmatrix} ae & af & be & bf \\ ag & ah & bg & bh \\ ce & cf & de & df \\ cg & ch & dg & dh \end{pmatrix}.$$

Properties

1.

$$A \otimes (B + C) = A \otimes B + A \otimes C.$$

Proof. By definition,

$$A \otimes (B + C) = \begin{pmatrix} a_{11}(B + C) & \cdots & a_{1n}(B + C) \\ \vdots & \ddots & \vdots \\ a_{m1}(B + C) & \cdots & a_{mn}(B + C) \end{pmatrix}.$$

Distribute each block:

$$a_{ij}(B + C) = a_{ij}B + a_{ij}C.$$

So

$$\begin{aligned} A \otimes (B + C) &= \begin{pmatrix} a_{11}B + a_{11}C & \cdots & a_{1n}B + a_{1n}C \\ \vdots & \ddots & \vdots \\ a_{m1}B + a_{m1}C & \cdots & a_{mn}B + a_{mn}C \end{pmatrix} \\ &= \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix} + \begin{pmatrix} a_{11}C & \cdots & a_{1n}C \\ \vdots & \ddots & \vdots \\ a_{m1}C & \cdots & a_{mn}C \end{pmatrix} \\ &= A \otimes B + A \otimes C. \end{aligned}$$

□

2.

$$(A + B) \otimes C = A \otimes C + B \otimes C.$$

Proof. Write $(A + B) = (a_{ij} + b_{ij})$. Then

$$(A + B) \otimes C = \begin{pmatrix} (a_{11} + b_{11})C & \cdots & (a_{1n} + b_{1n})C \\ \vdots & \ddots & \vdots \\ (a_{m1} + b_{m1})C & \cdots & (a_{mn} + b_{mn})C \end{pmatrix}.$$

Distribute $(a_{ij} + b_{ij})C = a_{ij}C + b_{ij}C$ block-by-block to obtain

$$(A + B) \otimes C = \begin{pmatrix} a_{11}C & \cdots & a_{1n}C \\ \vdots & \ddots & \vdots \\ a_{m1}C & \cdots & a_{mn}C \end{pmatrix} + \begin{pmatrix} b_{11}C & \cdots & b_{1n}C \\ \vdots & \ddots & \vdots \\ b_{m1}C & \cdots & b_{mn}C \end{pmatrix} = A \otimes C + B \otimes C.$$

□

3.

$$(A \otimes B)' = A' \otimes B'.$$

Proof. Start from the block definition:

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix}.$$

Taking transpose swaps block rows and columns, and transposes each block:

$$(A \otimes B)' = \begin{pmatrix} (a_{11}B)' & \cdots & (a_{m1}B)' \\ \vdots & \ddots & \vdots \\ (a_{1n}B)' & \cdots & (a_{mn}B)' \end{pmatrix} = \begin{pmatrix} a_{11}B' & \cdots & a_{m1}B' \\ \vdots & \ddots & \vdots \\ a_{1n}B' & \cdots & a_{mn}B' \end{pmatrix}.$$

But the right-hand side is exactly $A' \otimes B'$, since $A' = (a_{ji})$:

$$A' \otimes B' = \begin{pmatrix} a_{11}B' & \cdots & a_{m1}B' \\ \vdots & \ddots & \vdots \\ a_{1n}B' & \cdots & a_{mn}B' \end{pmatrix}.$$

Hence $(A \otimes B)' = A' \otimes B'$. □

4.

$$(A \otimes B)(C \otimes D) = AC \otimes BD,$$

for conformable products AC and BD .

Proof. Write $A = (a_{ij})$ with $i = 1, \dots, m$ and $j = 1, \dots, n$, and write $C = (c_{jk})$ with $k = 1, \dots, p$ (so AC is defined). Then

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix}, \quad C \otimes D = \begin{pmatrix} c_{11}D & \cdots & c_{1p}D \\ \vdots & \ddots & \vdots \\ c_{n1}D & \cdots & c_{np}D \end{pmatrix}.$$

Multiply these block matrices. The (i, k) block of the product is

$$\sum_{j=1}^n (a_{ij}B)(c_{jk}D) = \sum_{j=1}^n a_{ij}c_{jk}(BD).$$

Factor out BD (it does not depend on j):

$$\sum_{j=1}^n a_{ij}c_{jk}(BD) = \left(\sum_{j=1}^n a_{ij}c_{jk} \right) BD.$$

But $\sum_{j=1}^n a_{ij}c_{jk} = (AC)_{ik}$, so the (i, k) block equals $(AC)_{ik} BD$. Hence the full block matrix equals

$$\begin{pmatrix} (AC)_{11}BD & \cdots & (AC)_{1p}BD \\ \vdots & \ddots & \vdots \\ (AC)_{m1}BD & \cdots & (AC)_{mp}BD \end{pmatrix} = (AC) \otimes (BD).$$

Therefore $(A \otimes B)(C \otimes D) = AC \otimes BD$. $\square$

5.

$$(A \otimes B)^{-1} = A^{-1} \otimes B^{-1},$$

when A and B are invertible.

Proof. Using Property 4 with $C = A^{-1}$ and $D = B^{-1}$,

$$(A^{-1} \otimes B^{-1})(A \otimes B) = (A^{-1}A) \otimes (B^{-1}B) = I \otimes I = I,$$

where I denotes the identity matrix of the appropriate dimension. Thus $A^{-1} \otimes B^{-1}$ is a (two-sided) inverse of $A \otimes B$, so

$$(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}.$$

$\square$

GLS, Kronecker Product, and Seemingly Unrelated Regressions

42 GLS reminder

Let the linear model be

$$y = X\beta + \varepsilon, \quad y \in \mathbb{R}^{n \times 1}, \quad X \in \mathbb{R}^{n \times k}, \quad \beta \in \mathbb{R}^{k \times 1}, \quad \varepsilon \in \mathbb{R}^{n \times 1}.$$

Assume the conditional second moment matrix is

$$\mathbb{E}[\varepsilon\varepsilon' \mid X] = \Sigma, \quad \Sigma \text{ nonsingular.}$$

Then the GLS estimator is

$$\hat{\beta}_{GLS} = (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}y.$$

43 Kronecker product $\otimes$

Let $A = (a_{ij})$ be an $m \times n$ matrix and let $B = (b_{ij})$ be an $m' \times n'$ matrix. The Kronecker product $A \otimes B$ is the $(mm') \times (nn')$ block matrix

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix}.$$

In particular, each block in row i and column j equals $a_{ij}B$.

44 System of M equations and stacking

Consider M linear regressions (equations):

$$Y_1 = X_1\beta_1 + \varepsilon_1, \quad Y_2 = X_2\beta_2 + \varepsilon_2, \quad \dots, \quad Y_M = X_M\beta_M + \varepsilon_M,$$

where for each equation $i = 1, \dots, M$,

$$Y_i \in \mathbb{R}^{T \times 1}, \quad X_i \in \mathbb{R}^{T \times K_i}, \quad \beta_i \in \mathbb{R}^{K_i \times 1}, \quad \varepsilon_i \in \mathbb{R}^{T \times 1}.$$

Each β_i can be estimated by OLS equation-by-equation by regressing Y_i on X_i . However, under cross-equation error correlation, we can estimate the system more efficiently.

Stacked representation

Define the stacked dependent variable:

$$Y \equiv \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_M \end{pmatrix} \in \mathbb{R}^{(MT) \times 1}.$$

Define the block-diagonal regressor matrix:

$$X \equiv \begin{pmatrix} X_1 & 0 & \cdots & 0 \\ 0 & X_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & X_M \end{pmatrix} \in \mathbb{R}^{(MT) \times (\sum_{i=1}^M K_i)}.$$

Define the stacked coefficient vector:

$$\beta \equiv \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_M \end{pmatrix} \in \mathbb{R}^{(\sum_{i=1}^M K_i) \times 1}.$$

Define the stacked error:

$$\varepsilon \equiv \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_M \end{pmatrix} \in \mathbb{R}^{(MT) \times 1}.$$

Then the whole system can be written compactly as

$$Y = X\beta + \varepsilon.$$

Error indexing within each equation

Write the $T \times 1$ error vector in equation i as

$$\varepsilon_i \equiv \begin{pmatrix} \varepsilon_{i1} \\ \varepsilon_{i2} \\ \vdots \\ \varepsilon_{iT} \end{pmatrix}.$$

Then the stacked error vector is

$$\varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_M \end{pmatrix} = \begin{pmatrix} \varepsilon_{11} \\ \varepsilon_{12} \\ \vdots \\ \varepsilon_{1T} \\ \varepsilon_{21} \\ \varepsilon_{22} \\ \vdots \\ \varepsilon_{2T} \\ \vdots \\ \varepsilon_{M1} \\ \varepsilon_{M2} \\ \vdots \\ \varepsilon_{MT} \end{pmatrix}.$$

45 Seemingly unrelated regressions: covariance structure

Define the conditional covariance matrix of the stacked error:

$$\Omega \equiv \mathbb{E}[\varepsilon\varepsilon' \mid X].$$

Assume the errors are serially uncorrelated across time:

$$\mathbb{E}[\varepsilon_{it}\varepsilon_{js} \mid X] = 0 \quad \text{if } t \neq s.$$

Assume contemporaneous covariance across equations is constant over t :

$$\mathbb{E}[\varepsilon_{it}\varepsilon_{jt} \mid X] = \sigma_{ij}, \quad i, j = 1, \dots, M, \quad t = 1, \dots, T.$$

Let the $M \times M$ contemporaneous covariance matrix be

$$\Sigma \equiv \begin{pmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1M} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{M1} & \sigma_{M2} & \cdots & \sigma_{MM} \end{pmatrix}.$$

Block form of Ω

Under the assumptions above, the (i, j) block of Ω (a $T \times T$ matrix) is

$$\Omega_{ij} = \mathbb{E}[\varepsilon_i \varepsilon_j' \mid X].$$

Its (t, s) entry equals $\mathbb{E}[\varepsilon_{it} \varepsilon_{js} \mid X]$, which is 0 when $t \neq s$ and equals σ_{ij} when $t = s$. Therefore,

$$\Omega_{ij} = \sigma_{ij} I_T,$$

and so the full $(MT) \times (MT)$ matrix Ω is the $M \times M$ block matrix

$$\Omega = \begin{pmatrix} \sigma_{11} I_T & \sigma_{12} I_T & \cdots & \sigma_{1M} I_T \\ \sigma_{21} I_T & \sigma_{22} I_T & \cdots & \sigma_{2M} I_T \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{M1} I_T & \sigma_{M2} I_T & \cdots & \sigma_{MM} I_T \end{pmatrix}.$$

Kronecker representation of Ω

By the definition of the Kronecker product,

$$\Sigma \otimes I_T = \begin{pmatrix} \sigma_{11} I_T & \sigma_{12} I_T & \cdots & \sigma_{1M} I_T \\ \sigma_{21} I_T & \sigma_{22} I_T & \cdots & \sigma_{2M} I_T \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{M1} I_T & \sigma_{M2} I_T & \cdots & \sigma_{MM} I_T \end{pmatrix}.$$

Hence,

$$\boxed{\Omega = \Sigma \otimes I_T.}$$

46 GLS for SUR and simplification using Kronecker properties

The GLS estimator for the stacked system is

$$\hat{\beta}_{GLS} = (X' \Omega^{-1} X)^{-1} X' \Omega^{-1} Y.$$

Substitute $\Omega = \Sigma \otimes I_T$:

$$\hat{\beta}_{GLS} = \left(X' (\Sigma \otimes I_T)^{-1} X \right)^{-1} X' (\Sigma \otimes I_T)^{-1} Y.$$

Use the Kronecker inverse property: if A and B are invertible, then

$$(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}.$$

Apply this with $A = \Sigma$ and $B = I_T$:

$$(\Sigma \otimes I_T)^{-1} = \Sigma^{-1} \otimes I_T^{-1}.$$

Since $I_T^{-1} = I_T$,

$$(\Sigma \otimes I_T)^{-1} = \Sigma^{-1} \otimes I_T.$$

Therefore,

$$\hat{\beta}_{GLS} = \left(X' (\Sigma^{-1} \otimes I_T) X \right)^{-1} X' (\Sigma^{-1} \otimes I_T) Y.$$

Panel Data: Random Effects vs Fixed Effects and the Within Transformation

47 Baseline model and error condition

For individuals $i = 1, \dots, n$ and periods $t = 1, \dots, T$, consider

$$y_{it} = x'_{it}\beta + z'_i\theta + c_i + v_{it}.$$

Here $z'_i\theta$ is observed through z_i , while c_i is unobserved (time-invariant).

Assume the idiosyncratic component is unrelated to the regressors:

$$v_{it} \text{ is unrelated to } x_{it} \text{ and } z_i.$$

48 Random effects

Random effects assumes the individual effect is unrelated to regressors:

$$c_i \text{ is unrelated to } x_{it} \text{ and } z_i.$$

(Then OLS is consistent; GLS is used for efficiency because of within- i correlation.)

Define the combined individual effect

$$\alpha_i \equiv z'_i\theta + c_i.$$

Assume the random-effects mean restriction (written in the notes' style):

$$\mathbb{E}[\alpha_i \mid x_{it}] = \alpha.$$

Define

$$u_i \equiv \alpha - \alpha_i \implies \mathbb{E}[u_i \mid x_{it}] = 0.$$

Rewrite the model using $\alpha_i = \alpha - u_i$:

$$\begin{aligned} y_{it} &= x'_{it}\beta + \alpha_i + v_{it} \\ &= x'_{it}\beta + (\alpha - u_i) + v_{it} \\ &= x'_{it}\beta + \alpha + (v_{it} - u_i). \end{aligned}$$

Equivalently, with the composite error written as $(u_i + v_{it})$ (up to sign convention),

$$y_{it} = x'_{it}\beta + \alpha + (u_i + v_{it}).$$

49 Fixed effects

Fixed effects allows correlation:

$$c_i \text{ is related to } x_{it} \text{ and } z_i.$$

Define again

$$\alpha_i \equiv z'_i\theta + c_i,$$

and write the fixed-effects regression form

$$y_{it} = x'_{it}\beta_t + \alpha_i + v_{it}.$$

49.1 Dummy-variable representation

For a given i , the contribution of α_i can be written as

$$0 \cdot \alpha_1 + 0 \cdot \alpha_2 + \cdots + 0 \cdot \alpha_{i-1} + 1 \cdot \alpha_i + 0 \cdot \alpha_{i+1} + \cdots + 0 \cdot \alpha_n.$$

Equivalently, using indicator functions,

$$y_{it} = x'_{it}\beta_t + \mathbf{1}(i=1)\alpha_1 + \mathbf{1}(i=2)\alpha_2 + \cdots + \mathbf{1}(i=n)\alpha_n + v_{it}.$$

Thus one can use OLS by regressing y_{it} on x_{it} and the full set of individual dummies $\mathbf{1}(i=1), \dots, \mathbf{1}(i=n)$.

49.2 Stacking over time and Kronecker form

For each t , define the $n \times 1$ outcome vector

$$Y_t \equiv \begin{pmatrix} y_{1t} \\ y_{2t} \\ \vdots \\ y_{nt} \end{pmatrix},$$

and the $n \times k$ regressor matrix

$$X_t \equiv \begin{pmatrix} x'_{1t} \\ x'_{2t} \\ \vdots \\ x'_{nt} \end{pmatrix}.$$

Let the time- t slope be

$$\beta_t \equiv \begin{pmatrix} \beta_{t1} \\ \beta_{t2} \\ \vdots \\ \beta_{tk} \end{pmatrix}.$$

Then the model at time t is

$$Y_t = X_t\beta_t + \alpha + v_t, \quad \alpha \equiv \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix}, \quad v_t \equiv \begin{pmatrix} v_{1t} \\ \vdots \\ v_{nt} \end{pmatrix}.$$

Stack over t :

$$Y \equiv \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_T \end{pmatrix}, \quad \beta \equiv \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_T \end{pmatrix}, \quad v \equiv \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_T \end{pmatrix}.$$

Define the block-diagonal regressor matrix

$$X \equiv \begin{pmatrix} X_1 & 0 & \cdots & 0 \\ 0 & X_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & X_T \end{pmatrix}.$$

Define the dummy matrix D (stacked identity blocks):

$$D \equiv \begin{pmatrix} I_{n \times n} \\ I_{n \times n} \\ \vdots \\ I_{n \times n} \end{pmatrix} \quad (T \text{ blocks, each } n \times n).$$

Let ι_T be the $T \times 1$ vector of ones. Then

$$D = \iota_T \otimes I_n.$$

The stacked model is

$$Y = X\beta + D\alpha + v.$$

49.3 FWL / within transformation via the annihilator of D

Let

$$M \equiv I - D(D'D)^{-1}D'.$$

Let $(\hat{\beta}, \hat{\alpha})$ be the OLS estimates from regressing Y on (X, D) , and define the OLS residual

$$\hat{v} \equiv Y - X\hat{\beta} - D\hat{\alpha}.$$

Then

$$Y = X\hat{\beta} + D\hat{\alpha} + \hat{v}.$$

Multiply both sides by M :

$$MY = MX\hat{\beta} + MD\hat{\alpha} + M\hat{v}.$$

Since $MD = 0$ by construction, this becomes

$$MY = MX\hat{\beta} + M\hat{v}.$$

Using the normal equations from the OLS regression on (X, D) :

$$D'\hat{v} = 0,$$

we obtain

$$X'M\hat{v} = X'(I - D(D'D)^{-1}D')\hat{v} = X'\hat{v} - X'D(D'D)^{-1}D'\hat{v} = 0 - X'D(D'D)^{-1} \cdot 0 = 0.$$

Therefore,

$$X'MY = X'MX\hat{\beta},$$

so

$$(X'MX)\hat{\beta} = X'MY \quad \implies \quad \boxed{\hat{\beta} = (X'MX)^{-1}X'MY}.$$

49.4 Compute MY explicitly

Start from

$$MY = Y - D(D'D)^{-1}D'Y.$$

Kronecker identities used.

$$(A \otimes B)' = A' \otimes B', \quad (A \otimes B)(C \otimes D) = (AC) \otimes (BD), \quad (A \otimes B)^{-1} = A^{-1} \otimes B^{-1}.$$

Compute $D'D$. Since $D = \iota_T \otimes I_n$,

$$\begin{aligned} D'D &= (\iota_T \otimes I_n)'(\iota_T \otimes I_n) \\ &= (\iota'_T \otimes I'_n)(\iota_T \otimes I_n) \\ &= (\iota'_T \iota_T) \otimes (I'_n I_n) \\ &= (\iota'_T \iota_T) \otimes I_n. \end{aligned}$$

Because $\iota'_T \iota_T = T$ (a scalar),

$$D'D = T \otimes I_n = TI_n, \quad (D'D)^{-1} = (TI_n)^{-1} = \frac{1}{T}I_n.$$

Compute $D'Y$.

$$D'Y = (\iota_T \otimes I_n)'Y = (\iota'_T \otimes I_n)Y.$$

If the stacked vector is ordered as

$$Y = \begin{pmatrix} y_{11} \\ y_{21} \\ \vdots \\ y_{n1} \\ y_{12} \\ y_{22} \\ \vdots \\ y_{n2} \\ \vdots \\ y_{1T} \\ y_{2T} \\ \vdots \\ y_{nT} \end{pmatrix},$$

then $(\iota'_T \otimes I_n)Y$ sums across t for each i :

$$D'Y = \begin{pmatrix} \sum_{t=1}^T y_{1t} \\ \sum_{t=1}^T y_{2t} \\ \vdots \\ \sum_{t=1}^T y_{nt} \end{pmatrix}.$$

Compute the projection term.

$$D(D'D)^{-1}D'Y = D \left(\frac{1}{T}I_n \right) D'Y = \frac{1}{T}DD'Y.$$

Let

$$\bar{y}_i \equiv \frac{1}{T} \sum_{t=1}^T y_{it}.$$

Then $\frac{1}{T}DD'Y$ is the stacked vector that repeats $(\bar{y}_1, \dots, \bar{y}_n)'$ exactly T times:

$$\frac{1}{T}DD'Y = \begin{pmatrix} \bar{y}_1 \\ \bar{y}_2 \\ \vdots \\ \bar{y}_n \\ \bar{y}_1 \\ \bar{y}_2 \\ \vdots \\ \bar{y}_n \\ \vdots \\ \bar{y}_1 \\ \bar{y}_2 \\ \vdots \\ \bar{y}_n \end{pmatrix}.$$

Therefore MY is within-demeaned.

$$MY = Y - D(D'D)^{-1}D'Y = \begin{pmatrix} y_{11} - \bar{y}_1 \\ y_{21} - \bar{y}_2 \\ \vdots \\ y_{n1} - \bar{y}_n \\ y_{12} - \bar{y}_1 \\ y_{22} - \bar{y}_2 \\ \vdots \\ y_{n2} - \bar{y}_n \\ \vdots \\ y_{1T} - \bar{y}_1 \\ y_{2T} - \bar{y}_2 \\ \vdots \\ y_{nT} - \bar{y}_n \end{pmatrix}.$$

49.5 Within equation (demeaning in scalar form)

Start from

$$y_{it} = x'_{it}\beta_t + \alpha_i + v_{it}.$$

Define time averages

$$\bar{y}_i \equiv \frac{1}{T} \sum_{t=1}^T y_{it}, \quad \bar{x}_i \equiv \frac{1}{T} \sum_{t=1}^T x_{it}, \quad \bar{v}_i \equiv \frac{1}{T} \sum_{t=1}^T v_{it}.$$

Take the time average of the model:

$$\begin{aligned}\bar{y}_i &= \frac{1}{T} \sum_{t=1}^T (x'_{it}\beta_t + \alpha_i + v_{it}) \\ &= \frac{1}{T} \sum_{t=1}^T x'_{it}\beta_t + \alpha_i + \frac{1}{T} \sum_{t=1}^T v_{it} \\ &= \frac{1}{T} \sum_{t=1}^T x'_{it}\beta_t + \alpha_i + \bar{v}_i.\end{aligned}$$

Subtract this from the original equation:

$$\begin{aligned}y_{it} - \bar{y}_i &= (x'_{it}\beta_t + \alpha_i + v_{it}) - \left(\frac{1}{T} \sum_{s=1}^T x'_{is}\beta_s + \alpha_i + \bar{v}_i \right) \\ &= x'_{it}\beta_t - \frac{1}{T} \sum_{s=1}^T x'_{is}\beta_s + (v_{it} - \bar{v}_i).\end{aligned}$$

In the common case where β_t is constant across t (write $\beta_t = \beta$), the regressor part simplifies to

$$x'_{it}\beta - \bar{x}'_i\beta = (x_{it} - \bar{x}_i)'\beta,$$

so

$$y_{it} - \bar{y}_i = (x_{it} - \bar{x}_i)'\beta + (v_{it} - \bar{v}_i).$$

This yields the fixed-effects (within) regression: regress $y_{it} - \bar{y}_i$ on $x_{it} - \bar{x}_i$.

Program Evaluation: Potential Outcomes and Difference-in-Differences

50 Potential outcomes and the missing counterfactual

Potential outcomes

For each individual i :

Y_{1i} : outcome of i with the program, Y_{0i} : outcome of i without the program.

The individual treatment effect is

$$Y_{1i} - Y_{0i},$$

but $Y_{1i} - Y_{0i}$ is not observed for any i .

Treatment indicator and the observed outcome

Define the participation indicator

$$D_i = \begin{cases} 0 & i \text{ does not join the program,} \\ 1 & i \text{ joins the program.} \end{cases}$$

The observed outcome satisfies

$$Y_i = D_i Y_{1i} + (1 - D_i) Y_{0i}.$$

Hence

$$Y_i = \begin{cases} Y_{1i} & \text{if } D_i = 1, \\ Y_{0i} & \text{if } D_i = 0. \end{cases}$$

Parameter of interest

Want to know

$$\mathbb{E}(Y_{1i} - Y_{0i}).$$

By linearity of expectation,

$$\mathbb{E}(Y_{1i} - Y_{0i}) = \mathbb{E}(Y_{1i}) - \mathbb{E}(Y_{0i}).$$

What is observed in the data

From $Y_i = Y_{1i}$ when $D_i = 1$ and $Y_i = Y_{0i}$ when $D_i = 0$, we can observe (or estimate)

$$\mathbb{E}(Y_i \mid D_i = 1) = \mathbb{E}(Y_{1i} \mid D_i = 1), \quad \mathbb{E}(Y_i \mid D_i = 0) = \mathbb{E}(Y_{0i} \mid D_i = 0), \quad \mathbb{P}(D_i = 1).$$

51 Randomized experiment

Random assignment

Experiment: D_i is randomly assigned. Equivalently,

$$(Y_{1i}, Y_{0i}) \perp D_i.$$

Identification under random assignment

Independence implies

$$\mathbb{E}(Y_{1i} \mid D_i = 1) = \mathbb{E}(Y_{1i}), \quad \mathbb{E}(Y_{0i} \mid D_i = 0) = \mathbb{E}(Y_{0i}).$$

Also,

$$\mathbb{E}(Y_i \mid D_i = 1) = \mathbb{E}(Y_{1i} \mid D_i = 1) = \mathbb{E}(Y_{1i}), \quad \mathbb{E}(Y_i \mid D_i = 0) = \mathbb{E}(Y_{0i} \mid D_i = 0) = \mathbb{E}(Y_{0i}).$$

Therefore the average treatment effect (ATE) is identified by the difference in mean outcomes:

$$\mathbb{E}(Y_{1i} - Y_{0i}) = \mathbb{E}(Y_{1i}) - \mathbb{E}(Y_{0i}) = \mathbb{E}(Y_i \mid D_i = 1) - \mathbb{E}(Y_i \mid D_i = 0).$$

52 Two-period difference-in-differences (DiD)

Potential outcomes with time

Let $t = 0$ denote “before the program” and $t = 1$ denote “after the program.” Define:

Y_{00} : income before the program without participation, Y_{01} : income after the program without participation

Y_{10} : income before the program with participation, Y_{11} : income after the program with participation.

Before the program starts, participation has no effect on the pre-period outcome, so

$$Y_{00} = Y_{10}.$$

Let $D \in \{0, 1\}$ denote group membership (treated group $D = 1$, control group $D = 0$), and let X denote covariates.

Parallel trends (conditional) assumption

$$\mathbb{E}(Y_{01} - Y_{00} \mid X, D = 1) = \mathbb{E}(Y_{01} - Y_{00} \mid X, D = 0).$$

Target: the treated group’s post-period treatment effect

The (conditional) effect on the treated in period 1 is

$$\mathbb{E}(Y_{11} - Y_{01} \mid X, D = 1).$$

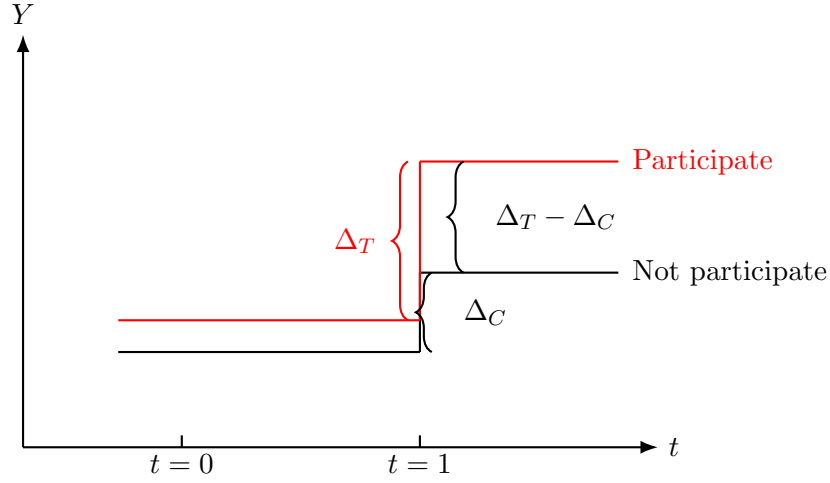
Algebraic decomposition and DiD

$$\begin{aligned} \mathbb{E}(Y_{11} - Y_{01} \mid X, D = 1) &= \mathbb{E}(Y_{11} - Y_{00} + Y_{00} - Y_{01} \mid X, D = 1) \\ &= \mathbb{E}(Y_{11} - Y_{00} \mid X, D = 1) - \mathbb{E}(Y_{01} - Y_{00} \mid X, D = 1) \\ &= \mathbb{E}(Y_{11} - Y_{00} \mid X, D = 1) - \mathbb{E}(Y_{01} - Y_{00} \mid X, D = 0) \\ &= \mathbb{E}(Y_{11} - Y_{10} \mid X, D = 1) - \mathbb{E}(Y_{01} - Y_{00} \mid X, D = 0). \end{aligned}$$

This last line is the DiD expression:

$$\text{DiD}(X) = \mathbb{E}(Y_{11} - Y_{10} \mid X, D = 1) - \mathbb{E}(Y_{01} - Y_{00} \mid X, D = 0).$$

53 Two-group, two-period picture (illustration)



Panel Data: Individual Effects, Stacking, and Dynamic Panels with IV

54 Baseline panel model with observed and unobserved components

For individuals $i = 1, \dots, n$ and periods $t = 1, \dots, T$:

$$y_{it} = x'_{it}\beta_t + z'_i\theta + c_i + v_{it},$$

where $z'_i\theta$ is *observed* (through z_i) and c_i is *unobserved* (time-invariant).

A baseline exogeneity statement in the notes is:

$$v_{it} \text{ is unrelated with } x_{it} \text{ and } z_i.$$

One convenient formalization is

$$\mathbb{E}[v_{it} \mid x_{it}, z_i, c_i] = 0, \quad \text{which implies} \quad \mathbb{E}[x_{it}v_{it}] = 0, \quad \mathbb{E}[z_iv_{it}] = 0.$$

55 Stacking over individuals and over time

Time- t stacking

Define the time- t outcome vector:

$$Y_t \equiv \begin{pmatrix} y_{1t} \\ y_{2t} \\ \vdots \\ y_{nt} \end{pmatrix}.$$

Define the time- t regressor matrix (row-stacking the x'_{it}):

$$X_t \equiv \begin{pmatrix} x'_{1t} \\ x'_{2t} \\ \vdots \\ x'_{nt} \end{pmatrix}.$$

Stacking over t

Stack the T time blocks:

$$Y \equiv \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_T \end{pmatrix} \in \mathbb{R}^{(nT) \times 1}.$$

If slopes are allowed to vary over t , define

$$\beta_t \equiv \begin{pmatrix} \beta_{t1} \\ \beta_{t2} \\ \vdots \\ \beta_{tk} \end{pmatrix}, \quad \beta \equiv \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_T \end{pmatrix} \in \mathbb{R}^{(kT) \times 1}.$$

Define the block-diagonal regressor matrix:

$$X \equiv \begin{pmatrix} X_1 & 0 & \cdots & 0 \\ 0 & X_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & X_T \end{pmatrix}.$$

Individual effects and the stacked representation

Let the (time-invariant) individual component be

$$\alpha_i \equiv z'_i \theta + c_i, \quad \alpha \equiv \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix}.$$

Let the stacked idiosyncratic error be

$$v \equiv \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_T \end{pmatrix}, \quad v_t \equiv \begin{pmatrix} v_{1t} \\ v_{2t} \\ \vdots \\ v_{nt} \end{pmatrix}.$$

Define the matrix that repeats α in each time block:

$$D \equiv \begin{pmatrix} I_n \\ I_n \\ \vdots \\ I_n \end{pmatrix} \in \mathbb{R}^{(nT) \times n}.$$

Then the stacked model can be written as

$$Y = X\beta + D\alpha + v.$$

Special case: common slope across time

If $\beta_t = \beta$ for all t , define the vertically stacked regressor matrix

$$X \equiv \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_T \end{pmatrix} \in \mathbb{R}^{(nT) \times k}, \quad \beta \equiv \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{pmatrix} \in \mathbb{R}^{k \times 1}.$$

Then the same stacked representation is

$$Y = X\beta + D\alpha + v.$$

A strict-exogeneity style condition in the notes is

$$\mathbb{E}[v_{it} \mid x_{i1}, x_{i2}, \dots, x_{iT}, \alpha_i] = 0.$$

56 Dynamic panel model and why OLS in differences is endogenous

Dynamic model with a lagged dependent variable

Consider

$$y_{it} = \alpha y_{i,t-1} + z'_{it}\beta + c_i + v_{it}.$$

Let

$$x_{it} \equiv \begin{pmatrix} y_{i,t-1} \\ z'_{it} \end{pmatrix}.$$

A strict-exogeneity condition of the form

$$\mathbb{E}[v_{it} \mid x_{i1}, x_{i2}, \dots, x_{iT}, c_i] = 0$$

fails in general, because the collection $(x_{i1}, \dots, x_{iT})$ contains variables such as y_{it} , $y_{i,t-1}$, and z'_{it} through the lag structure (e.g. y_{it} appears inside $x_{i,t+1}$).

Instead, the notes impose a weaker sequential (predeterminedness) condition:

$$\text{Assume} \quad \mathbb{E}[v_{it} \mid x_{i1}, x_{i2}, \dots, x_{it}, c_i] = 0,$$

where y_{it} is not contained in $(x_{i1}, \dots, x_{it})$.

First-differencing to eliminate c_i

Write the model at t :

$$y_{it} = \alpha y_{i,t-1} + z'_{it}\beta + c_i + v_{it},$$

and at $t - 1$:

$$y_{i,t-1} = \alpha y_{i,t-2} + z'_{i,t-1}\beta + c_i + v_{i,t-1}.$$

Subtract the $t - 1$ equation from the t equation:

$$y_{it} - y_{i,t-1} = \alpha (y_{i,t-1} - y_{i,t-2}) + (z'_{it} - z'_{i,t-1})\beta + (v_{it} - v_{i,t-1}).$$

Why OLS is not consistent after differencing

In the differenced equation, the regressor $(y_{i,t-1} - y_{i,t-2})$ is correlated with the error $(v_{it} - v_{i,t-1})$:

OLS is not consistent because $(v_{it} - v_{i,t-1})$ is correlated with $(y_{i,t-1} - y_{i,t-2})$.

A direct covariance decomposition makes the source of endogeneity explicit:

$$\text{Cov}(y_{i,t-1} - y_{i,t-2}, v_{it} - v_{i,t-1}) = \text{Cov}(y_{i,t-1} - y_{i,t-2}, v_{it}) - \text{Cov}(y_{i,t-1} - y_{i,t-2}, v_{i,t-1}).$$

Under standard no-serial-correlation conditions, the first term is 0, but the second term is typically nonzero because $y_{i,t-1}$ contains $v_{i,t-1}$ through the level equation.

Instrument choice

Use $y_{i,t-2}$ as an IV because

$y_{i,t-2}$ is uncorrelated with $(v_{it} - v_{i,t-1})$, and is correlated with $(y_{i,t-1} - y_{i,t-2})$.

A typical moment restriction corresponding to exogeneity is

$$\mathbb{E}[y_{i,t-2}(v_{it} - v_{i,t-1})] = 0,$$

and relevance follows since $y_{i,t-2}$ is mechanically related to $(y_{i,t-1} - y_{i,t-2})$ (it appears in the regressor).

OLS and IV as Method-of-Moments; Identification by Instrument Count

57 OLS as method-of-moments

Model and exogeneity

Start from the linear model

$$y_i = x'_i\beta + \varepsilon_i.$$

Assume exogeneity in conditional-mean form:

$$\mathbb{E}[\varepsilon \mid X] = 0.$$

From conditional mean to orthogonality

Let X be the regressor matrix stacking x'_i and let ε stack ε_i . Then

$$\mathbb{E}[X'\varepsilon] = \mathbb{E}[\mathbb{E}[X'\varepsilon \mid X]].$$

Inside the conditional expectation, X is fixed, so

$$\mathbb{E}[X'\varepsilon \mid X] = X'\mathbb{E}[\varepsilon \mid X].$$

Therefore,

$$\mathbb{E}[X'\varepsilon] = \mathbb{E}[X'\mathbb{E}[\varepsilon \mid X]] = \mathbb{E}\left[X' \cdot \underbrace{\mathbb{E}[\varepsilon \mid X]}_{=0}\right] = 0.$$

Equivalently, at the observation level,

$$\mathbb{E}[x_i\varepsilon_i] = 0.$$

Population moment condition and the implied equation for β

Using $\varepsilon_i = y_i - x'_i\beta$, the orthogonality condition becomes

$$\mathbb{E}[x_i(y_i - x'_i\beta)] = 0.$$

Expand:

$$\mathbb{E}[x_i y_i] - \mathbb{E}[x_i x'_i] \beta = 0,$$

so

$$\mathbb{E}[x_i y_i] = \mathbb{E}[x_i x'_i] \beta.$$

Sample analogue and MOM/OLS estimator

Set the sample analogue to zero:

$$\frac{1}{n} \sum_{i=1}^n x_i (y_i - x'_i \hat{\beta}_{\text{MOM}}) = 0.$$

Expand and collect $\hat{\beta}_{\text{MOM}}$:

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n x_i y_i - \left(\frac{1}{n} \sum_{i=1}^n x_i x'_i \right) \hat{\beta}_{\text{MOM}} &= 0, \\ \left(\frac{1}{n} \sum_{i=1}^n x_i x'_i \right) \hat{\beta}_{\text{MOM}} &= \left(\frac{1}{n} \sum_{i=1}^n x_i y_i \right). \end{aligned}$$

Assuming the matrix $(\frac{1}{n} \sum_{i=1}^n x_i x'_i)$ is invertible, left-multiply by its inverse:

$$\hat{\beta}_{\text{MOM}} = \left(\frac{1}{n} \sum_{i=1}^n x_i x'_i \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n x_i y_i \right).$$

Multiply numerator and denominator by n inside each term:

$$\hat{\beta}_{\text{MOM}} = \left(\sum_{i=1}^n x_i x_i' \right)^{-1} \left(\sum_{i=1}^n x_i y_i \right).$$

In matrix notation, $\sum_{i=1}^n x_i x_i' = X'X$ and $\sum_{i=1}^n x_i y_i = X'Y$, hence

$$\hat{\beta}_{\text{MOM}} = (X'X)^{-1} X'Y.$$

Therefore,

$$\hat{\beta}_{\text{OLS}} = (X'X)^{-1} X'Y.$$

58 IV moment conditions and identification

Instruments and core conditions

Let z_i be an $L \times 1$ instrument vector:

$$z_i \in \mathbb{R}^{L \times 1}.$$

Assume instrument exogeneity:

$$\mathbb{E}[z_i \varepsilon_i] = 0.$$

Assume an identification (relevance/rank) condition:

$$\text{rank}(\mathbb{E}[z_i x_i']) = k, \quad \mathbb{E}[z_i x_i'] \in \mathbb{R}^{L \times k}, \quad \beta \in \mathbb{R}^{k \times 1}.$$

Population IV moment equation

Using $\varepsilon_i = y_i - x_i' \beta$,

$$\mathbb{E}[z_i (y_i - x_i' \beta)] = 0.$$

Expand:

$$\mathbb{E}[z_i y_i] - \mathbb{E}[z_i x_i'] \beta = 0,$$

so

$$\mathbb{E}[z_i y_i] = \mathbb{E}[z_i x_i'] \beta.$$

Dimension check:

$$\mathbb{E}[z_i y_i] \in \mathbb{R}^{L \times 1}, \quad \mathbb{E}[z_i x_i'] \in \mathbb{R}^{L \times k}, \quad \beta \in \mathbb{R}^{k \times 1},$$

so the right-hand side is $(L \times k)(k \times 1) = (L \times 1)$, matching the left-hand side.

Given $\mathbb{E}[z_i y_i]$ and $\mathbb{E}[z_i x_i']$, there are L equations and k unknowns.

Identification cases by counting L vs k

- If $k > L$, there are fewer moment equations than unknowns, so β is not identified from these moments.
- If $k = L$, the system is exactly identified (just-identified).
- If $k < L$, the system is overidentified (more equations than unknowns).

59 Just-identified case: $k = L$

If $k = L$ and $\mathbb{E}[z_i x_i']$ is invertible, then from

$$\mathbb{E}[z_i y_i] = \mathbb{E}[z_i x_i'] \beta$$

we obtain

$$\beta = (\mathbb{E}[z_i x_i'])^{-1} \mathbb{E}[z_i y_i].$$

Sample analogue (MOM estimator)

Set the sample analogue to zero:

$$\frac{1}{n} \sum_{i=1}^n z_i (y_i - x_i' \hat{\beta}_{\text{MOM}}) = 0.$$

Expand and collect:

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n z_i y_i - \left(\frac{1}{n} \sum_{i=1}^n z_i x_i' \right) \hat{\beta}_{\text{MOM}} &= 0, \\ \left(\frac{1}{n} \sum_{i=1}^n z_i x_i' \right) \hat{\beta}_{\text{MOM}} &= \left(\frac{1}{n} \sum_{i=1}^n z_i y_i \right). \end{aligned}$$

Assuming $\left(\frac{1}{n} \sum_{i=1}^n z_i x_i' \right)$ is invertible (square $k \times k$), left-multiply by its inverse:

$$\hat{\beta}_{\text{MOM}} = \left(\frac{1}{n} \sum_{i=1}^n z_i x_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n z_i y_i \right).$$

In matrix notation, $Z'X = \sum_{i=1}^n z_i x_i'$ and $Z'Y = \sum_{i=1}^n z_i y_i$, so

$$\hat{\beta}_{\text{MOM}} = (Z'X)^{-1} Z'Y.$$

60 Overidentified case: $k < L$

When $k < L$, the sample moment equations

$$\frac{1}{n} Z'(y - X\beta) = 0$$

typically cannot be satisfied exactly in finite samples. A common approach is to choose $\hat{\beta}$ to make the sample moments as close to zero as possible.

Define the sample moment vector

$$g_n(\beta) \equiv \frac{1}{n} Z'(y - X\beta) \in \mathbb{R}^{L \times 1}.$$

Choose a positive definite weighting matrix $W_n \in \mathbb{R}^{L \times L}$ and minimize the quadratic form

$$Q_n(\beta) \equiv g_n(\beta)' W_n g_n(\beta).$$

With the common choice $W_n = (Z'Z/n)^{-1}$, the first-order condition yields the 2SLS estimator. Compute $g_n(\beta)$ explicitly:

$$g_n(\beta) = \frac{1}{n}Z'y - \frac{1}{n}Z'X\beta.$$

Then

$$Q_n(\beta) = \left(\frac{1}{n}Z'y - \frac{1}{n}Z'X\beta \right)' W_n \left(\frac{1}{n}Z'y - \frac{1}{n}Z'X\beta \right).$$

Differentiate with respect to β and set to zero (using that $Q_n(\beta)$ is a quadratic form):

$$\frac{\partial Q_n(\beta)}{\partial \beta} = -2 \left(\frac{1}{n}X'Z \right) W_n \left(\frac{1}{n}Z'y - \frac{1}{n}Z'X\beta \right) = 0.$$

Equivalently,

$$\left(\frac{1}{n}X'Z \right) W_n \left(\frac{1}{n}Z'X \right) \beta = \left(\frac{1}{n}X'Z \right) W_n \left(\frac{1}{n}Z'y \right).$$

With $W_n = (Z'Z/n)^{-1}$,

$$\left(\frac{1}{n}X'Z \right) \left(\frac{Z'Z}{n} \right)^{-1} \left(\frac{1}{n}Z'X \right) \beta = \left(\frac{1}{n}X'Z \right) \left(\frac{Z'Z}{n} \right)^{-1} \left(\frac{1}{n}Z'y \right).$$

Multiplying by n consistently gives the usual 2SLS form:

$$(X'Z)(Z'Z)^{-1}(Z'X)\hat{\beta} = (X'Z)(Z'Z)^{-1}(Z'y),$$

so

$$\hat{\beta} = (X'Z(Z'Z)^{-1}Z'X)^{-1}X'Z(Z'Z)^{-1}Z'y.$$

Overidentified IV Moments and GMM: From Moment Conditions to 2SLS

60.1 Setup and population moment conditions

Let the structural model be

$$y_i = x_i'\beta + \varepsilon_i,$$

where $x_i \in \mathbb{R}^{k \times 1}$ and $\beta \in \mathbb{R}^{k \times 1}$.

Let the instrument vector be

$$z_i \in \mathbb{R}^{L \times 1}.$$

Assume instrument exogeneity:

$$\mathbb{E}[z_i \varepsilon_i] = 0.$$

Assume the rank (relevance/identification) condition:

$$\text{rank}(\mathbb{E}[z_i z_i']) = k.$$

Using $\varepsilon_i = y_i - x_i'\beta$, instrument exogeneity implies the population moment condition

$$\mathbb{E}[z_i(y_i - x_i'\beta)] = 0.$$

Equivalently,

$$\mathbb{E}[z_i y_i] = \mathbb{E}[z_i x_i'] \beta.$$

Dimension check

$$\mathbb{E}[z_i y_i] \in \mathbb{R}^{L \times 1}, \quad \mathbb{E}[z_i x'_i] \in \mathbb{R}^{L \times k}, \quad \beta \in \mathbb{R}^{k \times 1},$$

so $(L \times k)(k \times 1) = (L \times 1)$ matches $\mathbb{E}[z_i y_i]$.

Given $\mathbb{E}[z_i y_i]$ and $\mathbb{E}[z_i x'_i]$, there are L equations and k unknowns.

Identification cases by counting

- If $k > L$, then β is not identified.
- If $k = L$, then

$$\beta = (\mathbb{E}[z_i x'_i])^{-1} \mathbb{E}[z_i y_i].$$

A sample analogue (method-of-moments) estimator is

$$\hat{\beta}_{\text{MOM}} = \left(\frac{1}{n} \sum_{i=1}^n z_i x'_i \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n z_i y_i \right) = (Z'X)^{-1} Z'Y.$$

- If $k < L$, then β is overidentified.

61 Sample moments and why exact MOM equations may fail when $k < L$

Assume (y_i, x_i, z_i) for $i = 1, 2, \dots, n$ are i.i.d.

Define the sample analogue of the population moment $\mathbb{E}[z_i(y_i - x'_i \beta)]$ by

$$\bar{m}_n(b) \equiv \frac{1}{n} \sum_{i=1}^n z_i (y_i - x'_i b), \quad b \in \mathbb{R}^{k \times 1}.$$

When $k < L$, the equation $\bar{m}_n(b) = 0$ typically has no solution (there are L sample moment equations but only k unknown parameters). This motivates choosing b to make $\bar{m}_n(b)$ “as close to zero as possible”.

62 GMM criterion and matrix representation

Let $\hat{W}$ be an $L \times L$ (symmetric positive definite) weighting matrix. Define

$$\hat{\beta}_{\text{GMM}} = \arg \min_b Q_n(b), \quad Q_n(b) = \bar{m}_n(b)' \hat{W} \bar{m}_n(b).$$

Stacked notation

Define

$$Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \quad X = \begin{pmatrix} x'_1 \\ \vdots \\ x'_n \end{pmatrix}, \quad Z = \begin{pmatrix} z'_1 \\ \vdots \\ z'_n \end{pmatrix}.$$

Then

$$\bar{m}_n(b) = \frac{1}{n} \sum_{i=1}^n z_i y_i - \frac{1}{n} \sum_{i=1}^n z_i x'_i b = \frac{1}{n} Z' Y - \frac{1}{n} Z' X b = \frac{1}{n} Z' (Y - X b).$$

Hence

$$Q_n(b) = \left(\frac{1}{n} Z' (Y - X b) \right)' \hat{W} \left(\frac{1}{n} Z' (Y - X b) \right).$$

Use $(AB)' = B' A'$:

$$\left(\frac{1}{n} Z' (Y - X b) \right)' = \frac{1}{n} (Y - X b)' Z.$$

Therefore

$$Q_n(b) = \frac{1}{n} (Y - X b)' Z \hat{W} \frac{1}{n} Z' (Y - X b) = \frac{1}{n^2} (Y - X b)' Z \hat{W} Z' (Y - X b).$$

Since $1/n^2$ is a constant factor, minimizing $Q_n(b)$ is equivalent to minimizing

$$\tilde{Q}_n(b) \equiv (Y - X b)' Z \hat{W} Z' (Y - X b).$$

Write

$$(Y - X b)' = Y' - b' X',$$

so

$$\tilde{Q}_n(b) = (Y' - b' X') Z \hat{W} Z' (Y - X b).$$

63 Expanding the quadratic form

Expand $\tilde{Q}_n(b)$ term-by-term:

$$\begin{aligned} \tilde{Q}_n(b) &= (Y' - b' X') Z \hat{W} Z' (Y - X b) \\ &= Y' Z \hat{W} Z' Y - Y' Z \hat{W} Z' X b - b' X' Z \hat{W} Z' Y + b' X' Z \hat{W} Z' X b. \end{aligned}$$

Note that $Y' Z \hat{W} Z' X b$ is a scalar and equals its transpose:

$$(Y' Z \hat{W} Z' X b)' = b' X' Z \hat{W} Z' Y = b' X' Z \hat{W} Z' Y \quad (\text{if } \hat{W}' = \hat{W}).$$

Hence the two middle terms are equal scalars, and we can combine them:

$$\tilde{Q}_n(b) = Y' Z \hat{W} Z' Y - 2 Y' Z \hat{W} Z' X b + b' X' Z \hat{W} Z' X b.$$

64 Matrix derivatives used

For a conformable vector z :

$$\frac{\partial(a'z)}{\partial z} = a, \quad \frac{\partial(z'Az)}{\partial z} = (A + A')z.$$

In particular, if $A' = A$, then $\frac{\partial(z'Az)}{\partial z} = 2Az$.

65 First-order condition and closed-form solution

Differentiate $\tilde{Q}_n(b)$ with respect to b .

Derivative of the linear term

The term $-2Y'Z\hat{W}Z'Xb$ is linear in b . Write it as

$$-2Y'Z\hat{W}Z'Xb = -2c'b, \quad c \equiv X'Z\hat{W}Z'Y.$$

Hence

$$\frac{\partial}{\partial b}(-2Y'Z\hat{W}Z'Xb) = -2X'Z\hat{W}Z'Y.$$

Derivative of the quadratic term

The term $b'X'Z\hat{W}Z'Xb$ is quadratic in b with

$$A \equiv X'Z\hat{W}Z'X.$$

If $\hat{W}$ is symmetric, then A is symmetric, so

$$\frac{\partial}{\partial b}(b'Ab) = 2Ab = 2(X'Z\hat{W}Z'X)b.$$

FOC

Set the gradient to zero:

$$-2X'Z\hat{W}Z'Y + 2(X'Z\hat{W}Z'X)b = 0.$$

Divide by 2:

$$(X'Z\hat{W}Z'X)b = X'Z\hat{W}Z'Y.$$

Assuming $X'Z\hat{W}Z'X$ is invertible,

$$\boxed{\hat{\beta}_{\text{GMM}} = (X'Z\hat{W}Z'X)^{-1} X'Z\hat{W}Z'Y.}$$

66 2SLS as a special case of GMM

If we choose

$$\hat{W} = (Z'Z)^{-1},$$

then

$$\hat{\beta}_{\text{GMM}} = (X'Z(Z'Z)^{-1}Z'X)^{-1}X'Z(Z'Z)^{-1}Z'Y = \hat{\beta}_{\text{2SLS}}.$$

GMM with Overidentifying Instruments

67 Model, instruments, and notation

For $i = 1, \dots, n$,

$$y_i = x_i'\beta + \varepsilon_i, \quad x_i \in \mathbb{R}^{k \times 1}, \quad z_i \in \mathbb{R}^{L \times 1}.$$

Instrument orthogonality (exogeneity) is

$$\mathbb{E}[z_i \varepsilon_i] = 0, \quad \text{equivalently} \quad \mathbb{E}[z_i(y_i - x_i'\beta)] = 0.$$

Let

$$Y \equiv \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad X \equiv \begin{pmatrix} x_1' \\ x_2' \\ \vdots \\ x_n' \end{pmatrix}, \quad Z \equiv \begin{pmatrix} z_1' \\ z_2' \\ \vdots \\ z_n' \end{pmatrix}.$$

Then $Y \in \mathbb{R}^{n \times 1}$, $X \in \mathbb{R}^{n \times k}$, $Z \in \mathbb{R}^{n \times L}$. The overidentified case is

$$L > k, \quad \text{rank}(\mathbb{E}[z_i z_i']) = k.$$

68 Sample moments and GMM objective

Define the sample moment vector

$$\bar{m}_n(b) \equiv \frac{1}{n} \sum_{i=1}^n z_i (y_i - x_i' b) \in \mathbb{R}^{L \times 1}.$$

In matrix notation,

$$\bar{m}_n(b) = \frac{1}{n} Z'(Y - Xb).$$

Let $\widehat{W}$ be an $L \times L$ (symmetric) weighting matrix. Define the criterion

$$Q_n(b) \equiv \bar{m}_n(b)' \widehat{W} \bar{m}_n(b).$$

The GMM estimator is

$$\hat{\beta}_{\text{GMM}} \equiv \arg \min_b Q_n(b).$$

69 Closed form of $\hat{\beta}_{GMM}$

Start from

$$Q_n(b) = \bar{m}_n(b)' \widehat{W} \bar{m}_n(b) = \left(\frac{1}{n} Z'(Y - Xb) \right)' \widehat{W} \left(\frac{1}{n} Z'(Y - Xb) \right).$$

Use $(AB)' = B'A'$:

$$\left(\frac{1}{n} Z'(Y - Xb) \right)' = \frac{1}{n} (Y - Xb)' Z.$$

Hence

$$Q_n(b) = \frac{1}{n^2} (Y - Xb)' Z \widehat{W} Z' (Y - Xb).$$

The constant factor $1/n^2$ does not affect the minimizer, so define the equivalent objective

$$\tilde{Q}_n(b) \equiv (Y - Xb)' Z \widehat{W} Z' (Y - Xb).$$

Expand $(Y - Xb)' = Y' - b'X'$:

$$\begin{aligned} \tilde{Q}_n(b) &= (Y' - b'X') Z \widehat{W} Z' (Y - Xb) \\ &= Y' Z \widehat{W} Z' Y - Y' Z \widehat{W} Z' Xb - b' X' Z \widehat{W} Z' Y + b' X' Z \widehat{W} Z' Xb. \end{aligned}$$

If $\widehat{W}$ is symmetric, then $Y' Z \widehat{W} Z' Xb$ is a scalar equal to its transpose, so the two middle terms are equal and combine:

$$\tilde{Q}_n(b) = Y' Z \widehat{W} Z' Y - 2 Y' Z \widehat{W} Z' Xb + b' X' Z \widehat{W} Z' Xb.$$

Differentiate with respect to b . Write

$$A \equiv X' Z \widehat{W} Z' X, \quad c \equiv X' Z \widehat{W} Z' Y.$$

Then

$$\tilde{Q}_n(b) = Y' Z \widehat{W} Z' Y - 2c'b + b'Ab.$$

Using $\frac{\partial(c'b)}{\partial b} = c$ and (when A is symmetric) $\frac{\partial(b'Ab)}{\partial b} = 2Ab$:

$$\frac{\partial \tilde{Q}_n(b)}{\partial b} = -2c + 2Ab.$$

Set the first-order condition to zero:

$$-2c + 2Ab = 0 \implies Ab = c.$$

Assuming A is invertible,

$$\boxed{\hat{\beta}_{GMM} = (X' Z \widehat{W} Z' X)^{-1} X' Z \widehat{W} Z' Y.}$$

70 Assumptions for consistency and asymptotic distribution

1. (y_i, x_i, z_i) , $i = 1, \dots, n$, are i.i.d.
2. $\mathbb{E}[z_i(y_i - x_i'\beta)] = 0$.
3. $\text{rank}(\mathbb{E}[z_i z_i']) = k$.
4. $\widehat{W} \xrightarrow{p} W$, where W is nonrandom, symmetric, and positive definite.

71 Consistency

Start from the closed form:

$$\hat{\beta}_{GMM} = (X'Z\widehat{W}Z'X)^{-1}X'Z\widehat{W}Z'Y.$$

Substitute $Y = X\beta + \varepsilon$:

$$\begin{aligned}\hat{\beta}_{GMM} &= (X'Z\widehat{W}Z'X)^{-1}X'Z\widehat{W}Z'(X\beta + \varepsilon) \\ &= (X'Z\widehat{W}Z'X)^{-1}X'Z\widehat{W}Z'X\beta + (X'Z\widehat{W}Z'X)^{-1}X'Z\widehat{W}Z'\varepsilon \\ &= \beta + (X'Z\widehat{W}Z'X)^{-1}X'Z\widehat{W}Z'\varepsilon.\end{aligned}$$

Therefore

$$\hat{\beta}_{GMM} - \beta = (X'Z\widehat{W}Z'X)^{-1}X'Z\widehat{W}Z'\varepsilon.$$

Insert n/n to express normalized sums:

$$\hat{\beta}_{GMM} - \beta = \left(\frac{X'Z}{n} \widehat{W} \frac{Z'X}{n} \right)^{-1} \left(\frac{X'Z}{n} \widehat{W} \frac{Z'\varepsilon}{n} \right).$$

Write the sample cross-products as sums:

$$X'Z = \sum_{i=1}^n x_i z_i', \quad Z'\varepsilon = \sum_{i=1}^n z_i \varepsilon_i.$$

Then

$$\frac{X'Z}{n} = \frac{1}{n} \sum_{i=1}^n x_i z_i', \quad \frac{Z'\varepsilon}{n} = \frac{1}{n} \sum_{i=1}^n z_i \varepsilon_i.$$

Define the probability limits

$$\frac{X'Z}{n} \xrightarrow{p} \mathbb{E}[x_i z_i'] \equiv C, \quad \frac{Z'\varepsilon}{n} \xrightarrow{p} \mathbb{E}[z_i \varepsilon_i] = 0, \quad \widehat{W} \xrightarrow{p} W.$$

Thus

$$\hat{\beta}_{GMM} - \beta \xrightarrow{p} (CWC')^{-1}CW \cdot 0 = 0,$$

so

$$\boxed{\hat{\beta}_{GMM} \xrightarrow{p} \beta.}$$

72 Asymptotic distribution

Define

$$\Lambda \equiv \text{Var}(z_i \varepsilon_i) = \mathbb{E}[(z_i \varepsilon_i)(z_i \varepsilon_i)'] = \mathbb{E}[z_i \varepsilon_i \varepsilon_i' z_i'] = \mathbb{E}[\varepsilon_i^2 z_i z_i'].$$

By the multivariate CLT,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n z_i \varepsilon_i \xrightarrow{d} N(0, \Lambda).$$

Since $Z' \varepsilon = \sum_{i=1}^n z_i \varepsilon_i$, this is equivalently

$$\sqrt{n} \frac{Z' \varepsilon}{n} = \frac{Z' \varepsilon}{\sqrt{n}} \xrightarrow{d} N(0, \Lambda).$$

Now rewrite

$$\sqrt{n}(\hat{\beta}_{GMM} - \beta) = \left(\frac{X'Z}{n} \widehat{W} \frac{Z'X}{n} \right)^{-1} \left(\frac{X'Z}{n} \widehat{W} \frac{Z' \varepsilon}{\sqrt{n}} \right).$$

Using $\frac{X'Z}{n} \xrightarrow{p} C$ and $\widehat{W} \xrightarrow{p} W$, we have

$$\left(\frac{X'Z}{n} \widehat{W} \frac{Z'X}{n} \right)^{-1} \xrightarrow{p} (CW C')^{-1}, \quad \left(\frac{X'Z}{n} \widehat{W} \right) \xrightarrow{p} CW.$$

Let

$$U_n \equiv \frac{Z' \varepsilon}{\sqrt{n}} \xrightarrow{d} N(0, \Lambda).$$

Then by Slutsky,

$$\sqrt{n}(\hat{\beta}_{GMM} - \beta) \xrightarrow{d} (CW C')^{-1} CW U, \quad U \sim N(0, \Lambda).$$

A linear transformation of a normal yields

$$\boxed{\sqrt{n}(\hat{\beta}_{GMM} - \beta) \xrightarrow{d} N\left(0, (CW C')^{-1} CW \Lambda W C' (CW C')^{-1}\right)}.$$

Optimal GMM Weighting Matrix and Efficiency

72.1 Assumptions

1. (y_i, x_i, z_i) , $i = 1, \dots, n$, are i.i.d.
2. $\mathbb{E}[z_i(y_i - x_i' \beta)] = 0$.
3. $\text{rank}(\mathbb{E}[z_i z_i']) = k$.
4. $\widehat{W} \xrightarrow{p} W$, where W is nonrandom, symmetric, and positive definite.

73 Definitions and asymptotic distribution

Let

$$C \equiv \mathbb{E}[z_i x_i'] \in \mathbb{R}^{L \times k}, \quad C' = \mathbb{E}[x_i z_i'] \in \mathbb{R}^{k \times L},$$

and

$$\Lambda \equiv \mathbb{E}[\varepsilon_i^2 z_i z_i'] \in \mathbb{R}^{L \times L}, \quad \Lambda \text{ positive definite.}$$

Let W be an $L \times L$ symmetric positive definite weighting matrix.

Then

$$\sqrt{n}(\hat{\beta}_{GMM} - \beta) \xrightarrow{d} N(0, V(W)),$$

where

$$V(W) \equiv (C'WC)^{-1} C'W\Lambda WC (C'WC)^{-1}.$$

74 Optimal choice $W = \Lambda^{-1}$

If $W = \Lambda^{-1}$, then

$$\begin{aligned} V(\Lambda^{-1}) &= (C'\Lambda^{-1}C)^{-1} C'\Lambda^{-1}\Lambda\Lambda^{-1}C (C'\Lambda^{-1}C)^{-1} \\ &= (C'\Lambda^{-1}C)^{-1} C'\Lambda^{-1}C (C'\Lambda^{-1}C)^{-1} \\ &= (C'\Lambda^{-1}C)^{-1}. \end{aligned}$$

Hence, when $W = \Lambda^{-1}$, the asymptotic covariance matrix is $(C'\Lambda^{-1}C)^{-1}$.

75 Efficiency: show $V(W) - V(\Lambda^{-1})$ is positive semidefinite

For arbitrary W satisfying Assumption 4, we want to show

$$V(W) - (C'\Lambda^{-1}C)^{-1} \text{ is positive semidefinite.}$$

Step 1: Rewrite $V(W)$ as the inverse of a $k \times k$ matrix

Define

$$B(W) \equiv C'WC (C'W\Lambda WC)^{-1} C'WC.$$

Then $B(W)$ is $k \times k$. Using the matrix identity

$$(MA^{-1}M)^{-1} = M^{-1}AM^{-1} \quad \text{for invertible } M, A,$$

with $M = C'WC$ and $A = C'W\Lambda WC$, we obtain

$$B(W)^{-1} = (C'WC)^{-1} C'W\Lambda WC (C'WC)^{-1} = V(W).$$

So it suffices to compare $B(W)^{-1}$ with $(C'\Lambda^{-1}C)^{-1}$.

Step 2: Inverse-ordering lemma

If A and B are symmetric positive definite and

$$A - B \succeq 0,$$

then

$$B^{-1} - A^{-1} \succeq 0.$$

One way to see it:

$$A - B \succeq 0 \implies B^{-1/2}(A - B)B^{-1/2} \succeq 0 \implies B^{-1/2}AB^{-1/2} \succeq I.$$

Inverting reverses the order (for symmetric p.d. matrices):

$$(B^{-1/2}AB^{-1/2})^{-1} \preceq I.$$

But $(B^{-1/2}AB^{-1/2})^{-1} = B^{1/2}A^{-1}B^{1/2}$, so

$$B^{1/2}A^{-1}B^{1/2} \preceq I \implies A^{-1} \preceq B^{-1} \iff B^{-1} - A^{-1} \succeq 0.$$

Step 3: Show $C'\Lambda^{-1}C - B(W)$ is positive semidefinite

Consider

$$C'\Lambda^{-1}C - B(W) = C'\Lambda^{-1}C - C'WC(C'W\Lambda WC)^{-1}C'WC.$$

Since Λ is positive definite, there exists $\Lambda^{1/2}$ such that

$$\Lambda = \Lambda^{1/2}\Lambda^{1/2}, \quad \Lambda^{-1/2} \equiv (\Lambda^{1/2})^{-1}, \quad \Lambda^{-1} = \Lambda^{-1/2}\Lambda^{-1/2}.$$

Rewrite the first term:

$$C'\Lambda^{-1}C = C'\Lambda^{-1/2}\Lambda^{-1/2}C.$$

Rewrite the second term by inserting $\Lambda^{1/2}\Lambda^{-1/2} = I$:

$$C'WC(C'W\Lambda WC)^{-1}C'WC = C'\Lambda^{-1/2}\left(\Lambda^{1/2}WC(C'W\Lambda WC)^{-1}C'W\Lambda^{1/2}\right)\Lambda^{-1/2}C.$$

Therefore

$$C'\Lambda^{-1}C - B(W) = C'\Lambda^{-1/2}\left[I - \Lambda^{1/2}WC(C'W\Lambda WC)^{-1}C'W\Lambda^{1/2}\right]\Lambda^{-1/2}C.$$

Note that

$$C'W\Lambda WC = C'W\Lambda^{1/2}\Lambda^{1/2}WC.$$

Define

$$G \equiv \Lambda^{1/2}WC.$$

Then

$$G' = C'W\Lambda^{1/2}, \quad G'G = C'W\Lambda WC.$$

Hence

$$\Lambda^{1/2}WC(C'W\Lambda WC)^{-1}C'W\Lambda^{1/2} = G(G'G)^{-1}G'.$$

So

$$C' \Lambda^{-1} C - B(W) = C' \Lambda^{-1/2} \left[I - G(G'G)^{-1} G' \right] \Lambda^{-1/2} C.$$

Define the residual-maker (projection) matrix

$$M_G \equiv I - G(G'G)^{-1} G'.$$

Then

$$C' \Lambda^{-1} C - B(W) = C' \Lambda^{-1/2} M_G \Lambda^{-1/2} C.$$

Step 4: M_G is symmetric and idempotent

Let $P_G \equiv G(G'G)^{-1} G'$. Then $M_G = I - P_G$.

Symmetry:

$$P_G' = (G(G'G)^{-1} G')' = G((G'G)^{-1})' G' = G(G'G)^{-1} G' = P_G,$$

so $M_G' = M_G$.

Idempotence:

$$\begin{aligned} P_G^2 &= G(G'G)^{-1} G' G(G'G)^{-1} G' \\ &= G(G'G)^{-1} (G'G) (G'G)^{-1} G' \\ &= G(G'G)^{-1} G' \\ &= P_G, \end{aligned}$$

so

$$M_G^2 = (I - P_G)^2 = I - 2P_G + P_G^2 = I - P_G = M_G.$$

Step 5: Positive semidefiniteness

For any conformable vector a ,

$$\begin{aligned} a' (C' \Lambda^{-1/2} M_G \Lambda^{-1/2} C) a &= (\Lambda^{-1/2} C a)' M_G (\Lambda^{-1/2} C a) \\ &= (\Lambda^{-1/2} C a)' M_G' M_G (\Lambda^{-1/2} C a) \\ &= (M_G \Lambda^{-1/2} C a)' (M_G \Lambda^{-1/2} C a) \\ &\geq 0. \end{aligned}$$

Hence

$$C' \Lambda^{-1} C - B(W) \succeq 0.$$

Step 6: Apply the inverse-ordering lemma

Take

$$A = C' \Lambda^{-1} C, \quad B = B(W).$$

We have shown $A - B \succeq 0$, so

$$B^{-1} - A^{-1} \succeq 0.$$

Using $B^{-1} = V(W)$ and $A^{-1} = (C' \Lambda^{-1} C)^{-1}$,

$$V(W) - (C' \Lambda^{-1} C)^{-1} \succeq 0.$$

Thus $W = \Lambda^{-1}$ is asymptotically efficient.

76 Optimal GMM and the 2SLS special case under homoskedasticity

If $\widehat{W} \xrightarrow{p} \Lambda^{-1}$, then

$$\hat{\beta}_{GMM} = (X'Z\widehat{W}Z'X)^{-1}X'Z\widehat{W}Z'Y$$

is called the (two-step) optimal GMM estimator.

If homoskedasticity holds, then

$$\Lambda = \mathbb{E}[\varepsilon_i^2 z_i z_i'] = \sigma^2 \mathbb{E}[z_i z_i'].$$

A natural estimator is

$$\widehat{W} = \frac{1}{\sigma^2} \left(\frac{1}{n} \sum_{i=1}^n z_i z_i' \right)^{-1} = \frac{1}{\sigma^2} \left(\frac{Z'Z}{n} \right)^{-1}.$$

Use $\left(\frac{Z'Z}{n} \right)^{-1} = n(Z'Z)^{-1}$ to rewrite

$$\widehat{W} = \frac{n}{\sigma^2} (Z'Z)^{-1}.$$

Substitute into $\hat{\beta}_{GMM}$:

$$\begin{aligned} \hat{\beta}_{GMM} &= \left(X'Z \frac{n}{\sigma^2} (Z'Z)^{-1} Z'X \right)^{-1} X'Z \frac{n}{\sigma^2} (Z'Z)^{-1} Z'Y \\ &= \frac{\sigma^2}{n} (X'Z(Z'Z)^{-1}Z'X)^{-1} \cdot \frac{n}{\sigma^2} X'Z(Z'Z)^{-1}Z'Y \\ &= (X'Z(Z'Z)^{-1}Z'X)^{-1} X'Z(Z'Z)^{-1}Z'Y. \end{aligned}$$

Let

$$P_Z \equiv Z(Z'Z)^{-1}Z',$$

so $X'P_ZX = X'Z(Z'Z)^{-1}Z'X$ and $X'P_ZY = X'Z(Z'Z)^{-1}Z'Y$. Hence

$$\hat{\beta}_{GMM} = (X'P_ZX)^{-1}X'P_ZY = \hat{\beta}_{2SLS}.$$

Two-Step (Optimal) GMM Weighting Matrix

76.1 Setup

For $i = 1, \dots, n$,

$$y_i = x_i'\beta + \varepsilon_i, \quad x_i \in \mathbb{R}^{k \times 1}, \quad z_i \in \mathbb{R}^{L \times 1}.$$

Instrument orthogonality:

$$\mathbb{E}[z_i \varepsilon_i] = 0 \quad \Longleftrightarrow \quad \mathbb{E}[z_i(y_i - x_i'\beta)] = 0.$$

Define the stacked objects:

$$Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \quad X = \begin{pmatrix} x_1' \\ \vdots \\ x_n' \end{pmatrix}, \quad Z = \begin{pmatrix} z_1' \\ \vdots \\ z_n' \end{pmatrix}.$$

77 GMM criterion and closed form

Define the sample moment vector

$$\bar{m}_n(b) \equiv \frac{1}{n} \sum_{i=1}^n z_i(y_i - x_i'b) = \frac{1}{n} Z'(Y - Xb) \in \mathbb{R}^{L \times 1}.$$

Let $\widehat{W}$ be an $L \times L$ symmetric positive definite weighting matrix. Define

$$Q_n(b) \equiv \bar{m}_n(b)' \widehat{W} \bar{m}_n(b), \quad \hat{\beta}_{GMM} \equiv \arg \min_b Q_n(b).$$

Expand:

$$Q_n(b) = \left(\frac{1}{n} Z'(Y - Xb) \right)' \widehat{W} \left(\frac{1}{n} Z'(Y - Xb) \right) = \frac{1}{n^2} (Y - Xb)' Z \widehat{W} Z' (Y - Xb).$$

The factor $1/n^2$ does not affect the minimizer, so define

$$\tilde{Q}_n(b) \equiv (Y - Xb)' Z \widehat{W} Z' (Y - Xb).$$

Use $(Y - Xb)' = Y' - b'X'$:

$$\begin{aligned} \tilde{Q}_n(b) &= (Y' - b'X') Z \widehat{W} Z' (Y - Xb) \\ &= Y' Z \widehat{W} Z' Y - Y' Z \widehat{W} Z' Xb - b' X' Z \widehat{W} Z' Y + b' X' Z \widehat{W} Z' Xb. \end{aligned}$$

If $\widehat{W}' = \widehat{W}$, the two middle terms are equal scalars, hence

$$\tilde{Q}_n(b) = Y' Z \widehat{W} Z' Y - 2 Y' Z \widehat{W} Z' Xb + b' X' Z \widehat{W} Z' Xb.$$

Let

$$A \equiv X' Z \widehat{W} Z' X, \quad c \equiv X' Z \widehat{W} Z' Y.$$

Then

$$\tilde{Q}_n(b) = Y' Z \widehat{W} Z' Y - 2c'b + b'Ab.$$

Using $\frac{\partial(c'b)}{\partial b} = c$ and (when A is symmetric) $\frac{\partial(b'Ab)}{\partial b} = 2Ab$,

$$\frac{\partial \tilde{Q}_n(b)}{\partial b} = -2c + 2Ab.$$

Set the FOC to zero:

$$-2c + 2Ab = 0 \implies Ab = c.$$

Assuming A is invertible,

$$\boxed{\hat{\beta}_{GMM} = (X' Z \widehat{W} Z' X)^{-1} X' Z \widehat{W} Z' Y.}$$

78 Key population matrices

Define

$$C \equiv \mathbb{E}[z_i x_i'] \in \mathbb{R}^{L \times k}, \quad \Lambda \equiv \mathbb{E}[\varepsilon_i^2 z_i z_i'] \in \mathbb{R}^{L \times L}.$$

Assume $\text{rank}(C) = k$ and Λ is positive definite.

If

$$\widehat{W} \xrightarrow{p} \Lambda^{-1},$$

then (optimal GMM)

$$\sqrt{n}(\hat{\beta}_{GMM} - \beta) \xrightarrow{d} N(0, (C' \Lambda^{-1} C)^{-1}).$$

79 Estimating Λ requires residuals

To estimate

$$\Lambda = \mathbb{E}[\varepsilon_i^2 z_i z_i'],$$

we need an estimator of ε_i , hence we construct residuals from an initial consistent GMM estimate.

80 Two-step GMM

Step 1: initial consistent weighting matrix

Choose an initial $\widehat{W}^{(1)}$ that is symmetric positive definite and satisfies the weighting-matrix regularity condition (e.g. $\widehat{W}^{(1)} \xrightarrow{p} W^{(1)}$ p.d.), but is not necessarily optimal. For example,

$$\widehat{W}^{(1)} = I_L, \quad \text{or} \quad \widehat{W}^{(1)} = \left(\frac{Z'Z}{n} \right)^{-1}.$$

Compute the initial GMM estimator

$$\hat{\beta}_{GMM}^{(1)} \equiv (X'Z\widehat{W}^{(1)}Z'X)^{-1}X'Z\widehat{W}^{(1)}Z'Y.$$

Define residuals

$$e_i \equiv y_i - x_i' \hat{\beta}_{GMM}^{(1)}.$$

Using $y_i = x_i' \beta + \varepsilon_i$,

$$e_i = x_i' \beta + \varepsilon_i - x_i' \hat{\beta}_{GMM}^{(1)} = \varepsilon_i - x_i' (\hat{\beta}_{GMM}^{(1)} - \beta).$$

Step 2: estimate Λ and set the optimal weight

Define the sample analogue

$$\widehat{\Lambda} \equiv \frac{1}{n} \sum_{i=1}^n e_i^2 z_i z_i'.$$

Then set

$$\widehat{W}^{(2)} \equiv \widehat{\Lambda}^{-1}.$$

Why $\widehat{W}^{(2)} \xrightarrow{p} \Lambda^{-1}$. Let $\Delta \equiv \hat{\beta}_{GMM}^{(1)} - \beta$ so $\Delta = o_p(1)$ by consistency of $\hat{\beta}_{GMM}^{(1)}$. Then

$$e_i = \varepsilon_i - x_i' \Delta, \quad e_i^2 = (\varepsilon_i - x_i' \Delta)^2 = \varepsilon_i^2 - 2\varepsilon_i x_i' \Delta + (x_i' \Delta)^2.$$

Multiply by $z_i z_i'$ and average:

$$\widehat{\Lambda} = \frac{1}{n} \sum_{i=1}^n \varepsilon_i^2 z_i z_i' - 2 \left(\frac{1}{n} \sum_{i=1}^n \varepsilon_i z_i z_i' x_i' \right) \Delta + \frac{1}{n} \sum_{i=1}^n (x_i' \Delta)^2 z_i z_i'.$$

Under i.i.d. sampling and finite moments, by LLN,

$$\frac{1}{n} \sum_{i=1}^n \varepsilon_i^2 z_i z_i' \xrightarrow{p} \mathbb{E}[\varepsilon_i^2 z_i z_i'] = \Lambda,$$

and the remaining terms vanish because $\Delta = o_p(1)$ while the sample averages in brackets are $O_p(1)$:

$$\left(\frac{1}{n} \sum_{i=1}^n \varepsilon_i z_i z_i' x_i' \right) \Delta = o_p(1), \quad \frac{1}{n} \sum_{i=1}^n (x_i' \Delta)^2 z_i z_i' = o_p(1).$$

Hence

$$\widehat{\Lambda} \xrightarrow{p} \Lambda.$$

Since Λ is positive definite, inversion is continuous at Λ , so

$$\widehat{W}^{(2)} = \widehat{\Lambda}^{-1} \xrightarrow{p} \Lambda^{-1}.$$

Second-step (optimal) estimator

Compute

$$\hat{\beta}_{GMM}^{(2)} \equiv (X' Z \widehat{W}^{(2)} Z' X)^{-1} X' Z \widehat{W}^{(2)} Z' Y.$$

With $\widehat{W}^{(2)} \xrightarrow{p} \Lambda^{-1}$, the asymptotic distribution is

$$\sqrt{n}(\hat{\beta}_{GMM}^{(2)} - \beta) \xrightarrow{d} N(0, (C' \Lambda^{-1} C)^{-1}).$$

With this choice of weighting matrix, $\hat{\beta}_{GMM}^{(2)}$ is asymptotically efficient among GMM estimators based on the same moment conditions.

GMM Setup: IV Moment Conditions and General Moment Conditions

81 IV example as a moment-condition problem

$$(x_i, y_i, z_i) \sim \text{i.i.d.}, \quad i = 1, 2, \dots, n.$$

$$\beta : \text{parameter}, \quad k \times 1, \quad \beta \in \mathbb{R}^k.$$

$L \times 1$ population moment condition.

$$\mathbb{E}[z_i (y_i - x_i' \beta)] = 0.$$

Sample analogue. Define the sample moment function (an $L \times 1$ vector)

$$m_n(b) \equiv \frac{1}{n} \sum_{i=1}^n z_i (y_i - x_i' b), \quad b \in \mathbb{R}^k.$$

GMM estimator. Let $\widehat{W}$ be an $L \times L$ symmetric positive definite weighting matrix. Define

$$\widehat{\beta}_{GMM} \equiv \arg \min_b m_n(b)' \widehat{W} m_n(b).$$

82 General GMM setup

$$w_i \sim \text{i.i.d.}, \quad i = 1, 2, \dots, n.$$

$$\theta : \text{parameter}, \quad k \times 1, \quad \theta \in \Theta.$$

$L \times 1$ population moment condition.

$$\mathbb{E}[m(w_i, \theta)] = 0.$$

Sample analogue. Define the sample moment function (an $L \times 1$ vector)

$$m_n(t) \equiv \frac{1}{n} \sum_{i=1}^n m(w_i, t), \quad t \in \Theta.$$

GMM estimator. Let $\widehat{W}$ be an $L \times L$ symmetric positive definite weighting matrix. Define

$$\widehat{\theta}_{GMM} \equiv \arg \min_t m_n(t)' \widehat{W} m_n(t).$$

83 Objectives $Q_n(t)$ and $Q(t)$

Sample objective.

$$Q_n(t) \equiv m_n(t)' \widehat{W} m_n(t).$$

Population objective. Define the population moment

$$m(t) \equiv \mathbb{E}[m(w_i, t)].$$

Let W be the probability limit of $\widehat{W}$ (or a fixed p.d. matrix). Then

$$Q(t) \equiv m(t)' W m(t) = \mathbb{E}[m(w_i, t)]' W \mathbb{E}[m(w_i, t)].$$

84 How $Q_n(t) \rightarrow Q(t)$ leads to $\widehat{\theta}_{GMM} \rightarrow \theta_0$

Step 1: LLN for the sample moment. For a fixed $t \in \Theta$, by i.i.d. LLN (componentwise),

$$m_n(t) = \frac{1}{n} \sum_{i=1}^n m(w_i, t) \xrightarrow{p} \mathbb{E}[m(w_i, t)] \equiv m(t).$$

Step 2: Convergence of the weighting matrix. Assume

$$\widehat{W} \xrightarrow{p} W, \quad W \text{ symmetric positive definite.}$$

Step 3: Pointwise convergence of the objective. Write

$$Q_n(t) = m_n(t)' \widehat{W} m_n(t).$$

Using $m_n(t) \xrightarrow{p} m(t)$ and $\widehat{W} \xrightarrow{p} W$, and continuity of quadratic forms,

$$Q_n(t) \xrightarrow{p} m(t)' W m(t) \equiv Q(t).$$

Step 4: The population minimizer. Define the population target as the minimizer of the population objective:

$$\theta_0 \equiv \arg \min_{t \in \Theta} Q(t).$$

Step 5: Argmin consistency (M-estimator logic). If:

- (a) $Q_n(t) \rightarrow Q(t)$ uniformly in $t \in \Theta$ in probability (e.g. via a uniform LLN),
- (b) Θ is compact (or the minimizer is well-behaved under coercivity),
- (c) $Q(t)$ has a unique minimizer θ_0 (identification),

then

$$\widehat{\theta}_{GMM} \xrightarrow{p} \theta_0.$$

85 Diagrammatic summary

$$\begin{array}{ccc}
 Q_n(t) & \xrightarrow{p} & Q(t) \\
 \uparrow \text{minimize} & & \uparrow \text{minimize} \\
 \widehat{\theta}_{GMM} = \arg \min_t Q_n(t) & \xrightarrow{p} & \theta_0 = \arg \min_t Q(t)
 \end{array}$$

86 Mapping the IV example into the general GMM notation

In the IV setup, define

$$w_i \equiv (x_i, y_i, z_i), \quad t \equiv b, \quad \theta \equiv \beta,$$

and define the moment function

$$m(w_i, b) \equiv z_i (y_i - x_i' b).$$

Then

$$\mathbb{E}[m(w_i, \beta)] = \mathbb{E}[z_i (y_i - x_i' \beta)] = 0$$

is exactly the IV population moment condition, and the GMM estimator

$$\hat{\beta}_{GMM} = \arg \min_b m_n(b)' \widehat{W} m_n(b)$$

is the IV-GMM estimator built from these moments.

Extremum Estimators and GMM: Consistency via Uniform Convergence

87 Moment conditions and GMM objectives

Data and parameter

Assume

$$w_i \sim \text{i.i.d.}, \quad i = 1, 2, \dots, n.$$

Let the parameter be

$$\theta : \text{parameter}, \quad k \times 1, \quad \theta \in \Theta.$$

$L \times 1$ population moment condition

Let $m(w_i, \theta) \in \mathbb{R}^{L \times 1}$ be a moment function. The population moment condition is

$$\mathbb{E}[m(w_i, \theta)] = 0.$$

Sample analogue

Define the sample moment function for any $t \in \Theta$:

$$m_n(t) \equiv \frac{1}{n} \sum_{i=1}^n m(w_i, t) \in \mathbb{R}^{L \times 1}.$$

GMM estimator

Let $\widehat{W}$ be an $L \times L$ symmetric positive definite weighting matrix. Define the GMM estimator by

$$\widehat{\theta}_{GMM} \equiv \arg \min_{t \in \Theta} m_n(t)' \widehat{W} m_n(t).$$

Define the sample objective function

$$Q_n(t) \equiv m_n(t)' \widehat{W} m_n(t).$$

Let

$$m(t) \equiv \mathbb{E}[m(w_i, t)].$$

If $\widehat{W} \xrightarrow{p} W$ for some (nonrandom) symmetric positive definite W , define the population objective

$$Q(t) \equiv m(t)' W m(t) = \mathbb{E}[m(w_i, t)]' W \mathbb{E}[m(w_i, t)].$$

Extremum-estimator notation

Define the population minimizer and the sample minimizer:

$$\theta_0 \equiv \arg \min_{t \in \Theta} Q(t), \quad \widehat{\theta} \equiv \arg \min_{t \in \Theta} Q_n(t).$$

In GMM, $\widehat{\theta}$ is $\widehat{\theta}_{GMM}$ and $Q_n(\cdot)$ is the GMM criterion.

$$\begin{array}{ccc} Q_n(t) & \xrightarrow{p} & Q(t) \\ \uparrow \text{minimize} & & \uparrow \text{minimize} \\ \widehat{\theta} = \arg \min_{t \in \Theta} Q_n(t) & \xrightarrow{p} & \theta_0 = \arg \min_{t \in \Theta} Q(t) \end{array}$$

88 Pointwise vs uniform convergence: an example

Let

$$\Theta = [0, 1].$$

Define a sequence of sample objective functions:

$$Q_n(t) = \begin{cases} -nt, & t \in [0, \frac{1}{n}], \\ -2 + nt, & t \in [\frac{1}{n}, \frac{2}{n}], \\ 0, & t \in [\frac{2}{n}, 1]. \end{cases}$$

Define the population objective:

$$Q(t) = 0, \quad t \in [0, 1].$$

Pointwise convergence

Fix any $t \in (0, 1]$. Choose N such that $\frac{2}{N} < t$ (for example, any $N > 2/t$). Then for all $n \geq N$, we have $t \in [2/n, 1]$, so $Q_n(t) = 0 = Q(t)$. Hence

$$Q_n(t) \rightarrow Q(t) \quad \text{for each fixed } t \in (0, 1].$$

At $t = 0$, $Q_n(0) = 0$ for all n , so $Q_n(0) \rightarrow Q(0)$ as well. Therefore $Q_n(t) \rightarrow Q(t)$ pointwise on Θ .

Failure of uniform convergence

Compute the supremum deviation:

$$\sup_{t \in \Theta} |Q_n(t) - Q(t)| = \sup_{t \in [0, 1]} |Q_n(t)|.$$

At $t = \frac{1}{n}$,

$$Q_n\left(\frac{1}{n}\right) = -n \cdot \frac{1}{n} = -1, \quad \left|Q_n\left(\frac{1}{n}\right)\right| = 1,$$

so

$$\sup_{t \in \Theta} |Q_n(t)| \geq 1.$$

Also, by construction, $Q_n(t) \in [-1, 0]$ for all $t \in [0, 1]$, so $\sup_{t \in \Theta} |Q_n(t)| \leq 1$. Thus

$$\sup_{t \in \Theta} |Q_n(t) - Q(t)| = 1 \quad \text{for all } n,$$

so the supremum does not converge to 0.

This shows pointwise convergence $Q_n(t) \rightarrow Q(t)$ does not imply the uniform convergence needed for consistency of argmin estimators.

89 Assumptions for consistency of extremum estimators

Assumption 1: (Uniform convergence in probability)

$$\sup_{t \in \Theta} |Q_n(t) - Q(t)| \xrightarrow{p} 0.$$

Assumption 2: (Unique minimizer / identification)

$$\theta_0 \text{ uniquely minimizes } Q(t) \text{ over } \Theta.$$

Equivalently: for every $\varepsilon > 0$,

$$\inf_{t \in \Theta: \|t - \theta_0\| \geq \varepsilon} Q(t) > Q(\theta_0).$$

Assumption 3: (Continuity)

$$Q(t) \text{ is continuous on } \Theta.$$

Assumption 4: (Compactness)

$$\Theta \text{ is compact.}$$

90 Consistency theorem and proof

Theorem. Under Assumptions 1–4,

$$\hat{\theta} = \arg \min_{t \in \Theta} Q_n(t) \xrightarrow{p} \theta_0 = \arg \min_{t \in \Theta} Q(t).$$

Proof. Fix $\varepsilon > 0$. Define the complement of an ε -ball around θ_0 :

$$A_\varepsilon \equiv \{t \in \Theta : \|t - \theta_0\| \geq \varepsilon\}.$$

Because Θ is compact and A_ε is closed in Θ , A_ε is compact. By Assumption 3 (continuity), $Q(t)$ attains its minimum on A_ε ; define

$$Q_\varepsilon \equiv \min_{t \in A_\varepsilon} Q(t).$$

By Assumption 2 (unique minimizer), $Q_\varepsilon > Q(\theta_0)$. Define the positive gap

$$\delta \equiv \frac{Q_\varepsilon - Q(\theta_0)}{2} > 0.$$

Define the event

$$E_n \equiv \left\{ \sup_{t \in \Theta} |Q_n(t) - Q(t)| < \delta \right\}.$$

By Assumption 1, $\Pr(E_n) \rightarrow 1$.

On the event E_n , for any $t \in A_\varepsilon$,

$$Q_n(t) \geq Q(t) - \delta \geq Q_\varepsilon - \delta = Q(\theta_0) + \delta,$$

and for $t = \theta_0$,

$$Q_n(\theta_0) \leq Q(\theta_0) + \delta.$$

Therefore, on E_n ,

$$\inf_{t \in A_\varepsilon} Q_n(t) > Q_n(\theta_0).$$

Since $\hat{\theta}$ minimizes $Q_n(t)$ over Θ , we have $Q_n(\hat{\theta}) \leq Q_n(\theta_0)$, so $\hat{\theta}$ cannot lie in A_ε on E_n . Hence, on E_n ,

$$\|\hat{\theta} - \theta_0\| < \varepsilon.$$

Thus

$$\Pr(\|\hat{\theta} - \theta_0\| \geq \varepsilon) \leq \Pr(E_n^c) \rightarrow 0.$$

Since $\varepsilon > 0$ is arbitrary, $\hat{\theta} \xrightarrow{p} \theta_0$.

Implication for GMM. Taking $Q_n(t) = m_n(t)' \widehat{W} m_n(t)$ and $Q(t) = m(t)' W m(t)$, the same argument yields

$$\hat{\theta}_{GMM} \xrightarrow{p} \theta_0,$$

provided the GMM criterion satisfies Assumptions 1–4.

Extremum Estimator

Objects

θ : parameter.

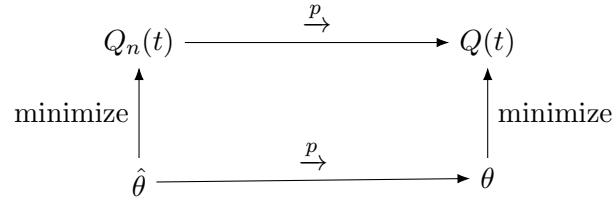
$Q_n(\cdot)$: objective function in the sample. $Q(\cdot)$: objective function in the population.

Define the population minimizer

$$\theta \equiv \arg \min_{t \in \Theta} Q(t),$$

and the sample minimizer

$$\hat{\theta} \equiv \arg \min_{t \in \Theta} Q_n(t).$$



Assumptions

1.

$$\sup_{t \in \Theta} |Q_n(t) - Q(t)| \xrightarrow{p} 0.$$

2. θ uniquely minimizes $Q(t)$ over Θ .

3. $Q(t)$ is continuous.

4. Θ is compact.

Theorem

Suppose Assumptions 1–4 hold. Then

$$\hat{\theta} \xrightarrow{p} \theta.$$

Proof

Fix any $\varepsilon > 0$. Consider the set

$$A_\varepsilon \equiv \{t \in \Theta : \|t - \theta\| \geq \varepsilon\}.$$

Since Θ is compact and A_ε is closed in Θ , A_ε is compact.

Because $Q(\cdot)$ is continuous and A_ε is compact, $Q(t)$ attains its minimum on A_ε . Hence there exists $t^* \in A_\varepsilon$ such that

$$\min_{t \in A_\varepsilon} Q(t) = Q(t^*), \quad \|t^* - \theta\| \geq \varepsilon.$$

Since θ uniquely minimizes $Q(t)$ on Θ and $t^* \neq \theta$ (because $t^* \in A_\varepsilon$),

$$Q(\theta) < Q(t^*).$$

Define

$$\delta_\varepsilon \equiv Q(t^*) - Q(\theta) > 0.$$

For any $t \in A_\varepsilon$ we have $Q(t) \geq \min_{u \in A_\varepsilon} Q(u) = Q(t^*)$, so

$$t \in A_\varepsilon \implies Q(t) - Q(\theta) \geq Q(t^*) - Q(\theta) = \delta_\varepsilon.$$

In particular,

$$\|\hat{\theta} - \theta\| \geq \varepsilon \implies \hat{\theta} \in A_\varepsilon \implies Q(\hat{\theta}) - Q(\theta) \geq \delta_\varepsilon.$$

Therefore,

$$\mathbb{P}(\|\hat{\theta} - \theta\| \geq \varepsilon) \leq \mathbb{P}(Q(\hat{\theta}) - Q(\theta) \geq \delta_\varepsilon).$$

Now decompose $Q(\hat{\theta}) - Q(\theta)$ by adding and subtracting $Q_n(\hat{\theta})$ and $Q_n(\theta)$:

$$Q(\hat{\theta}) - Q(\theta) = (Q(\hat{\theta}) - Q_n(\hat{\theta})) + (Q_n(\hat{\theta}) - Q_n(\theta)) + (Q_n(\theta) - Q(\theta)).$$

Hence

$$\mathbb{P}(Q(\hat{\theta}) - Q(\theta) \geq \delta_\varepsilon) = \mathbb{P}\left((Q(\hat{\theta}) - Q_n(\hat{\theta})) + (Q_n(\hat{\theta}) - Q_n(\theta)) + (Q_n(\theta) - Q(\theta)) \geq \delta_\varepsilon\right).$$

By definition of the supremum,

$$Q(\hat{\theta}) - Q_n(\hat{\theta}) \leq \sup_{t \in \Theta} |Q_n(t) - Q(t)|, \quad Q_n(\theta) - Q(\theta) \leq \sup_{t \in \Theta} |Q_n(t) - Q(t)|.$$

Also, since $\hat{\theta}$ minimizes $Q_n(t)$ over Θ ,

$$Q_n(\hat{\theta}) \leq Q_n(\theta) \implies Q_n(\hat{\theta}) - Q_n(\theta) \leq 0.$$

Therefore,

$$\begin{aligned} \mathbb{P}(Q(\hat{\theta}) - Q(\theta) \geq \delta_\varepsilon) &\leq \mathbb{P}\left(2 \sup_{t \in \Theta} |Q_n(t) - Q(t)| + \underbrace{(Q_n(\hat{\theta}) - Q_n(\theta))}_{\leq 0} \geq \delta_\varepsilon\right) \\ &\leq \mathbb{P}\left(2 \sup_{t \in \Theta} |Q_n(t) - Q(t)| \geq \delta_\varepsilon\right). \end{aligned}$$

Combining with the earlier bound,

$$\mathbb{P}(\|\hat{\theta} - \theta\| \geq \varepsilon) \leq \mathbb{P}\left(2 \sup_{t \in \Theta} |Q_n(t) - Q(t)| \geq \delta_\varepsilon\right).$$

By Assumption 1, $\sup_{t \in \Theta} |Q_n(t) - Q(t)| \xrightarrow{p} 0$, so the right-hand side converges to 0:

$$\mathbb{P}(\|\hat{\theta} - \theta\| \geq \varepsilon) \rightarrow 0.$$

Since this holds for every $\varepsilon > 0$, we conclude

$$\hat{\theta} \xrightarrow{p} \theta.$$

□

Asymptotic Normality of the GMM Estimator and Optimal Weighting

90.1 Setup

$$w_i \sim \text{i.i.d.}, \quad i = 1, 2, \dots, n.$$

$$\theta : \text{parameter}, \quad k \times 1, \quad \theta \in \Theta.$$

Let the moment function be

$$m(w_i, t) \in \mathbb{R}^{L \times 1}, \quad t \in \Theta.$$

Population moment condition:

$$\mathbb{E}[m(w_i, \theta)] = 0.$$

Sample analogue:

$$m_n(t) \equiv \frac{1}{n} \sum_{i=1}^n m(w_i, t).$$

Let $\widehat{W}$ be an $L \times L$ symmetric positive definite weighting matrix. Define the GMM objective:

$$Q_n(t) \equiv m_n(t)' \widehat{W} m_n(t).$$

GMM estimator:

$$\widehat{\theta}_{GMM} \equiv \arg \min_{t \in \Theta} Q_n(t) = \arg \min_{t \in \Theta} m_n(t)' \widehat{W} m_n(t).$$

90.2 Assumptions

1.

$$\widehat{\theta}_{GMM} \xrightarrow{p} \theta.$$

2.

$$\theta \in \text{Int}(\Theta).$$

3. $m(w_i, t)$ is continuously differentiable in t on a neighborhood $B_\delta(\theta)$.

4.

$$\sqrt{n} m_n(\theta) \xrightarrow{d} N(0, \Lambda), \quad \Lambda \equiv \mathbb{E}[m(w_i, \theta)m(w_i, \theta)'].$$

5.

$$\sup_{t \in B_\delta(\theta)} \left\| \frac{\partial}{\partial t'} m_n(t) - G(t) \right\| \xrightarrow{p} 0,$$

where

$$G(t) \equiv \mathbb{E} \left[\frac{\partial}{\partial t'} m(w_i, t) \right]$$

is continuous at θ , and

$$G \equiv G(\theta) \text{ has rank } k.$$

6.

$$\widehat{W} \xrightarrow{p} W, \quad W \text{ is symmetric positive definite.}$$

90.3 Theorem

Suppose Assumptions 1–6 hold. Then

$$\sqrt{n}(\widehat{\theta}_{GMM} - \theta) \xrightarrow{d} N(0, \Omega),$$

where

$$\Omega = (G'WG)^{-1} G'W \Lambda WG (G'WG)^{-1}.$$

90.4 Proof

By Assumption 1, $\widehat{\theta}_{GMM} \xrightarrow{p} \theta$. By Assumption 3, $m(w_i, t)$ is continuously differentiable in t on $B_\delta(\theta)$, so with probability approaching 1 the objective $Q_n(t)$ is differentiable at the interior minimizer $\widehat{\theta}_{GMM}$.

Step 1: First-order condition

Write

$$Q_n(t) = m_n(t)' \widehat{W} m_n(t).$$

Let the Jacobian be

$$\frac{\partial}{\partial t'} m_n(t) \in \mathbb{R}^{L \times k}, \quad \left(\frac{\partial}{\partial t'} m_n(t) \right)' \in \mathbb{R}^{k \times L}.$$

Using the chain rule for a quadratic form, the gradient of $Q_n(t)$ is

$$\frac{\partial}{\partial t} Q_n(t) = 2 \left(\frac{\partial}{\partial t'} m_n(t) \right)' \widehat{W} m_n(t).$$

At the minimizer $\widehat{\theta}_{GMM}$ (interior point), the first-order condition is

$$\frac{\partial}{\partial t} Q_n(\widehat{\theta}_{GMM}) = 2 \left(\frac{\partial}{\partial t'} m_n(\widehat{\theta}_{GMM}) \right)' \widehat{W} m_n(\widehat{\theta}_{GMM}) = 0. \quad (\text{FOC})$$

Step 2: Mean value theorem expansion for $m_n(\widehat{\theta}_{GMM})$

By the mean value theorem (componentwise), there exists a point $\tilde{\theta}$ lying between $\widehat{\theta}_{GMM}$ and θ such that

$$m_n(\widehat{\theta}_{GMM}) = m_n(\theta) + \frac{\partial}{\partial t'} m_n(\tilde{\theta}) (\widehat{\theta}_{GMM} - \theta). \quad (\text{MVT})$$

Step 3: Substitute (MVT) into (FOC)

Substituting (??) into (??) gives

$$\left(\frac{\partial}{\partial t'} m_n(\widehat{\theta}_{GMM}) \right)' \widehat{W} \left[m_n(\theta) + \frac{\partial}{\partial t'} m_n(\tilde{\theta}) (\widehat{\theta}_{GMM} - \theta) \right] = 0.$$

Distribute:

$$\left(\frac{\partial}{\partial t'} m_n(\hat{\theta}_{GMM})\right)' \widehat{W} m_n(\theta) + \left(\frac{\partial}{\partial t'} m_n(\hat{\theta}_{GMM})\right)' \widehat{W} \left(\frac{\partial}{\partial t'} m_n(\tilde{\theta})\right) (\hat{\theta}_{GMM} - \theta) = 0.$$

Move the second term to the right-hand side:

$$\left(\frac{\partial}{\partial t'} m_n(\hat{\theta}_{GMM})\right)' \widehat{W} m_n(\theta) = - \left(\frac{\partial}{\partial t'} m_n(\hat{\theta}_{GMM})\right)' \widehat{W} \left(\frac{\partial}{\partial t'} m_n(\tilde{\theta})\right) (\hat{\theta}_{GMM} - \theta).$$

Multiply both sides by $\sqrt{n}$:

$$\left(\frac{\partial}{\partial t'} m_n(\hat{\theta}_{GMM})\right)' \widehat{W} \sqrt{n} m_n(\theta) = - \left(\frac{\partial}{\partial t'} m_n(\hat{\theta}_{GMM})\right)' \widehat{W} \left(\frac{\partial}{\partial t'} m_n(\tilde{\theta})\right) \sqrt{n} (\hat{\theta}_{GMM} - \theta). \quad (\star)$$

Step 4: Limits of the pieces

By Assumption 4,

$$\sqrt{n} m_n(\theta) \xrightarrow{d} N(0, \Lambda).$$

By Assumption 6,

$$\widehat{W} \xrightarrow{p} W.$$

By Assumption 5 and Assumption 1, since $\hat{\theta}_{GMM} \xrightarrow{p} \theta$ and $\tilde{\theta}$ lies between $\hat{\theta}_{GMM}$ and θ , we have $\tilde{\theta} \xrightarrow{p} \theta$ and

$$\frac{\partial}{\partial t'} m_n(\hat{\theta}_{GMM}) \xrightarrow{p} G, \quad \frac{\partial}{\partial t'} m_n(\tilde{\theta}) \xrightarrow{p} G.$$

Therefore,

$$\left(\frac{\partial}{\partial t'} m_n(\hat{\theta}_{GMM})\right)' \widehat{W} \left(\frac{\partial}{\partial t'} m_n(\tilde{\theta})\right) \xrightarrow{p} G' W G.$$

By Assumption 5, G has rank k , and W is positive definite, hence $G' W G$ is invertible.

Also, by Slutsky and the continuous mapping theorem for linear transformations of asymptotic normals,

$$\left(\frac{\partial}{\partial t'} m_n(\hat{\theta}_{GMM})\right)' \widehat{W} \sqrt{n} m_n(\theta) \xrightarrow{d} G' W Z, \quad Z \sim N(0, \Lambda),$$

so

$$G' W Z \sim N(0, G' W \Lambda W G).$$

Step 5: Solve for $\sqrt{n}(\hat{\theta}_{GMM} - \theta)$

From (??), we have

$$\sqrt{n}(\hat{\theta}_{GMM} - \theta) = - \left[(\nabla_{\theta} m_n(\hat{\theta}_{GMM}))' \widehat{W} \nabla_{\theta} m_n(\tilde{\theta}) \right]^{-1} (\nabla_{\theta} m_n(\hat{\theta}_{GMM}))' \widehat{W} \sqrt{n} m_n(\theta),$$

where $\tilde{\theta}$ lies on the line segment between $\hat{\theta}_{GMM}$ and θ , and $\nabla_{\theta} m_n(\theta) \equiv \partial m_n(\theta) / \partial \theta'$ denotes the Jacobian of the sample moments.

Using the probability limits above and Slutsky,

$$\sqrt{n}(\widehat{\theta}_{GMM} - \theta) \xrightarrow{d} -(G'WG)^{-1}U, \quad U \sim N(0, G'W\Lambda WG).$$

Since multiplication by a constant matrix preserves normality and the sign does not affect covariance,

$$\sqrt{n}(\widehat{\theta}_{GMM} - \theta) \xrightarrow{d} N\left(0, (G'WG)^{-1}G'W\Lambda WG(G'WG)^{-1}\right).$$

91 Optimal GMM and two-step GMM

If the weighting matrix satisfies

$$\widehat{W} \xrightarrow{p} \Lambda^{-1},$$

then

$$\Omega = (G'\Lambda^{-1}G)^{-1}G'\Lambda^{-1}\Lambda\Lambda^{-1}G(G'\Lambda^{-1}G)^{-1} = (G'\Lambda^{-1}G)^{-1}.$$

This choice is called optimal (efficient) GMM.

Two-step GMM proceeds as:

1. Choose an initial $\widehat{W}^{(1)}$ (symmetric positive definite) and compute $\widehat{\theta}^{(1)}$ by minimizing

$$Q_n^{(1)}(t) = m_n(t)' \widehat{W}^{(1)} m_n(t).$$

2. Form residual-like moments at $\widehat{\theta}^{(1)}$ and estimate

$$\widehat{\Lambda} \equiv \frac{1}{n} \sum_{i=1}^n m(w_i, \widehat{\theta}^{(1)}) m(w_i, \widehat{\theta}^{(1)})'.$$

3. Set the second-step weight

$$\widehat{W}^{(2)} \equiv \widehat{\Lambda}^{-1},$$

and compute $\widehat{\theta}^{(2)}$ by minimizing

$$Q_n^{(2)}(t) = m_n(t)' \widehat{W}^{(2)} m_n(t).$$

Optimal GMM and the Overidentifying Restrictions (J) Test

91.1 Setup

Assume

$$w_i \text{ i.i.d.}, \quad i = 1, 2, \dots, n,$$

and let

$$\theta \in \Theta \subset \mathbb{R}^k$$

be the parameter of interest.

Let the moment function be

$$m(w_i, \theta) \in \mathbb{R}^{L \times 1}.$$

Population moment condition.

$$\mathbb{E}[m(w_i, \theta)] = 0.$$

Sample analogue. For any $t \in \Theta$, define

$$m_n(t) \equiv \frac{1}{n} \sum_{i=1}^n m(w_i, t).$$

92 GMM objective and estimator

Let $\widehat{W}$ be an $L \times L$ symmetric positive definite weighting matrix. Define the sample objective

$$Q_n(t) \equiv m_n(t)' \widehat{W} m_n(t).$$

The GMM estimator is

$$\widehat{\theta} \equiv \arg \min_{t \in \Theta} Q_n(t) = \arg \min_{t \in \Theta} m_n(t)' \widehat{W} m_n(t).$$

93 Optimal GMM weight

Define the (population) second moment matrix of the moment function at the true parameter θ :

$$\Lambda \equiv \mathbb{E}[m(w_i, \theta) m(w_i, \theta)'] \in \mathbb{R}^{L \times L}.$$

The optimal GMM weight (in the sense of minimizing the asymptotic variance of $\widehat{\theta}$ among GMM estimators based on the same moments) is

$$W^* = \Lambda^{-1}.$$

Accordingly, an “optimal GMM” implementation uses a weighting matrix satisfying

$$\widehat{W} \xrightarrow{p} \Lambda^{-1} = (\mathbb{E}[m(w_i, \theta) m(w_i, \theta)'])^{-1}.$$

94 Hypotheses for the overidentifying restrictions test

When $L > k$, there are more moment conditions than parameters. The overidentifying restrictions test is:

$$\begin{aligned} H_0 : \mathbb{E}[m(w_i, \theta)] &= 0 & (\text{all moment conditions are correct}) \\ H_1 : \mathbb{E}[m(w_i, \theta)] &\neq 0 & (\text{at least one moment condition fails}). \end{aligned}$$

95 CLT for the sample moments

Assume the central limit theorem applies at the true parameter θ :

$$\sqrt{n} m_n(\theta) \xrightarrow{d} N(0, \Lambda).$$

96 The J statistic and its asymptotic distribution

Rewriting $nQ_n(\hat{\theta})$

By definition,

$$nQ_n(\hat{\theta}) = n m_n(\hat{\theta})' \widehat{W} m_n(\hat{\theta}).$$

Rewrite n as $\sqrt{n} \cdot \sqrt{n}$ and regroup:

$$nQ_n(\hat{\theta}) = (\sqrt{n} m_n(\hat{\theta}))' \widehat{W} (\sqrt{n} m_n(\hat{\theta})).$$

Linearization of $m_n(\hat{\theta})$

Let

$$G(\theta) \equiv \mathbb{E} \left[\frac{\partial}{\partial \theta'} m(w_i, \theta) \right] \in \mathbb{R}^{L \times k}, \quad G \equiv G(\theta), \quad \text{rank}(G) = k.$$

A mean-value expansion (componentwise) gives: there exists $\tilde{\theta}$ between $\hat{\theta}$ and θ such that

$$m_n(\hat{\theta}) = m_n(\theta) + \frac{\partial}{\partial t'} m_n(\tilde{\theta}) (\hat{\theta} - \theta).$$

Under standard regularity conditions (uniform convergence of $\partial m_n / \partial t'$ to $G(t)$ and $\hat{\theta} \xrightarrow{p} \theta$),

$$\frac{\partial}{\partial t'} m_n(\tilde{\theta}) \xrightarrow{p} G,$$

so in $\sqrt{n}$ -scale,

$$\sqrt{n} m_n(\hat{\theta}) = \sqrt{n} m_n(\theta) + G \sqrt{n} (\hat{\theta} - \theta) + o_p(1). \quad (1)$$

FOC representation and the “moment residual-maker”

The first-order condition for minimizing $Q_n(t) = m_n(t)' \widehat{W} m_n(t)$ is

$$\frac{\partial}{\partial t} Q_n(t) = 2 \left(\frac{\partial}{\partial t'} m_n(t) \right)' \widehat{W} m_n(t) = 0 \quad \text{at } t = \hat{\theta}.$$

Thus,

$$\left(\frac{\partial}{\partial t'} m_n(\hat{\theta}) \right)' \widehat{W} m_n(\hat{\theta}) = 0.$$

Replacing $\frac{\partial}{\partial t'} m_n(\hat{\theta})$ by its probability limit G and $\widehat{W}$ by its probability limit W , the limiting FOC is

$$G' W m_n(\hat{\theta}) \approx 0.$$

Using the standard asymptotic linear representation,

$$\sqrt{n}(\hat{\theta} - \theta) = -(G' W G)^{-1} G' W \sqrt{n} m_n(\theta) + o_p(1). \quad (2)$$

Substitute (2) into (1):

$$\begin{aligned} \sqrt{n} m_n(\hat{\theta}) &= \sqrt{n} m_n(\theta) + G [-(G' W G)^{-1} G' W \sqrt{n} m_n(\theta)] + o_p(1) \\ &= [I_L - G(G' W G)^{-1} G' W] \sqrt{n} m_n(\theta) + o_p(1). \end{aligned}$$

Define

$$M_W \equiv I_L - G(G'WG)^{-1}G'W.$$

Then

$$\sqrt{n} m_n(\hat{\theta}) = M_W \sqrt{n} m_n(\theta) + o_p(1). \quad (3)$$

Specialize to the optimal weight $W = \Lambda^{-1}$

Set $W = \Lambda^{-1}$ and define

$$\tilde{G} \equiv \Lambda^{-1/2}G \quad \Rightarrow \quad \tilde{G}'\tilde{G} = G'\Lambda^{-1}G.$$

Let

$$P_{\tilde{G}} \equiv \tilde{G}(\tilde{G}'\tilde{G})^{-1}\tilde{G}'$$

be the projection matrix onto $\text{col}(\tilde{G})$, and define

$$M_{\tilde{G}} \equiv I_L - P_{\tilde{G}}.$$

One can verify

$$\Lambda^{-1/2}M_W\Lambda^{1/2} = M_{\tilde{G}}.$$

(Indeed, using $\tilde{G} = \Lambda^{-1/2}G$ and $\tilde{G}'\tilde{G} = G'\Lambda^{-1}G$,

$$\Lambda^{-1/2}G(G'\Lambda^{-1}G)^{-1}G'\Lambda^{-1}\Lambda^{1/2} = \tilde{G}(\tilde{G}'\tilde{G})^{-1}\tilde{G}' = P_{\tilde{G}}.$$

)

Now define

$$U_n \equiv \Lambda^{-1/2}\sqrt{n} m_n(\theta).$$

Since $\sqrt{n} m_n(\theta) \xrightarrow{d} N(0, \Lambda)$, we have

$$U_n \xrightarrow{d} N(0, I_L).$$

Using (3) and $\Lambda^{-1/2}M_W\Lambda^{1/2} = M_{\tilde{G}}$:

$$\Lambda^{-1/2}\sqrt{n} m_n(\hat{\theta}) = \Lambda^{-1/2}M_W\sqrt{n} m_n(\theta) + o_p(1) = M_{\tilde{G}}U_n + o_p(1). \quad (4)$$

Limit distribution of the J statistic

Under $W = \Lambda^{-1}$ and $\widehat{W} \xrightarrow{p} \Lambda^{-1}$,

$$nQ_n(\hat{\theta}) = (\sqrt{n} m_n(\hat{\theta}))' \widehat{W} (\sqrt{n} m_n(\hat{\theta})) = (\Lambda^{-1/2}\sqrt{n} m_n(\hat{\theta}))' (\Lambda^{-1/2}\sqrt{n} m_n(\hat{\theta})) + o_p(1).$$

Using (4) and $M'_{\tilde{G}} = M_{\tilde{G}}$, $M_{\tilde{G}}^2 = M_{\tilde{G}}$,

$$nQ_n(\hat{\theta}) = (U'_n M'_{\tilde{G}} M_{\tilde{G}} U_n) + o_p(1) = U'_n M_{\tilde{G}} U_n + o_p(1).$$

Let $U \sim N(0, I_L)$ be the limit of U_n . Then

$$U' M_{\tilde{G}} U \sim \chi^2(\text{rank}(M_{\tilde{G}})).$$

Because $\text{rank}(P_{\tilde{G}}) = \text{rank}(\tilde{G}) = k$, we have

$$\text{rank}(M_{\tilde{G}}) = \text{rank}(I_L - P_{\tilde{G}}) = L - k.$$

Therefore,

$$\boxed{nQ_n(\hat{\theta}) \xrightarrow{d} \chi^2(L - k).}$$

Likelihood, Log-Likelihood, and the Maximum Likelihood Estimator

Sampling model

Suppose $Y_1, \dots, Y_n$ are i.i.d. with probability density function $f(y | \theta)$.

Let $(y_1, \dots, y_n)$ denote a realized sample from $(Y_1, \dots, Y_n)$.

Joint density

The joint density of $(Y_1, \dots, Y_n)$ for a given value of θ is

$$f(y_1, \dots, y_n | \theta) = \prod_{i=1}^n f(y_i | \theta).$$

This factorization follows from independence:

$$f(y_1, \dots, y_n | \theta) = f(y_1 | \theta) f(y_2 | \theta) \cdots f(y_n | \theta) = \prod_{i=1}^n f(y_i | \theta).$$

Likelihood function

We can view the joint density as a function of θ given $(Y_1, \dots, Y_n)$, called the likelihood function:

$$L(\theta | Y_1, \dots, Y_n) = \prod_{i=1}^n f(Y_i | \theta).$$

Equivalently, for the realized data $(y_1, \dots, y_n)$,

$$L(\theta | y_1, \dots, y_n) = \prod_{i=1}^n f(y_i | \theta).$$

Maximum likelihood estimator (MLE)

The maximum likelihood estimator (MLE) is the value of θ that maximizes $L(\theta | y_1, \dots, y_n)$:

$$\hat{\theta}_{\text{MLE}} \in_{\theta} L(\theta | y_1, \dots, y_n) =_{\theta} \prod_{i=1}^n f(y_i | \theta).$$

(If the maximizer is unique, we can write $\hat{\theta}_{\text{MLE}} =_{\theta} L(\theta | y_1, \dots, y_n)$.)

Log-likelihood

Define the log-likelihood function:

$$\log L(\theta \mid Y_1, \dots, Y_n) = \sum_{i=1}^n \log f(Y_i \mid \theta).$$

This comes from applying $\log(\cdot)$ to the product form:

$$\begin{aligned} \log L(\theta \mid y_1, \dots, y_n) &= \log \left(\prod_{i=1}^n f(y_i \mid \theta) \right) \\ &= \log (f(y_1 \mid \theta) \cdots f(y_n \mid \theta)) \\ &= \log f(y_1 \mid \theta) + \cdots + \log f(y_n \mid \theta) \\ &= \sum_{i=1}^n \log f(y_i \mid \theta). \end{aligned}$$

Equivalence of maximizing likelihood and log-likelihood

Because $\log(\cdot)$ is strictly increasing, maximizing $L(\theta \mid y_1, \dots, y_n)$ is equivalent to maximizing $\log L(\theta \mid y_1, \dots, y_n)$. Therefore,

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \log L(\theta \mid Y_1, \dots, Y_n).$$